

Merged research report for the Surreal project

Polish Models of Omnific Arithmetic

Polish Presburger models for Glazer's Question 2, Borel presentations
and support barriers, continuous Presburger arithmetic, local
compactness, and Polish series fields

Built from seven manuscripts dated October 3 and 4, 2026

Abstract

Elliot Glazer's *A Topological Tennenbaum Theorem* shows that an uncountable Polish model of $Q + IOpen$ cannot have continuous addition and Borel multiplication, and asks (Question 2) whether some uncountable Polish space supports a model of Presburger arithmetic with continuous addition. Part I answers that question, as printed, affirmatively. The nonnegative cone $M_{\mathbb{R}} = (\mathbb{R}_{\geq 0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N})$ of the lexicographic group $\mathbb{R} \vec{\times} \mathbb{Z}$, with the usual topology on the real coordinate and the discrete one on the integer coordinate, is a perfect, locally compact, uncountable Polish model of the full theory $\text{Th}(\mathbb{N}; 0, 1, +, <)$ whose addition is continuous. Replacing $\mathbb{R}$ by the lexicographically ordered group $\mathbb{Q}^{\mathbb{N}}$ with discrete coordinates gives a model homeomorphic to Baire space, with Type-2 computable addition and division by standard integers. Every definable relation is Δ_2^0 , and this is optimal for the order; the truncated subtraction and the predecessor are discontinuous exactly on explicit loci; no unital semiring multiplication exists on either model; the self-embeddings are classified (positive dilations in the real case, row-finite matrices with increasing positive pivots in the Baire case), and both models embed additively in Conway's omnific integers $\mathbf{Oz}$ as $r\omega + n$ and $\sum_j a_j \omega^{1/(j+1)} + n$. Part I also isolates the coset-invariance hypothesis needed to carry the repository's integer Cooper lemmas to nonstandard groups. Glazer himself speculated that the answer to Question 2 would be negative; the affirmative answer is given for the question as printed, and its priority has not been established.

Part II is the other side of the same boundary. For a countable linear order L , its well-ordered subsets form a Borel set exactly when L is scattered, and a Π_1^1 -complete set otherwise; the same condition decides whether a full real Hahn field $\mathbb{R}((t^\Gamma))$ has a standard Borel parametrization with Borel coefficient maps. The finite Puiseux principal-part ring $\mathcal{A}_{\text{fin}}$, of which $M_{\mathbb{R}}$ is one stratum, satisfies open induction and has a Polish presentation on which addition and multiplication are continuous off explicit closed nowhere dense cancellation loci; by Glazer's Corollary 1 no Borel-compatible Polish recoding makes its addition continuous everywhere.

Part III merges two further manuscripts on continuous Presburger arithmetic. A Borel invariant order on a Polish abelian group has Δ_2^0 cones, so every divisible Polish group with Borel positive cone yields a Polish model; the cones $(\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$, α countably infinite, give $\aleph_1$ pairwise nonisomorphic models on Baire space, other families give exactly $2^{\aleph_0}$, and one order formula has unbounded oscillation ranks. The cone of $\mathbb{Z} + t\mathbb{R}[t]$ (t infinite) carries a locally compact Polish topology with continuous addition and multiplication and fails open induction at $x^2 < t$; and in every Polish topological cone of a discretely ordered ring with both operations continuous, 0 is isolated, which is Glazer's Theorem 1 with PA weakened to PA^- . Part IV shows that a discretely ordered ring whose additive group is Hausdorff, locally compact and second countable with Borel order is countable and discrete, with no regularity assumed of multiplication, and classifies the locally compact Borel $\mathbb{Z}$ -groups as $\mathbb{R}^d \times E$. Part V shows that a full Hahn field $k((t^\Gamma))$ admits a Polish additive topology with Baire order and Borel scalar maps exactly when k is countable and $\Gamma \cong \mathbb{Z}$; the Puiseux field over a countable field admits none, its Levi-Civita completion exactly one; and every integer part of such a field is countable and closed discrete.

The report merges seven manuscripts written independently on these subjects; it states which source proved each result, prints shared results once or as marked second routes, keeps every source's limitations, and cites the collection's earlier results instead of reprinting them as new. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized.

Sources: manuscripts 05 (the base text of Part I), 04 (Part I, marked [04]) and 03 (Part II) of batch 86, arrival commit 31fdc6571, placement commit 3d2177df4; pinned to repository commits a11efab09 (05), 3ad5f878c (04) and c5b4881fa (03). Manuscripts 07 (the base text of Part III), 06 (Part III, marked [06]), 08 (Part IV) and 09 (Part V) of batch 86, arrival commit fb8414869, placement commit 0bd0e5527; pinned to 0f95145cb (07), fa2f3e419 (06), 883e0b3b2 (08) and 4af956fe0 (09). Their titles and contributions are in [section 1.2](#); the provenance of every section is in [section E](#), and the non-claims of every source are collected in [section F](#).

Contents

1	Scope, sources and conventions	7
1.1	Glazer’s paper and its two questions	7
1.2	Seven manuscripts, one report	9
1.3	Notation	11
1.4	Relation to the repository	13
1.5	The status statements of the sources	14
I	Polish Presburger models	15
2	The research overlap and the precise target	15
2.1	Repository interfaces actually inspected	16
2.2	What is established, and what is not	16
2.3	Source 04’s framing	16
3	The construction in one theorem	17
4	Full Presburger arithmetic, not merely a few axioms	18
4.1	Discretely ordered groups with standard residues	18
4.2	Coset invariance versus standard-shift periodicity	20
4.3	Second route [04]: finite witnesses at a boundary	21
4.4	A self-contained elimination argument	22
4.5	Second route [04]: normalization and elimination	23
5	Polishness and a locally compact example	24
5.1	A useful exact criterion	25
5.2	Admissible coefficient groups and continuous division [04]	25
5.3	The real-coordinate model, including a complete metric	26
5.4	The real-coordinate model, source 04’s account [04]	27
6	An explicit Baire-space realization	27
6.1	The zero branch and its explicit tree code	28
6.2	A total code for every sequence of natural numbers	29
6.3	Second route [04]: explicit charts	29
7	Definability, effective truth, and the optimal order rank	32
7.1	A formula-by-formula bound on changes of mind	33
7.2	Why rank one is impossible for the order	34
7.3	A general obstruction from closed order [04]	34
8	Which definable operations are continuous?	35
8.1	Polish completeness need not be translation-invariant [04]	37
9	Explicit set-sized fragments of the omnific integers	38
9.1	What this realization does and does not preserve	38
9.2	Why no proper-class difficulty arises	39
10	An absolute obstruction to multiplication	40
10.1	A variation with no largest growth class [04]	41
11	Endomorphisms, elementary submodels, and a contrast	43
11.1	The real model: endomorphisms and elementary submodels	43
11.2	Reduction to the divisible part [04]	44
11.3	The Baire model: all self-embeddings [04] and clopen self-copies	45
12	A compactness obstruction independent of multiplication	47

13 Formalization and reproducibility	48
13.1 A minimal theorem dependency graph	48
13.2 The nonstandard Cooper boundary must be explicit	48
13.3 Executable material included with this manuscript	49
13.4 The formalization interface, source 04's audit [04]	50
13.5 Relation to the formal project <code>Logic/PresburgerArithmetic [write]</code>	51
14 Further research questions	52
14.1 Questions asked by both sources	52
14.2 Further questions of source 05	53
14.3 Further questions of source 04	54
15 Conclusions of Part I	55
 II Borel presentations and support barriers	 55
16 The intersection and the questions answered	56
16.1 Main theorem map	56
16.2 A precise source boundary	57
17 Conventions and the meaning of a presentation	57
17.1 Two opposite exponent conventions	57
17.2 Borel spaces, completeness, and observability	58
18 The exact support dichotomy	58
18.1 A complete obstruction on a dense countable alphabet	58
18.2 Why scattered alphabets are different	59
19 Exact benchmarks: integers, finite lexicographic powers, and rationals	60
20 An observable-presentation dichotomy for full Hahn fields	61
20.1 Root closure forces the non-Borel side for full fields	63
21 Every observable family has a countable support bound	63
22 The obstruction already lies in one omnific interval	64
23 A real closed Borel subfield and its finite omnific integer part	66
23.1 Why the Puiseux field itself has Borel codes	67
24 A Polish finite-normal-form presentation	68
24.1 The topology and its Borel structure	68
24.2 A general support-normalization lemma	68
24.3 The exact arithmetic cancellation loci	69
25 Glazer's obstruction survives every compatible Polish recoding	70
26 Where the two parts meet [merge]	71
27 Consequences for exact notation systems and formalization	72
27.1 A practical separation of three design goals	72
27.2 No paradoxes are needed	73
27.3 A formalization plan with an explicit dependency boundary	73
28 Further research questions and concrete next targets	74
29 Conclusions of Part II	75

III Continuous Presburger arithmetic on Polish spaces	75
30 The question and the research intersection	77
30.1 An exact question with a concrete answer	77
30.2 Why this is a substantive overlap with ProveIt	78
30.3 What is proved, and what is a novelty claim	78
30.4 The question, the overlap, and the contribution [06]	78
31 Conventions and classical inputs	80
31.1 Language, order, and induction	80
31.2 Topological and descriptive notation	80
31.3 Algebraic and topological preliminaries [06]	81
32 The real-coordinate model	82
32.1 A complete metric, rather than an incomplete inherited metric	83
32.2 Division and the meaning of nonstandard induction	83
32.3 A perfect locally compact model [06]	84
33 Borel orders on Polish abelian groups	85
33.1 A countable ordinal invariant of the order and topology	86
34 A general Polish realization theorem	87
34.1 Definable Borelness for arbitrary continuous presentations	88
35 A family of models on Baire space	89
35.1 The countable-ordinal coefficient groups	89
35.2 Residue classes as copies of the whole space	90
35.3 An uncountable ordinal spectrum of models	90
35.4 The optimal number of models is the continuum	91
35.5 An explicit model on Baire space [06]	92
35.6 How many continuous models fit on Baire space? [06]	93
36 Unbounded oscillation at Borel level two	96
36.1 The derivative and its exact convention	96
36.2 Explicit continuous approximations	97
37 What continuity permits and forbids	98
37.1 The arithmetic order is not the Polish topology	98
37.2 Compactness remains impossible	99
37.3 A second topology makes predecessor continuous	100
37.4 The explicit models have no ordered-semiring expansion	100
37.5 Definable regularity and an algebraic obstruction [06]	101
38 A Polish semiring with full additive induction [06]	103
38.1 The ring and its order [06]	103
38.2 The topology by degree and constant coefficient [06]	103
38.3 An exact failed induction instance [06]	104
39 Zero must be isolated when both operations are continuous [06]	105
39.1 Borel order from continuous addition [06]	105
39.2 A uniformly convergent subset-sum map [06]	105
39.3 Bounded complementary differences have probability zero [06]	106
39.4 The sign is a tail event [06]	107
40 Surreal and omnific realizations	107
40.1 The real model as an omnific subgroup	107
40.2 Normal forms for the entire ordinal family	108
40.3 Set size and formal foundations	109
40.4 Surreal realizations and a formalization route [06]	109

41 Formalization scope and a concrete development plan	110
41.1 What the existing repository actually proves	110
41.2 A small first formalization target	111
41.3 Extending the verified boundary	111
42 Where Part III meets Parts I and II [merge]	112
43 Further research questions and topics	113
43.1 Published reverse-mathematical question	113
43.2 Classification and topology	114
43.3 Definability and descriptive invariants	115
43.4 Arithmetic expansions and formal verification	115
43.5 Further projects of source 06 [06]	116
IV Local compactness forces arithmetic rigidity	117
44 The intersection and the contribution	118
44.1 The central theorem	118
44.2 Other results proved here	119
44.3 What is classical, and what remains uncertified	119
45 Conventions, topology, and foundations	119
46 An algebraic amplification theorem	120
47 Borel orders on locally compact additive groups	121
47.1 A compact subgroup cannot carry the restricted order	121
47.2 The connected component is an open real vector group	122
47.3 A Borel order on the real component is a linear flag	122
48 Proof and scope of arithmetic rigidity	123
49 Signed Presburger arithmetic and a uniform elimination lemma	124
50 An exact pointed topological-group classification	126
51 Positive cones: exact regularity and the dimension threshold	127
51.1 The order bound is genuinely rank two	128
52 Definable relations and continuous standard division	128
52.1 A model in every finite real dimension	129
52.2 Why ordinary scalar operations are not ring multiplication	129
53 Even unordered multiplication has few Borel scalars	130
53.1 Automatic continuity and component rigidity	130
53.2 The Borel-scalar subring	130
54 Examples isolating the hypotheses	131
54.1 The countable conclusion is optimal	131
54.2 Second countability cannot simply be deleted	131
54.3 Borelness of the order is also essential	132
54.4 Other boundaries	132
55 Consequences for surreal and omnific presentations	133
55.1 A set-sized no-go theorem for actual omnific rings	133
55.2 Finite additive fragments have excellent topology but no ring expansion	133
55.3 Implications for exact symbolic systems	134

56 A formalization plan with explicit trust boundaries	134
57 Further research questions and concrete next targets	135
58 Conclusions of Part IV	137
V Polish topologies at the Hahn–Puiseux–Levi-Civita boundary	137
59 The research intersection and the questions answered	137
59.1 The central conclusions	138
59.2 What is and is not claimed as new	139
60 Conventions: four different notions of presentation	139
61 Category forces local order bounds	140
62 Discretely ordered rings and integer parts	142
63 A complete classification for full Hahn fields	143
64 The Puiseux obstruction: a Borel field that is not Polishable	144
65 The Levi-Civita completion: a positive and unique presentation	145
66 A prefix criterion for separability	148
67 Exact descriptive complexity of the three code domains	148
68 Real closedness and completion without a defect argument	150
69 Local Polish hulls, unbounded support ranks, and an exact cover number	152
69.1 Real-closed, truncation-closed hulls	153
69.2 An exact global invariant	154
70 A countable omnific integer part and its exact floor regularity	154
70.1 Floors in the larger Polish hulls	156
70.2 The common integer part satisfies open induction	157
70.3 Why the finite integer part does not extend to the full Hahn field	158
71 Surreal realization and a formalization interface	158
71.1 A set-sized embedding, not a topology on a proper class	158
71.2 A modular Lean development plan	158
72 Consequences for exact symbolic arithmetic	159
73 Further research questions and concrete targets	160
74 Conclusions of Part V	162
A Worked examples and an audit of the hypotheses (source 05)	162
A.1 A negative constant term is compatible with positivity	162
A.2 A nonstandard finite-candidate calculation	163
A.3 A Baire convergence that order cannot detect continuously	163
A.4 Hypothesis checklist	163
B Reproduction and the scope of the executable checks (source 04)	164
C A compact proof-dependency map (source 04)	165

D	Source audit, contribution ledger, and reproducibility (source 03)	166
D.1	Repository material inspected	166
D.2	Contribution ledger	166
D.3	Finite verification artifact	167
E	Provenance of this report	167
F	Limitations and non-claims of the sources	171
G	An explicit induction check after quantifier elimination (source 07)	175
H	Source audit, proof dependencies, and scope (source 07)	176
H.1	Primary source checks	176
H.2	Repository inspection and duplication boundary	176
H.3	Dependency structure of the new arguments	177
H.4	Review status	177
I	A self-contained Presburger elimination argument (source 06)	178
I.1	Division in a $\mathbb{Z}$ -group	178
I.2	Eliminating one quantified variable	178
I.3	Completeness and induction in the nonnegative cone	179
J	Provenance and verification record (source 06)	179
K	Contribution ledger, source audit, and reproducibility (source 08)	180
K.1	Contribution ledger	180
K.2	What was actually inspected	181
K.3	Finite verification	181
K.4	Build instructions	182
L	Contribution ledger and proof dependencies (source 09)	182
M	Source audit and reproducibility (source 09)	183

1 Scope, sources and conventions

This report collects seven manuscripts, dated October 3 and 4, 2026, on one subject: where Elliot Glazer’s topological Tennenbaum programme meets the repository’s omnific arithmetic. Their common spine is the question which arithmetic operations of a Polish or standard Borel presentation of a substructure of $\mathbf{Oz}$ can be continuous. Three of them (03, 04, 05) were written first; the four later ones (06, 07, 08, 09) arrived together, and 08 and 09 cite 05 and 03 as earlier companion manuscripts. [Part I](#) (label prefix `pma:pr:`) constructs Polish models of Presburger arithmetic with continuous addition and studies them; its base text is manuscript 05, and the further results of manuscript 04 are printed as marked sections. [Part II](#) (label prefix `pma:bor:`) is manuscript 03: Borel presentations of Hahn fields, support barriers, and the finite principal-part ring on which Glazer’s theorem forbids continuous addition. Parts I and II meet in [section 26](#). [Part III](#) (label prefix `pma:cp:`) merges manuscripts 07 (base) and 06 (marked [06]) on continuous Presburger arithmetic: Borel orders on Polish groups, a general realization theorem, ordinal and continuum families on Baire space, a Polish semiring cone, and the isolated-zero theorem; it meets Parts I and II in [section 42](#). [Part IV](#) (prefix `pma:lc:`) is manuscript 08: local compactness forces arithmetic rigidity. [Part V](#) (prefix `pma:ser:`) is manuscript 09, which continues Part II: Polish topologies on Hahn, Puiseux and Levi-Civita fields and their integer parts.

1.1 Glazer’s paper and its two questions

Glazer’s paper [1] is an eight-page preprint, dated November 20, 2023 on its first page and submitted to arXiv on November 22, 2023 as arXiv:2311.13699. On October 3, 2026 the arXiv record listed one version (v1) and no journal reference. It answers negatively a question of Enayat, Hamkins and Wcisło [2]: no model of Peano arithmetic on a perfect Polish space has continuous addition and multiplication (its Theorem 1). Its Theorem 2 says that if $(M, +, \cdot) \models Q + I\Delta_0 + \text{Exp}$ lives on an uncountable Polish space with Borel addition and multiplication, then neither operation is continuous. Glazer notes that the axiom Exp is used only to obtain the function $x \mapsto 2^x$, and only for the discontinuity of multiplication; this gives his Corollary 1: *there is no model of $Q + \text{IOpen}$ on an uncountable Polish space in which addition is continuous and multiplication is Borel*. The paper ends with two questions. Question 1 asks whether Theorem 2 is provable in ATR_0 . Question 2 is quoted in [section 2](#): does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

Glazer’s speculation. Immediately after Question 2, Glazer writes that he speculates that an affirmative answer to Question 1 “would lead to a negative answer to Question 2” [1, p. 8]. His reason is an analogy: removing the exponential function from the proof that multiplication is discontinuous would parallel removing multiplication from a proof that addition is discontinuous. Neither source of [Part I](#) mentions this sentence; it is recorded here because [Part I](#) goes the other way.

How this report reads it [write]. [Part I](#) answers Question 2 *as printed* affirmatively ([theorem 3.1](#)). The question asks only for an uncountable Polish space, a model of Presburger arithmetic and continuous addition; it does not ask that the topology be the order topology, and it mentions no multiplication. The speculation is a heuristic about proofs, not a stated implication, and nothing here bears on Question 1, which stays open; no conclusion about Question 1 is drawn from the affirmative answer. Three facts locate the difference. First, the

affirmative models are not an artefact of a missing definability condition: every definable relation in them is Δ_2^0 ([theorem 7.1](#)), so the order and every other definable relation are Borel, as in the Borel models of Glazer’s theorems. Second, what fails in them is multiplication: no unital semiring multiplication exists on them at all ([theorem 10.4](#)), so Corollary 1 does not apply. Third, on a ring that does carry Borel multiplication and open induction, Corollary 1 does forbid continuous addition under every Borel-compatible Polish recoding ([theorem 25.2](#)), and [section 26](#) shows that the real model of [Part I](#) sits inside that ring. We do not know whether Glazer intended Question 2 with a further hypothesis, for instance an expansion to a model of a stronger theory, under which his speculation could hold; that would be a different question.

Parts III–V and the speculation [write]. None of sources 06–09 quotes the sentence either. Source 07 comes closest: after restating Question 1 it observes that, because the affirmative Presburger construction stands, a route from a weaker metatheory to a *negative* Presburger conclusion cannot yield that conclusion as stated ([section 43](#)). Read literally, the outcome the speculation predicts is therefore excluded for Question 2 as printed, whatever the answer to Question 1; only a variant of Question 2 with further hypotheses could carry it. Source 06’s isolated-zero theorem ([theorem 39.1](#)) removes a hypothesis from one of Glazer’s proofs, in the spirit of his analogy, but from his Theorem 1: with Glazer’s subset-sum and zero-one method, and without his injectivity claim, it weakens Peano arithmetic to PA^- in “no perfect Polish space carries a model with continuous addition and multiplication” ([corollary 39.2](#)). It does not bear on Question 1, which concerns his Theorem 2; Glazer himself remarks [[1](#), p. 7] that Theorem 1 and Corollary 1 can be proven in ATR_0 , and source 06’s project 6 asks for the strength of the isolated-zero principle. Sources 08 and 09 concern rings, where multiplication is present, and both state that they answer neither of Glazer’s questions. Priority of source 06’s strengthening is not established.

Priority. Priority of the affirmative answer is **not established**. Source 05 searched for the exact question and combinations of its terms and found no earlier answer, and says that this does not prove that none exists; source 04 says that historical priority has not been established. The intake re-checked only that the arXiv record has one version and no journal reference (October 3, 2026). The answer is an elementary construction; an earlier publication of it, or a published variant of the question, may exist.

1.2 Seven manuscripts, one report

Source	Manuscript (delivered pages)	Pin	Printed as
05	<i>Polish Presburger Arithmetic inside the Omnific Integers: An affirmative construction for Glazer’s Question 2, effective Baire-space models, and structural obstructions</i> (24, A4); archive <code>polish_presburger_glazer</code>	<code>a11efab09</code>	base text of Part I ; worked examples in section A
04	<i>Polish Presburger Arithmetic: A positive construction for Glazer’s question, an effective Baire-space model, and exact regularity boundaries</i> (25, letter); archive <code>Polish_Presburger_Glazer_ProveIt</code>	<code>3ad5f878c</code>	Part I , every statement marked [04]; second proofs; sections B and C
03	<i>Borel Presentations and Support Barriers in Omnific Arithmetic: Scattered exponent orders, Hahn fields, and the exact discontinuities of finite normal forms</i> (23, letter); archive <code>glazer_proveit_research</code>	<code>c5b4881fa</code>	Part II and section D
07	<i>Continuous Presburger Arithmetic on Polish Spaces: An affirmative construction for Glazer’s question, Baire-space models, and omnific realizations</i> (27, letter); archive <code>polish_presburger_research</code>	<code>0f95145cb</code>	base text of Part III ; sections G and H
06	<i>Continuous Presburger Arithmetic on Polish Spaces: Lexicographic models, an isolated-zero obstruction, and a boundary between additive and semiring arithmetic</i> (21, letter); archive <code>polish_presburger_article</code>	<code>fa2f3e419</code>	Part III , every passage marked [06]; sections I and J
08	<i>Local Compactness Forces Arithmetic Rigidity: Borel ordered groups, Presburger models, and multiplication without regularity assumptions</i> (23, A4); archive <code>glazer_proveit_local_compactness</code>	<code>883e0b3b2</code>	Part IV and section K
09	<i>Polishability at the Hahn–Puisseux–Levi–Civita Boundary: Automatic order bounds, local Polish hulls, and countable omnific integer parts</i> (30, letter); archive <code>Polish_Hahn_Puisseux_Levi_Civita_Research</code>	<code>4af956fe0</code>	Part V and sections L and M

The numbers are the manuscript numbers of batch 86 of the repository’s intake (manuscripts 01 and 02 of that batch are unrelated OEIS reports; manuscript 10, on the standard cut in one infinite interval, was written into the report `Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization`). Manuscripts 03–05 arrived in commit `31fdc6571` and were placed in `3d2177df4`; manuscripts 06–09 arrived in `fb8414869` and were placed in `0bd0e5527`. Every pin is an ancestor of its placement commit, and the repository files the sources inspected are unchanged between their pins and the placements.

Why one report. 04 and 05 prove the same main theorem independently: they share 0.6% of their word 8-grams, essentially only Glazer’s quoted question and his bibliography entry, and have different pins, notation and supplementary theorems. 03 answers the same brief from the other side: its finite principal-part ring contains the real model of 04 and 05 as one stratum (section 26), and its Theorem 10.2 (theorem 25.2) is Glazer’s obstruction on that ring. No earlier report of the collection names Glazer, his questions, Polish models of Presburger arithmetic, or Borel presentations of Hahn fields.

Why Parts III–V are here. Sources 06 and 07 answer Glazer’s Question 2 again and go on from Part I’s models; neither is an edition of the other or of 04 and 05 (at most 0.74% shared word 8-grams among manuscripts 03–10). Source 08 names 05 and 03 as its prior work and asks what local compactness forces on every Borel ordered presentation; source 09 names them too and asks the Polish-group question that follows 03’s Borel classification. No other report of the collection treats Polish topologies on Puiseux or Levi-Civita fields. Source 07 is the base of Part III because its realization theorem has the weakest hypotheses (Part III).

Why base 05. On the shared theorems 05 has the weaker hypotheses: its Polish criterion (theorem 5.2) is an equivalence for every Polish group topology on the coefficient group, where 04’s transfer theorem (theorem 5.4) is a sufficient condition under the stronger hypothesis that the positive cone be open and division continuous; its Δ_2^0 theorem holds under the same weaker hypothesis; and it treats the models explicitly as fragments of **Oz**. Results of 04 not in 05 are printed in full and marked [04]: continuity of fixed division under its hypothesis, the general closed-order obstruction and the invariant-metric obstruction, the reduction of all unital additive maps to the divisible part, the matrix classification of the self-embeddings of the Baire model with automatic continuity, and the Laurent-series model with its growth-order obstruction. Where 04 proves a result of 05 by a different argument, its proof is printed as a marked second proof.

Reading conventions. In Part I, unmarked text is source 05’s; text from source 04 is marked [04] (statements in their titles, inserted passages in the sentence that introduces them). In Part II all unmarked text is source 03’s; in Part III unmarked text is source 07’s and text from source 06 is marked [06]; in Parts IV and V all unmarked text is that of source 08 and source 09 respectively. Text written for this report is marked [merge] (a mathematical observation combining the sources, with proof) or [write] (editorial notes: provenance, citations of the collection, corrections). Phrases such as “this manuscript” or “this article” inside a source’s text refer to that source. References to the collection’s other reports give their labels in typewriter type, for example `odg:thm:floor`; these labels are not hyperlinks.

Labels. Every label carries the prefix `pma:pr:`. Source 05’s labels are `pma:pr:<label>`, source 03’s are `pma:bor:<label>`. Source 04’s labels are `pma:pr:<label>`, except the ten that coincide with labels of 05 (`eq:cone`, `eq:division`, `sec:algebra`, `sec:baire`, `sec:embeddings`, `sec:formalization`, `sec:multiplication`, `sec:questions`, `thm:baire`, `thm:qe`), which take the sub-prefix `pma:pr:s04:.` Of these, `eq:cone` and `eq:division` denote displays identical to 05’s and are printed once, under 05’s labels; source 04’s displays `eq:divisionformula`, `eq:monuslocus`, `eq:surrealR` and `eq:surrealQ` are likewise printed once as 05’s (4.2), (8.2), (9.1) and (9.2), and its Definition 2.1 (`def:split`) is 05’s setup in section 3. Where 04 states a theorem that 05 also states, 04’s label is a second label of 05’s statement (for example `pma:pr:thm:monuslocus` is theorem 8.3). Labels written for the merge have the bare prefix

pma: or a part prefix. Sources 07 and 06 have labels pma:cp:(label), except six labels of 06 that coincide with labels of 07, which take pma:cp:s06: (Part III); source 08 has pma:lc:, source 09 pma:ser:. A few labels were added to unlabelled statements of these sources so that notes can cite them.

1.3 Notation

The sources use clashing symbols. The table fixes the convention of this report; renamed symbols are marked $\rightarrow$. Its second block, for Parts III–V, was added with them.

Notion	This report	In the sources
Presburger arithmetic	$\text{Th}(\mathbb{N}; 0, 1, +, <)$	05: the same; 04: $\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +)$, kept in 04's text. The two are definitionally equivalent: $x \leq y$ is $\exists z (x + z = y)$.
Lexicographic group, coefficient group D dominant	$D \overrightarrow{\times} \mathbb{Z} = G_D$	05: the same; 04: $D \oplus_{\text{lex}} \mathbb{Z} = G(D) \rightarrow G_D$
Nonnegative cone	$M_D; M_E$ for a subgroup E	04: $M(D), M(E) \rightarrow M_D, M_E$
Baire coefficient group	$D_B = \mathbb{Q}^{\mathbb{N}}$, lexicographic, each $\mathbb{Q}$ discrete	05: the same; 04: $D_{\omega} \rightarrow D_B$ (the same group, order and topology)
Baire model	$M_B = (D_B \overrightarrow{\times} \mathbb{Z})_{\geq 0}$	04: $M_{\omega} \rightarrow M_B$ (the same model)
Laurent model (04)	$M_L = M_{D_L}, D_L = \mathbb{Q}((t))$	04 only
Baire space	$\mathbb{N}^{\mathbb{N}}$	04's macro $\backslash B = \mathbb{N}^{\mathbb{N}}$; 03's $\backslash B = \mathcal{B}$ is defined but never used
Homeomorphisms with Baire space	05: $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$ (tag code); 04: $\Psi : M_B \rightarrow \mathbb{N}^{\mathbb{N}}$, charts $\Psi_{\geq}, \Psi_{>}$, integer code $\zeta : \mathbb{Z} \rightarrow \mathbb{N}$	04: $H, H_{\geq}, H_{>}, h \rightarrow \Psi, \Psi_{\geq}, \Psi_{>}, \zeta$. The two codings are different maps.
Lexicographic symbol	$\overrightarrow{\times}$ (05); $\mathbb{Z}_{\text{lex}}^d$ for lexicographic powers (03)	04's subscript lex in $\oplus_{\text{lex}} \rightarrow \overrightarrow{\times}$
Division by a standard m	$q_m(x) \in M,$ $r_m(x) \in \{0, \dots, m-1\}$ (05); $Q_m(x) \in M,$ $R_m(x) = (0, z \bmod m) \in M$ (04)	kept; R_m is r_m viewed as an element of the model
Congruence predicates	$D_m(t) \iff t \in mG$ (05); $u \equiv_m v \iff u - v \in mG$ (04)	kept; $u \equiv_m v \iff D_m(u - v)$
Coset invariance	" mG -invariant" (05) = " m -residue invariant" (04)	the same notion
Cofinal infinite element	"cofinal infinite additive unit" (05) = "arithmetically infinite order unit" (04)	the same notion; not a multiplicative unit
Truncated subtraction	$T(x, y)$ (05), $x \dot{-} y$ (04)	the same operation; $\text{Disc}(f)$ is the discontinuity set of f
Pivots, filtration (04)	$p(k), F_k$ in theorem 11.4 ; F_n in proposition 10.6	local; p elsewhere is the predecessor; 03's closed pieces are $F_{Z,k}$
Growth classes (04)	$\mathcal{A}(M)$	local to proposition 10.7 ; not the ring $\mathcal{A}_{\text{fin}}$
Finite principal-part ring (03)	$\mathcal{A}_{\text{fin}} = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{>0}} \mathbb{R}X^q$, cone $\mathcal{A}_{\text{fin}, \geq 0}$	03: $\mathcal{A}, \mathcal{A}_{\geq 0} \rightarrow \mathcal{A}_{\text{fin}}, \mathcal{A}_{\text{fin}, \geq 0}$. Not the collection's $\mathbf{Oz}$, which the base of the omnific Diophantine report once called A .
Rational Hahn ring (03)	$\mathcal{H}_{\mathbb{Q}}$, (17.1)	03: $H \rightarrow \mathcal{H}_{\mathbb{Q}}$

Notion	This report	In the sources
Kleene–Brouwer embedding (03)	$\mu_{\text{KB}} : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{Q} \cap (0, 1)$	03: $h \rightarrow \mu_{\text{KB}}$. Not the dyadic coding κ or the midpoint coding μ into $(1, 2)$ of the omnific-notations report.
Growth variable (03)	$X = t^{-1}$, read as ω	03 §2.1: increasing- t convention in sections 18 and 19 , reverse X -convention from section 22 on; $q \mapsto -q$ converts
Well-ordered subsets (03)	$\text{WO}(L) \subseteq 2^L$	not the well-orderings $\text{WO}_\mu(X)$ of type μ of the surreal-well-orders report
Local sets (03)	W (proof of theorem 23.2); $\mathcal{Z}$ (lemma 24.2); D_+ , $D_\times$ (theorem 24.3)	03: $D \rightarrow W$ and $\mathcal{Z}$; D_+ , $D_\times$ kept. In Part I , D is always the divisible coefficient group.
L	a countable linear order (03)	in 05's and 04's elimination proofs, a lower bound or a common multiple (local)
e_j, e	unit vectors of $\mathbb{Q}^{\mathbb{N}}$ (05, 04); computable enumerations $e, e_+, e_{\geq 0}$ (05, section 6)	as in 05
Omnific integers	$\mathbf{Oz} = \mathbb{Z} \oplus \Pi$, growth notation ω^γ	as in the collection's docs/NOTATION.md
<i>Parts III–V (sources 07, 06, 08, 09)</i>		
The variable t	Part III: t positive <i>infinite</i> (06's $\mathbb{Z} + t\mathbb{R}[t]$, $t \mapsto \omega$); Part V: t positive <i>infinitesimal</i> (09's Hahn convention, $t^q \mapsto \omega^{-q}$)	kept in each part; tempting false reading: 06's t and 09's t are reciprocal, not equal. 03 uses $X = t^{-1}$ (infinite); 08 writes X for 06's t
Polynomial ring $\mathbb{Z} + t\mathbb{R}[t]$	A , cone $P = A_{\geq 0}$, topology τ_P by degree and constant term (06)	$= 08$'s $A_* = \mathbb{Z} + X\mathbb{R}[X] =$ the ring $\backslash\text{Pmod}[X]$ of the omnific Diophantine report (odg:frac:sub:models) and, at $t = \omega$, $\mathbb{Z} + \omega\mathbb{R}[\omega]$; under $t \mapsto X$ the integer-exponent subring of 03's $\mathcal{A}_{\text{fin}}$. Not 03's cone $\mathcal{A}_{\text{fin}, \geq 0}$
Presburger arithmetic	$\text{Th}(\mathbb{N}; 0, 1, +, <)$; 06: Pres	07's macro for $\text{Th}(\mathbb{N}; 0, 1, +, <)$ is expanded; 06's Pres = $\text{Th}(\mathbb{N}, 0, 1, +, <)$ kept. 08: signed language $\mathcal{L}_{\text{Pres}}$
Lexicographic product	$\overrightarrow{\times}$	07's $A \times_{\text{lex}} \mathbb{Z}$, 06's $D \text{ lex } \mathbb{Z}$ and 08's $\mathbb{R}^d \times_{\text{lex}} \mathbb{Z} \rightarrow \overrightarrow{\times}$ (same order: left factor dominant)
Real model	$M_{\mathbb{R}} = M_{\mathbb{R}}$	the same set and topology in 06, 07, 08; 07's second topology $M_{\mathbb{R}}^{\sharp}$ is a different space on the same algebra
Model families (07)	M_A, G_A (general A); $M_\alpha = (\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}$; M_s with $B_s = \mathbb{Q} + \mathbb{Q}s$, $s \in (1, 2)$ irrational	M_ω is Part I's M_B ; 07's H_s is a convex subgroup
Model families (06)	$M(D), G(D)$; $M_\omega (= M_B)$; M_α (as 07); M'_s, G'_s, H'_s for $s \in 2^{\mathbb{N}}$	06: $M_s, G_s, H_s \rightarrow M'_s, G'_s, H'_s$ (07's M_s is another family); 06's $H = \mathbb{Q}^{\mathbb{N}}$ in section 35.5 is local
Models of Part IV	$G_d = \mathbb{R}^d \overrightarrow{\times} \mathbb{Z}$, M_d (so $M_1 = M_{\mathbb{R}}$); $H_d \subseteq \mathbf{Oz}$; $\mathbb{R}^d \times E$	08's D in theorem 47.2 is a countable discrete group, not Part I's divisible D

Notion	This report	In the sources
Series fields of Part V	$\mathcal{P}_k$ (Puisseux), $\mathcal{L}_k$ (Levi-Civita), $\mathcal{H}_k = k((t^{\mathbb{Q}}))$, integer part I_k ; hulls F_A, J_C	09's macros \PP, \LL, \HH, \II renamed in the source only; same symbols printed. $\mathcal{P}_k$ is not 03's cone $\mathcal{A}_{\text{fin}, \geq 0}$; 03's $\mathcal{P}$ is the real-coefficient Puisseux field
Well-founded trees	WF	09's WF $\rightarrow$ 03's WF (same set)
Typography	1, $\div$, IOpen, ProveIt	07's \ind, \dotminus; 08's IOpen; 07's ProveIt $\rightarrow$ these (same meaning)
Local letters	A (a Polish group in 07, a ring in 06 and 08); P (a cone); $H, D,$ B, J	local to the statement or proof where they are introduced

1.4 Relation to the repository

The formal project `Logic/PresburgerArithmetic`. Both sources of [Part I](#) inspected this project at their pins, 05 its Lean file `Lean/PresburgerArithmetic/Cooper.lean` and 04 its Rocq file `Coq/Cooper.v`. The project decides Presburger sentences over the ordinary integers. Its periodicity lemmas assume a predicate on $\mathbb{Z}$ invariant under one standard shift; [example 4.4](#) and [proposition 4.6](#) show that this hypothesis cannot be carried verbatim to nonstandard $\mathbb{Z}$ -groups, and [lemma 4.7](#) and [theorem 4.8](#) give the model-uniform replacement. This is a porting requirement, not a defect of the integer theorems. [Section 13.5](#) names the declarations. **Nothing in this report is formalized**, and its placement in the Surreal collection, with a pointer from the formal project, confers no formal status.

The Surreal collection. Results already in the collection are cited, not reprinted as new:

- `exr:prop:division` and the paragraph after it in the `exponential-relations-over-r-omnific-integers` report: $\mathbf{Oz}$ is a $\mathbb{Z}$ -group, hence a model of Presburger arithmetic in its ordered-group presentation. [Proposition 4.2](#) and [theorem 4.9](#) prove the classical facts again for the split groups G_D .
- `isg:cf:main:presburger` in the `discrete-initial-subgroups-and-omnific-normalization` report: a set-sized $\mathbb{Z}$ -group has an initial realization in $\mathbf{Oz}$ if and only if it splits as $G^{\text{dv}} \xrightarrow{\sim} \mathbb{Z}$ with $G^{\text{dv}} = \bigcap_n nG$. The groups G_D of [Part I](#) are split by construction, with $\bigcap_m mG_D = D \times \{0\}$ ([corollary 4.3](#)), so that theorem applies to them [merge]; whether 05's maps $\iota_{\mathbb{R}}, \iota_{\mathbb{B}}$ are initial realizations in its sense is not checked here.
- `onot:neg:lem:kbtree`, `onot:neg:thm:pionesone` and `onot:neg:rem:secondroute` in the `omnific-notations` report: the lightface Π_1^1 -completeness of support validity. 03's [lemma 18.2](#) and [theorem 22.1](#) are their boldface counterparts (notes after those statements).
- `odg:thm:floor` in the `omnific-diophantine-geometry` report, proved in Lean for every surreal: 03's [lemma 23.1](#) is its restriction to the Puisseux field (note after it).
- `dsn:cor:openinduction` in the `definable-surreals-and-omnific-integers` report, and the Shepherdson paragraph of the `omnific-diophantine-geometry` report: the route of 03's [theorem 23.2](#) (note after it).

All these labels exist at the repository's current state and already existed at the pins of sources 03–05. Of them only `odg:thm:floor` is formalized (`docs/FORMALIZATION.md`, row “Full `odg:thm:floor`”); the others are recorded there as pending.

Parts III–V. Sources 06 and 07 cite the formal Presburger project (its `Lean Syntax.lean`, `Decision.lean`, `Audit.lean`) and source 06 also `Logic/Interpretability/PAHF/Lean/PAHF/PASyntax.lean`; their descriptions are correct at the current state ([section 41](#)). Source 06 cites `isg:thm:suspension`, `isg:cf:main:presburger` and `isg:cf:thm:PA`, and the omnific Diophantine report's `odg:frac:sub:models`, `odg:frac:prop:intersection` and `odg:frac:ex:workspace` for the ring $\mathbb{Z} + t\mathbb{R}[t]$; the notes of Parts III–V also cite `odg:thm:induction`, `odg:rem:ringinduction`, `odg:thm:floor`, `dsn:cor:openinduction`, `onot:neg:lem:kbtrees` and `onot:neg:thm:pionesone`. All exist at the current state; of them only `odg:thm:floor` is formalized. Placement beside the formal Presburger project confers no formal status on Parts III–V either.

1.5 The status statements of the sources

The seven manuscripts state their status as follows; these statements apply to the parts of this report taken from them.

Source 05: status and scope

The mathematical assertions designated as theorems are accompanied by proofs. The construction answers the cited question *as written*; independent bibliographic priority has not been established. Classical Presburger elimination and elementary topological-group facts are not claimed as new. This AI-assisted manuscript is not independently refereed or Lean-verified. The accompanying executable tests are finite checks, not a replacement for the proofs.

Source 04: from its abstract

The construction gives an affirmative answer to Question 2 as printed in Elliot Glazer's 2023 paper *A Topological Tennenbaum Theorem*. The historical priority of this answer has not been established. The new results are supported here by conventional proofs and executable exact tests, not by a claimed Lean or Rocq kernel check.

Source 03: status and provenance

Status and provenance. Complete mathematical arguments are provided for the constructions and deductions, with classical inputs identified. ProveIt's existing coanalytic support lower bound and omnific floor formula are credited, not claimed as new. The main contribution of this report is the presentation dichotomy and its algebraic consequences, together with the exact finite-support cancellation analysis. Global novelty and independent correctness review remain open; no established famous conjecture or Lean verification is claimed.

Source 07: status

Status. The main constructions and their consequences are proved below. Classical model theory and standard descriptive set theory are identified as inputs. No new result in this report has been checked by Lean or Rocq, and no claim of publication priority or external peer review is made.

Source 06: from its abstract

The article connects these constructions to ProveIt’s existing Presburger elimination development and its polynomial omnific subrings. All new assertions are proved in ordinary mathematics here; no machine-checked certification or unrestricted publication-priority claim is made.

Source 08: status

Status. The central results below have complete mathematical proofs, with classical inputs identified. Their bibliographic priority has not been established. This is an AI-assisted, unrefereed research manuscript, not a claim of an independently certified breakthrough or a Lean-checked proof. The included exact-arithmetic tests are supplementary checks only.

Source 09: status

Status. The central deductions are proved in this manuscript. Classical category theorems, Hahn and Puiseux constructions, real-closedness inputs, and previous ProveIt support and floor results are explicitly credited. Historical novelty has not been established by an exhaustive literature search. This is an unrefereed research contribution, not a claimed solution of a famous conjecture and not a Lean- or Rocq-verified development.

Every limitation and non-claim of the seven sources is also listed, source by source, in [section F](#).

Part I**Polish Presburger models**

This part answers Glazer’s Question 2 as printed. Its base text is source 05; source 04’s further results and second proofs are marked [04]. Repository pins: 05 inspected commit `a11efab09`, 04 commit `3ad5f878c`; the files they name are unchanged at the placement.

2 The research overlap and the precise target

The relevant intersection is *topological and effective model theory of weak arithmetic*. Glazer’s work supplies a sharply formulated question; ProveIt supplies two concrete mathematical interfaces: an executable Cooper quantifier-elimination development and a normal-form construction of the omnific integers. This makes it possible to connect an external Polish topology, internal induction, and exact surreal coordinates without appealing to the repository’s unreviewed research claims.

In the final section of [1], Glazer asks:

Does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

The source is Question 2 on page 8 of arXiv:2311.13699v1. We use precisely that formulation.

The topology need not be the arithmetic order topology, and no multiplication is required. These two distinctions are essential.

Glazer’s obstruction for stronger arithmetic includes the impossibility of an uncountable Polish model of $\mathcal{Q} + I\text{Open}$ with continuous addition and Borel multiplication [1, Corollary 1]. The earlier study of Enayat, Hamkins, and Wcisło [2] motivates the broader problem of finding topological presentations of arithmetic. Our construction does not contradict those results: it supports the additive language, and its additive structure prevents a semiring expansion.

2.1 Repository interfaces actually inspected

The repository snapshot used here is

a11efab09a2d08f692a09829cb6d3440ddd1795a.

The Presburger README [14] describes a complete Lean integer quantifier eliminator and an independent Rocq/Coq proof of the normalized one-variable step. The inspected file `PresburgerArithmetic/Cooper.lean` [15] states `periodic_interval` and `periodic_has_residue` for predicates on the ordinary type `Int`. Their hypotheses are entirely appropriate there. They must not be transferred unchanged to a nonstandard ordered group; section 4.2 gives an explicit counterexample and the correct replacement.

The surreal README [16] records an omnific ring whose normal forms have nonnegative exponents and integer constant term. Only this normal-form interface and the standard arithmetic of its additive group are needed below. We do not assume that the present topology or any of this article’s theorems has already been formalized in that repository.

2.2 What is established, and what is not

The principal result is an explicit affirmative construction, rather than a claim that a difficult negative theorem has been overturned. The lexicographic algebra is classical. The contribution developed here is the precise topology and its application to the stated question, together with an effective Baire presentation and a collection of sharp structural consequences.

The literature search verified the question in the source PDF and checked the arXiv record; it did not establish the absence of an earlier solution. Consequently, “an affirmative answer to the stated question” is the appropriate conclusion, not an independently certified first solution. No answer to Glazer’s separate reverse-mathematical Question 1 is claimed.

All constructions and proofs take place in ordinary ZFC with set-sized carriers. The notation $\mathbf{Oz}$ is used only to identify where explicit set-sized images lie. No topology on an unrestricted proper class, and no set of all surreal numbers, is used.

2.3 Source 04’s framing

Source 04 opens with the same question and the same two repository connections (its Section 1). It adds three points, in its own words; “this article” and “this package” there mean source 04.

The interpretation of the signature matters. Throughout this article, Presburger arithmetic is

$$\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +),$$

with its definable order, and with no multiplication of two variables. Continuity of addition does *not* mean continuity of every definable function. Indeed, the difference between these requirements is central to our result. Adding order and fixed congruence predicates to the language does not change which first-order sets are definable, but the predicates need not be open or closed in the topology.

The characterization of Presburger groups as discretely ordered abelian groups with divisible quotient by the standard integers, and their quantifier elimination after naming congruences, are classical. A convenient modern reference is Jeřábek's preliminaries [4, Section 2]. We include proofs of the facts needed here so the construction can be audited independently of that characterization.

The specific contribution is the topological application to the printed question, strengthened by an effective model on literal Baire space and an analysis of definable operations and self-embeddings. The finite-residue argument also identifies an exact hypothesis needed when transporting a standard-integer proof to arbitrary models. It is not a claim that the repository's integer proof is defective.

This is an unrefereed research manuscript. The statements called theorems below have proofs in the manuscript; the Python program is an independent finite diagnostic and a usable stream implementation. No new theorem in this package is represented as machine-checked. The literature check verified the wording of Glazer's question and the relevant background; it did not establish that the affirmative construction has never appeared elsewhere. Thus an answer to the specified question is claimed at the mathematical level, but first-publication priority is not claimed.

3 The construction in one theorem

We include 0 in $\mathbb{N}$. For an ordered abelian group D , let

$$G_D = D \overrightarrow{\times} \mathbb{Z}$$

mean the direct-product additive group with the D coordinate more significant in the order. Thus

$$(d, n) < (e, m) \iff d < e \text{ or } (d = e \text{ and } n < m). \quad (3.1)$$

Write $\mathbf{1} = (0, 1)$ and let

$$M_D = (G_D)_{\geq 0} = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (3.2)$$

The ambient product topology, whenever specified, is separate from (3.2)'s arithmetic order.

Theorem 3.1 (Polish and effective Presburger models). *The following assertions hold.*

- (1) *With $D = \mathbb{R}$ in its usual topology, $M_{\mathbb{R}}$ is an uncountable, locally compact Polish space without isolated points. Coordinatewise addition is continuous, and*

$$(M_{\mathbb{R}}; 0, \mathbf{1}, +, <) \models \text{Th}(\mathbb{N}; 0, 1, +, <).$$

- (2) *With $D = \mathbb{Q}^{\mathbb{N}}$, ordered by the first differing coordinate and with each rational coordinate discrete, the space $M_{\mathbb{B}} = M_D$ is homeomorphic to $\mathbb{N}^{\mathbb{N}}$. There is an explicit such homeomorphism under which addition, successor, and division by each positive standard integer are Type-2 computable.*
- (3) *In both models, every first-order definable relation, with parameters, is Δ_2^0 . The order is neither open nor closed. In the Baire presentation, each formula has a computable bound on the number of changes in a convergent truth approximation, relative to its parameter codes.*

- (4) Both models embed as ordered additive submonoids of $\mathbf{Oz}$, fixing 0 and 1. Neither admits any unital semiring multiplication with its given addition, zero, and unit.

The proofs occupy [sections 4 to 7, 9 and 10](#). The first assertion alone answers the source question. The later assertions explain how much regularity is possible and where it necessarily stops.

For orientation, an element $(r, -17)$ with $r > 0$ is a *positive* element of $M_{\mathbb{R}}$. Indeed it exceeds every standard numeral $(0, k)$. Nevertheless,

$$(1/j, 0) \longrightarrow (0, 0) \quad (j \longrightarrow \infty). \quad (3.3)$$

in the chosen topology. Arithmetic size and topological proximity are different notions. The construction would fail if one silently demanded that this topology be the order topology.

Source 04's statement [04]. Source 04 (its Section 1.2) sets up the same real model:

Let $\mathbb{Z}$ have the discrete topology and put

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (3.4)$$

Set $0 = (0, 0)$, $1 = (0, 1)$, and

$$(r, z) + (s, w) = (r + s, z + w).$$

Order the ambient group $\mathbb{R} \times \mathbb{Z}$ lexicographically, with the real coordinate dominant. Then $M_{\mathbb{R}}$ is exactly its nonnegative cone.

Remark 3.2 (Source 04's statement of [theorem 3.1](#) [04]). The structure $(M_{\mathbb{R}}; 0, 1, +)$ is a model of $\text{Pr}_{\mathbb{N}}$. With the topology inherited from $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$, its universe is an uncountable perfect locally compact Polish space, and addition is jointly continuous. There is also a model of $\text{Pr}_{\mathbb{N}}$ on $\mathbb{N}^{\mathbb{N}}$ with jointly continuous addition. In both models, quotient and remainder by each fixed positive standard integer are continuous, and every first-order definable relation with parameters is Δ_2^0 .

The algebraic assertion is proved in [section 4](#) and [section 4.3](#); the topology and the explicit Baire-space realization are proved in [sections 5.2, 5.4 and 6.3](#). The proof uses familiar $\mathbb{Z}$ -groups, not an alteration of what Presburger arithmetic means.

[merge] The two statements agree. Source 04's adds continuity of quotient and remainder by each fixed standard integer, which it proves for every admissible coefficient group ([theorem 5.4](#)) and which 05 proves for its two models ([proposition 8.1](#)); source 05's adds the optimal order rank, the finite-mind-change truth approximation and the omnific realization. Both prove the no-semiring and self-embedding results.

4 Full Presburger arithmetic, not merely a few axioms

4.1 Discretely ordered groups with standard residues

Definition 4.1. A $\mathbb{Z}$ -group in this article is a linearly ordered abelian group G with a distinguished least positive element $\mathbf{1}$, such that for every positive standard integer m and every $x \in G$ there exist $q \in G$ and $r \in \{0, \dots, m-1\}$ satisfying

$$x = mq + r\mathbf{1}. \quad (4.1)$$

Here mq denotes repeated addition by a *standard* integer, not a new multiplication operation on G .

Ordered abelian groups are torsion-free. The quotient and remainder in (4.1) are unique: if $m(q - q') = (r' - r)\mathbf{1}$ and $q > q'$, then $q - q' \geq \mathbf{1}$, whereas $r' - r < m$; the other inequality is similar. We suppress $\mathbf{1}$ in standard integer constants when no confusion is possible.

Proposition 4.2 (The split lexicographic group). *If D is a divisible ordered abelian group, then $D \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group. Its standard division algorithm is*

$$(d, n) = m(d/m, \lfloor n/m \rfloor) + (0, n \bmod m). \quad (4.2)$$

If $(d, n) \geq 0$, the displayed quotient is nonnegative.

Proof. Coordinatewise addition preserves the lexicographic order. A positive (d, n) either has $d > 0$, in which case it exceeds $(0, 1)$, or has $d = 0$ and $n \geq 1$. Thus $(0, 1)$ is the least positive element. Divisibility and torsion-freeness of D give the unique element d/m . Ordinary Euclidean division of n now proves (4.2). If $d > 0$, then $d/m > 0$, regardless of the integer quotient. If $d = 0$ and $n \geq 0$, the integer quotient is nonnegative. $\square$

Source 04 proves the same statement (its Lemma 2.2) with the same explicit quotient and the same nonnegativity clause; its uniqueness argument compares coordinates instead of using least positivity.

Second proof (source 04). If $(d, z) > 0$ and $d > 0$, then $(d, z) > (0, 1)$. If $d = 0$, positivity says $z \geq 1$. This proves discreteness. Existence in (4.1) follows from divisibility of D and ordinary integer Euclidean division, including for negative z .

For uniqueness, equality of $mq + r\mathbf{1}$ and $mq' + r'\mathbf{1}$ first gives equality of the D -coordinates of q, q' . Equality of the integer coordinates then gives ordinary uniqueness of quotient and remainder. Finally, if $d > 0$, then $d/m > 0$. If $d = 0$ and $z \geq 0$, then $\lfloor z/m \rfloor \geq 0$. $\square$

Corollary 4.3 ([04]). *For $m \geq 1$,*

$$mG_D = D \times m\mathbb{Z}, \quad (d, z) \equiv (e, w) \pmod{mG_D} \iff z \equiv w \pmod{m}.$$

Moreover,

$$\bigcap_{m \geq 1} mG_D = D \times \{0\}. \quad (4.3)$$

Proof. Every D -coordinate is an m -multiple. The only ordinary integer divisible by every positive integer is zero. $\square$

The standard subgroup $\mathbb{Z}\mathbf{1}$ is convex, and $G_D/\mathbb{Z}\mathbf{1}$ is the ordered divisible group D . This gives the usual quick recognition of G_D as a Presburger group. For the precise model-theoretic conclusion, we next give a direct elimination argument.

The implication from these axioms to *full* Presburger arithmetic is important enough to prove. We use the classical elimination architecture of Presburger and Cooper; [3] supplies historical and algorithmic context. The proof below makes explicit which algebraic hypotheses are needed in a nonstandard group.

4.2 Coset invariance versus standard-shift periodicity

For $m \geq 1$, write $mG = \{mh : h \in G\}$. A predicate P is mG -invariant if

$$y - z \in mG \implies (P(y) \leftrightarrow P(z)). \quad (4.4)$$

This is stronger than invariance under the single standard shift $m\mathbf{1}$.

Example 4.4 (A false generalization of an integer lemma). In $G = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$, let $P(d, n)$ mean $d > 0$. For every positive standard m ,

$$P(x + m\mathbf{1}) \leftrightarrow P(x)$$

holds. Yet P has witnesses and is false at every standard residue $0, 1, \dots, m-1$. Thus

$$(\exists x P(x)) \not\iff \bigvee_{0 \leq r < m} P(r\mathbf{1}).$$

It also fails the naive finite-interval reduction with lower bound 0 and upper bound $(1, 0)$. No flaw in the repository's Int theorem is involved: its domain is the ordinary integers, where every multiple is obtained by a standard finite iteration.

The general counterexample [04]. Source 04 phrases coset invariance as a definition and gives the counterexample for every nonzero divisible D , with the predicate $d \neq 0$ in place of $d > 0$. In its text G is a linearly ordered abelian group with a least positive element $\mathbf{1}$ and unique division by positive standard integers as in (4.1), that is, a $\mathbb{Z}$ -group.

Definition 4.5 ([04]). A predicate $P \subseteq G$ is m -residue invariant if

$$u - v \in mG \implies (P(u) \iff P(v)).$$

No definability hypothesis is imposed in this definition.

The condition is stronger than invariance under the single translation $x \mapsto x + m\mathbf{1}$. This distinction is invisible over the ordinary group $\mathbb{Z}$, but is essential over nonstandard groups.

Proposition 4.6 (One-step periodicity is insufficient [04]). If $D \neq 0$, the external predicate

$$P(d, z) \iff d \neq 0$$

on G_D satisfies $P(x + m\mathbf{1}) \leftrightarrow P(x)$ for every standard $m \geq 1$. Nevertheless P is nonempty and is false at every standard residue $0, 1, \dots, m-1$.

Proof. A standard integer translation does not change the D -coordinate. A nonzero D -coordinate exists, whereas every standard integer has D -coordinate zero. In particular P is not m -residue invariant: $(d, 0) - (0, 0) \in mG_D$ for every d . $\square$

Lemma 4.7 (The coset-invariant finite search). Let G be a $\mathbb{Z}$ -group, $m \geq 1$ standard, and P an mG -invariant predicate. For $L, U \in G$,

$$\begin{aligned} \exists y (L \leq y \leq U \wedge P(y)) \\ \iff \bigvee_{0 \leq r < m} (L + r \leq U \wedge P(L + r)). \end{aligned} \quad (4.5)$$

With only an upper bound, candidates $U - r$ suffice; with only a lower bound, candidates $L + r$ suffice. With neither bound, the candidates are $0, \dots, m-1$.

Proof. Given a witness $y \geq L$, write $y - L = mh + r$ with $0 \leq r < m$. One has $h \geq 0$: otherwise $h \leq -1$, and then $mh + r \leq -m + (m - 1) = -1$, contradicting $y - L \geq 0$. Consequently

$$L \leq L + r \leq y \leq U.$$

The difference $y - (L + r) = mh$ lies in mG , so $P(L + r)$ holds. Conversely every displayed candidate is already a witness. The same argument applied to $U - y$ proves the upper-bound version. Without bounds, use the standard remainder of y itself. $\square$

Congruence predicates have exactly the stronger invariance (4.4). If M is a common multiple of finitely many standard moduli, any Boolean combination of congruences with those moduli is MG -invariant. This remains true with arbitrary nonstandard parameters: adding Mh to a variable changes an affine congruence term by an element of each relevant modulus subgroup.

4.3 Second route [04]: finite witnesses at a boundary

Source 04's form of lemma 4.7 works with strict bounds and an explicit finite candidate set. It is the same theorem with a different normalization of the bounds, and it is the step that 04 uses in its elimination proof below. Here G is a $\mathbb{Z}$ -group, and a standard integer r also denotes $r1$.

Theorem 4.8 (Finite boundary criterion [04]). *Let A, B be finite lists of elements of G , let $m \geq 1$, and let P be m -residue invariant. Put*

$$\Phi(x) := P(x) \wedge \bigwedge_{a \in A} (a < x) \wedge \bigwedge_{b \in B} (x < b).$$

Define a finite candidate set C by

$$C = \begin{cases} \{a_* + 1, \dots, a_* + m\}, & A \neq \emptyset, \quad a_* = \max A, \\ \{b_* - 1, \dots, b_* - m\}, & A = \emptyset, B \neq \emptyset, \quad b_* = \min B, \\ \{0, \dots, m - 1\}, & A = B = \emptyset. \end{cases} \quad (4.6)$$

Then

$$(\exists x \in G) \Phi(x) \iff \bigvee_{c \in C} \Phi(c). \quad (4.7)$$

Proof. The reverse direction is immediate. Suppose first that $A \neq \emptyset$ and x is a witness. Divide $x - a_* - 1$ by m to obtain

$$x - a_* = mq + s, \quad 1 \leq s \leq m.$$

Here q is an element of G , not necessarily a standard integer. If $q < 0$, discreteness gives $q \leq -1$, whence $mq + s \leq 0$, contradicting $x > a_*$. Thus $q \geq 0$ and $c = a_* + s \leq x$. This c is larger than every lower bound and smaller than every upper bound because $c \leq x < b$. Also $x - c = mq \in mG$, so $P(c)$ holds.

If A is empty and B is nonempty, divide $b_* - x - 1$ instead. The result is $b_* - x = mq + s$ with $q \geq 0$ and $1 \leq s \leq m$. The candidate $c = b_* - s$ satisfies $x \leq c < b_* \leq b$ for every upper bound and is in the same residue class as x . Finally, when both lists are empty, take the unique standard representative of x modulo mG . $\square$

This proof uses no induction through a nonstandard interval. Its only finite searches have standard lengths fixed by the formula being eliminated. Weak bounds can be converted to strict ones using discreteness: $a \leq x$ is $a - 1 < x$, and $x \leq b$ is $x < b + 1$.

4.4 A self-contained elimination argument

Expand the ordered-group language by predicates

$$D_m(t) \iff t \in mG \quad (m \geq 2),$$

and regard D_1 as always true. All of these are definable by existential formulas in the group language.

Theorem 4.9 (Classical Presburger elimination over $\mathbb{Z}$ -groups). *In the expanded language*

$$\{0, 1, +, -, \leq, (D_m)_{m \geq 2}\},$$

the axioms of $\mathbb{Z}$ -groups eliminate quantifiers and form a complete theory. In particular every $\mathbb{Z}$ -group is elementarily equivalent to $(\mathbb{Z}; 0, 1, +, -, \leq, (m \mid \cdot)_m)$.

Proof. It suffices to eliminate one existential quantifier from a quantifier-free formula. Rewrite the formula in disjunctive normal form. Negated inequalities are replaced using discreteness, for example $\neg(s \leq t) \leftrightarrow t + 1 \leq s$. An equality is the conjunction of two inequalities; a disequality is a disjunction of strict inequalities. Finally, standard residue exhaustiveness gives

$$\neg D_m(t) \iff \bigvee_{r=1}^{m-1} D_m(t - r).$$

Thus it is enough to eliminate $\exists x$ from a conjunction of affine inequalities and positive divisibility atoms, along with conjuncts not involving x .

Let A be a positive common multiple of every nonzero absolute coefficient of x occurring in that conjunction; take $A = 1$ if there are none. Set $y = Ax$ and add the condition $D_A(y)$. An atom

$$ax + t \leq 0, \quad a \neq 0,$$

can be multiplied by the positive standard integer $k = A/|a|$, giving

$$\text{sgn}(a)y + kt \leq 0.$$

It is now an upper or lower bound on y . For a congruence, torsion-free cancellation gives the equivalence

$$D_m(ax + t) \iff D_{km}(\text{sgn}(a)y + kt). \quad (4.8)$$

Indeed $kz \in kmG$ if and only if $z \in mG$: one direction multiplies a witness, and the other cancels k from $kz = kmh$. The additional atom $D_A(y)$ is exactly what ensures that every candidate y comes from a unique x .

After normalizing signs, we have finitely many lower bounds $L_i \leq y$, finitely many upper bounds $y \leq U_j$, and a conjunction $P(y)$ of congruence conditions. Let M be a positive common multiple of all their moduli, including A . Then P is MG -invariant.

If lower bounds exist, choose their maximum L . By [lemma 4.7](#), a witness exists exactly when one of $L, L + 1, \dots, L + M - 1$ satisfies all the normalized conjuncts. The maximum need not be introduced as a new function symbol: take a finite disjunction over the indices i , require $L_h \leq L_i$ for every h , and substitute $L_i + r$ for y . This produces a quantifier-free formula in the parameters. If there are upper bounds but no lower bounds, choose a minimal U_j and substitute $U_j - r$. If no bounds exist, substitute the standard residues r . Conjuncts independent of x are retained throughout.

This eliminates one existential quantifier. Induction on formula complexity, replacing universal quantification by negated existential quantification, eliminates all quantifiers.

A quantifier-free sentence uses only standard integer multiples of 1. Their order and their residues are the same in every $\mathbb{Z}$ -group, by least positivity and uniqueness of standard remainders. Therefore all $\mathbb{Z}$ -groups satisfy the same sentences. The ordinary integers are one such group, proving completeness and the final assertion. $\square$

4.5 Second route [04]: normalization and elimination

Source 04 proves [theorem 4.9](#) by a differently organized argument: it keeps negated congruences instead of rewriting them through residues, substitutes away equalities in the eliminated variable, and uses strict bounds with [theorem 4.8](#). Its statement is uniform over all groups G of [section 4.3](#).

Use the group language

$$L_{Pr} = \{0, 1, +, -, <, (\equiv_m)_{m \geq 2}\},$$

where $u \equiv_m v$ means $u - v \in mG$. Affine terms have integer coefficients. The congruence predicates are definable by existential quantification in the underlying additive language.

Theorem 4.10 (Quantifier elimination, with a uniform proof [04]). *Every L_{Pr} formula is equivalent, uniformly in all groups G satisfying the hypotheses above, to a quantifier-free Boolean combination of affine inequalities, affine equalities, and congruences. Consequently every such G is elementarily equivalent to $(\mathbb{Z}; 0, 1, +, -, <)$ in the reduct language.*

Proof. It is enough to eliminate one existential quantifier from a conjunction of literals, after writing the quantifier-free part in disjunctive normal form. Negated equality is split into two strict inequalities, and negated inequalities are converted by discreteness. Negated congruences can be kept as negated congruences.

Let x be the variable to eliminate. If it occurs with no nonzero coefficient, the quantifier is vacuous. Otherwise let L be the least common multiple of the positive absolute values of its nonzero coefficients. Introduce $y = Lx$ and impose $y \in LG$. There is a unique x for every such y because G is torsion-free.

For an atom with term $ax + t$, $a \neq 0$, put $B = L/|a|$. Multiplying an inequality or equality by the positive standard integer B replaces its x -coefficient by $\text{sgn}(a)$ on y . For divisibility use

$$ax + t \in nG \iff \text{sgn}(a)y + Bt \in nBG. \quad (4.9)$$

Indeed the forward implication multiplies a witness by B , and the reverse implication cancels B in the torsion-free group. Atoms with zero x -coefficient remain independent conditions.

The resulting conjunction has only coefficient 1 or -1 on y in its inequalities, equalities, and congruences. An equality involving y permits substitution. In the absence of such an equality, the inequalities form finite strict lower and upper lists. The remaining congruences and negated congruences form a predicate $P(y)$ invariant modulo mG , where m is the least common multiple of their moduli, including L . For example, if $n \mid m$, replacing y by $y + mg$ cannot change membership of $\pm y + t$ in nG .

Apply [theorem 4.8](#). The maximum or minimum appearing there is not an extra function symbol: take a finite disjunction over choices of a largest lower bound, or a smallest upper bound, and add the finitely many inequalities that express that choice. Substitution of

the finite candidates leaves a quantifier-free formula in the remaining variables. Negation handles universal quantifiers, so induction on the external formula tree proves elimination.

Finally a closed quantifier-free formula has the same truth value in every such G as in $\mathbb{Z}$. Its terms are standard numerals. Their equality and order are fixed, and their congruences have the usual residues by division. Eliminating all quantifiers therefore proves elementary equivalence. $\square$

Corollary 4.11 (The entire induction scheme). *For every divisible ordered abelian group D ,*

$$(M_D; 0, 1, +, <) \models \text{Th}(\mathbb{N}; 0, 1, +, <).$$

In particular it satisfies induction for every first-order formula of this additive language, with arbitrary parameters from M_D .

Proof. By [proposition 4.2](#) and [theorem 4.9](#), $G_D \equiv \mathbb{Z}$. The nonnegative cone is uniformly definable by $x \geq 0$. Relativizing every quantifier to that cone translates any sentence about M_D into a sentence about G_D . The same translation in $\mathbb{Z}$ gives the ordinary nonnegative integers. Each universally closed induction instance is therefore true in M_D . $\square$

Proposition 4.12 (Definable least elements). *Every nonempty definable subset of M_D , allowing parameters, has a least element.*

Proof. Eliminate quantifiers and take a finite disjunction of conjunctions as in [theorem 4.9](#). Include $x \geq 0$ explicitly. For each conjunction, the substitution $y = Ax$ introduces a lower bound $y \geq 0$. Hence a nonempty conjunction has a least solution among the finitely many valid candidates at its maximum lower bound; the residue argument shows that one of these candidates lies below every witness. Dividing by A is order-preserving. Finally take the minimum of the least solutions of the nonempty disjuncts. $\square$

This also gives a direct induction check after elimination: a nonempty set of counterexamples to an induction instance would have a least member. It cannot be 0, and its predecessor would contradict the induction step. The external standard part $\{(0, n) : n \in \mathbb{N}\}$ is a proper inductive subset when $D \neq 0$, but is not definable, even with parameters. Induction quantifies over definable sets, not over every external subset.

Source 04 proves [corollary 4.11](#) (its Corollary 3.5, for $\text{Pr}_{\mathbb{N}}$) by the same relativization, and adds [04]:

The same argument shows that every definable subset of M_D^k is a Boolean combination of restrictions of group affine inequalities, equalities, and standard congruences. This observation will control its Borel complexity without an induction over projections of Borel sets.

5 Polishness and a locally compact example

A Polish space is a separable space whose topology is induced by some complete metric. A particular convenient metric may be incomplete without the space ceasing to be Polish. This distinction matters for open rays.

5.1 A useful exact criterion

We first record the standard subspace criterion with a proof, so that the completeness issue is visible.

Lemma 5.1 (Polish subspaces). *A subspace A of a Polish space X is Polish if and only if A is a G_δ subset of X .*

Proof. Fix a compatible complete metric d on X . If $A = \bigcap_j U_j$ with U_j open, consider on A the continuous functions $f_j(x) = 1/d(x, X \setminus U_j)$; when $U_j = X$, set $f_j = 0$. The graph of $x \mapsto (f_j(x))_j$ is closed in $X \times \mathbb{R}^\mathbb{N}$. Indeed a limit outside some U_j would force f_j to tend to infinity, not to a real number. Projection identifies this closed graph homeomorphically with A . Countable products and closed subspaces of complete separable metric spaces are complete and separable.

Conversely, let δ be a complete metric inducing the subspace topology on A . For each n , let W_n be the union of all ambient open sets U that meet A and satisfy

$$\text{diam}_d(U) < 2^{-n}, \quad \text{diam}_\delta(U \cap A) < 2^{-n}.$$

These unions cover A . If $x \in \bigcap_n W_n$, choose $x \in U_n$ and $a_n \in U_n \cap A$. Then $a_n \rightarrow x$ in d . For each fixed n , all sufficiently late a_k belong to U_n , so (a_k) is δ -Cauchy. Its δ -limit lies in A , and agreement of the two topologies identifies that limit with x . Thus $A = \bigcap_n W_n$. $\square$

Theorem 5.2 (The positive-cone criterion). *Let D be a divisible ordered abelian group equipped with a Polish topological-group topology, not necessarily its order topology. Give $\mathbb{Z}$ the discrete topology. Then the subspace $M_D \subseteq D \times \mathbb{Z}$ is Polish if and only if $D_{>0}$ is G_δ in D . When this holds, $D_{>0}$ is in fact Δ_2^0 , and addition on M_D is continuous.*

Proof. If $D_{>0}$ is G_δ , so is its union with the closed singleton $\{0\}$. On each clopen integer slice, the carrier (3.2) is either $D_{>0}$ or $D_{\geq 0}$. A countable topological sum of G_δ subspaces is G_δ in the corresponding sum: its complement is a countable union of F_σ subsets of the clopen slices. Apply lemma 5.1.

Conversely, the slice with integer coordinate -1 is closed in M_D and homeomorphic to $D_{>0}$. If M_D is Polish, so is that slice, and lemma 5.1 makes $D_{>0}$ a G_δ subset of D . Negation is a homeomorphism on D , so $D_{<0}$ is G_δ as well. The complement of $D_{>0}$ is $D_{<0} \cup \{0\}$, also G_δ . Thus $D_{>0}$ is both F_σ and G_δ .

Finally the cone is closed under coordinatewise addition. Restricting continuous addition on $D \times \mathbb{Z}$ to a subspace with its image in that subspace gives a continuous map $M_D^2 \rightarrow M_D$. $\square$

5.2 Admissible coefficient groups and continuous division [04]

Source 04 proves a sufficient version of theorem 5.2 under a stronger hypothesis, and obtains in return the continuity of division by standard integers for every such D ; source 05 proves that continuity only for its two explicit models (proposition 8.1). The Polish clause of theorem 5.4 is the case of theorem 5.2 in which $D_{>0}$ is open.

Definition 5.3 ([04]). *An admissible topological divisible group is a divisible linearly ordered abelian group D with a Polish group topology such that $D_{>0}$ is open and each map $d \mapsto d/m$, $m \geq 1$, is continuous. The specified topology is additional data; it is not silently identified with the order topology of G_D or M_D .*

Theorem 5.4 (Transfer [04]). *Let D be an admissible topological divisible group, and give G_D the product topology with discrete $\mathbb{Z}$. Then M_D , with its subspace topology, is Polish. Addition on M_D is jointly continuous. The functions*

$$Q_m(d, z) = (d/m, \lfloor z/m \rfloor), \quad R_m(d, z) = (0, z \bmod m)$$

are continuous maps from M_D to itself for each fixed $m \geq 1$.

Proof. The ambient group is Polish. Its complement of M_D is

$$(D_{<0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{<0}).$$

The first set is open and the second is closed. In a metrizable space every open set is F_σ , so this complement is F_σ and M_D is G_δ . A G_δ subspace of a Polish space is Polish. Equivalently, the integer-coordinate slices are a countable disjoint sum of copies of the Polish spaces $D_{\geq 0}$ and $D_{>0}$.

The set M_D is closed under addition; restriction of the continuous ambient addition is therefore continuous as a map into the subspace. The displayed formulas for Q_m, R_m are continuous in the ambient coordinates: division on D is continuous, and every function on the discrete coordinate is continuous. Their images belong to M_D by [proposition 4.2](#). $\square$

5.3 The real-coordinate model, including a complete metric

For $D = \mathbb{R}$, the positive cone is open. More explicitly, the slice

$$X_n = \{r : (r, n) \in M_{\mathbb{R}}\}$$

is $[0, \infty)$ when $n \geq 0$ and $(0, \infty)$ when $n < 0$. Hence

$$M_{\mathbb{R}} \cong \bigsqcup_{n \geq 0} [0, \infty) \sqcup \bigsqcup_{n < 0} (0, \infty). \quad (5.1)$$

This proves separability, local compactness, and absence of isolated points directly.

For a fully explicit compatible complete metric, define

$$d_n(r, s) = \begin{cases} \min\{1, |r - s|\}, & n \geq 0, \\ \min\{1, |\log r - \log s|\}, & n < 0, \end{cases}$$

and set

$$d((r, n), (s, m)) = \begin{cases} d_n(r, s), & n = m, \\ 2, & n \neq m. \end{cases} \quad (5.2)$$

Every Cauchy sequence is eventually contained in one slice. The closed ray with the first metric is complete; the open ray with the logarithmic metric is isometric, before truncation, to $\mathbb{R}$. Thus (5.2) is complete. It induces exactly the topology in (5.1), which is the original product-subspace topology. Continuity of addition follows from that topology, not from a claim of uniform continuity in this metric.

The map $r \mapsto (r, 0)$ on $r > 0$ establishes uncountability. Together with [corollary 4.11](#), this completes the simplest proof of the affirmative answer.

Remark 5.5 (Why the arithmetic order topology is not used). The least positive element is **1**. In G_D , the order interval $(x - 1, x + 1)$ is the singleton $\{x\}$. The positive cone has the same discreteness at its minimum. Consequently its arithmetic order topology is discrete, and an uncountable such topology cannot be Polish. The topology of (5.1) is different, as (5.2) and the convergence in (3.3) explicitly demonstrate.

5.4 The real-coordinate model, source 04's account [04]

Source 04 gives the same slice decomposition and the same complete metric as (5.2), except that distinct slices are at distance 1; it adds two arithmetic illustrations and a remark.

Take $D = \mathbb{R}$ with the usual order and topology. It is admissible, so [corollary 4.11](#) and [theorems 5.4](#) and [7.2](#) apply to (3.4).

Topology, perfection, and a complete metric. For $z \geq 0$, the slice of integer coordinate z is a copy of $[0, \infty)$; for $z < 0$, it is a copy of $(0, \infty)$. Each slice is clopen in $M_{\mathbb{R}}$. Thus $M_{\mathbb{R}}$ is uncountable, locally compact, and has no isolated points. In particular every standard numeral $(0, n)$ is nonisolated.

It is important not to assert that the inherited Euclidean metric is complete: $(0, \infty)$ with that metric is not. Here is a compatible complete metric, with different integer slices distance 1 apart:

$$\rho((r, z), (s, w)) = \begin{cases} 1, & z \neq w, \\ \min\{1, |r - s|\}, & z = w \geq 0, \\ \min\{1, |\log r - \log s|\}, & z = w < 0. \end{cases} \quad (5.3)$$

The last line only occurs for positive r, s . A Cauchy sequence is eventually in one slice. On a nonnegative slice use completeness of $[0, \infty)$; on a negative slice use the homeomorphism $r \mapsto \log r$ onto $\mathbb{R}$. The truncated metrics are metrics, and the inter-slice distance preserves the triangle inequality. Countable dense subsets in the slices give a countable dense subset of the whole space.

Arithmetically huge elements converging to zero. Put $a_j = (1/j, 0)$ for $j \geq 1$. Then

$$a_j > n\mathbf{1} \quad \text{for every standard } n, \quad a_j \longrightarrow 0 \quad \text{topologically.}$$

There is no contradiction. The arithmetic order and the chosen topology are distinct structures. The arithmetic order topology on M_D would be discrete, since every nonzero point has a predecessor and a successor, and 0 is isolated by its immediate successor. That discrete topology is *not* the topology used in the construction.

For a concrete division example, in $M_{\mathbb{R}}$ one has

$$(2, -7) = 3(2/3, -3) + (0, 2).$$

The quotient $(2/3, -3)$ is positive because its real coordinate is positive. This illustrates why restricting the integer coordinate to nonnegative values at all positive real coordinates would destroy the desired arithmetic: the negative integer sheets are necessary.

Remark 5.6 (Why a Cartesian product of two positive cones is not used [04]). The universe is not $\mathbb{R}_{\geq 0} \times \mathbb{N}$. The latter does not have the required predecessor property at points $(r, 0)$ with $r > 0$. By contrast, in $M_{\mathbb{R}}$ every positive (r, z) has predecessor $(r, z - 1)$, except that the ordinary zero remains the least element as required.

6 An explicit Baire-space realization

Let

$$D_B = \mathbb{Q}^{\mathbb{N}}$$

with coordinatewise addition. Each factor $\mathbb{Q}$ is discrete, *not* given its Euclidean subspace topology. Two distinct sequences are compared at the least index at which they differ. This lexicographic order makes D_B a divisible ordered abelian group. Its product topology is Polish and zero-dimensional; for example,

$$d_D(a, b) = 2^{-\min\{j: a_j \neq b_j\}} \quad (a \neq b), \quad d_D(a, a) = 0$$

is a compatible complete metric.

The sets

$$C_{k,q} = \{a : a_0 = \dots = a_{k-1} = 0, a_k = q\}, \quad k \in \mathbb{N}, q \in \mathbb{Q}_{>0},$$

are clopen cylinders, and

$$(D_B)_{>0} = \bigsqcup_{k,q>0} C_{k,q}. \quad (6.1)$$

No restriction is placed on later coordinates. Each cylinder is homeomorphic to $\mathbb{Q}^{\mathbb{N}}$. The nonnegative cone $(D_B)_{\geq 0}$ is closed, since the negative cone is open. [theorem 5.2](#) already proves that

$$M_B = (D_B \overrightarrow{\times} \mathbb{Z})_{\geq 0}$$

is Polish. We now identify its homeomorphism type without relying on a space-classification theorem.

6.1 The zero branch and its explicit tree code

Fix computable bijections

$$e : \mathbb{N} \rightarrow \mathbb{Q}, \quad e_+ : \mathbb{N} \rightarrow \mathbb{Q}_{>0}, \quad e_{\geq 0} : \mathbb{N} \rightarrow \mathbb{Q}_{\geq 0},$$

with computable inverses. A code $s \in \mathbb{N}^{\mathbb{N}}$ determines a sequence $a \in (D_B)_{\geq 0}$ as follows. As long as all previously chosen coefficients are zero, set $a_j = e_{\geq 0}(s_j)$. Once the first positive coefficient has appeared, use $a_j = e(s_j)$ at all later stages. This is a bijection: the inverse uses the corresponding enumeration at each step. A finite output prefix depends on exactly a finite input prefix, in both directions. Thus it is a homeomorphism

$$\mathbb{N}^{\mathbb{N}} \cong (D_B)_{\geq 0}. \quad (6.2)$$

The all-zero rational sequence is not omitted; it is the branch that continues choosing zero indefinitely.

Theorem 6.1 (Baire-space carrier). *There is an explicit homeomorphism $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$. In particular M_B is uncountable, has no isolated points, and is nowhere locally compact.*

Proof. Partition M_B into the following countably many clopen blocks. For each $n \geq 0$, take the entire slice $(D_B)_{\geq 0} \times \{n\}$. For $n < 0$, partition that slice further into $C_{k,q} \times \{n\}$ using (6.1). By (6.2), every nonnegative slice is homeomorphic to Baire space; every negative-slice block is so by coding its unrestricted rational tail. A countable disjoint sum of copies of $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to $\mathbb{N}^{\mathbb{N}}$: use the first natural coordinate to choose a block and the remaining coordinates to choose a point in it.

For completeness, no nonempty basic cylinder of $\mathbb{N}^{\mathbb{N}}$ is relatively compact. Its projection to the next unrestricted discrete coordinate has infinite image, whereas the continuous image of a compact set in a discrete space is finite. Every neighborhood contains such a cylinder. The other two assertions follow from the elementary properties of the product $\mathbb{N}^{\mathbb{N}}$. $\square$

6.2 A total code for every sequence of natural numbers

Here is a fully specified block indexing, also used in the accompanying program. Let $\langle a, b \rangle$ be Cantor's pairing function. The first digit of a Baire code is a tag. An even tag $2n$ specifies the nonnegative integer coordinate n and the tree code (6.2). An odd tag

$$2\langle a, \langle k, b \rangle \rangle + 1 \quad (6.3)$$

specifies integer coordinate $-a - 1$, first positive rational coordinate k , and value $e_+(b)$ there. The remaining digits code the unrestricted rational tail using e .

Every natural tag is of exactly one of these forms. Every infinite natural sequence therefore codes a valid element; no hidden domain test is required. Conversely, a point with negative integer coordinate has a first positive rational coefficient, so its tag and tail are uniquely recoverable. This describes h and h^{-1} explicitly.

Theorem 6.2 (Computable continuous arithmetic). *Under this presentation, the operations*

$$x + y, \quad x + \mathbf{1}, \quad q_m(x) = \lfloor x/m \rfloor, \quad r_m(x) = x \bmod m$$

are Type-2 computable for every positive standard integer m . Here $q_m(x)$ is the unique quotient in the model's Euclidean division by m , not a real-valued floor operation. The remainders have their finite discrete topology.

Proof. A Type-2 computation reads finitely many input digits to produce each requested output digit. Decode the integer coordinates n, z of the two inputs. Each rational coefficient $a_j + b_j$ is computable from finite prefixes. If $n + z \geq 0$, output tag $2(n + z)$ and encode the coefficient sequence by the nonnegative-cone tree. The tree decisions only inspect finite prefixes.

If $n + z < 0$, at least one input has a negative integer coordinate and hence a strictly positive coefficient sequence. Its tag supplies an index k of a positive leading coefficient. Since both coefficient sequences are lex-nonnegative, their sum is strictly positive and its first nonzero coordinate occurs at or before k . Search these finitely many coordinates, output the negative tag (6.3), and then encode the tail. Thus the tag itself is computable; the algorithm does not attempt to decide whether an arbitrary sequence is identically zero.

For division use

$$q_m(a, n) = (a/m, \lfloor n/m \rfloor), \quad r_m(a, n) = n \bmod m.$$

If the quotient's integer coordinate is negative, the input integer coordinate was negative as well, and the same known leading index can be used. Otherwise the nonnegative tree code applies. Constants and successor are special cases of these finite-prefix procedures. $\square$

The adjective "computable" here concerns operations on infinite sequence codes. It does not assert that every element is computable, or that equality and order of arbitrary streams can be decided in finite time. The distinction will be quantified next.

6.3 Second route [04]: explicit charts

Source 04 builds the same space: its M_ω is M_B , with the same group $\mathbb{Q}^{\mathbb{N}}$, the same discrete coordinates and the same lexicographic order. Its charts differ from the tag coding of [theorem 6.1](#): they go from the model to Baire space and code the nonnegative and the negative integer slices separately. Source 04's homeomorphism H , its charts $H_{\geq}$, $H_{>}$ and its integer coding h are written Ψ , $\Psi_{\geq}$, $\Psi_{>}$ and ζ here ([section 1.3](#)).

6.3.1 The lexicographic rational product

Let

$$D_B = \mathbb{Q}^{\mathbb{N}}$$

with coordinatewise addition, the product topology in which *each copy of $\mathbb{Q}$ is discrete*, and lexicographic order by the first unequal coordinate. A nonzero sequence has a least nonzero index, so the order is defined on all sequences, not merely on finite-support ones. It is a linear order compatible with addition. Division by standard integers is coordinatewise.

The metric

$$d(u, v) = \begin{cases} 0, & u = v, \\ 2^{-k}, & k = \min\{i : u_i \neq v_i\}, \end{cases}$$

is a complete compatible ultrametric. Finite-support rational sequences form a countable dense set. The positive cone is

$$(D_B)_{>0} = \bigcup_{k \in \mathbb{N}} \bigcup_{q \in \mathbb{Q}_{>0}} \{d : d_0 = \dots = d_{k-1} = 0, d_k = q\},$$

an open union of clopen cylinders. Hence D_B is admissible. Write

$$M_B = M_{D_B}.$$

It is already an uncountable Polish Presburger model with continuous addition. We now identify its space explicitly with $\mathbb{N}^{\mathbb{N}}$.

6.3.2 Two elementary coding homeomorphisms

Fix bijections

$$c : \mathbb{Q} \longrightarrow \mathbb{N}, \quad b : \mathbb{Q}_{>0} \longrightarrow \mathbb{N}_{>0}, \quad a : \mathbb{N} \times \mathbb{Q}_{>0} \longrightarrow \mathbb{N}, \quad \zeta : \mathbb{Z} \longrightarrow \mathbb{N}.$$

Since their domains and codomains are discrete, these are homeomorphisms between discrete spaces.

For $d \in (D_B)_{\geq 0}$ define $\Psi_{\geq}(d) \in \mathbb{N}^{\mathbb{N}}$ as follows. Send the zero sequence to the zero sequence. If k is the first nonzero index of d , then $d_k > 0$, and set

$$\Psi_{\geq}(d)_i = \begin{cases} 0, & i < k, \\ b(d_k), & i = k, \\ c(d_i), & i > k. \end{cases} \quad (6.4)$$

Every nonzero element of $\mathbb{N}^{\mathbb{N}}$ has a first positive entry, so this coding has an evident inverse. Both it and its inverse preserve information in finite initial segments: to determine the first n coordinates, only the first n coordinates are needed. It is therefore a homeomorphism, including at the zero sequence.

For $d \in (D_B)_{>0}$, let k again be its first nonzero index and put

$$\Psi_{>}(d)_0 = a(k, d_k), \quad \Psi_{>}(d)_{j+1} = c(d_{k+1+j}) \quad (j \geq 0). \quad (6.5)$$

The pair (k, d_k) identifies a clopen positive cylinder; within that cylinder the tail is an arbitrary rational sequence. This is another homeomorphism onto $\mathbb{N}^{\mathbb{N}}$.

Theorem 6.3 (A concrete Baire homeomorphism [04]). *The map*

$$\Psi : M_B \longrightarrow \mathbb{N}^{\mathbb{N}}, \quad \Psi(d, z) = \begin{cases} \zeta(z) \frown \Psi_{\geq}(d), & z \geq 0, \\ \zeta(z) \frown \Psi_{>}(d), & z < 0, \end{cases} \quad (6.6)$$

is a homeomorphism. Here $\zeta(z) \frown u$ means prepend the one entry $\zeta(z)$ to the sequence u .

Proof. The integer slices of M_B are clopen. Their images are exactly the clopen cylinders of $\mathbb{N}^{\mathbb{N}}$ specified by the first coordinate. On each slice the map is one of the two homeomorphisms just proved. These slice maps assemble to a bijection whose map and inverse are continuous. $\square$

Transporting addition by Ψ gives a binary operation on the *literal* set $\mathbb{N}^{\mathbb{N}}$:

$$u \oplus v = \Psi(\Psi^{-1}(u) + \Psi^{-1}(v)). \quad (6.7)$$

This proves the Baire-space assertion of [remark 3.2](#); it does not merely produce a Borel-isomorphic universe. The space is perfect and nowhere locally compact. For the latter assertion, every nonempty basic cylinder in $\mathbb{N}^{\mathbb{N}}$ has infinitely many disjoint next-coordinate cylinders covering it, with no finite subcover. If a compact neighborhood existed, a closed basic cylinder inside it would be compact, a contradiction.

6.3.3 Effective operations on infinite streams

A map on Baire space is *Type-2 computable* here if an algorithm, given oracle access to the input sequences, returns every requested output coordinate after reading finitely many input coordinates. The number of queries may depend on the input. This is a statement about computations with infinite input oracles, not an enumeration of an uncountable model by ordinary natural-number indices.

Choose the bijections above effectively. For example, index positive rationals by the binary tree starting at $1/1$, with children

$$\frac{p}{q} \mapsto \frac{p}{p+q}, \quad \frac{p+q}{q}.$$

Every positive rational in lowest terms has a unique finite path: repeatedly subtract the smaller numerator or denominator from the larger. The sum strictly decreases until $(1, 1)$ is reached. Recording whether the larger entry was the numerator or denominator and reversing the record recovers the path. This proves both bijectivity and effective invertibility. Enumerate $\mathbb{Q}$ by interleaving positive and negative rationals and adding zero. Combine this enumeration with an effective pairing of two natural numbers to obtain a and with a signed-integer coding to obtain ζ .

Theorem 6.4 (Effective strengthening [04]). *The homeomorphism Ψ and its inverse can be chosen Type-2 computable. With this choice, the addition $\oplus$ in (6.7), and quotient and remainder by each fixed positive integer, are Type-2 computable on $\mathbb{N}^{\mathbb{N}}$.*

Proof. For $\Psi_{\geq}$, a requested coordinate i requires only the first $i + 1$ coefficients: either all are zero, the first nonzero is at i , or it occurred earlier. These are precisely the three cases in (6.4). The same finite-prefix procedure computes its inverse. No decision whether an entire infinite sequence is zero is made.

For $\Psi_{>}$, search for the first nonzero coefficient. The search terminates because this chart has a positive, not merely nonnegative, domain. Once found, all output coordinates are

computable; its inverse reads the encoded pair in the first coordinate and then the tail. Reading the integer coordinate chooses the appropriate chart of Ψ and Ψ^{-1} .

Coordinatewise rational addition and division are computable, as are integer addition, quotient, and remainder. Compose these algorithms with Ψ and Ψ^{-1} . Whenever the output has negative integer coordinate its D_B -coordinate is strictly positive, so the positive-chart search is guaranteed to terminate. For addition, for instance, a negative sum of integer coordinates requires at least one negative input integer coordinate; that input has strictly positive D_B -coordinate, and the other has nonnegative D_B -coordinate. Their sum is strictly positive. $\square$

The companion `verification.py` implements these maps as memoized indexed streams. A caller supplies a function returning the i th natural number of a sequence. The module exports `baire_add`, `baire_quotient`, and `baire_remainder`. Its round-trip checks include long initial zero blocks, which exercise the boundary chart at the zero coefficient sequence.

7 Definability, effective truth, and the optimal order rank

A subset of a metrizable space is Δ_2^0 when it is both F_σ and G_δ . Finite Boolean combinations of such sets are again Δ_2^0 . Closed sets, open sets, and clopen sets belong to this class.

Theorem 7.1 (All definable relations have low Borel rank). *Under the hypotheses of [theorem 5.2](#), every first-order definable subset of M_D^k , with parameters, is Δ_2^0 .*

Proof. In $G_D = D \times \mathbb{Z}$, the strict positive cone is

$$(D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{>0}).$$

It is Δ_2^0 , as is its complement. Affine group terms are continuous maps to G_D , so their inequalities have Δ_2^0 truth sets. Equalities have closed truth sets because the ambient group is Hausdorff.

Divisibility is especially simple. Since D is divisible,

$$mG_D = D \times m\mathbb{Z}. \quad (7.1)$$

This is clopen. Thus congruence atoms have clopen truth sets, including when their affine terms contain parameters. Translate a cone formula into the group and apply [theorem 4.9](#). The result is a finite Boolean combination of these atoms. Finally restrict to M_D^k . $\square$

Second route [04]. Source 04 proves [theorem 7.1](#) for its admissible groups ([section 5.2](#)), a special case of the hypotheses of [theorem 5.2](#), by an explicit description of the order relation:

Theorem 7.2 (Low complexity of all definable relations [04]). *For every admissible D , every first-order definable relation with parameters on M_D , in every finite arity, is simultaneously F_σ and G_δ .*

Proof. It suffices by [theorem 4.10](#) to consider affine atoms. Affine group maps are continuous, so an affine equality defines a closed set. A congruence condition depends only on finitely many discrete integer coordinates and is clopen, by [corollary 4.3](#).

For the order, write two group elements as $(d, z), (e, w)$. The strict comparison is

$$(d, z) < (e, w) \iff e - d > 0 \quad \text{or} \quad (e = d \text{ and } z < w). \quad (7.2)$$

The first condition is open; the second is closed because equality of D -coordinates is closed and the integer comparison is clopen. Thus the order relation is F_σ . Its complement has the identical form with the opposite strict comparison and weak integer comparison, hence is also F_σ . Pulling back by affine maps preserves both classes. Finite Boolean combinations of Δ_2^0 sets remain Δ_2^0 , which proves the claim. $\square$

This is stronger than merely obtaining a Borel presentation. The proof also explains why quantification does not raise the complexity in this particular presentation: its apparent projections are replaced by finite residue searches and finite Boolean operations.

7.1 A formula-by-formula bound on changes of mind

Theorem 7.3 (Effective finite-mind-change truth). *For each first-order Presburger formula $\varphi(\bar{x}; \bar{p})$, one can compute a natural number K_φ and an algorithm producing*

$$A_s(\bar{x}; \bar{p}) \in \{0, 1\} \quad (s \in \mathbb{N})$$

from finite prefixes of the Baire codes, such that

$$\lim_s A_s(\bar{x}; \bar{p}) = \mathbf{1}_{M_B \models \varphi(\bar{x}; \bar{p})}.$$

For every input tuple the displayed sequence changes value at most K_φ times. Parameter codes are supplied as oracles; they are not assumed computable.

Proof. Compute a quantifier-free group formula equivalent to the relativized cone formula, by [theorem 4.9](#). An affine group term has the form (c, n) , where n is an exactly computable integer and each rational coefficient c_j is computable from finite prefixes of the inputs.

To approximate the sign of (c, n) , first use the sign of n . At stage s , inspect $c_0, \dots, c_{s-1}$. If a nonzero coefficient has been found, use the sign of the first such coefficient instead. If $c = 0$, the initial answer was correct. Otherwise the answer becomes correct when the first nonzero coefficient is reached and never changes again. An equality atom can be handled in the same way: it starts with the answer determined by n and becomes permanently false if a nonzero coefficient is found. A congruence atom is decided immediately from n by [\(7.1\)](#).

Evaluate the fixed Boolean combination of these approximations at each stage. Its value can change only when an equality or inequality atom changes. The number of those atom occurrences is a computable bound K_φ . Each stage reads only finitely many digits. The truth set is effectively F_σ because it is

$$\bigcup_N \bigcap_{s \geq N} \{\bar{x} : A_s(\bar{x}; \bar{p}) = 1\},$$

where the stage sets are effectively clopen relative to the parameter codes. The same expression with 0 proves effective G_δ status. $\square$

No effective stopping time is asserted: a later rational coordinate may be the first nonzero one. Deciding sentences and deciding arbitrary parameter instances are different problems. The former is ordinary Presburger decidability; the latter is represented here by a convergent, uniformly bounded-change process.

7.2 Why rank one is impossible for the order

Theorem 7.4 (Open or closed arithmetic order forces discreteness). *Let X carry a model of Presburger arithmetic and a topology in which successor $x \mapsto x + 1$ is continuous. If either of the relations $<$ or $\leq$ is open or closed in X^2 , then X is discrete. Consequently, in any uncountable Polish model with continuous addition, the arithmetic order is neither open nor closed.*

Proof. If $\leq$ is open, then for fixed a both sets $\{x : x \leq a\}$ and $\{x : a \leq x\}$ are open. Their intersection is $\{a\}$, so every point is isolated.

If $<$ is closed, taking the complement and reversing coordinates makes $\leq$ open. If $<$ is open, the discrete-order equivalence

$$x \leq y \iff x < y + 1$$

and continuity of successor make $\leq$ open. Finally, if $\leq$ is closed, then

$$x < y \iff x + 1 \leq y$$

makes $<$ closed, reducing to the previous case. An uncountable discrete space is not separable, hence not Polish. $\square$

Together, [theorems 7.1](#) and [7.4](#) make the Δ_2^0 order bound optimal. This is a statement about the order relation, not about every individual definable set: many congruence sets are clopen. It also rules out a continuous Boolean-valued order test on the Baire model, despite its computable addition.

7.3 A general obstruction from closed order [04]

In a Presburger model M , define truncated subtraction by

$$x \dot{-} y = \begin{cases} z, & y + z = x \text{ for some } z \in M, \\ 0, & x < y. \end{cases}$$

The witness in the first case is unique. Equivalently this is $\max\{x - y, 0\}$ in the ordered group completion.

Theorem 7.5 (Closed order forces discreteness [04]). *Let a model of $\text{Pr}_{\mathbb{N}}$ carry a Hausdorff topology. If its weak order relation is closed in M^2 , then every point of M is isolated. If the topology is also separable, then M is countable.*

Proof. By totality, the complement of the converse weak order is strict order. Thus $<$ is open in M^2 . Fixing either argument gives open order rays. For $x > 0$, let $x - 1$ be its predecessor. Discreteness of the arithmetic order gives

$$\{x\} = \{y : x - 1 < y < x + 1\}.$$

At zero one has $\{0\} = \{y : y < 1\}$. Each singleton is open, so the topology is discrete. A dense subset of a discrete space must contain every point; separability therefore implies countability. $\square$

Corollary 7.6 (No continuous monus on an uncountable separable model [04]). *If truncated subtraction on a Hausdorff Presburger model is jointly continuous, the topology is discrete. In particular no uncountable separable Hausdorff Presburger model has continuous truncated subtraction.*

Proof. The relation $x \leq y$ is exactly $x \dot{-} y = 0$. A singleton is closed in a Hausdorff space, so continuity would make weak order closed. Apply [theorem 7.5](#). $\square$

No continuity hypothesis on addition was needed for this obstruction. Consequently, extending the signature of our examples by their definable monus operation destroys the assertion that all named operations are continuous. Definitional equivalence of first-order theories does not by itself preserve this topological property.

Remark 7.7 (Two incomparable order obstructions [merge]). [Theorem 7.4](#) (source 05) and [theorem 7.5](#) (source 04) both show that the order of a nondiscrete model cannot be too simple, under incomparable hypotheses. 05 assumes that successor is continuous, assumes no separation axiom, and treats four cases ($<$ or $\leq$, open or closed). 04 assumes only that the weak order $\leq$ is closed and the topology Hausdorff, with no continuity at all. Neither implies the other: 04 covers a closed $\leq$ in a topology where successor is discontinuous, 05 covers open orders and topologies that are not Hausdorff. Both are therefore printed. For the metrizable models of this part, where successor is continuous, both give the same conclusion, and so does [corollary 7.6](#) for truncated subtraction, whose exact discontinuity locus is [theorem 8.3](#).

8 Which definable operations are continuous?

The coordinate topology makes several arithmetic operations exceptionally regular, but not every definable operation. This section identifies exact boundaries in the two explicit models.

Proposition 8.1 (Euclidean self-similarity). *In $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$, for each positive standard integer m the map*

$$x \mapsto (q_m(x), r_m(x)) \quad (8.1)$$

is a homeomorphism from the model onto $M \times \{0, \dots, m-1\}$, where the finite factor is discrete. Its inverse is $(q, r) \mapsto mq + r\mathbf{1}$.

Proof. Uniqueness and existence are [proposition 4.2](#). In both chosen coefficient topologies, division of d by m is continuous. The integer maps $n \mapsto \lfloor n/m \rfloor$ and $n \mapsto n \bmod m$ are continuous on a discrete space. Hence (8.1) is continuous into the product. Repeated addition and addition of a fixed standard numeral make its inverse continuous. $\square$

The same formula also explains why all standard congruences ignore the entire coefficient sequence. Two elements have the same remainder modulo every positive standard integer if and only if their integer coordinates agree. The forward direction uses the elementary fact that an ordinary integer divisible by every positive integer is zero. Their infinite coordinates can still be arbitrary and different.

Proposition 8.2 (The total predecessor has exactly one discontinuity). *Define*

$$p(0) = 0, \quad p(x) = x - \mathbf{1} \quad (x > 0).$$

In either explicit model, p is continuous exactly at the points other than 0.

Proof. Away from 0, this is a restriction of translation by $-\mathbf{1}$ in the ambient topological group. Since $\{0\}$ is closed, the nonzero domain is open, and translation has its image in the cone.

In the real model let $t_j = 1/j$; in the Baire model let $t_j = e_j$, the rational sequence with 1 in coordinate j and zeros elsewhere. In either case $t_j > 0$ and $t_j \rightarrow 0$ in D . Then

$$(t_j, 0) \rightarrow 0, \quad p(t_j, 0) = (t_j, -1).$$

The latter sequence does not converge to $p(0) = 0$, because its integer coordinate is always -1 and integer coordinates are discrete. $\square$

Source 04 proves [proposition 8.2](#) (its Proposition 7.4), treating the numerals $(0, n)$, $n \geq 1$, separately:

Second proof (source 04). At points with positive D -coordinate, $p(d, z) = (d, z - 1)$ locally. At an ordinary numeral $(0, n)$ with $n \geq 1$, the same formula holds on its integer-coordinate slice and maps into M_D , so it is continuous there. At zero, the sequence $(s_j, 0) \rightarrow 0$ used above has $p(s_j, 0) = (s_j, -1)$, which cannot converge to zero. $\square$

Theorem 8.3 (Exact discontinuity locus of truncated subtraction). *For $M = M_{\mathbb{R}}$ or $M_{\mathbb{B}}$, let*

$$T(x, y) = \max\{x - y, 0\}$$

using the arithmetic order in the ambient group. Its discontinuity locus is exactly

$$\{((d, n), (d, z)) \in M^2 : n \neq z\}. \quad (8.2)$$

Proof. Both positive and negative cones in the coefficient group D are open for our two examples. If the coefficient difference $d - e$ is nonzero, its sign is locally fixed. Thus T locally agrees either with the continuous ambient difference or with the constant zero map.

Suppose instead that $x = y$. Along inputs sufficiently near this pair, the integer difference is zero. The output is either their ambient difference or zero. For any neighborhood V of zero in the ambient group, both possible outputs lie in V when the inputs are sufficiently near (x, x) . This proves continuity there.

It remains to consider equal coefficient coordinates and $k = n - z \neq 0$. Use the positive null sequence t_j from [proposition 8.2](#). If $k > 0$, replace y by $y + (t_j, 0)$. These inputs converge to (x, y) , but their truncated differences are zero, whereas $T(x, y) = (0, k)$. If $k < 0$, replace x by $x + (t_j, 0)$. The outputs are (t_j, k) , which cannot converge to the original value 0 because their integer coordinate is the nonzero integer k . All perturbed inputs remain in the positive cone. This proves exactly (8.2). $\square$

Source 04 states [theorem 8.3](#) as its Theorem 7.3, for the same two models and with the same locus, written $\text{Disc}(\dot{-}) = \{((d, z), (e, w)) : d = e \text{ and } z \neq w\}$, and adds [04]: *this locus is a closed nowhere dense subset of M^2 , and truncated subtraction is continuous on its dense open complement*. Its proof covers the addition.

Second proof (source 04), with the nowhere-density clause. Suppose first that $d \neq e$. The sign of $d - e$ is locally constant because both strict cones of D are open. On a neighborhood the monus operation is therefore either the continuous ambient difference or the constant zero map. Its values lie in the positive cone, so it is continuous as a map to that subspace.

Next let $d = e$ and $z = w$. In a sufficiently small neighborhood the two integer coordinates stay equal to these fixed integers. Monus has integer coordinate zero and D -coordinate $\max\{u - v, 0\}$. This positive-part map is continuous at zero in both groups under consideration. Over $\mathbb{R}$ this follows from the usual absolute-value estimate. Over $D_{\mathbb{B}}$, if $u - v$ has its first n coordinates zero, so does its positive part: it is either $u - v$ or zero. These finite-prefix neighborhoods prove continuity at the point.

For the converse, choose $s_j > 0$ in D with $s_j \rightarrow 0$: use $1/j$ in $\mathbb{R}$ or the j th unit vector in $D_{\mathbb{B}}$. Let $x = (d, z)$, $y = (d, w)$ and $z \neq w$. If $z < w$, replace x by $x_j = (d + s_j, z) \in M_D$. Then $x_j \rightarrow x$ but

$$x_j \dot{-} y = (s_j, z - w), \quad x \dot{-} y = 0.$$

The output integer coordinate is constantly negative and cannot converge to the zero integer coordinate. If $z > w$, instead replace y by $y_j = (d + s_j, w)$. Then $x \dot{-} y_j = 0$ for every j , whereas $x \dot{-} y = (0, z - w) \neq 0$. This proves discontinuity in both cases, including at the boundary $d = 0$.

The set in (8.2) is closed: equality of the D -coordinates is closed and inequality of the discrete integer coordinates is clopen. It has empty interior. At any of its points, perturbing the first D -coordinate upward by a sufficiently small positive element stays in M_D^2 , lies in any prescribed neighborhood, and makes the two D -coordinates unequal. Thus the closed set is nowhere dense. $\square$

The standard numerals form a closed, relatively discrete subset of both models, yet zero is not isolated in the whole space. This causes no conflict with [proposition 8.2](#). Nor does continuity of standard division imply continuity of every definable piecewise operation: the branch boundary is decisive.

Source 04 continues [04]:

The failure is not an indiscriminate loss of regularity: the graph of monus is still Δ_2^0 by [theorem 7.2](#), because monus is definable. Its ordinary predecessor specialization has a smaller exceptional set.

8.1 Polish completeness need not be translation-invariant [04]

Proposition 8.4 (A complete-metric obstruction [04]). *If zero is nonisolated in a metrizable Presburger model, there is no compatible complete metric invariant under all additive translations. More generally, for any compatible complete metric, the successor map cannot be a uniform homeomorphism onto its image.*

Proof. Successor is a bijection from the model onto $M \setminus \{0\}$. If it were a uniform homeomorphism onto this subspace, every Cauchy sequence in the image would pull back to a Cauchy sequence in the complete domain and would therefore converge in the image. Thus the image would be a complete subspace of a metric space, hence closed. Since zero is nonisolated, $M \setminus \{0\}$ is not closed, a contradiction. A translation-invariant metric would make successor an isometry, which is a special case. $\square$

This applies to both main examples. Their Polishness and continuous addition therefore cannot be replaced by a claim of completeness in the uniform structure of an invariant additive metric.

Structure or operation	Real-coordinate model	Baire-coordinate model
Underlying space	Perfect, locally compact, Polish	Homeomorphic to $\mathbb{N}^{\mathbb{N}}$
Addition	Continuous	Type-2 computable
Fixed quotient/remainder	Continuous	Type-2 computable
All definable relations	Δ_2^0	Δ_2^0
Monus	Discontinuous exactly on (8.2)	Same exact locus
Total predecessor	Discontinuous only at 0	Discontinuous only at 0
Arithmetic order topology	Discrete; not used	Discrete; not used

9 Explicit set-sized fragments of the omnific integers

We use Conway normal-form notation

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma,$$

where S is reverse well-ordered and all coefficients are real. In the omnific ring the occurring exponents are nonnegative, and the coefficient at exponent zero is an ordinary integer. This is the interface recorded in the repository's surreal development [16]. Addition is coefficientwise, and a nonzero normal form has the sign of its leading coefficient.

Theorem 9.1 (One-scale and countable-scale realizations). *There are injective ordered additive maps, fixing 0 and 1,*

$$\iota_{\mathbb{R}} : \mathbb{R} \overrightarrow{\times} \mathbb{Z} \longrightarrow \mathbf{Oz}, \quad \iota_{\mathbb{R}}(r, n) = r\omega + n, \quad (9.1)$$

$$\iota_{\mathbb{B}} : \mathbb{Q}^{\mathbb{N}} \overrightarrow{\times} \mathbb{Z} \longrightarrow \mathbf{Oz}, \quad \iota_{\mathbb{B}}(a, n) = \sum_{j=0}^{\infty} a_j \omega^{1/(j+1)} + n. \quad (9.2)$$

Their positive images, equipped with the transported coefficient topologies, are respectively the Polish models $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$.

Proof. The first formula is a finite normal form. For the second, the fixed exponents

$$1, \frac{1}{2}, \frac{1}{3}, \dots$$

strictly decrease and are positive. Every nonempty subset of these exponents has a largest element, namely the one with least index. Thus every possible support in (9.2) is reverse well-ordered, and adjoining exponent zero preserves that property. The expression is a legitimate normal form, even when infinitely many coefficients are nonzero. All positive-exponent coefficients are real, and the constant term is the integer n , so the value is omnific.

Uniqueness of normal forms gives injectivity. Addition of two such expressions is coordinatewise on a common fixed support, giving additivity. The leading nonzero coefficient is the first nonzero a_j ; if all are zero, the sign is that of n . Therefore the map preserves and reflects the lexicographic order. The same reasoning proves the one-scale case. Transporting a topology through a bijection onto its image preserves all the asserted topological properties. $\square$

9.1 What this realization does and does not preserve

These are ordered additive embeddings, *not* ring embeddings. Both images contain ω , but neither contains ω^2 . Their multiplicative nonclosure is therefore immediate from the normal forms. The stronger theorem in [section 10](#) will show that replacing surreal multiplication by some other multiplication cannot repair the problem while retaining the same carrier, addition, and unit.

The topologies in [theorem 9.1](#) are precisely specified coefficient topologies. The real coordinate is Euclidean in (9.1); the rational coordinates are discrete and combined by a product topology in (9.2). Neither is the discrete arithmetic order topology. No assertion that they are a canonical topology on all omnific integers is intended.

The ordinary constant-term map becomes simply

$$\text{ct}(\iota(d, n)) = n.$$

It is continuous into the discrete integers for these models. Standard congruence data see only this coordinate, matching (7.1). The standard part itself remains externally specified and nondefinable in the Presburger structure.

9.2 Why no proper-class difficulty arises

The source sets $\mathbb{R} \times \mathbb{Z}$ and $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ are ordinary sets. The normal-form constructions above are definable assignments on those sets. Replacement therefore gives set-sized images. The set of birthdays of their elements has an ordinal supremum; choosing a larger ordinal places both images inside a set-sized birthday cutoff of **No**. No explicit optimal cutoff is needed for any theorem here.

Accordingly, one can conduct the entire argument in the product-group models and use [theorem 9.1](#) only as a final realization theorem. This is also the most economical formalization strategy. It avoids making a class-sized object the carrier of a Polish space or a first-order structure in ordinary set-based semantics.

Source 04's account [04]. Source 04 proves [theorem 9.1](#) as its Proposition 10.1 with the same proof; its second map is written ι_{ω} there and ι_B here. It adds the following remarks on the normal-form rule and on the transported topologies.

The construction is purely set-sized, but it has a natural expression in the surreal normal-form language. We use only the classical normal-form rule: a reverse-well-ordered set of surreal exponents with real coefficients defines a surreal sum; addition is coefficientwise, and order is decided by the greatest exponent with nonzero coefficient. Omnific integers have positive growth exponents and an ordinary integer constant term. This is the normal-form description relevant to the ordered-group study in [5] and to the repository's omnific development [19].

We transfer the topologies from the coefficient models to these images. This is not a claim that the whole proper class **Oz** has a Polish topology, nor that a canonical global surreal topology is being restricted. For example, ω/j converges to zero in the transported real-coefficient topology on the first image, despite each term being arithmetically infinite. Likewise $\omega^{1/(k+1)}$ converges to zero in the second coefficient topology.

These are additive subgroups, not subrings. Squaring ω yields ω^2 , which belongs to neither displayed image. The absence of inherited multiplicative closure is consistent with the stronger nonexistence result [corollary 10.5](#) for *any* compatible semiring multiplication on these positive cones.

For foundations, the domains $\mathbb{R} \times \mathbb{Z}$ and $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ are sets. One may work entirely with their normal-form codes in ZFC and interpret the notation inside a sufficiently large set-sized portion of the surreal construction. Alternatively one may use a class theory for the ambient surreal field while retaining these set-sized domains. None of the proofs quantifies over a set of all surreals, a set of all class maps, or a Polish space whose points form a proper class.

10 An absolute obstruction to multiplication

A Borel obstruction would not rule out a wildly discontinuous multiplication. For the models constructed here, the obstruction is stronger and purely algebraic.

Definition 10.1. In a Presburger model M , a positive element u is a *cofinal infinite additive unit* if

$$u > n\mathbf{1} \quad \text{for every standard } n,$$

and for every $x \in M$ there is a positive standard integer N with $x \leq Nu$. The word “unit” here refers only to cofinality under repeated addition; it is not a multiplicative unit.

Source 04 splits the hypothesis of [theorem 10.4](#) into two notions and proves a positivity lemma first:

Definition 10.2 ([04]). An element h in an ordered additive monoid is an *order unit* if for every x there is a positive standard integer N with $x \leq Nh$. It is *arithmetically infinite* if $h > n\mathbf{1}$ for every standard n .

Lemma 10.3 (Positivity is automatic in a semiring expansion [04]). *Suppose a Presburger monoid is expanded to a unital semiring with the same addition, zero, and one. Then the product of two positive elements is positive, and multiplication by a positive element is strictly increasing on either side.*

Proof. If $x, y > 0$, write $x = 1 + a$, $y = 1 + b$ with $a, b \geq 0$. Distributivity and the unit laws give $xy = 1 + a + b + ab \geq 1$. If $x > y$, write $x = y + c$ with $c > 0$. For $z > 0$, one has $xz = yz + cz > yz$ and similarly on the other side. Here the order is the additive natural order of the Presburger model, so no separate order axiom for the multiplication has been assumed. $\square$

Theorem 10.4 (Cofinal-infinite-element obstruction). *A Presburger model with a cofinal infinite additive unit cannot be expanded to a unital semiring with the same addition, zero, and multiplicative unit $\mathbf{1}$. No continuity, measurability, commutativity of multiplication, or induction involving multiplication is assumed.*

Proof. Suppose such a multiplication exists. Cofinality applied to u^2 gives a positive standard N such that $u^2 \leq Nu$. Since u exceeds every standard numeral, it has a decomposition

$$u = (N + 1)\mathbf{1} + v \quad (v \in M).$$

Distributivity and the unit law give

$$u^2 = (N + 1)u + vu \geq (N + 1)u > Nu,$$

a contradiction. The first inequality uses only that vu belongs to the nonnegative carrier. The last one follows from $u > 0$ and additive cancellation. In any semiring expansion of this additive monoid, its natural order $x \leq y \leftrightarrow \exists z (x + z = y)$ is already the Presburger order, so no additional order-compatibility axiom was hidden in the argument. $\square$

Corollary 10.5. *Neither $M_{\mathbb{R}}$ nor $M_{\mathbb{B}}$ admits a unital semiring multiplication with the displayed addition and unit.*

Proof. In $M_{\mathbb{R}}$ take $u = (1, 0)$. It exceeds every standard numeral, and an integer $N > r$ gives $(r, n) < Nu$. In $M_{\mathbb{B}}$ take $u = (e_0, 0)$, where e_0 has first coefficient 1 and all later coefficients zero. For any (a, n) choose an integer $N > a_0$; comparison at coordinate zero gives $(a, n) < Nu$. Apply [theorem 10.4](#). $\square$

In their omnific realizations, this cofinal element is ω . The missing larger scale ω^2 is not merely a defect of one chosen multiplication: every possible semiring multiplication would have to produce an element growing faster than every standard multiple of ω , which the additive carrier cannot contain.

For a different uncountable Polish Presburger model that does allow a semiring expansion algebraically, Glazer's cited Borel obstruction for $Q + I\text{Open}$ would still matter. Our argument is not a replacement for that general theorem; it identifies a stronger obstruction in these particular additive structures.

An arithmetically infinite order unit in 04's sense is exactly a cofinal infinite additive unit in 05's sense, so [theorem 10.4](#) is 04's Theorem 9.3 and [corollary 10.5](#) its Corollary 9.4. Source 04's proof routes the strict inequality through [lemma 10.3](#):

Second proof of theorem 10.4 (source 04). Suppose such a multiplication exists and let h be the order unit. Choose a standard $N \geq 1$ with $h^2 \leq Nh$. Since $h > N1$, [lemma 10.3](#) gives $h^2 > Nh$, a contradiction. $\square$

This algebraic argument is specific to the supplied models and does not replace the general topological theorem about stronger arithmetic. Presburger models need not be additive reducts of models of Peano arithmetic.

10.1 A variation with no largest growth class [04]

Let $D_L = \mathbb{Q}((t))$ be the additive group of formal Laurent series

$$f = \sum_{k \geq k_0} a_k t^k, \quad k_0 \in \mathbb{Z}, \quad a_k \in \mathbb{Q},$$

ordered by the sign of the coefficient at the least nonzero exponent. Use the t -adic topology, with rational coefficients discrete. Its basic neighborhoods at zero are

$$F_n = \{f : v(f) \geq n\}, \quad n \in \mathbb{N},$$

where $v(0) = +\infty$. A bounded compatible metric is $\min\{1, 2^{-v(f-g)}\}$ for $f \neq g$.

Proposition 10.6 (A Laurent-series Polish construction [04]). *The group D_L is admissible, and $M_L = M_{D_L}$ is a Polish Presburger model homeomorphic to Baire space, with continuous addition and fixed quotient and remainder. Unlike the two main models, M_L has no arithmetically infinite order unit.*

Proof. A Cauchy sequence of Laurent series eventually has the same entire negative part, because its pairwise differences eventually lie in F_0 . That common negative part is finite. Every subsequent coefficient stabilizes in turn, giving a Laurent-series limit. Thus the metric is complete. Laurent polynomials with rational coefficients form a countable dense set. The leading sign is determined on a basic neighborhood, so the strict positive cone is open. Addition and division by fixed positive integers are coefficientwise continuous.

For the asserted homeomorphism, decompose D_L into the clopen cosets $p + \mathbb{Q}[[t]]$, where p is a finite negative-power rational polynomial. There are countably many such cosets, and each is a rational product. If $p \neq 0$, the sign on its coset is constant, determined by its leading coefficient. The coset $p = 0$ contributes respectively $(D_B)_{>0}$ or $(D_B)_{\geq 0}$ to the positive or nonnegative cone. Thus both Laurent cones are countable disjoint sums of spaces explicitly homeomorphic to $\mathbb{N}^{\mathbb{N}}$, by the codings in [section 6.3](#). Prepending an index codes such a sum by $\mathbb{N}^{\mathbb{N}}$. The integer slices of M_L are another such sum, proving the claim.

Finally, for $f > 0$ with least exponent k , the element $(t^{k-1}, 0)$ is larger than every standard multiple of (f, z) : its leading exponent is strictly earlier. Standard finite elements also fail to be order units. $\square$

The absence of an order unit does not establish the existence of a semiring expansion. In this example another simple obstruction decides that issue negatively.

10.1.1 A growth-order obstruction

For positive arithmetically infinite elements, define

$$x \asymp y \iff (\exists m, n \in \mathbb{N}_{>0})(x \leq my \wedge y \leq nx).$$

Let $\mathcal{A}(M)$ be the set of these Archimedean growth classes, ordered so that $[x] < [y]$ means $nx < y$ for every positive standard n . These are external set-sized classes of elements of our set-sized model; no first-order definability is being asserted.

Proposition 10.7 (Growth classes must embed into their strict final segments [04]). *If a Presburger monoid M has a unital semiring expansion, then for every arithmetically infinite h , the map*

$$[x] \mapsto [hx]$$

is an order embedding of $\mathcal{A}(M)$ into its strict final segment above $[h]$.

Proof. Multiplication by $h > 0$ preserves and reflects every comparison $x \leq ny$. Preservation follows from distributivity and positivity. For reflection, if $x > ny$ then strict monotonicity gives $hx > n(hy)$. It follows that multiplication induces an order embedding on growth classes. Because x is infinite, $x > n\mathbf{1}$ for every standard n , so $hx > nh$ for every n . Thus its class lies strictly above $[h]$. $\square$

Corollary 10.8 ([04]). *The Laurent-series model M_L also has no unital semiring expansion.*

Proof. Its infinite growth classes are determined exactly by the least Laurent exponent, with increasing growth corresponding to decreasing exponent. Hence $\mathcal{A}(M_L)$ has order type $\mathbb{Z}$. A strict final segment of $\mathbb{Z}$ is well-ordered, while $\mathbb{Z}$ contains an infinite strictly decreasing sequence. There is no order embedding of $\mathbb{Z}$ into such a final segment. Apply [proposition 10.7](#). $\square$

The obstruction is not a statement that every split Presburger group forbids multiplication. For example, take $D = X\mathbb{Q}[X]$ ordered by its highest-degree nonzero coefficient, and let the constant term be an ordinary integer. The ordered ring

$$\mathbb{Z} + X\mathbb{Q}[X]$$

has this split additive group, least positive element 1, and a positive cone closed under its polynomial multiplication. Its growth classes have order type $\mathbb{N}_{>0}$, which does embed in each

of its strict final segments. This countable comparison makes the scope of the obstruction explicit.

11 Endomorphisms, elementary submodels, and a contrast

The two models have markedly different self-embedding behavior. All homomorphisms in the next theorem preserve addition, zero, and **1**; continuity and injectivity are *not* assumed.

11.1 The real model: endomorphisms and elementary submodels

Theorem 11.1 (Rigidity of the real-coordinate model). *Every unital additive endomorphism $f : M_{\mathbb{R}} \rightarrow M_{\mathbb{R}}$ has the form*

$$f(r, n) = (cr, n) \quad (c \in \mathbb{R}_{>0}). \quad (11.1)$$

Conversely each such formula is an endomorphism. All of them are homeomorphisms and Presburger automorphisms. In particular $M_{\mathbb{R}}$ has no proper elementary self-embedding.

Proof. The group completion of the cancellative monoid $M_{\mathbb{R}}$ is $G = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$. Thus f extends uniquely to an additive group map $F : G \rightarrow G$ fixing **1**. Inside G ,

$$\bigcap_{m \geq 1} mG = \mathbb{R} \times \{0\}.$$

One inclusion follows from real divisibility; the other follows because the integer coordinate of an element of the intersection is divisible by every positive integer. Every additive group map preserves this intersection. Hence

$$F(r, 0) = (h(r), 0), \quad F(0, n) = (0, n)$$

for some additive map $h : \mathbb{R} \rightarrow \mathbb{R}$.

Since f maps the positive cone into itself, $h(r) \geq 0$ whenever $r \geq 0$. Thus h is monotone. If $c = h(1)$, rational comparisons and additivity give $h(r) = cr$ for every real r : sandwich r between rational numbers and use the corresponding inequalities for h . Here $c \geq 0$. But $c = 0$ would send the positive element $(1, -1)$ to $(0, -1)$, outside the cone. Thus $c > 0$.

Formula (11.1) and its inverse with scalar $1/c$ preserve the cone, addition, order, and unit, and are continuous in the product coordinates. This proves all assertions. $\square$

Theorem 11.2 (All elementary submodels of the real model). *The elementary submodels of $M_{\mathbb{R}}$ are exactly*

$$M_E = (E \overrightarrow{\times} \mathbb{Z})_{\geq 0}, \quad E \text{ a } \mathbb{Q}\text{-linear subspace of } \mathbb{R}. \quad (11.2)$$

Among them, the only ones Polish in the inherited topology are the standard copy $\mathbb{N}$ and the whole model $M_{\mathbb{R}}$. Every nonzero proper choice of E gives a dense, non-Polish elementary submodel.

Proof. If E is a $\mathbb{Q}$ -linear subspace, then $E \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group. Its inclusion in $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ preserves and reflects the expanded quantifier-free language, including all congruences, since a standard modulus depends only on the integer coordinate. Quantifier elimination makes this inclusion elementary. Relativizing to positive cones proves $M_E \preccurlyeq M_{\mathbb{R}}$.

Conversely, let $S \preccurlyeq M_{\mathbb{R}}$, and let $H = S - S$ be its group completion inside G . If $a, b \in S$ and $a - b \geq 0$, the unique solution of $b + x = a$ lies in S by elementarity. Therefore

$S = H \cap M_{\mathbb{R}}$. Since H contains every standard integer, subtracting the integer coordinate from any $(r, n) \in H$ shows that $H = E \times \mathbb{Z}$ for an additive subgroup $E \leq \mathbb{R}$. For $r \in E_{\geq 0}$, the model's unique division of $(r, 0)$ by a positive standard m has remainder zero and quotient $(r/m, 0)$. Elementarity puts that quotient in S . Applying the same observation to negatives in H shows E is divisible, hence $\mathbb{Q}$ -linear.

If $E \neq 0$, it contains all rational multiples of a nonzero real and is dense in $\mathbb{R}$. Thus M_E is dense in every slice of $M_{\mathbb{R}}$. Suppose M_E is also Polish in the inherited topology. Its closed slice of integer coordinate zero is homeomorphic to $E \cap [0, \infty)$. By [lemma 5.1](#), this is G_δ in $\mathbb{R}$. Its negative is also G_δ , so their union E is G_δ in $\mathbb{R}$. For any $r \in \mathbb{R}$, the dense G_δ sets E and $r + E$ intersect by the Baire category theorem. Hence $r \in E - E = E$, giving $E = \mathbb{R}$. Finally $E = 0$ gives the relatively discrete standard copy $\mathbb{N}$, which is Polish, and $E = \mathbb{R}$ gives the whole model. $\square$

For example, $E = \mathbb{Q}$ gives a countable dense elementary submodel. Its inherited topology is not Polish; there is no inconsistency between its density and the complete metric on the ambient model.

11.2 Reduction to the divisible part [04]

This section considers embeddings in the arithmetic language containing $0, 1, +$. Naming 1 is explicit. Order is already definable, so its preservation is not an extra independent assumption.

Proposition 11.3 (All unital additive maps [04]). *Let D, E be divisible linearly ordered abelian groups. Every map $f : M_D \rightarrow M_E$ preserving $0, 1, +$ is injective and has the unique form*

$$f(d, z) = (T(d), z), \quad (11.3)$$

where $T : D \rightarrow E$ is a $\mathbb{Q}$ -linear order embedding. Conversely every such T defines a map of this kind. Every one of these maps is an elementary embedding of Presburger models.

Proof. If $x > y$, then $x = y + 1 + u$ for some $u \in M_D$. Thus $f(x) = f(y) + 1 + f(u) > f(y)$, so f strictly preserves order and is injective. Every group element is a difference of nonnegative elements. The rule

$$\bar{f}(x - y) = f(x) - f(y)$$

therefore extends f uniquely to an additive order embedding of group completions. It is well defined because $x - y = x' - y'$ implies $x + y' = x' + y$, and f preserves that equality.

For each $d \in D$ and every standard $m \geq 1$, the element $(d, 0)$ is an m -multiple. Its image is consequently in every mG_E , whose intersection is $E \times \{0\}$ by (4.3). Hence $\bar{f}(d, 0) = (T(d), 0)$. Since $f(1) = 1$, the map fixes every standard integer, giving (11.3). Divisibility and torsion-freeness make T $\mathbb{Q}$ -linear; order preservation and injectivity are inherited. The converse follows directly from the formula.

Such a group map preserves and reflects order, equality, and all congruences, the last because integer coordinates are unchanged. Quantifier elimination in [theorem 4.10](#) makes it elementary on the groups. Relativizing to the nonnegative cones makes f elementary as well. $\square$

Source 04 proves [theorem 11.1](#) (its Theorem 8.2) from [proposition 11.3](#):

Second proof (source 04). By [proposition 11.3](#), it remains to classify increasing additive maps $T : \mathbb{R} \rightarrow \mathbb{R}$. Put $c = T(1) > 0$. For rational q one has $T(q) = cq$. For a real r , compare r with rational numbers from below and above and use monotonicity. The resulting inequalities force $T(r) = cr$. The inverse dilation is $f_{1/c}$, and both dilations are continuous in the specified product topology. $\square$

Here 1 in $T(1)$ denotes the real vector $1 \in D$, not the arithmetic unit $(0, 1)$. The distinction prevents a mistaken conclusion that c must equal one. The model is co-Hopfian for unital additive maps but is not rigid.

11.3 The Baire model: all self-embeddings [04] and clopen self-copies

Source 04 classifies every unital additive self-map of M_B ; source 05's shift theorem follows.

For $d \in D_B \setminus \{0\}$ define

$$v(d) = \min\{k : d_k \neq 0\}.$$

Let e_k be the rational sequence with a single 1 in coordinate k , and put

$$F_k = \{d : d_0 = \dots = d_{k-1} = 0\}.$$

Thus $F_0 = D_B$; the subgroups F_k form a neighborhood basis at zero.

Theorem 11.4 (Complete matrix classification [04]). *Every $\mathbb{Q}$ -linear order embedding $T : D_B \rightarrow D_B$, without any assumed continuity, has the form*

$$(Td)_i = \sum_{k:p(k) \leq i} a_{ik} d_k, \quad (11.4)$$

where $p : \mathbb{N} \rightarrow \mathbb{N}$ is strictly increasing and

$$a_{ik} = 0 \ (i < p(k)), \quad a_{p(k),k} > 0. \quad (11.5)$$

The coefficients below each pivot, a_{ik} with $i > p(k)$, are arbitrary rationals. Conversely every such matrix defines an order embedding. All the sums in (11.4) are finite. Every such map is continuous and has closed image. It is surjective exactly when $p(k) = k$ for all k .

Proof. Let T first be an arbitrary $\mathbb{Q}$ -linear order embedding. Since $Te_k > 0$, its first nonzero coefficient is positive. Define $p(k) = v(Te_k)$. The inequalities

$$ne_{k+1} < e_k \quad (n = 1, 2, \dots)$$

remain true after applying T . If $p(k+1) < p(k)$, the image on the left already dominates the right for $n = 1$. If the pivots were equal, a large enough standard n would make the positive leading rational coefficient on the left larger than the one on the right. Both are impossible. Thus $p(k+1) > p(k)$, and in particular $p(k) \geq k$.

Every $d \in F_k$ satisfies $|d| < Ne_k$ for some standard integer $N > 0$: choose $N > |d_k|$, with a further unit of margin. Order preservation implies $|Td| < NTe_k$, hence

$$T(F_k) \subseteq F_{p(k)} \subseteq F_k. \quad (11.6)$$

This already proves continuity at zero, and therefore everywhere by additivity. It also excludes noncontinuous linear maps arising from a choice of a vector-space basis.

Let $a_{ik} = (Te_k)_i$. To compute coordinate i of Td , subtract the finite initial sum $\sum_{k=0}^i d_k e_k$. The remainder belongs to F_{i+1} , whose image has zero coordinates through i by (11.6). Thus $(Td)_i = \sum_{k=0}^i a_{ik} d_k$. Terms with $p(k) > i$ vanish, yielding (11.4) and (11.5).

Conversely, a strictly increasing p satisfies $p(k) \geq k$, so each row in (11.4) has finite support. The formula defines a linear continuous map on all rational sequences. If $d \neq 0$ has first nonzero index k , then all rows before $p(k)$ vanish and

$$(Td)_{p(k)} = a_{p(k),k} d_k.$$

Its sign is the sign of d_k . This proves order preservation and injectivity, including comparison of arbitrary pairs by their difference.

To describe the range, for any proposed target sequence y recursively recover

$$d_k = \frac{y_{p(k)} - \sum_{j < k} a_{p(k),j} d_j}{a_{p(k),k}}. \quad (11.7)$$

Each recovered coordinate is a continuous function of finitely many coordinates of y . The target belongs to the range exactly when, for every nonpivot row i ,

$$y_i = \sum_{k:p(k) < i} a_{ik} d_k. \quad (11.8)$$

These are closed, indeed clopen, finite-coordinate conditions. Their intersection is closed. If every row is a pivot, the recovery solves the system for every target. This happens exactly when $p(k) = k$. If a row j is omitted, no image has its first nonzero coordinate equal to j , so surjectivity fails. $\square$

Corollary 11.5 (Automatic continuity and the topology of self-images [04]). *Every unital additive self-map of M_B is elementary, continuous, and a homeomorphism onto a closed image. Its image is clopen when the complement of $p(\mathbb{N})$ is finite, and closed nowhere dense when that complement is infinite. The automorphisms are exactly the infinite lower triangular rational matrices with strictly positive diagonal, acting on the divisible coordinate and fixing the integer coordinate.*

Proof. Combine proposition 11.3 and theorem 11.4. The inverse on the image is continuous by the coordinate-recovery formula. The group image is $T(D_B) \times \mathbb{Z}$, and its intersection with M_B is exactly the monoid image, because T reflects order.

When finitely many rows are not pivots, (11.8) gives a finite intersection of clopen conditions, so the group and monoid images are clopen. When infinitely many rows are not pivots, they occur arbitrarily far out. At an image point whose D_B -coordinate is positive, keep an initial segment containing its leading coordinate and change a later nonpivot coordinate so that its constraint fails. The result remains positive and lies in any sufficiently specified cylinder. At a point $(0, z)$ in the image, necessarily $z \geq 0$. In any given zero-prefix cylinder first insert a positive pivot far beyond that prefix, using a small-in-topology image vector Te_k . Then violate a still later nonpivot constraint. Positivity is unchanged and the resulting point is outside the image. Thus the image has empty interior even at the boundary. It is closed, hence nowhere dense. $\square$

Example 11.6 (Two contrasting proper elementary copies [04]). The shift

$$S(d)_0 = 0, \quad S(d)_{k+1} = d_k$$

has pivots $p(k) = k + 1$ and produces a proper clopen elementary self-copy of M_B . The insertion map

$$E(d)_{2k} = d_k, \quad E(d)_{2k+1} = 0$$

has pivots $p(k) = 2k$ and produces a closed nowhere dense elementary self-copy. Both are homeomorphisms onto their images. Their order-theoretic and model-theoretic behavior is identical at the level of elementary embedding, while their image interiors are different.

Theorem 11.7 (Clopen elementary self-copies of the Baire model). *Let*

$$\sigma((a_0, a_1, a_2, \dots), n) = ((0, a_0, a_1, a_2, \dots), n).$$

Then $\sigma : M_B \rightarrow M_B$ is a proper elementary embedding and a homeomorphism onto a clopen subspace. Its iterates satisfy

$$\bigcap_{k \geq 0} \sigma^k(M_B) = \{(0, n) : n \in \mathbb{N}\}.$$

Thus a descending sequence of proper clopen elementary self-copies has intersection exactly the standard elementary submodel.

Proof. Inserting an initial zero preserves and reflects the lexicographic order, addition, and every standard congruence. It maps the divisible coefficient group into the subgroup with first coordinate zero. Quantifier elimination gives elementarity, first for the groups and then for their cones. The image is exactly the subset with $a_0 = 0$, which is clopen because the first rational coordinate is discrete. Deleting that zero is a continuous inverse on the image. It is proper, since elements with $a_0 > 0$ are omitted. The k -th image has its first k rational coordinates zero. Membership in all images therefore means that every rational coordinate is zero; positivity then forces $n \geq 0$. $\square$

These clopen submodels are external subsets, not Presburger-definable subsets, even with parameters: a proper definable subset containing zero and closed under successor would violate induction. Topological simplicity and internal definability are different properties.

[merge] The map σ of [theorem 11.7](#) (source 05) is the shift S of [example 11.6](#) (source 04), with pivots $p(k) = k + 1$. Source 05 adds the intersection of the iterated images; source 04 adds the insertion map E , whose image is closed and nowhere dense.

12 A compactness obstruction independent of multiplication

The positive answer on Baire space does not extend to every familiar uncountable Polish space. In particular compact carriers are impossible. The underlying fact is the standard compact-cancellative-monoid argument, which we include in its commutative form.

Lemma 12.1 (Compact cancellation produces inverses). *Every compact Hausdorff commutative cancellative topological monoid is a group algebraically.*

Proof. Write the identity as 0 and fix a . Let S be the closure of $\{na : n \geq 1\}$. Continuity of addition makes S a nonempty compact subsemigroup. By the compact intersection property and Zorn's lemma, S has a minimal nonempty compact subsemigroup K .

For $x \in K$, the set $x + K$ is a nonempty compact subset of K and is a subsemigroup: for $u, v \in K$,

$$(x + u) + (x + v) = x + (x + u + v) \in x + K.$$

Minimality gives $x + K = K$. Since $x \in K$, there is $y \in K$ with $x + y = x = x + 0$, and cancellation gives $y = 0$. Thus $0 \in K \subseteq S$. But $a + X$ is compact and hence closed, and contains every positive multiple of a . Therefore it contains S and in particular 0 . So $a + b = 0$ for some $b \in X$. $\square$

Corollary 12.2 (No compact positive-cone model). *No model of Presburger arithmetic has a compact Hausdorff topology with continuous addition. In particular Cantor space cannot be a carrier, although Baire space can.*

Proof. Its addition is a commutative cancellative monoid. The lemma would give an additive inverse of $\mathbf{1}$, but no nonnegative element can sum with $\mathbf{1}$ to zero. $\square$

This obstruction uses neither Borel definability nor multiplication. It is separate from [theorem 10.4](#): the real model is locally compact but not compact, and the Baire model is nowhere locally compact.

13 Formalization and reproducibility

13.1 A minimal theorem dependency graph

The construction naturally splits into independently checkable layers. The core algebra does not require real analysis or surreal numbers. First define a discretely ordered abelian group with standard residue representatives. Prove the coset-invariant finite-search lemma. Then prove coefficient normalization, elimination, completeness, and the positive-cone interpretation. Only afterward supply the real or rational-sequence carrier and its topology. The omnific realization can be verified last, as an independent bridge.

A practical Lean development can use the following *proposed* modules. These names are a plan, not existing checked files:

Proposed module	Required mathematical content
<code>ZGroup.Basic</code>	Least positive element, standard residue witnesses, uniqueness, and order properties of the quotient.
<code>ZGroup.CosetCooper</code>	Full mG -invariance and the finite-search criterion; the counterexample to weaker periodicity.
<code>ZGroup.Elimination</code>	Affine-term semantics over an arbitrary $\mathbb{Z}$ -group, coefficient scaling, quantifier elimination.
<code>PolishPresburger.Real</code>	The positive cone of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ and its complete slice metric.
<code>PolishPresburger.Baire</code>	The rational-sequence cone and the explicit block/tree homeomorphism.
<code>PolishPresburger.Structure</code>	Order-rank obstruction, absence of semiring expansions, and self-embedding theorems.
<code>PolishPresburger.Omnific</code>	The fixed-support normal-form maps and their additive/order laws.

13.2 The nonstandard Cooper boundary must be explicit

The inspected repository theorem `periodic_interval` has an ordinary-integer predicate and a one-step periodicity hypothesis [15]. Its proof can iterate by any ordinary integer

quotient. Over a general $\mathbb{Z}$ -group, the quotient in $x - L = mq + r$ is an element of that group, not necessarily a standard numeral. A proof by ordinary natural-number induction cannot supply invariance under that shift for an arbitrary external predicate.

The correct interface is therefore a predicate on the quotient G/mG , or equivalently the hypothesis (4.4). Affine congruence atoms satisfy it by algebra, and Boolean combinations preserve it. This makes the generalized arithmetic lemma reusable and prevents an incorrect formal transfer from being hidden by notation. The original integer theorem should be retained as an instance; it does not need to be weakened or retracted.

The existing decision procedure's semantics are over integers. Reusing its syntax and transformations is plausible, but proving soundness over all $\mathbb{Z}$ -groups is an additional obligation. The repository's existence of an executable integer eliminator, by itself, does not certify [corollary 4.11](#) for a new carrier.

13.3 Executable material included with this manuscript

`polish_presburger.py` implements exact finite-rank rational arithmetic and the Baire codec of [section 6](#). The codec uses a computable positive-rational enumeration and Cantor pairing, supplies addition and standard division on infinite-sequence oracles, and gives a one-change comparison approximation. Every Baire input code is valid. The lower-level encoding routine takes a positive-cone promise, documented in its interface; the public arithmetic operations preserve that promise.

`verify.py` performs deterministic tests using exact fractions.Fraction arithmetic and a fixed seed. The executed run produced **17,930 successful checks**. They cover pairing and rational-enumeration round trips, additive and order identities, Euclidean division, cone closure, coset-invariant finite candidates, Baire-code prefix round trips, arithmetic output prefixes, the comparison change bound, and the predecessor boundary witness. The deliberately false weak-periodicity generalization is tested as a counterexample, not asserted as a theorem.

Reproduction

With Python 3.10 or later, run `python verify.py`. No external Python package is needed. The script writes `verification_results.json`. Compile the article twice with `pdflatex -interaction=nonstopmode -halt-on-error polish_presburger_glazer.tex` to resolve references.

Tests on finite-rank elements do not quantify over arbitrary elements of $\mathbb{Q}^{\mathbb{N}}$, and finite prefixes do not decide equality of infinite streams. Polishness, induction, and universal semantic statements are established by the proofs, not by the test count. No new Lean or Rocq formalization is included in this archive.

[write] The box above gives source 05's delivery names. In this report 05's programs are `code/05-omnific-presburger-polish_presburger.py` and `code/05-omnific-presburger-verify.py`, its recorded run is `data/05-omnific-presburger-verification_results.json`, and its manuscript is merged into `this_article.tex`. The runner imports `polish_presburger` by its delivery name and writes its record beside itself, so it is rerun on a copy with the delivery names restored (the report's README gives the commands). The intake reran it on such a copy with Python 3.14.4: 17,930 checks passed, and the record differed from the delivered one only in the interpreter version and line endings.

13.4 The formalization interface, source 04's audit [04]

Source 04 inspected the Rocq file of the formal project rather than the Lean file, at its own pin.

13.4.1 A pinned, limited source audit

The repository files used in this manuscript were checked at commit

3ad5f878c26d2d82c092190af2f3edc50d6feb06.

The relevant Presburger source is `Logic/PresburgerArithmetic/Coq/Cooper.v`; the project description is `Logic/PresburgerArithmetic/README.md`. The latter describes the Lean pipeline through syntax, normal form, one-variable elimination, and sentence decision [17]. The present manuscript does not claim to have rebuilt that repository or audited its unrelated mathematical projects.

In the inspected Coq source, the predicate hypothesis is defined over *ordinary* integers by one-step translation invariance. The lemma `periodic_shift` then proves invariance under arbitrary integer multiples, using induction on ordinary positive and negative integer representations. This supports `periodic_has_residue` and the interval criterion [18]. That is sound for its stated standard-integer domain.

For a model-uniform version, replacing the type of integers by an arbitrary Presburger group while leaving the predicate hypothesis unchanged would be invalid: [proposition 4.6](#) is a counterexample. The quotient in $x = mq + r$ may have a nonzero divisible coordinate, and iteration through standard integer shifts does not reach mq . The repair is conceptual, not a missing induction tactic: use [definition 4.5](#). Congruence literals satisfy that stronger property directly, so [theorem 4.8](#) supplies the needed uniform lemma.

13.4.2 A concrete implementation sequence

A formalization can be divided into independently reviewable stages. The following are proposed new components, not assertions about existing repository declarations.

Abstract arithmetic semantics. Define a class of linearly ordered abelian groups with a named least positive unit and a standard finite quotient–remainder principle. Interpret the existing affine syntax in an arbitrary group of this class. Keep the syntactic integers as *coefficients*; they must not be confused with the semantic carrier.

Finite boundary theorem. Formalize m -residue invariance as invariance under differences in mG . Prove [theorem 4.8](#), separating existence of a finite candidate from any decidability hypothesis on its predicate. Then specialize to a Boolean evaluator when the residue conditions are decidable. This keeps the mathematical statement valid even when the carrier has undecidable equality or order in a computational representation.

Model-uniform elimination. Reuse the finite syntax transformations, with a fresh semantic correctness proof for coefficient normalization, cancellation in (4.9), bound selection, and finite substitution. The correctness endpoint should quantify over all groups satisfying the abstract assumptions. Specialization to $\mathbb{Z}$ must recover the current integer semantics rather than replace it.

Split models and topological structure. Construct G_D and its nonnegative cone, prove explicit division, and obtain the complete Presburger theory by relativization. Next formalize the G_δ argument, continuity of restricted addition, and the coordinate formulas for fixed quotient and remainder. For D_B , construct the two coding homeomorphisms rather than invoking a black-box classification of zero-dimensional Polish spaces.

Self-embeddings. First prove extension from the positive cone to the group completion and preservation of the divisible subgroup. For D_B , prove the filtration lemma (11.6) before introducing matrices. This exposes the reason for automatic continuity and avoids an unproved assumption that an arbitrary linear map is determined by its values on the coordinate vectors. Those vectors are *not* a vector-space basis of the full product; the finite-row conclusion requires the order argument.

13.4.3 Trust and proof-status ledger

Component	Status in this package	Dependence
Presburger characterization and QE	Classical; self-contained argument supplied	Ordered groups and finite syntax
Affirmative Polish construction	Conventional proof supplied	Explicit split group and topology
Baire homeomorphism and stream operations	Proof and executable implementation	Effective rational coding
Monus and embedding theorems	Conventional proofs supplied	Earlier lemmas in this manuscript
Finite arithmetic diagnostics	Executed successfully	Python standard library, exact fractions
New Lean/Rocq kernel certification	Not supplied	Proposed work above
Historical priority of the answer	Not established	Further literature and expert review

The distinction is especially important here because the repository already contains machine-checked material. Reusing its mathematical strategy does not automatically confer its certification status on the new semantic carrier or on the topology.

13.5 Relation to the formal project `Logic/PresburgerArithmetic` [write]

At the placement, the project's Lean development (namespace `PresburgerArithmetic`) and its Rocq file are unchanged from the two pins. The declarations the sources discuss are:

- Lean, `Lean/PresburgerArithmetic/Cooper.lean`: `periodic_shift_of_mod`, `periodic_interval`, `periodic_has_residue`, `periodic_unbounded_above`, `periodic_unbounded_below` and `cooper_finite_criterion`, all for a predicate P on `Int` with the one-step hypothesis $\forall x (P(x + m) \leftrightarrow P(x))$;
- Lean, `Lean/PresburgerArithmetic/Decision.lean`: the decision procedure `PresburgerArithmetic.Formula.presburgerArithmetic_decidable` and its correctness theorem `Formula.decideSentence_eq_true_iff`;

- Rocq, Coq/Cooper.v: `periodic`, `periodic_shift`, `periodic_has_residue`, `periodic_interval`, `periodic_unbounded_above`, `periodic_unbounded_below`, `cooper_finite_criterion` and `cooper_step_decidable`, all over $\mathbb{Z}$.

These statements are about the ordinary integers, where they are correct. [Example 4.4](#) and [proposition 4.6](#) show that their periodicity hypothesis cannot be read over a nonstandard $\mathbb{Z}$ -group, and [lemma 4.7](#) and [theorem 4.8](#) state the coset-invariant replacement that a model-uniform version would need. Neither source changed the project, and neither reports a defect in it. **No statement of this report is formalized:** the proposed modules of [section 13](#) and the stages of [section 13.4](#) are plans, and the collection's docs/FORMALIZATION.md maps no label of this report.

14 Further research questions

Sources 05 and 04 each end with a list of questions (ten and eleven). Four questions occur in both lists and are merged in [section 14.1](#); the remaining six of 05 and seven of 04 follow, so that this part has seventeen questions. Source 05 introduces its list as follows, and source 04 its own list as in the second paragraph. None of these questions is answered in this report; [Part II](#) bears on [question 14.4](#) and on the last questions of both lists ([section 26](#)).

The following questions are proposed directions arising from the proved results. Except where explicitly linked to the source question, their status is not presented as a certification that the literature has no answer.

The construction answers the printed existence question, but it leaves several sharper problems. The questions below distinguish implementation work, structural problems, and possible new obstructions. No solution to them is asserted merely by listing them.

14.1 Questions asked by both sources

Question 14.1 (A formal certificate for the affirmative answer [05, 04]). *Source 05:* Can the one-scale model, the full $\mathbb{Z}$ -group elimination theorem, and its Polish topology be formalized with a compact audit boundary in ProveIt? The key initial milestone is the generic coset-invariant Cooper lemma, rather than merely a finite test of arithmetic identities. *Source 04:* Can the complete proof of [remark 3.2](#), including the Baire homeomorphism and the full induction scheme via model-uniform elimination, be added to ProveIt with no new axioms? The first milestone should be the finite boundary lemma with residue invariance, followed by the semantic specialization checks described in [section 13.4](#).

Question 14.2 (Weak metatheory [05, 04]). *Source 05:* What weak foundational theory suffices to certify the affirmative construction, including the chosen representation of Polishness? The real model has an explicit complete metric, and the Baire model has an explicit sequence code, so a general model-existence theorem is unnecessary. This is distinct from Glazer's Question 1 concerning the strength needed for his stronger-arithmetic obstruction. *Source 04:* What is the weakest useful subsystem of second-order arithmetic in which the Baire-space construction, its stream algorithms, and its formula-by-formula model correctness can be coded and proved? Distinguish coding all points of Baire space from asserting a set of all such points. This question concerns the positive construction, not a resolution of the separate reverse-mathematical problem posed in Glazer's Question 1 [1].

Question 14.3 (Connected carriers [05, 04]). *Source 05:* What additional obstruction prevents or permits a continuous-addition model on a connected Polish space? Our locally compact

model has countably many ray components, and the arithmetic order cannot be open or closed. A proof for a prescribed connected carrier must use more than the order-topology observation alone. *Source 04*: Can a connected uncountable Polish space support a Presburger model with continuous addition? Our examples do not decide this: their integer coordinate is a continuous map onto a countable discrete space. Requiring all fixed remainders to be continuous would already force nontrivial clopen partitions, so the connected problem must separate continuity of addition from continuity of these definable functions.

Question 14.4 (Multiplication and growth orders [05, 04]). *Source 05*: Can one give an equally explicit Polish continuous-addition Presburger model whose underlying additive structure admits a unital semiring expansion? Any candidate must avoid a cofinal infinite additive unit. If one also requires $Q + IOpen$, the multiplication cannot be Borel by the cited obstruction. Separating algebraic expandability from Borel expandability is the central issue. *Source 04*: Find admissible uncountable topological groups D whose infinite growth order embeds into each of its strict final segments, thereby passing [proposition 10.7](#). For their positive cones, determine whether any semiring multiplication exists, and, separately, which descriptive regularity it can have. Glazer's obstruction to Borel multiplication under stronger arithmetic remains relevant, but it does not by itself classify bare semiring expansions.

[write] (Batch 86C.) The first sentence of source 05's part is answered affirmatively in [Part III](#): source 06's cone of $\mathbb{Z} + t\mathbb{R}[t]$ ([theorem 38.1](#)) is an explicit uncountable, locally compact Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous addition whose additive structure carries a unital semiring multiplication, continuous for the same topology. It has no cofinal infinite element and fails open induction ([theorem 38.2](#)). Whether such a model can also satisfy $Q + IOpen$ with non-Borel multiplication, and source 04's growth-order question, remain open.

[merge] For [question 14.4](#): the real model is the additive monoid of one stratum of 03's ring $\mathcal{A}_{\text{fin}, \geq 0}$, which does carry a semiring multiplication, Borel and continuous off a closed nowhere dense set ([theorem 24.3](#)); but $\mathcal{A}_{\text{fin}, \geq 0}$ is not a model of Presburger arithmetic, and on it continuous addition is impossible ([theorem 25.2](#)). 03's [question 28.1](#) asks the multiplicative counterpart on $\mathcal{A}_{\text{fin}, \geq 0}$.

14.2 Further questions of source 05

Question 14.5 (Prescribed zero-dimensional locally compact carriers). Which uncountable zero-dimensional locally compact Polish spaces support continuous-addition Presburger models? Compact spaces are excluded and Baire space is realized, but Baire space is nowhere locally compact. This leaves a meaningful topological region not settled by our examples.

Question 14.6 (Classifying admissible coefficient groups). Which divisible ordered Polish abelian groups have G_δ positive cone? [Theorem 5.2](#) turns this into an exact criterion for the split construction. Can their possible underlying topologies, order ranks, and continuous order-preserving endomorphisms be classified under useful additional hypotheses?

Question 14.7 (Continuity of a richer definable language). How much of Presburger definable choice can be made continuous in one uncountable Polish presentation? Standard quotients and remainders are continuous here, but total predecessor and truncated subtraction are not everywhere continuous. Is there a natural maximal collection of definable operations admitting simultaneous continuity?

Question 14.8 (Optimal finite-mind-change complexity). For a given Presburger formula, what is the smallest possible uniform change bound in the Baire presentation? The proof

gives the number of order/equality atom occurrences after elimination. That bound need not be optimal and may be much larger than necessary. Relating the minimal bound to the finite difference hierarchy or to formula structure would refine the Δ_2^0 result.

Question 14.9 (Polish elementary-submodel lattices). Which lattices of closed or clopen elementary submodels can occur in a Polish continuous-addition Presburger model? The real model has only two Polish elementary submodels with inherited topology, whereas the Baire model has infinitely many proper clopen elementary self-copies with standard intersection. What intermediate behaviors can be realized?

Question 14.10 (Canonical surreal cutoffs and topologies). Find useful explicit birthday bounds for the two fixed-support omnific images and formalize the coefficient-topology transport inside those cutoffs. More generally, which support sets admit Polish coefficient topologies with continuous addition and full Presburger induction on the resulting positive cone? Such a classification should keep ring closure separate from additive closure.

14.3 Further questions of source 04

The last question below and the last question of [section 14.2](#) (canonical surreal cutoffs and topologies) overlap in their second halves: both ask which set-sized supports or subgroups of **Oz** carry Polish coefficient topologies with continuous addition. They are printed separately because their first halves differ. [Theorem 25.2](#) gives a negative datum for one ring-closed family ([section 26](#)).

Question 14.11 (Beyond split topological models). Which Polish presentations of Presburger groups admit a continuous integer-coordinate retraction and a topological decomposition $D \xrightarrow{\sim} \mathbb{Z}$ of the kind used here? What can occur when the algebraic extension or the topology does not split? Any characterization must separate the standard subgroup, the divisible subgroup, and the full system of residues modulo standard integers.

Question 14.12 (Rigidity with continuous addition). Is there an uncountable rigid Presburger model with a Polish topology making addition continuous and order Borel? Rigid models are studied by Jeřábek [4]; our split examples are not rigid, because positive rational scaling of their divisible part gives nontrivial automorphisms. One must therefore move beyond the present easy symmetry.

Question 14.13 (Exact discontinuity profiles of definable functions). For each definable function $f : M_D^k \rightarrow M_D$, can its discontinuity set be computed from a quantifier-free description of its graph? For the two main models, determine whether every such set admits a finite description in terms of equality of divisible-coordinate affine expressions and residue conditions. Keep graph complexity separate from target-valued Baire class; they are not interchangeable statements for arbitrary target spaces.

Question 14.14 (Embedding categories for other coefficient groups). Extend [theorem 11.4](#) to products of ordered coefficient groups other than $\mathbb{Q}$, and to countable support systems with more complicated growth orders. Which hypotheses force every order embedding to be continuous? Which image topologies replace the finite-versus-infinite omitted-pivot dichotomy?

Question 14.15 (Effective truth versus effective operations). The Baire presentation has computable addition and fixed division, but not a computable test for every definable relation on arbitrary input streams. Give sharp oracle bounds for evaluation of formulas in this presentation, including uniform bounds under parameter coding and quantifier elimination.

Distinguish the decidability of closed Presburger sentences from the discontinuous evaluation of an atomic relation on arbitrary infinite names.

Question 14.16 (Metrics compatible with arithmetic translations). What useful uniform continuity properties of addition can be retained by compatible complete metrics after the invariant-metric obstruction in [proposition 8.4](#)? Determine whether addition can be made jointly uniformly continuous, or all translations uniformly continuous in one direction, and quantify the unavoidable distortion of their inverses. This would clarify completion arguments for nonclosed positive cones.

Question 14.17 (Coefficient topologies inside surreal arithmetic). Classify set-sized omnific subgroups carrying Polish coefficient topologies with continuous addition, and determine how those topologies behave under changes of normal-form support. The natural comparison is between coefficient topology, arithmetic order, and model-theoretic elementary embedding, not between a Polish topology and the whole proper class of surreals.

15 Conclusions of Part I

The affirmative construction can be summarized by one change of viewpoint: use the lexicographic order to interpret arithmetic, and a coefficient topology to interpret continuity. Divisibility of the leading-coordinate group supplies all standard residues; quantifier elimination then supplies full Presburger induction. Neither countability of the standard model nor discreteness of the arithmetic order forces the external presentation to be countable or discrete.

The real model gives a locally compact answer, while the rational-sequence model gives a genuinely effective answer on Baire space. Their order relations have the smallest possible nontrivial Borel rank, yet they exhibit different elementary self-embedding behavior. Their realization inside **Oz** connects these results directly to the repository's normal-form arithmetic. Finally, the cofinal-infinite-element argument shows exactly why these additive examples cannot be mistaken for topological models of full arithmetic.

Source 04's conclusion [04]. The separation needed for the construction is simple: place a divisible ordered group above a discrete integer coordinate algebraically, while retaining the ordinary or product topology on that divisible group. The arithmetic becomes discretely ordered without forcing the topology to be discrete. Unique standard residues give the whole Presburger theory, not just a fragment.

The real-coordinate construction supplies a perfect locally compact example; the rational product supplies literal Baire space and effective stream operations. Their regularity is sharply delimited: every definable relation has Borel rank at most two, whereas monus is discontinuous exactly on an explicitly described geometric boundary. Their self-embeddings range from unavoidable surjective dilations to closed nowhere dense elementary copies, but all are automatically continuous.

For Provell, the most immediate next step is precise and finite: generalize the Cooper residue lemma with the correct subgroup-invariance hypothesis, then connect the already familiar syntax to these new semantics. That would turn the conventional answer presented here into a kernel-checked one without changing the mathematical construction.

Part II

Borel presentations and support barriers

This part is source 03, pinned to repository commit c5b4881fa. It keeps 03’s definitions, in particular coefficient observability ([definition 17.1](#)) and its two exponent conventions: $X = t^{-1}$ is read as ω , and the change $q \mapsto -q$ converts the increasing t -convention into the reverse X -convention ([section 17](#)). Four symbols are renamed ([section 1.3](#)): 03’s ring $\mathcal{A}$ is $\mathcal{A}_{\text{fin}}$, its rational Hahn ring H is $\mathcal{H}_{\mathbb{Q}}$, its Kleene–Brouwer embedding h is μ_{KB} , and two local sets called D are W and $\mathcal{Z}$. Several of 03’s results restate results already in the collection; they keep their proofs here, because [theorem 25.2](#) needs their exact hypotheses, and a note after each names the collection’s statement. In 03’s text, “this report” means source 03.

16 The intersection and the questions answered

Glazer’s paper *A Topological Tennenbaum Theorem* studies the interaction between arithmetic, Borel definability, and complete separable topologies. The part used here is particularly well matched to omnific arithmetic: an uncountable Polish model of Robinson arithmetic plus open induction cannot have continuous addition and Borel multiplication [[1](#), Corollary 1]. This is weaker than the hypothesis needed in his separate multiplication obstruction; we preserve that distinction throughout.

The relevant ProveIt material is the surreal-number project, especially the reports on omnific notations and computer algebra, and its construction of the omnific ring and floor. The inspected repository guides explicitly distinguish research drafts, finite verification programs, and statements mapped to Lean proofs [[20](#), [21](#)]. That distinction is retained here. The repository already reports a Π_1^1 classification of unrestricted support validity, using Kleene–Brouwer coding. Thus merely repeating a well-founded-tree reduction would not constitute a new advance on this material.

Instead, we ask for an exact boundary and a positive arithmetic counterpart. For which countable exponent orders is support validity Borel? Does the answer also determine whether the whole Hahn field can be given a standard Borel presentation with readable coefficients? Can a concrete uncountable integer part have arithmetic that is continuous almost everywhere in the category sense, even though Glazer rules out global continuity? Precisely where does continuity fail?

16.1 Main theorem map

The report proves the following connected package.

1. **Support and field-presentation dichotomy.** For countable L , $\text{WO}(L)$ is Borel if and only if L contains no copy of $\mathbb{Q}$. Otherwise it is coanalytic-complete ([Theorem 18.4](#)). For countable ordered abelian Γ , the same condition characterizes coefficient-observable standard Borel presentations of $\mathbb{R}((t^\Gamma))$ ([Theorem 20.2](#)).
2. **Exact benchmark complexities and a root-closure obstruction.** The support predicates for $\mathbb{Z}$ and $\mathbb{Z}_{\text{lex}}^d$, $d \geq 2$, are respectively Σ_2^0 -complete and Π_3^0 -complete. A nontrivial countable exponent group in a square-root-closed full real Hahn field cannot lie on the Borel side ([Theorem 19.1](#), [Corollary 20.4](#)).

3. **Uniform bounds and localization.** Every Borel coefficient-observable family of valid series has its support order types bounded below one countable ordinal. Arbitrarily large countable support types, and coanalytic-complete validity, already occur in a fixed 0–1 coefficient slice inside $[X, 2X)$ (Theorems 21.2 and 22.1).
4. **A positive model with exact negative boundaries.** The finite principal-part model $\mathcal{A}_{\text{fin}, \geq 0}$ satisfies open induction and has a Polish presentation with explicit closed nowhere dense discontinuity sets for both operations. Yet no Polish topology with the same Borel sets makes addition continuous everywhere (Theorems 23.2, 24.3, and 25.2).

The order-theoretic dichotomy is deduced from classical Hausdorff theory and well-founded-tree completeness. The uniform bound is an application of classical analytic boundedness, for which we include a proof. The integer-part model is of Shepherdson type, not a newly discovered model of arithmetic. The report's claims of contribution concern the explicit combined classification, the observable-field criterion, the benchmark calculations, and the exact topological analysis. These are proved here; bibliographic priority is not asserted.

16.2 A precise source boundary

The source audit used selected repository guides at commit

c5b4881fa603339c7ff90f5e1098ede0651243d8.

The inspected paths are recorded in Appendix D. This was not an exhaustive audit of the repository, nor a rebuild of its Lean development. No new theorem below depends on an unreviewed repository theorem: the required normal-form and integer-part arguments are supplied independently. The repository is used to identify the research overlap and to avoid claiming its previously recorded results as new.

17 Conventions and the meaning of a presentation

17.1 Two opposite exponent conventions

For an ordered abelian group Γ , a Hahn series is written

$$f = \sum_{\gamma \in \Gamma} a_{\gamma} t^{\gamma}, \quad a_{\gamma} \in \mathbb{R},$$

with $\text{supp}(f) = \{\gamma : a_{\gamma} \neq 0\}$ well-ordered in the increasing order of Γ . Its sign is the sign of the coefficient at the least exponent. Thus t is infinitesimal when $\Gamma = \mathbb{Q}$ and $t = t^1$.

For the omnific applications we instead write $X = t^{-1}$, so X is positive infinite. In

$$f = \sum_{q \in \mathbb{Q}} b_q X^q,$$

the support must be *reverse* well-ordered: every nonempty subset has a greatest exponent. The change $q \mapsto -q$ converts one convention into the other. All descriptive-set-theoretic statements in Sections 18 and 19 use the increasing convention. Section 22 explicitly reverses it.

The ring

$$\mathcal{H}_{\mathbb{Q}} = \left\{ n + \sum_{q \in S} b_q X^q : n \in \mathbb{Z}, S \subseteq \mathbb{Q}_{>0} \text{ reverse well-ordered} \right\} \quad (17.1)$$

is a set-sized rational-exponent omnific ring. Replacing X^q by the Conway monomial ω^q identifies it with the corresponding normal-form subring of $\mathbf{Oz}$. Hahn addition and multiplication agree with those normal-form operations. The normal-form viewpoint and its ordered-algebraic context are standard; see [11, 5].

17.2 Borel spaces, completeness, and observability

A Polish space is a separable topological space admitting a compatible complete metric. A standard Borel space is a measurable space isomorphic to a Borel subset of a Polish space. We use the usual boldface pointclasses: Σ_2^0 consists of countable unions of closed sets, Π_3^0 of countable intersections of Σ_2^0 sets, and Π_1^1 denotes coanalytic sets. An analytic set is a projection of a Borel set in a product of Polish spaces. Completeness is with respect to continuous reductions from zero-dimensional Polish spaces. Standard references are [8, 9].

Definition 17.1. Let K be a set of series over a countable exponent alphabet. A *coefficient-observable standard Borel parametrization* of K consists of a standard Borel space Y and a surjection $\theta : Y \rightarrow K$ such that, for every exponent γ , the map

$$y \mapsto [t^\gamma]\theta(y)$$

is Borel as a map to $\mathbb{R}$. Repetitions of names are allowed.

No Borel structure on K is presupposed in this definition. Since the alphabet is countable, coefficient observability says exactly that the composite code map $Y \rightarrow \mathbb{R}^\Gamma$ is Borel. This is stronger than the mere existence of a set-theoretic bijection with the reals, and different from an abstract Borel presentation that is allowed to make every canonical coefficient unobservable. Our negative results explicitly retain the observability hypothesis.

Definition 17.2. For a countable linear order L , put

$$\text{WO}(L) = \{S \in 2^L : (S, <_L) \text{ is well-ordered}\}.$$

The empty support is allowed. The order L is *scattered* if it contains no order-isomorphic copy of $\mathbb{Q}$.

A subset of 2^L is a set of codes, not a new collection of proper-class objects. In particular, asking whether $\text{WO}(L)$ is Borel is an ordinary set-sized descriptive-set-theoretic question.

18 The exact support dichotomy

18.1 A complete obstruction on a dense countable alphabet

Lemma 18.1. *For every countable linear order L , $\text{WO}(L)$ is coanalytic.*

Proof. A support S fails to be well-ordered exactly when there exists $(x_n)_{n \in \mathbb{N}} \in L^{\mathbb{N}}$ with $x_n \in S$ and $x_{n+1} <_L x_n$ for every n . The relation on $(S, (x_n))$ specified by these countably many coordinate conditions is closed, with L discrete. Its projection is analytic. Its complement is $\text{WO}(L)$. $\square$

Let Tr be the closed space of trees $T \subseteq \mathbb{N}^{<\mathbb{N}}$, where trees are closed under initial segments, and let $\text{WF} \subseteq \text{Tr}$ be the well-founded trees. Recall briefly why WF is coanalytic-complete. If A is coanalytic in Baire space, its complement is the projection of the body of a tree R on pairs of finite sequences. For $x \in \mathbb{N}^{\mathbb{N}}$, let

$$T_x = \{s : (x \upharpoonright |s|, s) \in R\}.$$

Membership of a fixed s depends on finitely many coordinates of x , so $x \mapsto T_x$ is continuous, and $x \in A$ exactly when T_x has no branch. This is the standard universal-tree proof [9, Section 16.B].

Lemma 18.2 (Kleene–Brouwer coding). *There is a fixed injection $\mu_{\text{KB}} : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{Q} \cap (0, 1)$ such that*

$$T \in \text{WF} \iff \mu_{\text{KB}}[T] \in \text{WO}(\mathbb{Q}).$$

Consequently $\text{WO}(\mathbb{Q})$ is Π_1^1 -complete.

Proof. Order the finite sequences by the Kleene–Brouwer order: a proper extension precedes its prefix, and otherwise the first differing entry determines the order, with the ordinary order on $\mathbb{N}$. This is a countable linear order. Embed it in $\mathbb{Q} \cap (0, 1)$ as follows. Enumerate its elements. At each finite stage, put the next element at a rational strictly between its nearest already assigned lower and upper neighbors, using 0 and 1 as external bounds when needed. A midpoint works. The resulting μ_{KB} is one fixed order embedding of the entire tree alphabet, not an embedding selected separately for each T .

If T has a branch, its successively longer prefixes form a strictly descending Kleene–Brouwer sequence. If T is well-founded, well-founded induction over its nodes shows that the Kleene–Brouwer order on each descendant subtree is a well-order: it is the increasing $\mathbb{N}$ -indexed sum of the child-subtree orders, followed by the root. Therefore $\text{KB}(T)$ is a well-order exactly when T is well-founded.

The map $T \mapsto \mu_{\text{KB}}[T]$ is continuous. At an exponent in the range of μ_{KB} , its characteristic function reads exactly one tree-membership bit; outside the range it is constantly zero. Thus it reduces WF to $\text{WO}(\mathbb{Q})$. Lemma 18.1 supplies membership in the pointclass. $\square$

[write] **Lemma 18.2** and part (ii) of **theorem 18.4** below are boldface counterparts of results already in the collection’s `omnific-notations` report: `onot:neg:lem:kbtree` (a tree is well-founded exactly when its Kleene–Brouwer order is a well-order), `onot:neg:thm:pionesone` (Π_1^1 -completeness, for recursive trees, of the validity of decidable dyadic supports in $[1, 2)$ with coefficient one) and `onot:neg:rem:secondroute` (the midpoint codings of that report’s sources 08 and 09, which are the construction of μ_{KB} up to the target interval). Source 03 reproves the reduction to make the continuous (boldface) form explicit, as its contribution ledger says ([section D](#)). What is new in [section 18](#) is the dichotomy between scattered and nonscattered orders.

18.2 Why scattered alphabets are different

We use the classical countable Hausdorff characterization: starting with finite linear orders, closing under $\mathbb{Z}$ -indexed sums and finite sums through countable ordinal stages produces precisely the countable scattered orders. A precise formulation is recalled in [10, Definition 1]. Only this characterization, not its more advanced rank consequences, is needed below.

Lemma 18.3. *Suppose $L = \sum_{i \in \mathbb{Z}} L_i$ is a lexicographic sum and each $\text{WO}(L_i)$ is Borel. Then $\text{WO}(L)$ is Borel.*

Proof. For $S \subseteq L$, let $S_i = S \cap L_i$ and $I(S) = \{i : S_i \neq \emptyset\}$. Then

$$S \in \text{WO}(L) \iff (\forall i \in \mathbb{Z} S_i \in \text{WO}(L_i)) \wedge (\exists N \in \mathbb{Z} \forall i < N S_i = \emptyset). \quad (18.1)$$

Necessity of the second condition follows because unboundedly negative occupied indices yield an infinite descending sequence, choosing one element from each of a decreasing sequence of such indices. Conversely, when the condition holds, S is a well-ordered sum of well-ordered fibers over a subset of an integer tail.

Each restriction map $S \mapsto S_i$ is continuous. The fiber-empty conditions are closed. The right-hand side of (18.1) is therefore Borel. For a finite sum the analogous criterion has only finitely many fiber conditions and no additional obstruction. $\square$

Theorem 18.4 (Support dichotomy). *For every countable linear order L :*

- (i) *if L is scattered, $\text{WO}(L)$ is Borel;*
- (ii) *if L is not scattered, $\text{WO}(L)$ is Π_1^1 -complete.*

In particular, Borelness of $\text{WO}(L)$ is equivalent to scatteredness of L .

Proof. For a finite order the support predicate is the whole finite code space. Induction through the Hausdorff construction, using Lemma 18.3 and the finite-sum observation, proves (i).

For (ii), fix an order embedding $j : \mathbb{Q} \rightarrow L$. Extending a support by zero outside $j[\mathbb{Q}]$ gives a continuous map $S \mapsto j[S]$ from $2^{\mathbb{Q}}$ to 2^L , preserving and reflecting well-orderedness. Compose this map with Lemma 18.2, and use Lemma 18.1. A coanalytic-complete set is not Borel, completing the equivalence. $\square$

Remark 18.5. The quantification here is over *all subsets of one fixed countable order*. It is not a statement about the Borel complexity of the class of all scattered orders, nor about finding a Hausdorff decomposition uniformly from an arbitrary code for L . The fixed decomposition used in the positive proof need not be computable.

19 Exact benchmarks: integers, finite lexicographic powers, and rationals

The dichotomy gives a qualitative boundary. On useful discrete alphabets one can calculate the precise Borel level.

Theorem 19.1. *With the product topology on the corresponding binary support spaces:*

- (i) $\text{WO}(\mathbb{Z})$ is Σ_2^0 -complete;
- (ii) for every integer $d \geq 2$, $\text{WO}(\mathbb{Z}_{\text{lex}}^d)$ is Π_3^0 -complete;
- (iii) every well-ordered subset of $\mathbb{Z}_{\text{lex}}^d$ has order type at most ω^d , and this bound is attained.

Proof. A subset of $\mathbb{Z}$ is well-ordered if and only if it is bounded below. Hence

$$\text{WO}(\mathbb{Z}) = \bigcup_{N \in \mathbb{Z}} \{S : \forall n < N (n \notin S)\},$$

a countable union of closed sets. The continuous map $A \subseteq \mathbb{N} \mapsto \{-n - 1 : n \in A\}$ reduces the familiar Σ_2^0 -complete set Fin of finite subsets of $\mathbb{N}$ to $\text{WO}(\mathbb{Z})$. This proves (i).

For completeness of the benchmark being used, Fin is countable and dense in $2^{\mathbb{N}}$. Were it G_δ , it would be comeager, while it is also meager; the Baire theorem rules this out. The usual completeness criterion for Borel sets in zero-dimensional Polish spaces then gives its Σ_2^0 -completeness [9, Section 16.A].

For $d = 2$, view the order as $\sum_{i \in \mathbb{Z}} \mathbb{Z}$. Formula (18.1) is the intersection of a Π_3^0 condition (every fiber is bounded below) and a Σ_2^0 condition (the occupied outer indices are bounded below). The result is Π_3^0 . For $d > 2$, the same formula and induction work: countable intersections of Π_3^0 sets remain Π_3^0 .

For hardness, use

$$\text{Fin}^{\mathbb{N}} = \{a \in 2^{\mathbb{N} \times \mathbb{N}} : \forall m \{n : a(m, n) = 1\} \text{ is finite}\}.$$

This set is Π_3^0 -complete. Indeed, any Π_3^0 subset of a zero-dimensional Polish space is a countable intersection of Σ_2^0 sets; reduce each of these sets to Fin and take the product of the reductions. Map a to the support

$$S_a = \{(m, -n - 1) : a(m, n) = 1\} \subseteq \mathbb{Z}_{\text{lex}}^2. \quad (19.1)$$

The outer indices are automatically bounded below by zero. A fiber is well-ordered exactly when its row is finite. Thus (19.1) is a continuous reduction. Embedding $\mathbb{Z}_{\text{lex}}^2$ into $\mathbb{Z}_{\text{lex}}^d$ gives hardness for all $d \geq 2$.

Finally, a well-ordered subset of $\mathbb{Z}$ has type at most ω . Inductively, a well-ordered support in $\mathbb{Z}^d$ is a sum of at most ω fibers of type at most ω^{d-1} , and so has type at most $\omega^{d-1} \cdot \omega = \omega^d$. The subset $\mathbb{N}^d$, with the same lexicographic convention (first coordinate most significant), has type exactly ω^d . $\square$

Exponent alphabet	Valid-support complexity	Support-type bound
$\mathbb{Z}$	Σ_2^0 -complete	$\leq \omega$
$\mathbb{Z}_{\text{lex}}^d, d \geq 2$	Π_3^0 -complete	$\leq \omega^d$
A fixed countable scattered order	Borel	Some countable bound
$\mathbb{Q}$, or any countable nonscattered order	Π_1^1 -complete	Unbounded in ω_1

The last two support-type assertions follow also from Theorem 21.2 below. An important feature of the table is that the finite lexicographic rank can grow while the exact Borel level stays at Π_3^0 . Support order type and Borel complexity measure different things. Conversely, passing from $\mathbb{Z}$ to its divisible hull $\mathbb{Q}$ is a genuine jump out of the entire Borel hierarchy.

20 An observable-presentation dichotomy for full Hahn fields

Fix a countable ordered abelian group Γ . Let

$$V_\Gamma = \{a \in \mathbb{R}^\Gamma : \text{supp}(a) \in \text{WO}(\Gamma)\}.$$

As a set, V_Γ is the canonical coefficient presentation of the full Hahn field $K_\Gamma = \mathbb{R}((t^\Gamma))$.

Lemma 20.1. *If Γ is scattered, V_Γ is Borel. If Γ is nonscattered, V_Γ is Π_1^1 -complete.*

Proof. The support map

$$\sigma : \mathbb{R}^\Gamma \rightarrow 2^\Gamma, \quad \sigma(a)(\gamma) = 1 \iff a_\gamma \neq 0$$

is Borel. Thus the scattered case follows from Theorem 18.4. In all cases, failure of validity is witnessed by an infinite descending sequence of nonzero coordinates, an analytic condition, so V_Γ is coanalytic. The binary-coefficient inclusion $2^\Gamma \hookrightarrow \mathbb{R}^\Gamma$ is continuous and reduces $\text{WO}(\Gamma)$ to V_Γ , proving completeness in the remaining case. $\square$

Theorem 20.2 (Observable Hahn-field criterion). *For countable ordered abelian Γ , the following are equivalent:*

- (i) Γ is scattered as a linear order;
- (ii) V_Γ is a Borel subset of $\mathbb{R}^\Gamma$;
- (iii) K_Γ admits a coefficient-observable standard Borel parametrization, allowing repeated names;
- (iv) K_Γ admits a standard Borel presentation with Borel coefficient maps, Borel addition and multiplication, Borel inversion on the nonzero elements, and Borel order.

Proof. Lemma 20.1 gives (i) $\Leftrightarrow$ (ii). Assume (ii), and use V_Γ itself as the standard Borel domain. Addition and negation are coordinatewise and therefore Borel. For $\Gamma = \{0\}$ all assertions reduce to those for $\mathbb{R}$. Otherwise, for multiplication enumerate $\Gamma = \{\delta_0, \delta_1, \dots\}$ without repetition. At a fixed exponent γ , the product coefficient is

$$(ab)_\gamma = \lim_{N \rightarrow \infty} \sum_{j < N} a_{\delta_j} b_{\gamma - \delta_j}. \quad (20.1)$$

For valid Hahn supports, only finitely many terms are nonzero. To see the finiteness, infinitely many contributing δ would contain a strictly increasing sequence in the well-ordered support of a ; the corresponding $\gamma - \delta$ would be a strictly decreasing sequence in the support of b . This is impossible. Thus the sums in (20.1) eventually stabilize. They are continuous on the ambient product, and their pointwise limit on the Borel domain is Borel.

The standard Hahn-field theorem supplies closure under multiplication and the existence of multiplicative inverses; its well-ordered-support lemma is the usual Neumann lemma [11]. The graph

$$\{(a, b) \in (V_\Gamma \setminus \{0\}) \times V_\Gamma : ab = 1\}$$

is Borel and has exactly one point in each vertical section. The Lusin–Novikov theorem gives Borel inversion. Equivalently, for an open set U the inverse image under inversion is the projection of this graph intersected with $V_\Gamma \times U$, whose sections have at most one point.

Positivity is Borel because it is expressed by

$$\exists \gamma \in \Gamma (a_\gamma > 0 \wedge \forall \delta < \gamma \ a_\delta = 0).$$

There are only countably many coordinates. Order follows by subtraction. This proves (iv), which immediately implies (iii).

Assume (iii). The coefficient code map from its standard Borel parameter space to $\mathbb{R}^\Gamma$ is Borel, and its image is exactly V_Γ . Consequently V_Γ is analytic. For nonscattered Γ , Lemma 20.1 makes it coanalytic-complete and hence nonanalytic. Thus Γ must be scattered, proving (i). $\square$

Remark 20.3 (What the negative direction does and does not exclude). The theorem excludes even a surjective system of Borel-observable names, not just unique or canonical names. It does not exclude abstract Borel models in which the coefficient functions fail to be Borel. Nor does it exclude useful Borel subfields of K_Γ , including real closed subfields. The word *full* is essential.

20.1 Root closure forces the non-Borel side for full fields

Corollary 20.4. *Suppose $\Gamma \neq \{0\}$ is countable and every positive element of the full Hahn field $\mathbb{R}((t^\Gamma))$ has a square root in that field. Then Γ is nonscattered, and the field admits no coefficient-observable standard Borel parametrization. In particular, this holds whenever the full Hahn field is real closed.*

Proof. For every $\gamma \in \Gamma$, the positive monomial t^γ has a root f with $f^2 = t^\gamma$. Taking least exponents gives $2 \min \text{supp}(f) = \gamma$. Thus Γ is 2-divisible. For $a < b$ in Γ , the element $a + (b - a)/2$ lies strictly between them. A nontrivial ordered group has no endpoints, so Γ is a countable dense linear order without endpoints and is order-isomorphic to $\mathbb{Q}$. Apply Theorem 20.2. $\square$

This is a necessary-condition argument; it does not use a classification of which Hahn fields are real closed. It also explains the source of the jump: arbitrary well-ordered supports over a densely divisible scale are the problem, not the mere availability of algebraic roots. The Puiseux field in Section 23 supplies the contrasting real closed Borel subfield.

21 Every observable family has a countable support bound

We now strengthen the nonexistence statement into a restriction on every individual Borel system of names. The essential input is the classical boundedness theorem for analytic families of well-founded trees [8]. The following proof makes the particular form we need explicit.

Lemma 21.1 (Analytic boundedness, with proof). *If $\mathcal{T} \subseteq \text{WF}$ is analytic, there is a countable ordinal β bounding the root ranks of all trees in $\mathcal{T}$.*

Proof. The empty tree is assigned rank zero and can be discarded. If no trees remain, take $\beta = 1$. Otherwise write the remaining analytic family $\mathcal{T}$ as the projection of a closed set $C \subseteq \mathbb{N}^\mathbb{N} \times \text{Tr}$. Fix a binary coding of trees using an enumeration of $\mathbb{N}^{<\mathbb{N}}$. Construct a countable tree U whose level- k nodes are triples (s, u, v) with $s, u \in \mathbb{N}^k$, $v \in 2^k$, for which some $(y, T) \in C$ satisfies

$$u = y \upharpoonright k, \quad v = \text{code}(T) \upharpoonright k, \quad s \in T.$$

Taking initial segments preserves these conditions, so U is a tree.

If U had an infinite branch, its three coordinates would give $b, y \in \mathbb{N}^\mathbb{N}$ and a binary code z . For each k choose a witness $(y_k, T_k) \in C$ for the level- k node. The witnesses converge to (y, T) , where T has code z : for every fixed number of coordinates the witnesses agree with (y, z) once k is large. Since C and Tr are closed, $(y, T) \in C$. For every j , the node $b \upharpoonright j$ belongs to T_k for all $k \geq j$, and hence belongs to T . Thus $T \in \mathcal{T}$ has a branch, a contradiction. Therefore U is well-founded.

For each $(y, T) \in C$, the assignment

$$s \mapsto (s, y \upharpoonright |s|, \text{code}(T) \upharpoonright |s|)$$

embeds the extension tree T into U , taking children to children. Induction on node ranks gives $\text{rk}(T) \leq \text{rk}(U)$. A well-founded countable tree has countable rank. Taking $\beta = \text{rk}(U) + 1$ gives a strict bound. $\square$

Theorem 21.2 (Uniform support bound). *Let L be countable, Y standard Borel, and $c : Y \rightarrow \mathbb{R}^L$ Borel. Suppose every $\text{supp}(c(y))$ is well-ordered in L . Then there is $\alpha < \omega_1$ such that*

$$\text{otp}(\text{supp}(c(y))) < \alpha \quad \text{for every } y \in Y.$$

The same conclusion holds for reverse well-ordered supports, with the reverse support order used to compute the type.

Proof. For $S \subseteq L$, form the descending-sequence tree D_S . Its root is the empty sequence, and its other nodes are finite strictly decreasing sequences of members of S . The map $S \mapsto D_S$ is continuous on binary support codes; the composite $y \mapsto D_{\text{supp}(c(y))}$ is Borel, since each nonzero coefficient test is Borel. Its image is an analytic family of well-founded trees. Apply Lemma 21.1.

For a well-ordered S of type ξ , the root rank of D_S is exactly ξ . Indeed, the rank of a node ending in $s \in S$ is the order type of the initial segment of S below s ; induction verifies the recursion. The root rank is the supremum of these ranks plus one, namely ξ . Reversing L proves the last assertion. $\square$

Corollary 21.3. *For a countable linear order L , the following are equivalent:*

- (i) L is scattered;
- (ii) the order types of all well-ordered subsets of L have a common countable upper bound;
- (iii) $\text{WO}(L)$ is analytic;
- (iv) $\text{WO}(L)$ is Borel.

Proof. Theorem 18.4 gives (i) $\Leftrightarrow$ (iv) and excludes (iii) in the nonscattered case. If (iv) holds, apply Theorem 21.2 to the identity binary-coefficient coding of $\text{WO}(L)$ to obtain (ii). If L is nonscattered, it contains $\mathbb{Q}$, and every countable ordinal embeds in $\mathbb{Q}$ by the finite-stage rational insertion construction used in Lemma 18.2. Its well-ordered suborders therefore have unbounded countable types, contradicting (ii). $\square$

The bound is uniform over the entire parameter space, even if it is uncountable. It is not asserted to be computable from an arbitrary code for the Borel map. In particular, this is a boldface descriptive theorem, not an unqualified statement that every effective system has a computable ordinal bound.

22 The obstruction already lies in one omnific interval

Return to the X -convention. Let $I = \mathbb{Q} \cap (1/3, 2/3)$ and consider the compact space of candidate coefficient arrays

$$C = \left\{ X + \sum_{q \in I} \varepsilon_q X^q : \varepsilon_q \in \{0, 1\} \right\}. \quad (22.1)$$

Here the displayed sums are only *candidate codes*. A member of $\mathcal{C}$ denotes a Hahn or surreal normal form only when its nonzero support is reverse well-ordered.

Theorem 22.1 (Bounded-interval localization). *The valid-code subset $\mathcal{C}_{\text{valid}}$ of (22.1) is Π_1^1 -complete. Every valid value lies in the omnific interval $[X, 2X)$. The reverse support order types in this slice are unbounded in ω_1 .*

Proof. Take the fixed increasing Kleene–Brouwer embedding μ_{KB} from Lemma 18.2 and put

$$e(s) = \frac{2}{3} - \frac{\mu_{\text{KB}}(s)}{3} \in I.$$

This reverses the Kleene–Brouwer order. Define the candidate code

$$z_T = X + \sum_{s \in T} X^{e(s)}. \quad (22.2)$$

The map $T \mapsto z_T$ is continuous in the coefficient product topology. The support below the leading exponent 1 is reverse well-ordered exactly when $\text{KB}(T)$ is well-ordered, exactly when $T \in \text{WF}$. Appending the single larger exponent 1 changes none of these equivalences. Coanalytic membership follows from the descending-sequence characterization applied to the reverse exponent order.

For every valid code, the lower part is nonnegative because all its nonzero coefficients are positive, while $2X - z_T$ has greatest exponent 1 and coefficient 1. Hence $X \leq z_T < 2X$; the lower equality occurs when the lower support is empty.

Every countable ordinal ξ embeds into I in its reverse order, by the same rational-insertion principle. Taking that binary support gives total reverse support type $1 + \xi$. These types are unbounded below ω_1 . $\square$

Corollary 22.2. *There is no coefficient-observable standard Borel parametrization onto $\mathcal{C}_{\text{valid}}$, even with repeated names. More generally, no such parametrization can cover a family whose support types are unbounded in ω_1 .*

Proof. The general statement is the contrapositive of Theorem 21.2. Apply it to Theorem 22.1. $\square$

This localization removes several possible explanations for the obstruction. It is not due to large coefficients, an unbounded range of leading degrees, or a need to represent the proper class of all surreals. The leading monomial is fixed, the other coefficients are bits, the remaining exponents lie in one fixed rational band, and all actual values lie in one bounded omnific interval.

Remark 22.3 (Relation to ProveIt’s existing lower bound). The use of well-founded trees, and the fact that general support validation is coanalytic-hard, are already recorded in the omnific-notations report [21]. The present localization is included as an explicit corollary with a fixed coefficient cylinder. The more substantial extensions are the scattered/nonscattered classification and its observable-field and uniform-family formulations. None of these descriptions establishes global priority for a result outside the inspected material.

[write] Theorem 22.1 and the collection’s `onot:neg:thm:pionesone` are the same phenomenon in two slices, as the remark above says: there, lightface completeness for recursive trees, with decidable dyadic supports in $[1, 2)$ and coefficient one; here, boldface completeness with the fixed leading monomial X , bits on the band $(1/3, 2/3)$ and values in $[X, 2X)$. The unboundedness of the support types and corollary 22.2 are not in the omnific-notations report.

23 A real closed Borel subfield and its finite omnific integer part

The preceding obstructions do not make exact non-Archimedean algebra impossible. A particularly useful positive model separates finite support, bounded-denominator infinite support, and arbitrary Hahn support.

Let

$$\mathcal{P} = \bigcup_{n \geq 1} \mathbb{R}((t^{1/n})), \quad X = t^{-1}. \quad (23.1)$$

The union is a field: any two denominators have a common multiple. Newton–Puisseux says that the corresponding complex field $\bigcup_n \mathbb{C}((t^{1/n}))$ is algebraically closed [12, 13]. It is exactly $\mathcal{P}(i)$, and $\mathcal{P}$ is ordered by the leading coefficient. Hence $\mathcal{P}$ is real closed.

Define

$$\mathcal{A}_{\text{fin}} = \left\{ n + \sum_{q \in S} a_q X^q : n \in \mathbb{Z}, S \subseteq \mathbb{Q}_{>0} \text{ finite}, a_q \in \mathbb{R} \right\}. \quad (23.2)$$

Zero coefficients may be removed to obtain a unique finite normal form. The notation $\mathbb{Z} \oplus \bigoplus_{q>0} \mathbb{R}X^q$ describes an additive direct sum. It does not mean that $\mathcal{A}_{\text{fin}}$ is an $\mathbb{R}$ -algebra: its constant subring is $\mathbb{Z}$, not $\mathbb{R}$.

Lemma 23.1 (Exact principal-part floor). *The ring $\mathcal{A}_{\text{fin}}$ is an integer part of $\mathcal{P}$. More explicitly, write uniquely*

$$f = p + r + \varepsilon,$$

where p contains the positive powers of X , $r \in \mathbb{R}$ is the constant term, and ε has only negative powers of X . Then p is finite, ε is infinitesimal, and the unique $a \in \mathcal{A}_{\text{fin}}$ with $a \leq f < a + 1$ is

$$a = p + \begin{cases} \lfloor r \rfloor - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ \lfloor r \rfloor, & \text{otherwise.} \end{cases} \quad (23.3)$$

Proof. For $f \in \mathbb{R}((t^{1/n}))$, its exponents are bounded below in the lattice $\frac{1}{n}\mathbb{Z}$. Only finitely many are negative in the t -convention, so the positive- X principal part p is finite. Every remaining nonconstant term has negative X -exponent; if the tail is nonzero, its greatest exponent is negative, so it is smaller in absolute value than every positive real.

When $r \notin \mathbb{Z}$, the fractional part $r - \lfloor r \rfloor$ lies strictly between 0 and 1, and adding an infinitesimal preserves both inequalities. When $r \in \mathbb{Z}$ and $\varepsilon \geq 0$, the remainder is in $[0, 1)$. When $r \in \mathbb{Z}$ and $\varepsilon < 0$, subtracting one from the proposed floor makes the remainder $1 + \varepsilon$, which also lies in $[0, 1)$. This proves the inequalities.

The ring is discretely ordered. A positive element with nonempty positive support exceeds every real, in particular 1; a positive constant is an ordinary positive integer. Thus 1 is its least positive element, and two distinct elements of $\mathcal{A}_{\text{fin}}$ cannot both satisfy the floor inequalities. $\square$

The boundary correction in (23.3) is essential. For example, $\lfloor -X^{-1} \rfloor_{\mathcal{A}_{\text{fin}}} = -1$, not 0. This correction is also explicitly recorded in Provel’s current omnific-floor description [20]; it is not a new correction discovered by this report.

[write] Lemma 23.1 is the collection’s `odg:thm:floor` (omnific-diophantine-geometry report) read in the Puiseux field. Under 03’s identification of X with ω (section 17), $\mathcal{P}$ is a subfield of $\mathbf{No}$; p , r and ε are the positive-growth part, the real constant and the infinitesimal tail of `odg:eq:floor`; and $\mathbf{Oz} \cap \mathcal{P} = \mathcal{A}_{\text{fin}}$, because the positive-growth

part of a Puiseux series is finite. Formula (23.3) is therefore the same formula, and its negative-infinitesimal correction is the case $r \in \mathbb{Z}$, $\varepsilon < 0$ there. `odg:thm:floor` is proved in Lean for every surreal (`Algebra/SurrealNumbers/Surreal/Foundations/OmnificFloor.lean: omnificFloor_spec, existsUnique_omnific_integerPart, omnificFloor_of_integer_negative_tail`; row “Full `odg:thm:floor`” of docs/FORMALIZATION.md). The Lean development does not mention $\mathcal{P}$ or $\mathcal{A}_{\text{fin}}$, so the lemma as stated here, with the finiteness of p and the integer-part property of $\mathcal{A}_{\text{fin}}$ in $\mathcal{P}$, is not formalized. Source 03 credits the Surreal README for the correction.

Theorem 23.2. *The nonnegative cone $\mathcal{A}_{\text{fin}, \geq 0}$ is an uncountable model of Robinson arithmetic plus open induction in the ordered semiring language.*

Proof. A discretely ordered ring gives the elementary semiring and predecessor axioms on its nonnegative cone. In particular, if $x > 0$ then $x - 1 \geq 0$. Uncountability follows already from the distinct elements aX , $a > 0$ real.

We prove open induction directly, using the familiar integer-part direction of Shepherdson’s theorem [6, 7]. Let $W \subseteq \mathcal{A}_{\text{fin}, \geq 0}$ be a nonempty set defined by a quantifier-free formula with parameters from $\mathcal{A}_{\text{fin}}$. Interpret the same formula in the real closed field $\mathcal{P}$. Its truth set is a finite union of intervals and points: the finitely many roots of the polynomials appearing in the formula partition the line into regions of constant sign.

Intersect these finitely many cells with $[0, \infty)$ and choose the first one meeting $\mathcal{A}_{\text{fin}}$. A point cell contributes its own element. For an interval cell with lower endpoint $b \in \mathcal{P}$, the integer-part property gives the least integer-part element at or above b , or the least one strictly above b when the endpoint is excluded. This element is nonnegative; if it were above the interval, that interval would contain no element of $\mathcal{A}_{\text{fin}}$ at all. For a cell beginning at 0, the same argument starts with 0 or 1. Thus W has a least element.

If a quantifier-free induction instance failed while its base and successor steps held, the nonempty quantifier-free set of counterexamples would have a least member m . It would satisfy $m > 0$, so $m - 1 \in \mathcal{A}_{\text{fin}, \geq 0}$ would satisfy the formula. The successor step would then force the formula at m , a contradiction. $\square$

[write] The proof of [theorem 23.2](#) is the Shepherdson route of the collection’s `dsn:cor:openinduction (definable-surreals-and-omnific-integers report: open induction for the integer part  $I_A$  of a real closed field  $K_A$ )` and of the paragraph citing Shepherdson’s theorem in the subsection “Sums of squares and induction” of the `omnific-diophantine-geometry report (open induction for the omnific integers of a divisible Hahn workspace)`. Here it is applied to $\mathcal{A}_{\text{fin}}$ inside the real closed field $\mathcal{P}$; uncountability and the semiring axioms are 03’s. `dsn:cor:openinduction` is pending in docs/FORMALIZATION.md. As 03 says, $\mathcal{A}_{\text{fin}, \geq 0}$ is a Shepherdson-type model, not a new model of arithmetic.

23.1 Why the Puiseux field itself has Borel codes

In the ambient coefficient space $\mathbb{R}^{\mathbb{Q}}$, the field (23.1) is exactly the union, over $n \geq 1$ and $k \in \mathbb{Z}$, of the closed sets

$$\{a : a_q = 0 \text{ whenever } q \notin \frac{1}{n}\mathbb{Z} \text{ or } q < k/n\}.$$

It is therefore F_σ and standard Borel. The convolution and unique graph arguments of Theorem 20.2 make its field operations Borel. Every such support has increasing order type at most ω .

This gives a useful three-way contrast. Finite principal parts have finite support; the real closed Puiseux field has a uniform support bound ω ; the full rational-exponent Hahn field has supports of unbounded countable type. Real closedness alone does not prevent a Borel coefficient presentation. Full arbitrary-support closure is what makes Corollary 20.4 apply.

24 A Polish finite-normal-form presentation

24.1 The topology and its Borel structure

Write an element of $\mathcal{A}_{\text{fin}, \geq 0}$ uniquely as $(S, n, (a_q)_{q \in S})$, where all $a_q \neq 0$. If $S \neq \emptyset$, positivity means $a_{\max S} > 0$ and the integer n is unrestricted. If $S = \emptyset$, we require $n \geq 0$.

For finite nonempty $S \subseteq \mathbb{Q}_{>0}$ and $n \in \mathbb{Z}$, let

$$U_{S,n} = \{a \in (\mathbb{R} \setminus \{0\})^S : a_{\max S} > 0\}.$$

For $S = \emptyset$, let $U_{\emptyset,n}$ be a singleton when $n \geq 0$. Put on $\mathcal{A}_{\text{fin}, \geq 0}$ the topological disjoint union

$$(\mathcal{A}_{\text{fin}, \geq 0}, \tau_{\text{FS}}) = \bigsqcup_{S,n} U_{S,n}. \quad (24.1)$$

Support and integer constant are locally constant in this topology.

Proposition 24.1. *The topology τ_{FS} is Polish. Its Borel sets are exactly the Borel sets obtained from the canonical finite-support coefficient coding.*

Proof. Each nonsingleton component is open in a finite-dimensional Euclidean space, and hence Polish. There are countably many components. A countable topological disjoint union of Polish spaces is Polish: choose bounded complete metrics on the components and put distance 2 between distinct components. Taking the union of countable dense sets gives separability.

Every coefficient map is continuous for (24.1). Conversely, an exact support condition consists of nonzero tests at finitely many coordinates and zero tests at countably many others, so is Borel in the coefficient product. On each such Borel piece the coefficient coordinates give exactly the usual finite-dimensional Borel structure. Since there are countably many pieces, the two Borel structures agree. $\square$

This is not the order topology. The order topology on a discretely ordered uncountable arithmetic domain is discrete and therefore not separable. Nor is τ_{FS} the ambient coefficient subspace topology, where convergence need not make the exact support eventually constant. The distinction is responsible for the cancellation discontinuities below.

24.2 A general support-normalization lemma

Lemma 24.2. *Let $U \subseteq \mathbb{R}^d$ be nonempty and open, and let $(p_j)_{j \in J}$ be a finite family of real polynomials. Define a map whose output is the finite list of nonzero coordinates $p_j(u)$, with the target topologized as a disjoint union by exact nonzero support. Let $E = \{j : p_j \not\equiv 0\}$. Its discontinuity set is exactly*

$$\mathcal{Z} = \bigcup_{j \in E} \{u \in U : p_j(u) = 0\}.$$

This set is closed and nowhere dense in U . The map is continuous on countably many closed subsets covering U .

Proof. Outside $\mathcal{Z}$, all nonidentically-zero coordinates stay nonzero on a sufficiently small neighborhood, while the others are always zero. The output support is therefore constant and the coordinate formulas are continuous.

At $u \in \mathcal{Z}$, choose $j \in E$ with $p_j(u) = 0$. A nonzero real polynomial cannot vanish on a nonempty open set. Thus arbitrarily near u there are points where this coordinate is nonzero. Their output support differs from the support at u , so their images lie outside the clopen target component containing the image of u . Continuity fails. This also proves that the finite union of zero sets is nowhere dense; closedness is immediate.

For $Z \subseteq E$ and integer $k \geq 1$, use the closed set

$$F_{Z,k} = \{u \in U : p_j(u) = 0 \ (j \in Z), \quad |p_j(u)| \geq 1/k \ (j \in E \setminus Z)\}. \quad (24.2)$$

These sets cover U : at any point choose the actual zero set and then a large enough k for the finitely many remaining coordinates. On each piece, the output support is constant and the map is continuous. For a zero-dimensional component consisting of one point, the same conclusion holds with empty discontinuity set. $\square$

24.3 The exact arithmetic cancellation loci

Fix an input component $U_{S,n} \times U_{T,m}$. Extend the variable coefficient families by $a_q = 0$ for $q \notin S$ and $b_q = 0$ for $q \notin T$. For addition, the output coefficient at $r > 0$ is

$$A_r = a_r + b_r, \quad (24.3)$$

and its constant is $n + m$. For multiplication, the output constant is nm and the coefficient at $r > 0$ is

$$M_r = nb_r + ma_r + \sum_{p+q=r} a_p b_q. \quad (24.4)$$

Only finitely many exponents need to be considered: $S \cup T$ for addition and $S \cup T \cup (S + T)$ for multiplication.

For multiplication define

$$E_{S,T,n,m} = \{r \in S \cup T \cup (S + T) : M_r \not\equiv 0\},$$

where identity means polynomial identity in the independent input coefficient variables after fixing the integers n, m . This removes absent terms and zero-factor components correctly.

Theorem 24.3 (Exact cancellation theorem). *On the Polish presentation (24.1), addition and multiplication are Borel. Their discontinuity sets are given componentwise by*

$$D_+ \cap (U_{S,n} \times U_{T,m}) = \bigcup_{r \in S \cap T} \{a_r + b_r = 0\}, \quad (24.5)$$

$$D_\times \cap (U_{S,n} \times U_{T,m}) = \bigcup_{r \in E_{S,T,n,m}} \{M_r = 0\}. \quad (24.6)$$

Globally, both sets are closed and nowhere dense. Both operations are continuous on countably many closed pieces. In particular, both are continuous at every point of the common dense open set $\mathcal{A}_{\text{fin}, \geq 0}^2 \setminus (D_+ \cup D_\times)$.

Proof. On an input component, the actual normal form is obtained from (24.3) or (24.4) by deleting the zero coefficients. The integer output constant is fixed. Apply Lemma 24.2. For

addition, a coefficient occurring in only one input support is constrained to be nonzero, so the possible zero sets are precisely those in (24.5). For multiplication the expected polynomial support is exactly $E_{S,T,n,m}$.

Every input component is clopen, and these components partition the domain. A set whose intersection with each component is closed is globally closed; a set whose intersection with each nonempty open component is nowhere dense has empty interior and, when closed, is nowhere dense globally. This gives the assertions about D_+ and $D_\times$.

The pieces (24.2) are closed relative to a clopen component, hence closed in the full domain. Taking them over countably many components, zero patterns, and integers k gives countable closed covers for both operations. To see Borelness directly, the inverse image of an open set is a countable union of sets open relative to these closed pieces. Such sets are Borel (indeed F_σ in a metric space). $\square$

Example 24.4 (Addition: cancellation below the leading term). For real δ sufficiently near 0, set

$$x = X^2 + X, \quad y_\delta = X^2 + (-1 + \delta)X.$$

All these elements are positive and remain in fixed input components. But

$$x + y_\delta = 2X^2 + \delta X.$$

At $\delta = 0$ the output support loses exponent 1, so the outputs do not converge to $2X^2$ in τ_{FS} . Leading coefficients of positive inputs cannot cancel each other; the lower coefficient is what fails here.

Example 24.5 (Multiplication: a vanished middle coefficient). Let $x = X + 1$ and $y_\delta = (1 + \delta)X - 1$. Near $\delta = 0$ both inputs are positive, their supports and integer constants are fixed, and

$$xy_\delta = (1 + \delta)X^2 + \delta X - 1.$$

The middle term disappears at $\delta = 0$. Thus $(X + 1, X - 1) \in D_\times$. Multiplication by 0, on the other hand, is constant and continuous; this explains the exclusion of identically zero coefficient polynomials.

Remark 24.6 (A measurability qualification). Countable closed-piece continuity is the statement proved here. We do not identify it without further hypotheses with the Baire-one class of maps into this disconnected target. Nor does the dense-open conclusion say that an algebraic cancellation can be ignored by exact computation: deleting a single zero coefficient changes the canonical component.

25 Glazer's obstruction survives every compatible Polish recoding

We use precisely the following published result.

Theorem 25.1 (Glazer, Corollary 1 of [1]). *There is no model of Robinson arithmetic plus !Open on an uncountable Polish space in which addition is continuous and multiplication is Borel.*

Theorem 25.2 (No removal of the addition obstruction). *No Polish topology on the set $\mathcal{A}_{\text{fin}, \geq 0}$ with the same Borel sets as τ_{FS} makes addition continuous. More generally, there is no Polish presentation Borel-isomorphic to the canonical one in which transported addition is continuous.*

Proof. By Theorem 23.2, the algebraic structure is an uncountable model of the required arithmetic. By Theorem 24.3, multiplication is Borel in the canonical Borel structure. Changing the Polish topology without changing its Borel sets leaves multiplication Borel on the product; finite products of standard Borel spaces have the same product Borel structure. Continuous addition would contradict Theorem 25.1. Transporting the operations through a Borel isomorphism changes none of these hypotheses. $\square$

Corollary 25.3. *The distinction between global continuity and continuity off a closed nowhere dense set is genuine for this explicit model: the latter is attained for both operations, while the former is impossible for addition under any Borel-compatible Polish presentation.*

This is a sharp distinction between an empty discontinuity set and an explicit closed nowhere dense one. It is not a proof that our particular zero loci have minimal cardinality, minimal dimension, or a minimal descriptive rank among all recodings.

Glazer's Theorem 2 also excludes continuous multiplication for uncountable Borel models of the stronger theory $\mathbf{Q} + I\Delta_0 + \text{Exp}$ [1]. We have proved only open induction for $\mathcal{A}_{\text{fin}, \geq 0}$. The multiplication obstruction from that stronger result is therefore *not* transferred to every compatible topology on $\mathcal{A}_{\text{fin}, \geq 0}$. Multiplication fails to be continuous in our displayed topology by Theorem 24.3; whether a different compatible Polish topology can make multiplication continuous remains a separate question here.

26 Where the two parts meet [merge]

The real model of Part I is one stratum of the cone $\mathcal{A}_{\text{fin}, \geq 0}$ of section 24. Read ω as X and let

$$\iota : M_{\mathbb{R}} \longrightarrow \mathcal{A}_{\text{fin}}, \quad \iota(r, n) = rX + n,$$

which is 05's map $\iota_{\mathbb{R}}$ of theorem 9.1 in 03's notation. Put

$$C_{\mathbb{R}} = \{a \in \mathcal{A}_{\text{fin}, \geq 0} : \text{supp } a \subseteq \{1\}\} = \bigcup_{n \in \mathbb{Z}} U_{\{1\}, n} \cup \bigcup_{n \geq 0} U_{\emptyset, n}.$$

Proposition 26.1 (The real model as a stratum [merge]). (i) ι is an injective ordered additive map onto $C_{\mathbb{R}}$, fixing 0 and 1. The set $C_{\mathbb{R}}$ is clopen in $(\mathcal{A}_{\text{fin}, \geq 0}, \tau_{\text{FS}})$ and closed under addition, but not under multiplication.

- (ii) On $\{(r, n) \in M_{\mathbb{R}} : r > 0\}$, ι is a homeomorphism onto $\bigcup_n U_{\{1\}, n}$, for the topology of $M_{\mathbb{R}}$.
- (iii) At the standard numerals the two topologies differ: every ordinary integer $n \geq 0$ is isolated in τ_{FS} , but $(0, n)$ is not isolated in $M_{\mathbb{R}}$. Thus τ_{FS} induces on $C_{\mathbb{R}}$ a topology strictly finer than the one transported from $M_{\mathbb{R}}$.
- (iv) Addition is continuous on $(C_{\mathbb{R}}, \tau_{\text{FS}})$. Consequently $(C_{\mathbb{R}}, \tau_{\text{FS}})$ is another uncountable, locally compact Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous addition; it is not perfect, its isolated points being exactly the standard numerals.

Proof. (i) An element of $\mathcal{A}_{\text{fin}}$ with support in $\{1\}$ is $n + aX$; by (24.1) it lies in $\mathcal{A}_{\text{fin}, \geq 0}$ when $a > 0$, or $a = 0$ and $n \geq 0$. These are exactly the values of ι . Addition is coordinatewise and the order is decided by the leading coefficient, as in (3.1). The set $C_{\mathbb{R}}$ is a union of components of the topological sum (24.1), hence open, and its complement is the union of the other components, hence also open. Finally $X \cdot X = X^2 \notin C_{\mathbb{R}}$.

(ii) Both $U_{\{1\},n}$ and the part $r > 0$ of the slice of integer coordinate n in $M_{\mathbb{R}}$ are the open ray $(0, \infty)$ with its Euclidean topology, and ι is the identity on the coefficient.

(iii) $U_{\emptyset,n}$ is a singleton component of (24.1). In $M_{\mathbb{R}}$, $(1/j, n) \rightarrow (0, n)$ by (5.1).

(iv) On a component $U_{S,n} \times U_{T,m}$ with $S, T \subseteq \{1\}$, (24.5) is a union over $r \in S \cap T \subseteq \{1\}$ of the sets $\{a_1 + b_1 = 0\}$, which are empty because both leading coefficients are positive. So $C_{\mathbb{R}}^2$ does not meet D_+ , and addition maps $C_{\mathbb{R}}^2$ into $C_{\mathbb{R}}$. A clopen subspace of a Polish space is Polish, the components are points and open rays, and the algebraic structure is that of $M_{\mathbb{R}}$, so [corollary 4.11](#) applies. $\square$

The boundary. Glazer’s Corollary 1 concerns a model of $Q+I\text{Open}$ with Borel multiplication. The ring cone $\mathcal{A}_{\text{fin}, \geq 0}$ is such a model ([theorems 23.2](#) and [24.3](#)), so no Borel-compatible Polish topology on $\mathcal{A}_{\text{fin}, \geq 0}$ makes its addition continuous ([theorem 25.2](#)). The stratum $C_{\mathbb{R}}$ is not such a model: as an additive monoid it is $M_{\mathbb{R}}$, which carries no unital semiring multiplication at all ([corollary 10.5](#)), and the multiplication of $\mathcal{A}_{\text{fin}, \geq 0}$ leaves it. On $C_{\mathbb{R}}$ addition is continuous for τ_{FS} itself; the discontinuities of addition on $\mathcal{A}_{\text{fin}, \geq 0}$ lie on the cancellation loci (24.5), which need two inputs whose supports share an exponent below their leading ones, and so lie outside $C_{\mathbb{R}}^2$. Thus [Part I](#) and [theorem 25.2](#) delimit, inside one ring, where Corollary 1 applies: the additive stratum admits continuous addition for both topologies, while the ring, with its Borel multiplication and open induction, admits it for none.

The Baire model. The image $\iota_B(M_B)$ of [theorem 9.1](#) lies in 03’s rational Hahn ring $\mathcal{H}_{\mathbb{Q}}$ of (17.1): its supports are contained in $\{1, \frac{1}{2}, \frac{1}{3}, \dots\} \cup \{0\}$ and have reverse order type at most $\omega + 1$. Its elements with infinitely many nonzero coefficients lie neither in $\mathcal{A}_{\text{fin}}$ nor in the Puiseux field $\mathcal{P}$, whose denominators are bounded. Each coefficient of $\iota_B(a, n)$ is a_j , n or 0, a continuous function on M_B , so the composite of 05’s homeomorphism $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$ ([theorem 6.1](#)) with ι_B is a coefficient-observable standard Borel parametrization of $\iota_B(M_B)$ in the sense of [definition 17.1](#). This is consistent with [theorem 21.2](#), whose bound can be taken to be $\omega + 2$ here, while the full field $\mathbb{R}((t^{\mathbb{Q}}))$ lies on the non-Borel side of [theorem 20.2](#). The Baire model is one of the restricted support families that [section 27](#) calls for.

A correction to the placement record [write]. The placement commit 3d2177df4 and the intake dossier said that 03’s topology τ_{FS} restricts on the stratum to the topology of 04 and 05. Parts (ii) and (iii) of [proposition 26.1](#) give the exact statement: the two topologies agree on the elements with positive real coefficient and differ at the standard numerals, which τ_{FS} isolates. Both make addition continuous on the stratum, so the conclusion drawn there stands.

27 Consequences for exact notation systems and formalization

27.1 A practical separation of three design goals

The results separate three tasks that can otherwise be conflated. *Arithmetic* asks whether operations can be performed on valid inputs. *Validation* asks whether an arbitrary code is a valid well-based normal form. *Topological presentation* asks whether the operations respect a complete separable topology on the names or values. A positive answer to one is not a positive answer to the others.

In the finite-support model, every code carries an explicit finite list, and arithmetic is polynomial coefficient arithmetic followed by exact zero tests. The canonical domain is Polish, the operations are Borel, and their failure of continuity is exactly the support-changing cancellation calculated above. No general theorem of computability over arbitrary real coefficients is claimed: a real coefficient supplied by an unrestricted approximation oracle need not have decidable equality with zero.

For full Hahn fields on scattered exponent groups, support validation is a Borel predicate. This is not the same as decidability by a finite algorithm. The Π_3^0 examples already involve quantification over infinitely many rows. For dense rational exponents the obstruction is stronger: no standard Borel system of fully observable names covers every valid series.

A useful exact system must therefore limit its admissible support families, relax coefficient observability, or accept a nonstandard measurable domain. The Puiseux field demonstrates that a severe-looking support restriction may still permit real closedness. Theorem 21.2 says every observable Borel system necessarily has some uniform countable support bound, even when that bound is not built into the notation explicitly.

27.2 No paradoxes are needed

All main results are in ZFC and concern sets. For countable Γ , the ambient coefficient space $\mathbb{R}^\Gamma$ is a set of cardinality continuum. Its valid-code subset is a set whether or not it is Borel. The rational Hahn ring $\mathcal{H}_\mathbb{Q}$ and its finite subring $\mathcal{A}_{\text{fin}}$ are also sets. The construction never forms a set of all surreal numbers or quantifies over all class maps as a set.

The optional identification with Conway normal forms can be read as a set-indexed definable embedding into the proper class **No**, or carried out inside a sufficiently large set-sized normal-form universe. Standard Replacement handles the image of a fixed set of codes. No global choice, large cardinal, or class-sized Polish topology is used.

27.3 A formalization plan with an explicit dependency boundary

A Lean development can be organized into independent interfaces rather than starting from the entire proper-class surreal hierarchy.

- Layer 1. Finite algebra and normalization.** Use finitely supported rational-exponent coefficient maps with an integer constant. Establish convolution, leading-exponent sign, discreteness, and the coefficient identities (24.3)–(24.4). The exact cancellation examples are small executable tests at this layer.
- Layer 2. Series and arithmetic.** Use a set-sized Hahn or Puiseux field, prove the principal-part floor with its negative-infinitesimal case, and derive quantifier-free induction via the one-variable sign-cell argument. Newton–Puiseux is a named external dependency until its real-closedness interface is formally supplied.
- Layer 3. Topological support components.** Formalize the countable disjoint union, the polynomial normalization lemma, and the closed-piece covers. None of this requires descriptive ranks or proper-class surreals.
- Layer 4. Descriptive-set-theoretic structure.** Formalize the fixed Kleene–Brouwer embedding, Hausdorff-sum support criterion, analytic boundedness, and observable-field equivalence. The classical Hausdorff characterization and Glazer theorem must be proved or explicitly listed as hypotheses; importing them informally is not a checked proof.

This is a proposed architecture, not a claim that these layers have been implemented or that all required library APIs currently exist. The provided Python artifact checks finite rational instances, not Lean terms or infinitary assertions.

28 Further research questions and concrete next targets

The following questions are proposed here. Unless explicitly attributed, “open” means unresolved in this report, not certified as absent from the entire literature.

Question 28.1 (Continuous multiplication on the finite model). Does $\mathcal{A}_{\text{fin}, \geq 0}$ admit a Polish topology with the canonical Borel sets in which multiplication is continuous?

Theorem 25.2 settles the analogous addition question negatively. Glazer’s stronger multiplication theorem cannot be invoked without establishing its extra arithmetic hypotheses. A positive solution would isolate a concrete asymmetry between the two operations; a negative solution should identify an obstruction valid already for this particular open-induction model.

Question 28.2 (How small can the discontinuity set be?). Among all Borel-compatible Polish presentations of $\mathcal{A}_{\text{fin}, \geq 0}$, what restrictions hold on the discontinuity set of addition? Can it be countable? Can it be contained in a prescribed small-dimensional or otherwise canonical set?

The present model gives a nonempty closed nowhere dense set. Glazer excludes only its complete disappearance, not every possible reduction in complexity. This is a natural refinement of the topological theorem that can first be studied on explicit finite-support components.

Question 28.3 (Exact complexity beyond finite lexicographic rank). For a fixed countable scattered order L , compute the least Borel level of $\text{WO}(L)$ directly from an economical Hausdorff decomposition. Which Wadge degrees occur among these support predicates?

The qualitative dichotomy is settled by Theorem 18.4, while Theorem 19.1 shows that simply increasing finite order rank need not increase Borel level. A satisfactory answer should distinguish ordinal support capacity from descriptive complexity rather than treating them as interchangeable ranks.

Question 28.4 (Borel, algebraically rich subfields of the rational Hahn field). Classify Borel subfields $F \subseteq \mathbb{R}((t^{\mathbb{Q}}))$ containing all rational monomials and stable under specified truncations or algebraic-root operations. Which countable support bounds can such fields attain?

The Puiseux field is one real closed example with supports of type at most ω . The full Hahn field is excluded. An interesting target is a well-described hierarchy of truncation-closed real closed Borel subfields between them, together with provably sharp support bounds at each stage. No claim is made that every ordinal cutoff is preserved by inversion.

Question 28.5 (How much coefficient observability is necessary?). If only a selected family of coefficient combinations or truncations is required to be Borel, which versions of Theorem 20.2 survive?

The current proof uses all countably many coefficient maps to obtain one Borel map into the full product. This pinpoints a possible escape route and suggests investigating minimal observation families that still recover the binary support cylinder in Theorem 22.1.

Question 28.6 (Certified support languages). Construct useful exact languages whose valid supports carry explicit well-foundedness certificates, and determine their closure under arithmetic, root extraction, and named truncations.

The repository already contains effective restricted omnific calculi [21]. The present results suggest comparing them by their Borel parameter spaces and support bounds as well as by decidable equality. A certificate verifier for a fixed language is not a total validator for all raw rational-exponent supports; Theorem 22.1 forbids conflating those objectives.

Question 28.7 (A minimal arithmetic hypothesis for the presentation barrier). Which concrete consequences of open induction suffice to rule out continuous addition in the finite principal-part model or related integer parts?

A proof specialized to finite Puiseux principal parts might replace the full open-induction interface with explicitly stated root-approximation or division properties. This would clarify the boundary between algebraic normal-form obstructions and arithmetic induction.

Question 28.8 (Formalized transfer from support theory to arithmetic). Can the observable Hahn-field criterion and the finite-support cancellation theorem be formalized as reusable results, with the classical Hausdorff and Glazer inputs exposed as exact dependency interfaces?

The suggested first milestone is the support-sum criterion (18.1) together with the normalization Lemma 24.2. Both are modular and elementary enough to be reviewed independently before the larger descriptive-set-theoretic machinery is introduced.

[merge] Question 28.1 is the multiplicative counterpart, on $\mathcal{A}_{\text{fin}, \geq 0}$, of question 14.4. The question on the size of the discontinuity set of addition concerns all of $\mathcal{A}_{\text{fin}, \geq 0}$; on the clopen stratum $C_{\mathbb{R}}$ of proposition 26.1 that set is empty.

29 Conclusions of Part II

The intersection with Glazer’s research produces a precise hierarchy rather than a blanket impossibility theorem. Full Hahn support is Borel exactly on scattered countable exponent orders. Dense alphabets carry a coanalytic-complete validation problem, and no standard Borel family with observable coefficients can exhaust supports of unbounded countable type. This obstruction persists inside one fixed interval of omnific integers.

At the same time, a real closed Puiseux subfield and its finite principal-part integer part have explicit Borel presentations. The latter even admits a Polish topology where both arithmetic operations are continuous off closed nowhere dense sets, with exact polynomial equations describing every failure. Glazer’s theorem makes the remaining addition discontinuity unavoidable under all compatible Polish recordings. The positive construction and the negative result constrain one another: Borel arithmetic is attainable, globally continuous addition is not, and arbitrary well-based support introduces a separate obstruction before topological continuity can even be asked in the canonical standard-Borel setting.

Part III

Continuous Presburger arithmetic on Polish spaces

[write] This part merges two manuscripts of batch 86 that carry the same title, *Continuous Presburger Arithmetic on Polish Spaces*, but are independent texts: source 07 (subtitle *An affirmative construction for Glazer’s question, Baire-space models, and omnific realizations*, dated October 4, 2026, pinned to 0f95145cb) and source 06 (subtitle *Lexicographic models, an isolated-zero obstruction, and a boundary between additive and semiring arithmetic*, dated October 3, 2026, pinned to fa2f3e419). They share at most 0.74% of their word 8-grams, the longest shared runs being Glazer’s quoted question and bibliography entries, and their theorem sets differ.

Base and reading conventions. Source 07 is the base: its realization theorem ([theorem 34.1](#)) needs only a divisible Polish abelian group with a Borel positive cone and covers the coordinate families of source 06. Unmarked text in this part is source 07’s. Text from source 06 is marked [06]: in the titles of its sections, subsections and statements, and in the sentence that introduces an inserted passage. Text written for this report is marked [write] or [merge] as in [section 1.2](#). Section and subsection headings of source 06 were moved one level down where its sections are printed inside source 07’s.

What is printed again, and what is new. Both sources prove Part I’s main results once more; these statements are printed with their proofs, because the proofs differ in detail, and each says in its title which statement of Part I it repeats:

- the real model ([theorem 32.1](#) of 07 and [theorem 32.2](#) of 06 are [theorem 3.1\(1\)](#));
- the Baire model and its effective addition ([theorems 35.6](#) and [35.7](#) of 06 are [theorems 6.1](#), [6.2](#) and [6.4](#));
- Δ_2^0 definability ([theorem 37.7](#) of 06 and part (iii) of [theorem 34.1](#) extend [theorem 7.1](#));
- the order obstruction ([proposition 37.1](#) is [theorem 7.4](#)), compactness ([proposition 37.3](#) is [corollary 12.2](#)), and the semiring obstruction ([proposition 37.5](#) and [theorem 37.10](#) are [theorem 10.4](#), the latter without a prescribed unit);
- the omnific realizations ([theorem 40.1](#) extends [theorem 9.1](#) from ω to every countably infinite ordinal);
- the classical elimination theorem ([sections G](#) and [I](#), compare [theorem 4.9](#) and [corollary 4.11](#)).

The results new to the collection are 07’s Borel-cone theorem ([theorem 33.1](#)) and cone closure rank, the general realization theorem and the Borelness of definable sets in every continuous presentation ([theorem 34.1](#) and [proposition 34.3](#)), the ordinal family on Baire space ([theorems 35.1](#) and [35.4](#); also 06’s [theorem 35.9](#)), the count of exactly $2^{\aleph_0}$ models on Baire space (07’s [theorem 35.5](#), and 06’s [theorem 35.8](#) with topology, order, 0 and 1 fixed), the oscillation ranks ([theorem 36.1](#) and [corollary 36.2](#)), the discontinuity of min and max ([corollary 37.2](#)), a second topology with continuous predecessor ([proposition 37.4](#)), and source 06’s Polish semiring cone of $\mathbb{Z} + t\mathbb{R}[t]$ with continuous addition and multiplication that

fails open induction ([theorems 38.1](#) and [38.2](#)) and its isolated-zero theorem ([theorem 39.1](#)). [Section 42](#) locates these results with respect to Parts I and II.

A consequence for Part I’s criterion [merge]. Let D be a divisible ordered abelian group with a Polish group topology. By [theorem 5.2](#), M_D is Polish if and only if $D_{>0}$ is G_δ ; by [theorem 33.1](#), a Borel $D_{>0}$ is already Δ_2^0 . Hence

$$M_D \text{ is Polish} \iff D_{>0} \text{ is Borel} \iff D_{>0} \text{ is } \Delta_2^0.$$

The proof of [theorem 34.1](#) also shows that division by a standard integer is then automatically continuous (open mapping), a hypothesis that source 04’s [theorem 5.4](#) assumed.

Labels and notation. Labels of this part are `pma:cp:<label>` for both sources, except the six labels of source 06 that coincide with labels of source 07 (`eq:realmodel`, `sec:baire`, `sec:real`, `thm:baire`, `thm:continuum`, `thm:real`), which take the sub-prefix `pma:cp:s06:`. Source 06’s family M_s ($s \in 2^{\mathbb{N}}$) is renamed M'_s , with G'_s , H'_s , because source 07’s M_s is a different family; [section 1.3](#) lists this and the other conventions of Parts III–V. In particular t is a positive infinite element in this part (source 06’s $\mathbb{Z} + t\mathbb{R}[t]$), whereas in [Part V](#) t is a positive infinitesimal.

30 The question and the research intersection

30.1 An exact question with a concrete answer

Glazer’s *A Topological Tennenbaum Theorem* ends with the question:

Does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

This is Question 2 of the November 2023 version [1]. Here “Polish” means separable and completely metrizable. Presburger arithmetic is the complete first-order theory of addition on the natural numbers, with multiplication of two variables absent from the language.

The answer supplied by this report is affirmative. Its simplest witness is

$$M_{\mathbb{R}} = ((0, \infty) \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}), \quad (30.1)$$

with coordinatewise addition, $0 = (0, 0)$, $1 = (0, 1)$, and lexicographic arithmetic order that compares the real coordinate first. The topology is the subspace topology from $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$. The distinction between this topology and the arithmetic order is essential.

There is an even stronger realization on $\mathbb{N}^{\mathbb{N}}$, the usual Baire space. It is obtained by replacing $\mathbb{R}$ with a countable product of discrete copies of $\mathbb{Q}$, then taking the same lexicographic integer extension. No forcing, nonprincipal ultrafilter, or nonconstructive choice of an arithmetic operation is needed for either coordinate model.

Glazer also proves that certain stronger Borel arithmetics on uncountable Polish spaces cannot have either addition or multiplication continuous. His Corollary 1 already excludes continuous addition and Borel multiplication for $Q + I\text{Open}$ [1]. Our result concerns the additive theory. [Section 37](#) explains the precise boundary, including a stronger algebraic obstruction for our examples.

30.2 Why this is a substantive overlap with ProveIt

The repository snapshot examined is commit

0f95145cb65d5dc279e2053e8707a71bb490bb59.

Its timestamp is October 3, 2026, 18:50:05 in the America/Los_Angeles time zone. Two parts of that snapshot are especially relevant:

1. `Logic/PresburgerArithmetic` contains an executable Cooper quantifier-elimination development for formulas over ordinary integers. The main semantic correctness theorem is implemented in Lean; Rocq independently checks its normalized one-variable core [32].
2. The surreal research report on discrete initial subgroups and omnific normalization studies split ordered groups of the form $D \overrightarrow{\times} \mathbb{Z}$, including their relation to omnific integers [33].

The first supplies an existing computational and formalization framework; the second supplies the relevant algebraic shape. The new work here is to give that shape explicit Polish topologies, prove its model-theoretic and descriptive properties, and connect it to the exact published question.

The main positive theorems do not depend on accepting an unrefereed repository theorem. We prove the required ordered-group facts directly and use the classical characterization of Presburger models. The surreal connection is also given by an explicit normal-form map.

30.3 What is proved, and what is a novelty claim

The existence assertion is a theorem with a short proof. The assertion that it is the first published resolution would be a historical claim requiring different evidence. We make only the former claim. A targeted literature search found the question in Glazer's paper and found no later resolution; that is not an exhaustive priority search.

The same distinction applies to the general cone theorem, the family of Baire-space models, and the rank computations. They are proved here and are useful extensions of the construction. Their precise place in the literature remains suitable for specialist review. In particular, the underlying lexicographic construction of a $\mathbb{Z}$ -group is classical and is not presented as an invention.

30.4 The question, the overlap, and the contribution [06]

30.4.1 A precise research target [06]

Glazer's work on topological models of arithmetic gives a particularly direct intersection with ProveIt. His 2023 paper ends with the question [1, Question 2]:

Does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

Here *Presburger arithmetic* means the complete first-order theory of addition on the natural numbers, rather than full Peano arithmetic. We answer this formulation affirmatively, with

explicit models and proofs. The use of a topology different from the arithmetic order topology is essential.

The context is a sharp obstruction for stronger arithmetic. Glazer proves that neither arithmetic operation can be continuous in an uncountable Polish Borel model of $\mathbb{Q} + \text{I}\Delta_0 + \text{Exp}$; his additive obstruction already holds for $\mathbb{Q} + \text{I}\text{Open}$ when multiplication is Borel [1, Theorem 2 and Corollary 1]. The constructions below show why the induction and language hypotheses matter.

30.4.2 What is already in ProveIt [06]

We inspected the repository at commit

fa2f3e419e592a1e90c065b79d559865ec45d436.

The following connections are concrete, not merely thematic.

1. `Logic/PresburgerArithmetic` contains a Lean development of Cooper elimination and an independent Coq development of the normalized one-variable step [34]. Its current semantics use ordinary integers. The abstract $\mathbb{Z}$ -group semantics needed for the nonstandard models below are an additional formalization task.
2. The report *Discrete Initial Groups and Omnific Normalization* studies split groups $D \xrightarrow{\sim} \mathbb{Z}$ and their connection to omnific integers [35]. The algebraic form of our examples thus already has a close repository counterpart.
3. The report *Omnific Integers and Omnific–Diophantine Geometry* explicitly studies $\mathbb{Z} + t\mathbb{R}[t]$ and its realization $\mathbb{Z} + \omega\mathbb{R}[\omega]$ [36]. We equip its nonnegative cone with a useful Polish topology and prove continuity of both operations.

The lexicographic $\mathbb{Z}$ -group construction and its model theory are classical [4]; rational polynomial Presburger examples also appear in established lecture notes [24]. The contribution here is the explicit topological realization, its consequences for the stated question, and the accompanying obstruction and abundance results. We use the repository reports as documented points of contact, rather than as unverified premises for our main proofs.

30.4.3 A map of the results [06]

Model	Topology	Continuous operations	Arithmetic boundary
$(\mathbb{R} \xrightarrow{\sim} \mathbb{Z})_{\geq 0}$	Perfect, locally compact Polish	Addition; fixed standard division	Presburger; no unital semiring expansion
$(\mathbb{Q}^{\mathbb{N}} \xrightarrow{\sim} \mathbb{Z})_{\geq 0}$	Baire space	Addition, with an effective stream presentation	Presburger; no unital semiring expansion
$(\mathbb{Z} + t\mathbb{R}[t])_{\geq 0}$	Locally compact Polish; isolated standard points	Addition and multiplication	PA^- and Presburger; fails IOpen

The general obstruction in Section 39 says that the last row cannot be made perfect while retaining Polishness and continuity of both operations. The abundance theorem in Section 35.6 gives the exact possible cardinality of the collection of isomorphism types in the Baire-space setting: $2^{\aleph_0}$.

Scope of the literature check. The cited version of Glazer’s paper is arXiv:2311.13699v1. Searches of the paper, its available version record, related primary papers, and the selected repository materials did not locate a later answer or this topological construction. This is a bounded bibliographic finding, not a proof of universal priority. The mathematical assertion that our model answers the displayed question is independent of that priority issue.

31 Conventions and classical inputs

31.1 Language, order, and induction

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. A model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ is a model of the *complete first-order theory* of $(\mathbb{N}; 0, 1, +, <)$, including every instance of first-order induction with parameters. In this theory,

$$x \leq y \iff \exists z (x + z = y), \quad x < y \iff \exists z (x + z + 1 = y). \quad (31.1)$$

Thus naming $<$ is a definitional convenience. A topological presentation in this report requires $+$ to be jointly continuous. It does not require the definable relation $<$ to be open or closed.

If G is an ordered abelian group, write $G_{\geq 0}$ and $G_{>0}$ for its nonnegative and positive cones. A distinguished $e \in G$ is a *least positive element* if $0 < e \leq g$ whenever $g > 0$. A $\mathbb{Z}$ -group is an ordered abelian group with a least positive element e such that $G/\mathbb{Z}e$ is divisible.

We use the following classical facts about Presburger arithmetic; a convenient precise reference is Jeřábek’s preliminary section [4].

1. A group is a model of $\text{Th}(\mathbb{Z}; 0, 1, +, -, <)$ exactly when it is a $\mathbb{Z}$ -group. Its nonnegative cone is then a model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$.
2. In the language enlarged by fixed-modulus congruence predicates, formulas admit quantifier elimination: they reduce to finite Boolean combinations of integer-linear comparisons and congruences.

These inputs address *all* formulas, not merely the displayed division axioms. That is why our construction satisfies full first-order induction. We do not confuse it with second-order induction over every external subset.

31.2 Topological and descriptive notation

An F_σ set is a countable union of closed sets; a G_δ set is a countable intersection of open sets. A Δ_2^0 set is both. In metrizable spaces, open and closed sets are Δ_2^0 , finite Boolean combinations of Δ_2^0 sets are Δ_2^0 , and continuous inverse images preserve this class.

We use three standard descriptive-set-theoretic inputs: a subspace of a Polish space is Polish when it is G_δ ; a Borel injective map between standard Borel spaces has Borel image and inverse onto its image; and a nonempty zero-dimensional Polish space in which every compact subset has empty interior is homeomorphic to $\mathbb{N}^\mathbb{N}$. References and proofs of these standard tools can be found in [9, 8].

All metamathematics is carried out in ordinary ZFC. Every model constructed below is a set. Countable ordinals are actual ordinals of this metatheory, not nonstandard ordinal codes inside an arithmetic model.

31.3 Algebraic and topological preliminaries [06]

31.3.1 First-order arithmetic and induction [06]

Let

$$\text{Pres} = \text{Th}(\mathbb{N}, 0, 1, +, <).$$

Adding $0, 1, <$ to the addition-only language is harmless here because they are definable. A model of Pres satisfies every first-order induction instance in this additive language, including instances with parameters. It need not satisfy induction for formulas involving a subsequently added multiplication symbol.

We use PA^- in its standard algebraic sense: the theory of the nonnegative cones of discretely ordered commutative rings, with their inherited $0, 1, +, \cdot, <$. This convention agrees with that used in the topological-arithmetic literature [2]. We will describe all concrete models directly, so the results do not depend on choosing a particular printed list of PA^- axioms.

A discretely ordered abelian group is a linearly ordered abelian group with a least positive element, denoted by 1. The subgroup generated by 1 is a convex copy of $\mathbb{Z}$: the only elements between consecutive standard integers are the endpoints. A $\mathbb{Z}$ -group is such a group for which $G/\mathbb{Z}$ is divisible.

Lemma 31.1 (Classical Presburger characterization [06]; = [theorem 4.9](#) and [corollary 4.11](#)). *If G is a $\mathbb{Z}$ -group, then*

$$(G, 0, 1, +, -, <) \equiv (\mathbb{Z}, 0, 1, +, -, <), \quad G_{\geq 0} \models \text{Pres}.$$

Every formula in the group language is equivalent to a Boolean combination of integer-linear inequalities and congruences modulo positive standard integers.

This characterization is recalled in [4, Section 2]. Appendix I supplies a proof of the elimination step and of the transfer to the nonnegative cone. In particular, our use of this lemma includes full first-order additive induction; it does not merely check a few semiring-like equations.

Lemma 31.2 (Split construction [06]; the setup of [theorem 3.1](#)). *Let D be a divisible ordered abelian group. Give*

$$G(D) = D \times \mathbb{Z}$$

coordinatewise addition and lexicographic order with the D coordinate dominant. Then $G(D)$ is a $\mathbb{Z}$ -group with $1 = (0, 1)$, and

$$M(D) = G(D)_{\geq 0} = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \tag{31.2}$$

models Pres.

Proof. If $(d, z) > 0$, either $d > 0$, in which case $(d, z) > (0, 1)$, or $d = 0$ and $z \geq 1$. Thus $(0, 1)$ is least positive. The subgroup $\{0\} \times \mathbb{Z}$ is convex and the quotient is D , which is divisible. Apply Lemma 31.1.

For additional explicitness, if $n \geq 1$ is a standard integer, write $z = nq + r$ in $\mathbb{Z}$, with $0 \leq r < n$. Then

$$(d, z) = n(d/n, q) + (0, r). \tag{31.3}$$

The quotient is unique. If $(d, z) \geq 0$, then $(d/n, q) \geq 0$ as well: either $d > 0$, or $d = 0$ and $z \geq 0$, so $q \geq 0$. $\square$

31.3.2 The topologies under discussion [06]

A space is *Polish* if it is separable and admits a compatible complete metric. Its displayed metric need not itself be complete. We use the product topology for each product of topological groups, and the subspace topology on a cone unless a different topology is explicitly specified.

We write Δ_2^0 for the sets that are simultaneously F_σ and G_δ . This class is closed under finite Boolean operations and continuous inverse images. Every open or closed subset of a metrizable space belongs to this class. A G_δ subspace of a Polish space is Polish. For reference, if $X = \bigcap_n U_n$ in a complete metric space X_0 , the map

$$x \mapsto (x, (d(x, X_0 \setminus U_n)^{-1})_{n \in \mathbb{N}})$$

embeds X as a closed graph in a product of complete spaces, with the empty-complement coordinates omitted. This gives the usual complete metrizability argument.

32 The real-coordinate model

Theorem 32.1 (An affirmative construction; = [theorem 3.1\(1\)](#)). *The structure $M_{\mathbb{R}}$ of (30.1), with the operations and order described in Section 30, is an uncountable model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$. Its topology is Polish, locally compact, and has no isolated points. Addition is jointly continuous. Therefore Question 2 of [1], as stated there, has an affirmative answer.*

Proof. Consider the abelian group

$$G_{\mathbb{R}} = \mathbb{R} \times \mathbb{Z}, \quad (r, n) + (s, m) = (r + s, n + m).$$

Give it the order

$$(r, n) < (s, m) \iff r < s \text{ or } (r = s \text{ and } n < m). \quad (32.1)$$

This is a translation-invariant total order. Its least positive element is $e = (0, 1)$: a positive pair either has positive real coordinate and exceeds every $(0, n)$, or has real coordinate zero and a positive integer coordinate.

The subgroup $\mathbb{Z}e$ is $\{0\} \times \mathbb{Z}$, and $G_{\mathbb{R}}/\mathbb{Z}e \cong (\mathbb{R}, +)$ is divisible. Hence $G_{\mathbb{R}}$ is a $\mathbb{Z}$ -group, and its nonnegative cone is a model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$. Computing that cone gives exactly (30.1).

Topologically, $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$ is Polish, and

$$M_{\mathbb{R}} = ([0, \infty) \times \mathbb{Z}) \setminus (\{0\} \times \mathbb{Z}_{<0}). \quad (32.2)$$

The first set is closed, and the removed set is closed. Thus $M_{\mathbb{R}}$ is open in a closed Polish subspace, so is Polish. Equivalently, it is the countable topological disjoint sum of the slices

$$M_{\mathbb{R}} \cap (\mathbb{R} \times \{n\}) \cong \begin{cases} [0, \infty), & n \geq 0, \\ (0, \infty), & n < 0. \end{cases}$$

Each slice is locally compact and has no isolated points. Those properties therefore hold for the disjoint sum. The map $r \mapsto (r, 0)$ embeds $[0, \infty)$ into the model, proving uncountability.

Ambient coordinatewise addition is continuous. The sum of two nonnegative pairs is again nonnegative, so its restriction is a continuous map $M_{\mathbb{R}}^2 \rightarrow M_{\mathbb{R}}$. $\square$

[write] [Theorem 32.1](#) and its proof are those of [theorem 3.1\(1\)](#); source 07 adds the picture in [figure 1](#) and a complete metric that uses $\log r$ on the negative slices ([equation \(32.3\)](#)); compare (5.2)). Source 06's account follows in [section 32.3](#).

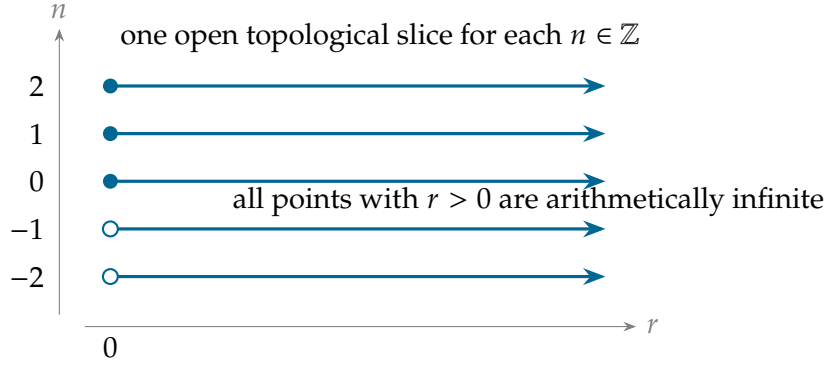


Figure 1: The space $M_{\mathbb{R}}$. Filled endpoints are present; hollow endpoints are absent. Different horizontal slices are topologically separated. Arithmetic order compares the horizontal coordinate before the slice label.

32.1 A complete metric, rather than an incomplete inherited metric

Being a Polish subspace does not mean that the restriction of the ambient Euclidean product metric is complete. For example, $(1/j, -1)$ approaches a missing endpoint in the ambient space. A compatible complete metric is easy to exhibit.

For the slice with integer coordinate n , put

$$h_n(r) = \begin{cases} r, & n \geq 0, \\ \log r, & n < 0. \end{cases}$$

Define

$$d((r, n), (s, m)) = \begin{cases} 1, & n \neq m, \\ \min\{1, |h_n(r) - h_n(s)|\}, & n = m. \end{cases} \quad (32.3)$$

The triangle inequality holds within each slice and also across slices. A Cauchy sequence is eventually in a single slice. In a nonnegative slice it is complete because $[0, \infty)$ is complete; in a negative slice the logarithm identifies it with a complete copy of $\mathbb{R}$. Thus d is complete and gives precisely the topology used in Theorem 32.1.

32.2 Division and the meaning of nonstandard induction

For a standard integer $d \geq 1$, divide the ordinary integer n as $n = dq + r$, where $0 \leq r < d$. Then

$$(a, n) = d(a/d, q) + (0, r). \quad (32.4)$$

When $(a, n) \in M_{\mathbb{R}}$, its quotient $(a/d, q)$ is also in $M_{\mathbb{R}}$. This remains true for negative n : such an input has $a > 0$, so its positive real coordinate determines positivity of the quotient.

For example,

$$(5/2, -7) = 3(5/6, -3) + (0, 2).$$

The integer coordinate is permitted to be negative in a positive element. Forbidding it would destroy predecessor and division properties and would give the wrong model.

Every (a, n) with $a > 0$ exceeds every standard numeral $(0, k)$. Nevertheless, each induction instance in the first-order additive language holds, by the $\mathbb{Z}$ -group characterization. The external set of standard numerals is proper and successor-closed, but is not definable in

this model, even with parameters: a definition would violate the corresponding induction instance. This is the usual distinction between first-order and full second-order induction, manifested in very elementary coordinates.

32.3 A perfect locally compact model [06]

Theorem 32.2 (An affirmative answer to Glazer's question [06]; = [theorem 3.1\(1\)](#)). *With the usual topology on $\mathbb{R}$ and the discrete topology on $\mathbb{Z}$, the space*

$$M_{\mathbb{R}} = ((0, \infty) \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (32.5)$$

is uncountable, perfect, locally compact, and Polish. With coordinatewise addition, $0 = (0, 0)$, $1 = (0, 1)$, and lexicographic order, it is a model of Pres with jointly continuous addition.

Proof. The algebra follows from Lemma 31.2 with $D = \mathbb{R}$. For each integer z , the slice with second coordinate z is clopen and is homeomorphic to

$$I_z = \begin{cases} [0, \infty), & z \geq 0, \\ (0, \infty), & z < 0. \end{cases}$$

Thus $M_{\mathbb{R}}$ is a countable topological disjoint union of locally compact Polish rays. It is second countable and has no isolated points.

One can verify complete metrizability without relying on the disjoint sum assertion. Put $\phi_z(r) = r$ for $z \geq 0$ and $\phi_z(r) = \log r$ for $z < 0$, and define

$$\rho((r, z), (s, w)) = \begin{cases} 2, & z \neq w, \\ \min\{1, |\phi_z(r) - \phi_z(s)|\}, & z = w. \end{cases}$$

This is a compatible complete metric: a Cauchy sequence eventually stays in one slice, where it corresponds to a Cauchy sequence in $[0, \infty)$ or $\mathbb{R}$. Rational points in all slices give a countable dense set.

Finally, $\mathbb{R} \times \mathbb{Z}$ is a topological group under coordinatewise addition, and its nonnegative cone is closed under addition. Restricting the ambient operation gives a continuous map $M_{\mathbb{R}} \times M_{\mathbb{R}} \rightarrow M_{\mathbb{R}}$. $\square$

Example 32.3 (Arithmetic order and topological proximity [06]). Every (r, z) with $r > 0$ is larger, arithmetically, than every standard natural number $(0, n)$. Nevertheless

$$(1/k, 0) \longrightarrow (0, 0).$$

There is no contradiction: the topology is the product topology, not the order topology. In fact, the order topology of any discretely ordered group or its nonnegative cone is discrete, since consecutive integers relative to any point isolate it.

The explicit quotient and remainder maps in (31.3) are continuous for every fixed standard n . In this example they are

$$Q_n(r, z) = (r/n, \lfloor z/n \rfloor), \quad R_n(r, z) = (0, z - n\lfloor z/n \rfloor).$$

The residue classes form a finite clopen partition. These are fixed standard-divisor operations; the assertion does not introduce division by a variable nonstandard element into Presburger arithmetic.

33 Borel orders on Polish abelian groups

The real construction can be extended whenever the positive cone of a divisible ordered Polish group is a Polish subspace. The following theorem shows that Borel definability of the order is already enough. We include the full proof because it is the descriptive engine for the general result.

Theorem 33.1 (Low complexity of an invariant Borel order). *Let A be a Polish abelian group with a translation-invariant total order. If $P = A_{>0}$ is Borel, then all four sign cones are Δ_2^0 . Both order relations $<$ and $\leq$ are Δ_2^0 subsets of A^2 . Moreover, there is a canonically defined countable decreasing chain of closed convex subgroups that decomposes P into countably many locally closed pieces.*

Proof. For a closed subgroup $H \leq A$, write

$$P_H = P \cap H, \quad C(H) = \text{cl}_H(P_H \cup \{0\}), \quad D(H) = C(H) \cap (-C(H)).$$

The set $C(H)$ is a closed additive monoid. Indeed, approximate two of its points by nonnegative points of H and use continuity of addition. Consequently $D(H)$ is a closed subgroup. Totality of the order gives

$$H = C(H) \cup (-C(H)). \quad (33.1)$$

We first show that $D(H) \subsetneq H$ whenever $H \neq \{0\}$. The strict cone P_H is nonmeagre in H . In the nondiscrete case this follows from the Baire category theorem and

$$H = P_H \dot{\cup} (-P_H) \dot{\cup} \{0\},$$

since $\{0\}$ is nowhere dense and negation is a homeomorphism. In the discrete case P_H is nonempty and hence nonmeagre.

As a Borel set, P_H has the Baire property. There is therefore a nonempty open set $V \subseteq H$ on which it is comeagre. For every $t \in V + V$, the set $V \cap (t - V)$ is nonempty and open. Both P_H and $t - P_H$ are comeagre there, so they intersect. Hence

$$V + V \subseteq P_H + P_H \subseteq P_H. \quad (33.2)$$

This is the sumset form of the category argument often associated with Piccard and Pettis; the displayed proof fixes the version needed here.

If $C(H) = H$, then $-P_H \cup \{0\}$ is dense in H . It would meet the nonempty open set $V + V \subseteq P_H$, which is impossible. Thus $C(H) \neq H$, and a fortiori $D(H) \neq H$.

Now form the transfinite recursion

$$H_0 = A, \quad H_{\xi+1} = D(H_\xi), \quad H_\lambda = \bigcap_{\xi < \lambda} H_\xi \quad (\lambda \text{ a limit}). \quad (33.3)$$

Every term is a closed subgroup, with a strict successor decrease whenever the current subgroup is nonzero. This process reaches $\{0\}$ at a countable ordinal η .

To prove the last claim, suppose instead that there were a strict decrease at every $\xi < \omega_1$. Fix a countable base $\mathcal{B}$ of A . For each ξ , choose the first basic set B_ξ meeting H_ξ and disjoint from $H_{\xi+1}$. Such a basic set exists because $H_{\xi+1}$ is closed and properly contained in H_ξ . If $\zeta > \xi$, then B_ξ misses H_ζ , whereas B_ζ meets it. The B_ξ are therefore distinct, contradicting countability.

For $\xi < \eta$, set $C_\xi = C(H_\xi)$ and

$$L_\xi = H_\xi \setminus (-C_\xi).$$

Every member of L_ξ is positive, since a nonpositive member would already lie in $-P_{H_\xi} \cup \{0\} \subseteq -C_\xi$. Equation (33.1) gives the disjoint decomposition

$$H_\xi \setminus H_{\xi+1} = L_\xi \dot{\cup} (-L_\xi).$$

A nonzero element leaves the decreasing chain at a successor stage: its first absent stage cannot be a limit, by the definition of intersection. It follows that

$$P = \bigcup_{\xi < \eta} L_\xi, \quad -P = \bigcup_{\xi < \eta} (-L_\xi). \quad (33.4)$$

Each layer is relatively open in the closed set H_ξ , and hence is F_σ in the ambient metrizable space. Both unions in (33.4) are countable. Thus P and $-P$ are F_σ , and their complements, obtained by adjoining $\{0\}$ to the opposite cone, are also F_σ . This proves all four Δ_2^0 assertions. Applying the continuous difference map $(x, y) \mapsto y - x$ proves the assertions for order relations.

Finally, every H_ξ is order-convex. The limit step is immediate. For a successor step, suppose H is convex, $h \in D(H)$, and $0 \leq x \leq h$. Then $x \in H$ and $x \in C(H)$. Also

$$-x = (-h) + (h - x) \in C(H) + C(H) \subseteq C(H).$$

Hence $x \in D(H)$, proving convexity. □

[write] The sumset step (33.2) is the Pettis argument that source 08 uses in [lemmas 47.3](#) and [53.1](#) and source 09 states as [lemma 61.1](#). For locally compact second-countable groups, part (1) of [theorem 51.2](#) in Part IV is the special case of [theorem 33.1](#), with a different proof through finite-dimensional cuts.

Remark 33.2 (The hypotheses that do the work). The topology must make A a Polish group. A standard Borel group without an established compatible Polish group topology is not enough for this proof. Borelness is used on every closed subgroup; merely knowing that the cone has the Baire property in the ambient space does not automatically give that hereditary regularity.

There is classical related work on measurable orders on locally compact abelian groups and their convex subgroup structure; see Asmar and Montgomery-Smith [22]. We use a direct category proof on Polish groups, including groups that are not locally compact. No historical priority claim for the precise formulation of Theorem 33.1 is needed for the positive construction.

33.1 A countable ordinal invariant of the order and topology

Definition 33.3. The *cone closure rank* $\rho(A, <)$ is the least η for which the recursion (33.3) reaches $H_\eta = \{0\}$. It is an invariant of the ordered topological group, not just of its abstract group.

Proposition 33.4. For every countable ordinal γ , the product group $\mathbb{R}^\gamma$ with its product topology and lexicographic order has cone closure rank γ . For infinite countable γ , these are different orders on topological groups all isomorphic to $\mathbb{R}^\mathbb{N}$.

Proof. For $\beta \leq \gamma$ put

$$K_\beta = \{x \in \mathbb{R}^\gamma : x_\xi = 0 \text{ for all } \xi < \beta\}.$$

These are closed subgroups. The positive cone is Borel, since it is the countable union, over $\beta < \gamma$, of the sets with zero prefix below β and positive β -coordinate. For $\beta < \gamma$, inside K_β , the closure of the nonnegative cone is exactly

$$\{x \in K_\beta : x_\beta \geq 0\}.$$

The inclusion from left to right follows by coordinate continuity. For the reverse inclusion, points with $x_\beta = 0$ are limits of $x + j^{-1}e_\beta$, which are positive. Intersecting with the negative closure gives $K_{\beta+1}$. At limit stages the intersection is the corresponding K_λ . Thus $H_\beta = K_\beta$ throughout, with the first zero subgroup at γ . Reindexing the countably infinite coordinate set gives the asserted topological group isomorphism. $\square$

Even very low Borel complexity therefore permits arbitrarily long countable closure recursions. The Δ_2^0 conclusion is sharp: the lexicographic positive cone of $\mathbb{R}^2$ is neither open nor closed.

34 A general Polish realization theorem

Theorem 34.1 (Divisible-group realization). *Let A be a divisible Polish abelian group carrying a translation-invariant total order with Borel positive cone. Define*

$$G_A = A \overrightarrow{\times} \mathbb{Z}, \quad M_A = (A_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (34.1)$$

Use coordinatewise addition, $0 = (0, 0)$, $1 = (0, 1)$, and the product subspace topology. Then:

- (i) M_A is a Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$, and addition is jointly continuous.
- (ii) Every fixed-standard-divisor quotient and remainder map is continuous. Partial subtraction on its arithmetic domain is continuous.
- (iii) Every relation definable with parameters in a finite Cartesian power of M_A is Δ_2^0 in the product topology.
- (iv) If A is uncountable, so is M_A .

Proof. The order on G_A is lexicographic with A the more significant coordinate. Its least positive element is $(0, 1)$, and its quotient by $\{0\} \times \mathbb{Z}$ is the divisible group A . Thus it is a $\mathbb{Z}$ -group and M_A is its nonnegative cone.

Theorem 33.1 says that $A_{>0}$ and $A_{\geq 0}$ are Δ_2^0 . In the clopen slice $A \times \{k\}$, membership in M_A means $a > 0$ if $k < 0$ and $a \geq 0$ if $k \geq 0$. Both M_A and its complement are therefore countable unions of F_σ sets. In particular M_A is G_δ in $A \times \mathbb{Z}$, so is Polish. Coordinatewise addition restricts continuously to it.

An ordered group is torsion-free. Consequently multiplication by a standard integer $d \geq 1$ is a continuous bijective homomorphism on the divisible Polish group A . Its inverse, division by d , is continuous. Here is a short justification of openness. Given a neighborhood U of zero, choose open V with $V - V \subseteq U$. Countably many translates of V cover A , so countably many translates of dV do as well. The analytic set dV has the Baire property and is nonmeagre. The difference form of the category argument then gives a neighborhood of zero inside $dV - dV \subseteq dU$. Thus multiplication by d is open.

Writing $k = dq + r$, $0 \leq r < d$, the desired maps are

$$Q_d(a, k) = (a/d, q), \quad R_d(a, k) = (0, r). \quad (34.2)$$

They are restrictions of continuous coordinate maps and take their values in M_A . They give the unique decomposition $x = dQ_d(x) + R_d(x)$. Partial subtraction is the restriction of the ambient continuous map $(x, y) \mapsto x - y$ to the subspace where $y \leq x$.

For (iii), consider affine integer-linear expressions on G_A . Their A -coordinate is a continuous affine map into A , and their integer coordinate is a continuous map into the discrete group $\mathbb{Z}$. The nonnegative cone of G_A is Δ_2^0 , by the same slice argument used for M_A . Consequently each linear comparison is Δ_2^0 ; equalities are closed. Since A is divisible, congruence modulo d depends only on the integer coordinate modulo d , so is clopen. Classical Presburger quantifier elimination reduces every parameter-definable relation to a finite Boolean combination of these atoms. Restricting to M_A^k preserves their complexity.

Finally, if A is uncountable, totality and negation show that $A_{>0}$ is uncountable. It embeds into M_A via $a \mapsto (a, 0)$. $\square$

[write] Part (ii) of [theorem 34.1](#) includes, for $A = \mathbb{R}$ and $A = \mathbb{Q}^{\mathbb{N}}$, the continuity statements of [proposition 8.1](#) and of source 04's [theorem 5.4](#); part (iii) extends [theorems 7.1](#) and [7.2](#) to every divisible Polish group with a Borel cone. Source 06 proves part (iii) for its own families in [theorem 37.7](#).

Remark 34.2. Only fixed standard divisors occur in (34.2). A function taking a *variable* divisor as input belongs to a stronger language. The theorem also distinguishes continuous partial subtraction from total truncated subtraction, which will be impossible in an uncountable Polish presentation.

34.1 Definable Borelness for arbitrary continuous presentations

The special Δ_2^0 estimate uses the ambient Polish group. A weaker conclusion holds for every Polish Presburger presentation with continuous addition, whether or not it comes from our construction.

Proposition 34.3. *Let $M \models \text{Th}(\mathbb{N}; 0, 1, +, <)$ have a Polish topology making addition continuous. Every relation definable with parameters is Borel. Every definable partial function between finite Cartesian powers has Borel domain and is Borel measurable on its domain.*

Proof. Equation (31.1) makes $<$ analytic: it is the projection of the closed equality relation $x + z + 1 = y$. Its complement $y \leq x$ is analytic for the same reason. Hence $<$ is Borel by the analytic–coanalytic characterization of Borel sets.

For a fixed $d \geq 1$ and $0 \leq r < d$, the map $x \mapsto dx + r$ is continuous and injective. Its image is Borel by the Borel injective-image theorem. These d residue classes partition M . Congruence relations are therefore Borel, as are integer-linear comparisons after negative coefficients have been moved to the other side. Quantifier elimination now gives every definable relation Borel.

The graph of a definable partial function is a Borel set. Projection from that graph to its input coordinates is Borel and injective. Its image, the domain, is Borel, and its inverse is Borel. Composing that inverse with projection to output coordinates gives the desired function. $\square$

35 A family of models on Baire space

35.1 The countable-ordinal coefficient groups

Fix a countably infinite ordinal α . Let

$$A_\alpha = \mathbb{Q}^\alpha, \quad G_\alpha = A_\alpha \overrightarrow{\times} \mathbb{Z}, \quad M_\alpha = (G_\alpha)_{\geq 0}. \quad (35.1)$$

Here $\mathbb{Q}^\alpha$ is the full Cartesian product, not the finite-support direct sum. Give each copy of $\mathbb{Q}$ the *discrete* topology. On A_α , the least coordinate at which two vectors differ determines their lexicographic comparison. Such a coordinate exists because α is well ordered. Addition and division by a nonzero integer are coordinatewise.

The product topology uses finite coordinate restrictions. In particular, when $\alpha > \omega$, it does not treat an entire transfinite prefix as one open condition. Keeping these two uses of coordinates separate is the main point in the topology proof.

Theorem 35.1 (Continuous arithmetic on Baire space; for $\alpha = \omega$, [theorem 6.1](#)). *For every countably infinite ordinal α , M_α is a continuum-sized model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ homeomorphic to $\mathbb{N}^\mathbb{N}$. Addition, every quotient and remainder by a fixed positive standard integer, and partial subtraction are continuous. Every parameter-definable relation in every finite arity is Δ_2^0 .*

Proof. The group A_α is divisible, ordered, and Polish: it is a countable product of countable discrete groups. Its topology is zero-dimensional. For each $\beta < \alpha$ define

$$C_\beta^+ = \{a : a_\gamma = 0 \ (\gamma < \beta), \ a_\beta > 0\}, \quad C_\beta^- = \{a : a_\gamma = 0 \ (\gamma < \beta), \ a_\beta < 0\}.$$

Each set is closed. The prefix-zero condition is an intersection of closed coordinate conditions, and the sign condition on a discrete rational coordinate is clopen. Therefore

$$(A_\alpha)_{>0} = \bigcup_{\beta < \alpha} C_\beta^+, \quad (A_\alpha)_{<0} = \bigcup_{\beta < \alpha} C_\beta^-$$

are F_σ , with F_σ complements after adjoining $\{0\}$. [Theorem 34.1](#) now gives Polishness and all the arithmetic and definability assertions. Cardinality is $\mathfrak{c}$ because the entire fiber $a_0 = 1$ lies in the positive cone and has countably infinitely many free rational coordinates.

It remains to identify the underlying Polish space. We prove that every nonempty open subset of M_α contains a closed noncompact cylinder. Let U be nonempty and choose $x = (a, n) \in U$. For some finite $F \subseteq \alpha$, the ambient cylinder

$$B = \{(b, n) : b_\gamma = a_\gamma \text{ for } \gamma \in F\}$$

satisfies $x \in B \cap M_\alpha \subseteq U$. Let j be the least natural number not in F . All indices $i < j$ are in F .

If the finite prefix $(a_0, \dots, a_{j-1})$ has a nonzero entry, its first nonzero entry is positive, since $a \geq 0$. Positivity is already forced; in this case additionally set $b_j = 0$. If that prefix is zero, additionally set $b_j = 1$. In both cases the resulting cylinder C fixes only finitely many coordinates, is nonempty, and has strictly positive A_α coordinate. Thus

$$C \subseteq B \cap M_\alpha \subseteq U.$$

The set C is clopen in the ambient product and closed in M_α . Infinitely many rational coordinates remain free. Projection onto any one of them maps C continuously onto the infinite discrete space $\mathbb{Q}$, so C is not compact.

If a compact subset $K \subseteq M_\alpha$ had nonempty interior U , the cylinder just constructed would be a closed subset of K and hence compact, a contradiction. Thus every compact subset has empty interior. The space is nonempty, zero-dimensional, and Polish. The Baire-space characterization gives $M_\alpha \cong \mathbb{N}^\mathbb{N}$. $\square$

[write] For $\alpha = \omega$, M_ω is Part I's Baire model M_B , and [theorem 35.1](#) is [theorem 6.1](#), where an explicit and computable homeomorphism is given ([theorems 6.2](#) and [6.4](#); also source 06's [theorems 35.6](#) and [35.7](#) below). For $\alpha > \omega$ the theorem is new to the collection, and the homeomorphism above comes from the topological characterization of Baire space.

Remark 35.2 (Why the transfinite cases require the full argument). For $\alpha = \omega$, both strict sign cones of A_α are open, since a finite prefix detects each nonzero vector's sign. When $\alpha > \omega$, a vector first nonzero at ω is not an interior point of its sign cone. The closed-piece proof works in both cases. Likewise, the cylinder argument above covers neighborhoods of such transfinite-leading vectors and of zero, not just the dense set of vectors with finite leading coordinate.

Choose a homeomorphism $h : \mathbb{N}^\mathbb{N} \rightarrow M_\alpha$ and transport the arithmetic:

$$x \oplus y = h^{-1}(h(x) + h(y)), \quad 0_{\mathbb{N}^\mathbb{N}} = h^{-1}(0), \quad 1_{\mathbb{N}^\mathbb{N}} = h^{-1}(1).$$

The operation $\oplus$ is jointly continuous. The coefficient model is explicit; the homeomorphism here is supplied by a topological characterization. We do not claim a computable homeomorphism for an arbitrary presentation of a countable ordinal.

35.2 Residue classes as copies of the whole space

Corollary 35.3 (extends [proposition 8.1](#)). *For each standard $d \geq 1$, the map*

$$x \mapsto (Q_d(x), r_d(x))$$

is a homeomorphism $M_A \rightarrow M_A \times \{0, \dots, d-1\}$, where $R_d(x) = r_d(x)1$. Each residue class is clopen and homeomorphic to M_A .

Proof. Continuity of the forward map follows from [Theorem 34.1](#). Its inverse is $(q, r) \mapsto dq + r1$, continuous because the second factor is finite discrete and addition is continuous. Division with remainder proves that the maps are inverse. $\square$

[write] For $M_A = M_\mathbb{R}$ and $M_A = M_B$ this is [proposition 8.1](#).

This is an arithmetic explanation of a topological self-similarity. It also applies to $M_\mathbb{R}$, whose countably many ray components can be rearranged into any positive finite number of copies.

35.3 An uncountable ordinal spectrum of models

For positive elements g, h of an ordered abelian group, define

$$g \asymp h \iff (\exists k, \ell \in \mathbb{N}_{>0}) (g \leq kh \text{ and } h \leq \ell g).$$

These are the Archimedean equivalence classes. We list the classes in *decreasing magnitude*: a class occurs earlier when its members exceed every standard multiple of a member of the later class.

Theorem 35.4 (An ordinal spectrum on one topological space). *The Archimedean classes of G_α , listed in decreasing magnitude, have order type $\alpha + 1$. Therefore the models M_α , for countably infinite α , are pairwise nonisomorphic, even as additive monoids. There are at least $\aleph_1$ pairwise nonisomorphic continuum-sized Presburger structures continuously presented on $\mathbb{N}^\mathbb{N}$. No continuum hypothesis is used.*

Proof. For a positive (a, n) with $a \neq 0$, let $\lambda(a, n)$ be the least $\beta < \alpha$ with $a_\beta \neq 0$. For $a = 0$, let $\lambda(0, n) = \alpha$. If two positive elements have the same leading index $\beta < \alpha$, their leading coefficients are positive rationals. Each can be made larger than the other by multiplication by a sufficiently large standard integer; lower coordinates do not affect that comparison. Thus they are Archimedean equivalent. The same holds in the integer class.

If $\lambda(g) < \lambda(h)$, then $g > kh$ for every standard positive integer k , since the leading coordinate of g is positive while that coordinate of kh is zero. Every index $\beta \leq \alpha$ occurs. This proves the asserted class order.

An isomorphism of additive positive monoids extends uniquely to an isomorphism of their group completions: represent a group element as a difference of two positives and use cancellation to check independence of the representation. The extension preserves positivity and hence order. It preserves the Archimedean class order. Thus an isomorphism between M_α and M_δ would imply $\alpha + 1 = \delta + 1$, so $\alpha = \delta$.

The set of countably infinite ordinals has cardinality $\aleph_1$. Theorem 35.1 places each corresponding model on Baire space. $\square$

The invariant uses addition. It does not by itself distinguish the pure underlying linear orders; no classification of those pure orders is claimed.

35.4 The optimal number of models is the continuum

The ordinal family gives an uncountable hierarchy with a transparent invariant. A different parameter reaches the largest possible number of isomorphism types on a fixed Baire space, without any hypothesis about the value of the continuum.

Theorem 35.5 (Sharp continuum multiplicity). *There are exactly $\mathfrak{c} = 2^{\aleph_0}$ isomorphism types of Presburger models continuously presentable on $\mathbb{N}^\mathbb{N}$. The lower bound can be realized with all parameter-definable relations Δ_2^0 and with continuous quotient and remainder by every fixed standard positive integer.*

Proof. For an irrational $s \in (1, 2)$ let $B_s = \mathbb{Q} + \mathbb{Q}s$, ordered as a subgroup of $\mathbb{R}$ but given the discrete topology. This is a countable divisible ordered abelian group. Put

$$A_s = B_s \overrightarrow{\times} \mathbb{Q}^\omega, \quad G_s = A_s \overrightarrow{\times} \mathbb{Z}, \quad M_s = (G_s)_{\geq 0}.$$

The product of the discrete coordinate topologies is Polish. The leading coordinate test makes the sign cones Borel. Theorem 34.1 gives all the arithmetic and definability properties. The finite-cylinder proof of Theorem 35.1 applies with B_s as the first coordinate, so M_s is homeomorphic to Baire space.

The subgroup

$$H_s = \{0\} \times \mathbb{Q}^\omega \times \mathbb{Z}$$

is the unique largest proper convex subgroup of G_s . Indeed it is proper and convex. If a convex subgroup contains an element g with nonzero B_s -coordinate, standard multiples of

$|g|$ dominate every element of G_s , because $B_s \subseteq \mathbb{R}$ is Archimedean. Convexity forces such a subgroup to be the whole group. Therefore an isomorphism $M_s \cong M_t$, extended to group completions, induces an ordered additive isomorphism $B_s \cong B_t$.

Every increasing additive isomorphism $f : B_s \rightarrow B_t$ is multiplication by a positive real scalar λ . To see this, let $\lambda = f(1) > 0$. Additivity and divisibility give $f(q) = q\lambda$ for every rational q . For any $x \in B_s$, comparison with every rational shows that $f(x)/\lambda$ and x have the same rational cut in $\mathbb{R}$, hence are equal. Surjectivity now supplies rational a, b, c, d with

$$1 = \lambda(c + ds), \quad t = \lambda(a + bs), \quad t = \frac{a + bs}{c + ds}.$$

The denominator is nonzero, and $ad - bc \neq 0$, since otherwise t would be rational. Thus for each fixed s only countably many parameters t can yield an isomorphic model. The continuum-sized set of irrational parameters therefore yields continuum many isomorphism types in ZFC.

For the upper bound, fix $X = \mathbb{N}^{\mathbb{N}}$ and a countable dense set $D \subseteq X^2$. Since X is Hausdorff, a continuous map $X^2 \rightarrow X$ is determined by its values on D . There are at most $|X|^{\aleph_0} = \mathfrak{c}$ possible such maps. Choosing finitely many constants does not increase this bound, and all Presburger relations are determined by the addition and constants. Consequently there are at most $\mathfrak{c}$ continuously presented structures and at most $\mathfrak{c}$ isomorphism types. $\square$

[write] Source 06 proves the same count with a different family and adds that the lower bound is attained with the topology, the arithmetic order, 0 and 1 fixed ([theorem 35.8](#)); source 07's statement adds the Δ_2^0 and continuous-division clauses, and [corollary 36.2](#) below adds every oscillation rank. The two refinements are independent, so both statements are printed.

35.5 An explicit model on Baire space [06]

Let $H = \mathbb{Q}^{\mathbb{N}}$, giving every rational coordinate the *discrete* topology. Order H lexicographically: for a nonzero sequence h , its sign is the sign of its first nonzero coordinate. Put

$$G_\omega = H \overrightarrow{\times} \mathbb{Z}, \quad M_\omega = (G_\omega)_{\geq 0}.$$

The coordinatewise group H is divisible. It follows at once that $M_\omega \models \text{Pres}$ and that addition is continuous in the product subspace topology. The remaining issue is to identify that space.

Theorem 35.6 (Baire-space realization [06]; = [theorem 6.1](#)). *The space M_ω is homeomorphic to $\mathbb{N}^{\mathbb{N}}$. Consequently Baire space supports a model of Presburger arithmetic with jointly continuous addition.*

Proof. We give homeomorphisms rather than merely a cardinality argument. The strict cone $H_{>0}$ is the disjoint union of cylinders

$$C_{j,q} = \{h : h(0) = \dots = h(j-1) = 0, h(j) = q\}, \quad j \in \mathbb{N}, q \in \mathbb{Q}_{>0}.$$

Each cylinder is clopen and is homeomorphic to $\mathbb{Q}^{\mathbb{N}}$ by deleting its fixed prefix. A coordinatewise enumeration of $\mathbb{Q}$ identifies $\mathbb{Q}^{\mathbb{N}}$ with $\mathbb{N}^{\mathbb{N}}$. A countable disjoint union of Baire spaces is again Baire space, by using the first coordinate to record the component. Hence $H_{>0} \cong \mathbb{N}^{\mathbb{N}}$.

The closed cone $H_{\geq 0}$ has an equally direct description. Consider the tree of finite rational words whose first nonzero entry, if present, is positive. An all-zero word can be extended by any nonnegative rational; a word already having positive first nonzero entry can be extended

by any rational. Every node therefore has countably infinitely many successors. Choose a bijection from $\mathbb{N}$ to each successor set. The resulting rooted-tree isomorphism with $\mathbb{N}^{<\mathbb{N}}$ induces a homeomorphism of branch spaces $H_{\geq 0} \cong \mathbb{N}^{\mathbb{N}}$.

The fixed- z slice of M_ω is clopen, and equals $H_{>0}$ when $z < 0$ and $H_{\geq 0}$ when $z \geq 0$. Their countable disjoint union is therefore Baire space. Transport the arithmetic structure through any of these explicit homeomorphisms. $\square$

35.5.1 A computable stream presentation [06]

Theorem 35.7 (Effective addition [06]; = [theorems 6.2](#) and [6.4](#)). *The homeomorphism in Theorem 35.6 can be chosen computable in both directions for exact rational-coordinate streams. Under this homeomorphism, addition on $\mathbb{N}^{\mathbb{N}}$ is Type-2 computable. The constants 0, 1 and the quotient and remainder maps for each fixed standard divisor are computable as well.*

Proof. Fix computable bijections, with computable inverses, between $\mathbb{N}$ and each of $\mathbb{Q}$, $\mathbb{Q}_{>0}$, $\mathbb{Q}_{\geq 0}$, and $\mathbb{Z}$. Canonical reduced fractions provide such enumerations. Choose the nonnegative enumeration to send 0 to 0. Also fix a computable pairing bijection $\mathbb{N} \leftrightarrow \mathbb{N}^2$.

For $H_{\geq 0}$, decode successive input coordinates using the nonnegative rational enumeration until the first positive output; after that, use the enumeration of all rationals. The inverse uses exactly the same finite-prefix state. Thus each coordinate in either direction requires only finitely much input.

For $H_{>0}$, decode the first input coordinate as (j, k) , emit j zeros and then the k th positive rational, and use the remaining input to produce an arbitrary rational tail. The inverse searches for the first nonzero coordinate. This search terminates on its promised domain $H_{>0}$.

Finally, the first coordinate of a Baire input selects $z \in \mathbb{Z}$; the remaining stream selects a point of $H_{>0}$ or $H_{\geq 0}$ according to the sign of z . This defines a computable homeomorphism

$$F : \mathbb{N}^{\mathbb{N}} \longrightarrow M_\omega.$$

Coordinatewise rational addition and integer addition are computable. Therefore

$$a \oplus b = F^{-1}(F(a) + F(b))$$

is Type-2 computable. If the output integer coordinate is negative, the output rational sequence is necessarily strictly positive: a sum of nonnegative ordered-group elements has zero rational part only when both rational parts vanish, in which case both input integers are nonnegative. Thus the search used by F^{-1} always terminates when it is invoked on that branch. Fixed standard division is coordinatewise rational division together with ordinary integer division, so the same argument applies to it and to remainders. $\square$

Here “computable” concerns functions on infinite streams: every requested output coordinate is produced after finitely much input has been inspected. It does not assert a computable enumeration of this uncountable model or decidability of equality between arbitrary streams. It therefore has no conflict with the classical countable-presentation Tennenbaum theorem.

35.6 How many continuous models fit on Baire space? [06]

The existence result has substantial room for variation. The next theorem counts abstract isomorphism types, not merely different coordinate descriptions of one model.

Theorem 35.8 (Sharp abundance [06]; the count of [theorem 35.5](#)). *There are exactly $2^{\aleph_0}$ isomorphism types of Presburger models on Baire space with jointly continuous addition. The lower bound can be realized while keeping the underlying topology, arithmetic order, 0, and 1 fixed and varying only addition.*

Proof. Let $A_0 = \mathbb{Q}$ and $A_1 = \mathbb{Q} + \mathbb{Q}\sqrt{2}$, as additive ordered subgroups of $\mathbb{R}$, and give both groups discrete topology. For $s \in 2^{\mathbb{N}}$ put

$$H'_s = \prod_{n \in \mathbb{N}} A_{s(n)}, \quad G'_s = H'_s \overrightarrow{\times} \mathbb{Z}, \quad M'_s = (G'_s)_{\geq 0},$$

using the first differing coordinate for the order on H'_s . Both coefficient groups are countable and divisible. Each has countably infinitely many positive and nonnegative elements. The proof of [Theorem 35.6](#), with its coordinate-dependent enumerations, therefore applies unchanged. Hence each M'_s is a Baire-space model with continuous addition.

For positive elements of an ordered abelian group, say $x \asymp y$ if $x \leq my$ and $y \leq nx$ for some positive standard integers m, n . This is Archimedean equivalence. In G'_s , two positive elements are equivalent exactly when they have the same first nonzero coordinate, or when both are ordinary integers. Indeed, equal-rank real leading coefficients dominate one another after sufficiently large standard integer multiplication; an earlier nonzero coordinate dominates every standard multiple of a later one. The classes, ordered from largest magnitude to smallest, consequently have order type $\omega + 1$.

For each n , define the convex subgroup

$$U_n = \{(h, z) : h(0) = \dots = h(n-1) = 0\}.$$

It is determined intrinsically by the Archimedean classes of rank at least n , and

$$U_n/U_{n+1} \cong A_{s(n)}. \tag{35.2}$$

Any additive-monoid isomorphism $f : M'_s \rightarrow M'_t$ extends to an ordered-group isomorphism of their difference groups by

$$\tilde{f}(a - b) = f(a) - f(b).$$

This is well-defined since $a - b = c - d$ means $a + d = c + b$; every signed group element is a difference of nonnegative ones. The order is preserved because $a \leq b$ in a cone is equivalent to $\exists c (a + c = b)$.

The ordered set $\omega + 1$ has no nontrivial order automorphism, so $\tilde{f}$ must preserve each rank and send U_n to its counterpart. The quotients in (35.2) then recover $s(n)$: A_0 has rational vector-space dimension 1, whereas A_1 has dimension 2. An additive isomorphism between divisible torsion-free groups is automatically $\mathbb{Q}$ -linear. Thus $M'_s \cong M'_t$ implies $s = t$. This gives $2^{\aleph_0}$ distinct isomorphism types.

For the fixed-order refinement, A_0 and A_1 are countable dense linear orders without endpoints. A back-and-forth construction gives an order isomorphism $\theta : A_0 \rightarrow A_1$ fixing 0: alternately place the next unused element of each enumerated order into the appropriate nonempty interval of the other. Since both topologies are discrete, θ is a homeomorphism. Apply either θ or the identity in coordinate n according to $s(n)$, and leave the integer coordinate unchanged. This gives an order-preserving homeomorphism $M_\omega \rightarrow M'_s$ fixing 0, 1. Pulling back all the additions, and then using one homeomorphism $M_\omega \cong \mathbb{N}^{\mathbb{N}}$, proves the refinement.

Finally, a continuous function $\mathbb{N}^{\mathbb{N}^2} \rightarrow \mathbb{N}^{\mathbb{N}}$ is determined by its values on a fixed countable dense subset. There are at most $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ such value assignments. Choices of the two

constants do not enlarge this bound, and in Presburger arithmetic the order is determined by addition. Thus no more than continuum many isomorphism types are possible. $\square$

The continuous family in this theorem is not asserted to be an uncountable family of absolutely computable operations. There are only countably many algorithms. The effective assertion proved above applies to the particular rational ω -indexed model; computable variations can be handled separately, and noncomputable sequences s may serve as oracles.

35.6.1 Countably transfinite Archimedean scales [06]

There is a different useful variation, relevant to the repository's interest in ordered groups and ordinal scales.

Theorem 35.9 (Prescribed well-ordered scales [06]; = [theorems 35.1](#) and [35.4](#)). *For every countable ordinal $\alpha \geq \omega$, the positive cone*

$$M_\alpha = (\mathbb{Q}^\alpha \xrightarrow{\rightarrow} \mathbb{Z})_{\geq 0},$$

with discrete rational coordinates and the product subspace topology, is a Baire-space Presburger model with continuous addition. Its Archimedean classes, in decreasing magnitude, have order type $\alpha + 1$. In particular distinct such ordinals give nonisomorphic additive models.

Proof. For $\beta < \alpha$, let

$$C_\beta^\pm = \{h : (\forall \gamma < \beta) h(\gamma) = 0, \pm h(\beta) > 0\}.$$

Each set is closed, including at limit indices, since the preceding zero conditions are closed and the sign condition is clopen in a discrete coordinate. Both

$$M_\alpha = \bigcup_{\beta < \alpha} (C_\beta^+ \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N})$$

and its complement in $\mathbb{Q}^\alpha \times \mathbb{Z}$ are F_σ . Therefore M_α is G_δ in a Polish space, hence Polish and zero-dimensional. Its algebra and addition follow from Lemma [31.2](#).

Every nonempty open set in M_α contains a closed copy of $\mathbb{Q}^\mathbb{N}$. To see this, choose a point and a neighborhood fixing its integer coordinate and finitely many rational coordinates. If the chosen point first becomes positive at a finite coordinate j , fix all its coordinates through j . If it vanishes on every finite coordinate, choose a finite j beyond all the finitely prescribed finite indices, put zeros below j , and put 1 at j . Keep the other prescribed coordinates. In either case, the resulting finite cylinder is clopen in the ambient product, lies inside the positive cone and the chosen neighborhood, and has countably infinitely many free coordinates. Hence no compact subset of M_α has nonempty interior.

For completeness, a nonempty zero-dimensional Polish space X with this last property is homeomorphic to Baire space. Use a complete compatible metric. Every nonempty clopen set is complete and noncompact, and hence is not totally bounded. It therefore has a countable clopen cover of arbitrarily small diameter with no finite subcover. Removing earlier members from each member disjointifies it into countably infinitely many nonempty clopen sets. Repeat within each piece, with diameters tending to zero. Each branch of the resulting $\mathbb{N}$ -branching tree determines exactly one point by completeness, and each point determines a branch. The pieces are clopen, so the resulting bijection in both directions is continuous. Apply this construction to M_α .

Finally, the same leading-coordinate argument as in Theorem 35.8 identifies the Archimedean class of a noninteger element with its least nonzero index $\beta < \alpha$; the integer class is last. Their ordered quotient is exactly $\alpha + 1$. An additive isomorphism preserves this quotient. Equality of the successor order types implies equality of α and β by removing their final elements. $\square$

The countable ordinals in this theorem describe *classes of magnitude*, not an increasing or decreasing ω_1 -sequence inside one arithmetic order. There is therefore no conflict with the uncountable-monotone-sequence obstruction in [2].

36 Unbounded oscillation at Borel level two

The Borel-level conclusion may suggest that all definable sets in these models have similarly simple topological behavior. A finer invariant shows that they do not. One fixed Presburger formula already has arbitrarily high countable boundary oscillation ranks across the models.

36.1 The derivative and its exact convention

Let X be a space and $E \subseteq X$. For a closed $F \subseteq X$, put

$$D_E(F) = \text{cl}_F(E \cap F) \cap \text{cl}_F(F \setminus E).$$

Starting with $F_0 = X$, set

$$F_{\xi+1} = D_E(F_\xi), \quad F_\lambda = \bigcap_{\xi < \lambda} F_\xi \quad (\lambda > 0 \text{ a limit}). \quad (36.1)$$

The *boundary oscillation rank* used here is the least ξ with $F_\xi = \emptyset$, when such a stage exists; otherwise we assign rank ∞ . For the characteristic function $\mathbf{1}_E$, this is the oscillation derivative at any threshold $0 < \varepsilon \leq 1$. The definition fixes the indexing convention; no identification with all possible conventions for difference-hierarchy ranks is intended.

In A_α , let u be the vector with $u_0 = 1$ and $u_\gamma = 0$ for $\gamma > 0$. Set

$$p = (u, 0), \quad E_\alpha = \{x \in M_\alpha : x < p\}.$$

This is the fixed formula $\varphi(x; y) \equiv x < y$, unary in x , with the indicated parameter p . Its chosen positive zeroth coordinate ensures that all boundary perturbations below stay inside the model.

Theorem 36.1 (Exact boundary oscillation ranks). *Write the countably infinite ordinal α uniquely as*

$$\alpha = \omega \cdot \beta + n, \quad 0 < \beta < \omega_1, \quad n < \omega.$$

For $0 < \mu \leq \alpha$, write

$$C_\mu = \{(a, z) \in M_\alpha : a_\gamma = u_\gamma \text{ for all } \gamma < \mu\}.$$

The recursion (36.1) for E_α satisfies

$$F_\xi = C_{\omega \cdot \xi} \quad (1 \leq \xi \leq \beta), \quad F_{\beta+1} = \emptyset. \quad (36.2)$$

Its rank is exactly $\beta + 1$. Consequently this one formula realizes every countable successor rank at least 2, and its ranks are unbounded below ω_1 .

Proof. Suppose first that $x = (a, z)$ differs from p at a finite first rational coordinate $k < \omega$. Fixing coordinates $0, \dots, k$ decides its comparison with p on a neighborhood of x . Such a point is not in F_1 .

Conversely, suppose that $a_k = u_k$ for every $k < \omega$. Every basic neighborhood of x fixes only finitely many coordinates. Choose an unfixed positive natural index k , change its value from 0 to -1 or to 1 , and leave everything else unchanged. All earlier coordinates still agree with u . The two resulting points are respectively below and above p , and remain positive because the zeroth coordinate stays 1. Thus $F_1 = C_\omega$.

For the successor step assume $1 \leq \xi < \beta$ and $F_\xi = C_\mu$, where $\mu = \omega \cdot \xi$. At least ω coordinates remain before α . Inside C_μ , a first residual difference at $\mu + k$, $k < \omega$, is decided by fixing the finite residual prefix $\mu, \dots, \mu + k$. These points leave the derivative. If a point agrees with u throughout the block $[\mu, \mu + \omega)$, every basic relative neighborhood leaves some $\mu + k$ unfixed. Perturbing this coordinate in the two directions gives points on both sides of p , still in C_μ and in M_α . Therefore

$$F_{\xi+1} = C_{\mu+\omega} = C_{\omega \cdot (\xi+1)}.$$

At a nonzero limit $\lambda \leq \beta$, intersecting the prefix conditions and using continuity of ordinal multiplication in its right argument gives

$$F_\lambda = \bigcap_{1 \leq \xi < \lambda} C_{\omega \cdot \xi} = C_{\omega \cdot \lambda}.$$

At stage β , only n rational coordinates and the integer coordinate are free. The resulting fiber is homeomorphic to the discrete space $\mathbb{Q}^n \times \mathbb{Z}$. Every subset of that fiber is relatively clopen, so the next derivative is empty. Each earlier displayed stage contains $\{u\} \times \mathbb{Z}$, proving that no earlier stage is empty.

Taking $\alpha = \omega \cdot \beta$ gives all stated successor ranks. □

Coordinate order α	Boundary oscillation rank of E_α
ω	2
$\omega \cdot 2$	3
ω^2	$\omega + 1$
$\omega \cdot (\omega + 1)$	$\omega + 2$
ω^ω	$\omega^\omega + 1$

The first derivative removes points where a finite prefix decides the comparison. Each later derivative removes one further block of ω coordinates. This is a topological calculation, although the defining formula itself never changes.

36.2 Explicit continuous approximations

There is also a direct approximation of $\mathbf{1}_{E_\alpha}$ by continuous functions. Choose an enumeration $\gamma_0, \gamma_1, \dots$ of α and put $S_N = \{\gamma_0, \dots, \gamma_N\}$. Sort this finite set in its ordinal order. Define $c_N(a, z)$ to be 1 exactly when comparison of the coordinates in S_N , followed by the integer coordinate, places (a, z) below $(u, 0)$.

Each $c_N : M_\alpha \rightarrow \{0, 1\}$ is continuous, because it depends on finitely many discrete coordinates. If $a \neq u$, its true first differing coordinate eventually belongs to S_N . From that

point on, every $c_N(a, z)$ is the correct answer; coordinates preceding the true first difference are equal and cannot change it. If $a = u$, every $c_N(a, z)$ already equals $\mathbf{1}_{\{z < 0\}}$. Hence

$$c_N(x) \longrightarrow \mathbf{1}_{E_\alpha}(x)$$

with eventual equality at each point.

Thus these characteristic functions are explicitly Baire class 1, while their boundary oscillation ranks have no countable uniform bound across the family. Low Borel level and finite formula length do not give a uniform bound on this finer topological invariant.

Corollary 36.2 (Optimal multiplicity at every specified rank). *For every countable $\beta > 0$, there are c pairwise nonisomorphic continuous Presburger models on Baire space with a parameter p for which the cut $x < p$ has boundary oscillation rank $\beta + 1$. All their definable relations are Δ_2^0 .*

Proof. In Theorem 35.5, replace the tail $\mathbb{Q}^\omega$ by $\mathbb{Q}^{\omega \cdot \beta}$ and choose $p = (1, 0, 0)$. The model remains Polish and homeomorphic to Baire space by the same closed-cylinder proof. An isomorphism still induces an isomorphism of the greatest Archimedean quotients B_s , so the continuum lower bound is unchanged.

For the cut at p , comparison at the discrete B_s -coordinate is locally decided unless that coordinate equals 1. On that fiber, the derivative removes successive blocks of ω rational coordinates, exactly as in Theorem 36.1. The first derivative fixes the B_s -coordinate to 1 and the first ω tail coordinates to zero; the ξ th derivative fixes the first $\omega \cdot \xi$ tail coordinates. The β th derivative is the discrete integer fiber and the next is empty. The rank is $\beta + 1$. $\square$

37 What continuity permits and forbids

37.1 The arithmetic order is not the Polish topology

In any Presburger model, the intrinsic order topology is discrete. For $a > 0$, the arithmetic interval $(a - 1, a + 1)$ is the singleton $\{a\}$; at zero the initial interval $[0, 1)$ is $\{0\}$. An uncountable discrete space is not second countable and hence not Polish. Accordingly, a positive answer to Glazer's question must distinguish continuity of addition from compatibility with the intrinsic order topology.

In $M_{\mathbb{R}}$,

$$x_j = (1/j, 0) \longrightarrow (0, 0), \quad x_j > (0, k) \quad \text{for every standard } k.$$

Topological convergence to zero is entirely compatible with being larger than every standard numeral. In M_α , let v_j be the coordinate unit vector at the natural index j . Then the analogous sequence $(v_j, 0)$ converges to zero and exceeds every standard numeral.

These sequences show explicitly that strict order is neither open nor closed. The ordered pair $(0, 1)$ belongs to $<$ but is approached by $(x_j, 1)$, which do not. The pairs $(1, x_j)$ belong to $<$ and converge to $(1, 0)$, which does not. The following general obstruction explains why this behavior is unavoidable.

Proposition 37.1 (Closed arithmetic order forces discreteness; = theorem 7.4). *Let M be a Hausdorff model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous successor. If $\leq$ is closed in M^2 , then the topology of M is discrete. If M is uncountable Polish with continuous addition, neither $<$ nor $\leq$ is open or closed.*

Proof. If $\leq$ is closed, then $<$ is open because its complement is the reverse non-strict order. Sections of $<$ are open. The singleton intervals just described are therefore open, so the topology is discrete.

If $<$ is open, the same singleton argument applies. If $<$ is closed, then $x \leq y \iff x < y + 1$ and continuity of successor make $\leq$ closed. If $\leq$ is open, then $x < y \iff x + 1 \leq y$ makes $<$ open. Each of these four possibilities forces discreteness. An uncountable second-countable space cannot be discrete. $\square$

[write] [Proposition 37.1](#) is [theorem 7.4](#) (source 05), and its first sentence also follows from source 04's [theorem 7.5](#), which needs no continuity of successor. The monus clause of [corollary 37.2](#) is source 04's [corollary 7.6](#); the min and max clauses are new.

Corollary 37.2 (Three individually forbidden operations). *On an uncountable Polish Presburger model with continuous addition, none of the following operations can be continuous:*

$$\min(x, y), \quad \max(x, y), \quad x \dot{-} y = \max\{x - y, 0\}.$$

Proof. In a Hausdorff space equalizers of continuous maps are closed. Each of the three operations, if continuous, would make $\leq$ closed through one of the equivalences

$$x \leq y \iff \min(x, y) = x \iff \max(x, y) = y \iff x \dot{-} y = 0.$$

Apply [Proposition 37.1](#). $\square$

This does not contradict continuity of partial subtraction: its domain is the subspace of pairs with $y \leq x$, and it is not extended across that domain's boundary by assigning the value zero.

37.2 Compactness remains impossible

The following obstruction is established in the earlier work of Enayat, Hamkins, and Wcisło [2]. We include the short jointly continuous case to locate our locally compact example relative to compact spaces.

Proposition 37.3 (= [corollary 12.2](#)). *No infinite compact Hausdorff space supports a Presburger model with jointly continuous addition. In particular, Cantor space does not.*

Proof. The set

$$C = \{(x, y, z) \in M^3 : x + z = y\}$$

is closed and compact. Its projection onto (x, y) is $\leq$, so this relation is compact and closed in the Hausdorff space M^2 . [Proposition 37.1](#) makes M discrete, contradicting compactness and infinitude. $\square$

[write] This is [corollary 12.2](#) of Part I, there with the compact-cancellative-monoid proof.

The model $M_{\mathbb{R}}$ is locally compact but not compact. Its separated ray components and its incomplete ambient-metric ends present no conflict with this obstruction.

37.3 A second topology makes predecessor continuous

In the topology of Theorem 32.1, total truncated predecessor

$$\text{pred}(x) = \begin{cases} 0, & x = 0, \\ x - 1, & x > 0 \end{cases}$$

is discontinuous at zero: $x_j = (1/j, 0) \rightarrow 0$, whereas $\text{pred}(x_j) = (1/j, -1)$ cannot converge to zero. That particular discontinuity is not forced by additive arithmetic.

Proposition 37.4. *The same algebraic model $M_{\mathbb{R}}$ admits a second Polish topology making addition, successor, and truncated predecessor continuous. Its isolated points are exactly the standard numerals.*

Proof. Use the topological disjoint sum

$$M_{\mathbb{R}}^{\sharp} = \mathbb{N}_{\text{discrete}} \sqcup ((0, \infty) \times \mathbb{Z}_{\text{discrete}}),$$

with the same underlying identification $(0, n) \leftrightarrow n$. It is a countable sum of Polish spaces. Standard points are isolated, and nonstandard points lie in open real rays, so have no isolated points.

The product splits into four clopen pieces according to whether each input is standard. On two standard inputs addition is a map on a discrete space. On a mixed piece it is, on each standard slice, translation of the integer coordinate. On two nonstandard inputs it is ordinary coordinate addition on positive real coordinates. Each restriction is continuous, and a sum involving a nonstandard input remains nonstandard. Hence addition is continuous on the whole product.

Successor and truncated predecessor act on the standard discrete part as the familiar maps, and on the nonstandard part as translations by ± 1 of the integer coordinate. They are continuous on the clopen pieces and therefore globally. $\square$

[merge] Under the identification $(a, n) \mapsto aX + n$ of section 26, $M_{\mathbb{R}}^{\sharp}$ is exactly the clopen stratum $(C_{\mathbb{R}}, \tau_{\text{FS}})$ of 03's cone in proposition 26.1, whose parts (iii) and (iv) found the same topology with continuous addition; proposition 37.4 adds that successor and truncated predecessor are continuous. It is also the subspace of polynomials of degree at most one in source 06's cone (proposition 42.1). Part I does not ask whether the discontinuity of predecessor in proposition 8.2 is forced; this proposition shows that it depends on the topology.

The two presentations are not homeomorphic: one has no isolated points and the other has countably many. Binary truncated subtraction still cannot be continuous in the second presentation, by Corollary 37.2.

37.4 The explicit models have no ordered-semiring expansion

Our examples are subject to an algebraic obstruction stronger than a mere failure of Borel regularity of multiplication.

Proposition 37.5 (Cofinal multiples obstruct multiplication; compare theorem 10.4). *Suppose $M \models \text{Th}(\mathbb{N}; 0, 1, +, <)$ has an element c such that*

- (i) $c > n1$ for every standard $n \in \mathbb{N}$;
- (ii) the standard multiples $\{nc : n \in \mathbb{N}\}$ are cofinal in M .

Then there is no multiplication, with the given addition and identity 1, making M the nonnegative cone of a discretely ordered ring. More generally, no semiring multiplication can satisfy distributivity, the given identity, and monotonicity under this arithmetic order.

Proof. Suppose such a multiplication exists. By cofinality choose a standard k with $c^2 < kc$; increasing a weak cofinal bound by one gives a strict bound. Since $c > (k + 1)1$, additive order gives $c = (k + 1)1 + u$ for some $u > 0$. Distributivity and nonnegativity of products give

$$c^2 = (k + 1)c + uc \geq (k + 1)c > kc,$$

a contradiction. $\square$

Corollary 37.6. *Neither $M_{\mathbb{R}}$ nor any M_{α} has a Peano-arithmetic expansion with the given addition, regardless of the regularity of multiplication.*

Proof. For $M_{\mathbb{R}}$ take $c = (1, 0)$. For M_{α} take $c = (u, 0)$ with $u_0 = 1$ and all other coordinates zero. In each case c exceeds every standard numeral. Given any element, choose a standard integer larger than its leading real or rational coordinate; that multiple of c strictly exceeds the element. Proposition 37.5 applies. $\square$

This proof shows exactly why the additive construction cannot be promoted to a model of full arithmetic by selecting a sufficiently irregular product. It also leaves room for studying other additive models, without the cofinal-multiple property, against Glazer's more general topological obstructions.

[write] Proposition 37.5 is a weaker form of theorem 10.4, which needs neither monotonicity nor commutativity; corollary 37.6 applies it to the whole ordinal family. Source 06's lemma 37.9 and theorem 37.10 remove the remaining hypothesis that the multiplicative unit be the named 1.

37.5 Definable regularity and an algebraic obstruction [06]

37.5.1 All additive definable relations are low in the Borel hierarchy [06]

Theorem 37.7 (Definable relations [06]; extends theorem 7.1). *In $M_{\mathbb{R}}$, in every M_{α} above, and in every M'_s of Theorem 35.8, every first-order Presburger-definable relation with parameters is Δ_2^0 in the corresponding finite product topology. Fixed-standard-modulus residue classes are clopen.*

Proof. In the ambient group $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$, the nonnegative cone and its complement are each a union of an open real-coordinate condition and a closed zero-coordinate condition. Thus the cone is Δ_2^0 . For the countable rational and mixed-coefficient products, the decompositions by first nonzero index used above give the same conclusion. Equality is closed in the ambient Hausdorff topological group.

Every integer-linear affine map from a finite power of the ambient group to that group is continuous. Consequently its comparison with zero defines a Δ_2^0 set. Congruence modulo n depends only on the discrete final integer coordinate, because the other factor is divisible. Such congruences are clopen, including for affine linear combinations and parameters.

Apply the quantifier-elimination conclusion of Lemma 31.1. A formula on the nonnegative cone is translated to the group by restricting its quantified variables to that cone. Elimination produces a finite Boolean combination of the described atomic conditions. Restrict it back to the cone and use closure of Δ_2^0 under finite Boolean operations. $\square$

Remark 37.8 (Graphs versus functions [06]). Low Borel complexity of a definable graph does not imply continuity of its function. In the real example the truncated predecessor

$$p(x) = \begin{cases} 0, & x = 0, \\ x - 1, & x > 0 \end{cases}$$

is definable but discontinuous at zero: $(1/k, 0) \rightarrow 0$, whereas $p(1/k, 0) = (1/k, -1)$ stays in a different integer slice from $p(0)$. In the Baire model, let e_k be the rational sequence with 1 in coordinate k and zeros elsewhere. Then $(e_k, 0) \rightarrow 0$ gives the identical obstruction.

The order relation itself is neither open nor closed in these two basic models. Write $x_k = (e_k, 0)$ in the Baire model and $x_k = (1/k, 0)$ in the real model. In both cases $x_k > 1$ and $x_k \rightarrow 0$. Thus the pairs $(x_k, 1)$ lie outside the relation $<$ but converge to $(0, 1)$ inside it. The pairs $(1, x_k)$ lie inside $<$ but converge to $(1, 0)$ outside it.

The standard cut $\{0\} \times \mathbb{N}$ is closed in these models and is not definable, even with parameters, in the Presburger language. If it were defined by $\varphi(x, a)$, it would contain zero and be closed under successor, while omitting nonstandard elements. The induction instance for φ would then fail. This supplies a concrete distinction between topological visibility and first-order definability.

37.5.2 Why the lexicographic models cannot acquire multiplication [06]

The failure of multiplication for our first examples is stronger than the absence of a continuous or Borel multiplication.

Lemma 37.9 (The unit is forced [06]). *Suppose a nontrivial Presburger additive monoid M , with least positive element e , is given any unital semiring multiplication, with the same addition and zero. Then its multiplicative identity is e .*

Proof. The natural order $x \leq y \iff \exists z (x + z = y)$ is preserved by multiplication on either side, by distributivity. Write u for the multiplicative identity. It is nonzero, so $e \leq u$ and $0 \leq e^2 \leq eu = e$. By discreteness, e^2 is either 0 or e . If $e^2 = 0$, use division by two in the Presburger monoid to write $u = 2q + re$, where $r = 0$ or 1. Multiplication by e gives $e = ue = 2(qe)$, impossible for the least positive element. Therefore $e^2 = e$. If $u > e$, write $u = e + v$ with $v \geq e$. Then $e = ue = e^2 + ve \geq 2e$, again impossible. Thus $u = e$. $\square$

Theorem 37.10 (No semiring expansion [06]; compare [theorem 10.4](#)). *None of $M_{\mathbb{R}}$, the nontrivial M_{α} , or the M'_s admits a unital semiring multiplication with its prescribed additive monoid. This remains true without imposing any topology or prescribing the multiplicative identity in advance.*

Proof. In each example choose a with leading coordinate 1 in the first divisible coordinate and zero elsewhere. Then $a > n \cdot 1$ for every standard n . On the other hand, for every $b \in M$ there is a standard N with $b < Na$, since an ordinary integer can be chosen larger than the real leading coefficient of b . Thus standard multiples of a are cofinal in M .

Any proposed multiplication has identity 1 by Lemma 37.9. Since $a \geq n \cdot 1$, distributivity and the natural order imply $a^2 \geq na$ for every standard n . Taking $b = a^2$ in cofinality gives $a^2 < Na$ for some standard N , a contradiction. $\square$

The existing repository report contains a more general obstruction to PA expansions of nonstandard split $\mathbb{Z}$ -groups, using residue types [35]. The argument here has a different and stronger conclusion for the displayed groups: their largest Archimedean magnitude class

already rules out every unital semiring structure. The next example escapes this obstruction by having polynomial degrees of arbitrarily large magnitude.

38 A Polish semiring with full additive induction [06]

38.1 The ring and its order [06]

Let

$$A = \mathbb{Z} + t\mathbb{R}[t] = \left\{ m + \sum_{i=1}^d a_i t^i : m \in \mathbb{Z}, a_i \in \mathbb{R}, d < \omega \right\}.$$

The indeterminate t is ordered as positive infinite: a nonzero polynomial is positive exactly when its leading coefficient is positive. This is an ordered subring of $\mathbb{R}[t]$ under its leading-term order. It is discrete because a positive constant in A is a positive integer, while every positive nonconstant polynomial exceeds every integer. Put $P = A_{\geq 0}$.

Algebraically, $A/\mathbb{Z} \cong t\mathbb{R}[t]$ is a divisible ordered abelian group. Thus P is a model of Pres as an additive structure, and its inherited polynomial multiplication makes it a model of PA^- . This underlying polynomial construction has classical precedents and occurs explicitly in the repository [24, 36]. Our task is to choose and verify a topology.

38.2 The topology by degree and constant coefficient [06]

For $d \geq 1$ and $m \in \mathbb{Z}$, let

$$P_{d,m} = \{m + a_1 t + \cdots + a_d t^d : a_d > 0\}.$$

Give $P_{d,m}$ the coefficient topology $\mathbb{R}^{d-1} \times (0, \infty)$, and make each $n \in \mathbb{N}$ a singleton component. Define P to be the topological disjoint union

$$P = \mathbb{N}_{\text{discrete}} \sqcup \bigsqcup_{d \geq 1, m \in \mathbb{Z}} P_{d,m}. \quad (38.1)$$

In particular, convergence within a nonconstant stratum eventually has fixed degree and constant coefficient.

Theorem 38.1 (A continuous weak arithmetic semiring [06]). *The space in (38.1) is uncountable, locally compact, and Polish. Both polynomial addition and polynomial multiplication are jointly continuous on it. Its additive reduct satisfies full Presburger arithmetic, and its semiring structure satisfies PA^- .*

Proof. Each $P_{d,m}$ is homeomorphic to $\mathbb{R}^d$ by applying log to the positive leading coefficient. The countably many components in (38.1) are locally compact Polish spaces. A complete metric is obtained by bounding complete component metrics by 1 and putting distance 2 between components. Countable dense sets in the components give separability. The algebraic assertions were verified above.

It remains to prove joint continuity rather than only separate continuity. Products of components form an open cover of P^2 . On $P_{d,m} \times P_{e,n}$, positivity of both leading coefficients prevents cancellation in addition, so

$$\deg(p + q) = \max(d, e), \quad (p + q)(0) = m + n.$$

Thus this whole product maps into the fixed component $P_{\max(d,e),m+n}$, and the coefficient map there is linear and continuous.

On the same product,

$$\deg(pq) = d + e, \quad (pq)(0) = mn,$$

and the leading coefficient is a positive product. It maps into the fixed component $P_{d+e,mn}$, with continuous polynomial coefficient maps. If one input is a positive standard integer, its component is a singleton and the corresponding operations are coefficient translation or scalar multiplication. If an input is zero, multiplication is constant on its open singleton component. Pairs of standard integers are handled by singleton products. This proves continuity on every member of an open cover of P^2 , hence joint continuity globally. $\square$

This theorem concerns the nonnegative cone. The analogous topology on all signed polynomials would not make the entire ring a topological ring: cancellation can reduce degree. Even partial subtraction on the cone fails continuity. For instance

$$((1 + 1/k)t, t) \longrightarrow (t, t), \quad (1 + 1/k)t - t = (1/k)t \not\longrightarrow 0,$$

because zero is isolated. This is consistent with the continuity just proved for addition and multiplication of nonnegative elements.

38.3 An exact failed induction instance [06]

Theorem 38.2 (Failure of open induction [06]). *In the polynomial model P , the quantifier-free formula*

$$\varphi(x; t) \equiv x^2 < t$$

defines exactly the standard natural numbers. Consequently P fails IOpen , despite satisfying every first-order additive induction instance.

Proof. If $x = n \in \mathbb{N}$, then $n^2 < t$. If x is a positive nonconstant polynomial of degree $d \geq 1$, then x^2 has degree $2d \geq 2$ and positive leading coefficient, so $x^2 > t$. Therefore $\varphi(P; t) = \mathbb{N}$. It contains zero and is closed under successor, but omits t . This falsifies the open-induction instance with parameter t . $\square$

The full omnific integer cone and the polynomial subring are different arithmetic structures. This counterexample concerns the subring $\mathbb{Z} + t\mathbb{R}[t]$ only; it does not transfer to the full omnific integer cone simply because the subring embeds there.

Proposition 38.3 (Additive definability in the polynomial topology [06]). *Every relation definable with parameters in the Presburger reduct of P is Δ_2^0 . Quotient and remainder by each fixed standard divisor are continuous.*

Proof. On a fixed product of coefficient components, an affine integer-linear expression has bounded degree and continuous coefficient functions, with fixed integer constant coefficient. Its positive and negative sign conditions are finite unions of sets where all higher coefficients vanish and a specified coefficient has the required sign; the residual constant case is also included. These sets and their complements are F_σ on each component. There are countably many clopen product components, so the conditions are Δ_2^0 globally. Congruences depend only on the integer constant term and are clopen. Apply Presburger elimination.

For standard $n \geq 1$, division sends the nonconstant coefficients a_i to a_i/n and the constant term m to $\lfloor m/n \rfloor$; the positive leading coefficient preserves degree. Remainder is the standard integer $m \bmod n$. These maps are continuous on each clopen component, hence on all of P . $\square$

39 Zero must be isolated when both operations are continuous [06]

The preceding polynomial example has isolated points, whereas our additive examples are perfect. There is an exact structural reason for this contrast. The essential subset-sum and probability method below comes from Glazer's proof of his first theorem [1]. An audit of its assumptions removes the arithmetic induction hypothesis. We also avoid any need to prove injectivity of the subset-sum limit map.

Theorem 39.1 (Isolated zero [06]). *Let A be a discretely ordered commutative unital ring, and put $M = A_{\geq 0}$. If M carries a Polish topology under which its restricted addition and multiplication are jointly continuous, then 0 is isolated.*

Corollary 39.2 ([06]). *No perfect Polish space supports a topological model of PA^- with continuous addition and multiplication. In particular Baire space supports the additive models constructed above, but cannot support such a weak-arithmetic semiring structure.*

39.1 Borel order from continuous addition [06]

We first record a point that prevents any hidden regularity assumption. For $a, b \in M$,

$$\begin{aligned} a < b &\iff \exists c \in M (b = a + c + 1), \\ a \geq b &\iff \exists c \in M (a = b + c). \end{aligned}$$

The equations define closed subsets of M^3 , because addition is continuous and M is Hausdorff. Their projections are analytic. They are complementary relations, so the analytic/coanalytic separation theorem gives that the order is Borel [8]. This argument uses no order-topology compatibility.

39.2 A uniformly convergent subset-sum map [06]

Suppose, toward a contradiction, that 0 is not isolated. Since $[0, 5] = \{0, 1, \dots, 5\}$ and finite sets are closed, one can choose $x_k > 5$ with $x_k \rightarrow 0$. Fix a complete compatible metric d on M . Set $y_0 = 1$. After choosing $y_0, \dots, y_{n-1}$, choose $k(n)$ so large that, with

$$y_n = x_{k(n)} y_{n-1},$$

we have

$$d(s, s + y_n) < 2^{-n} \quad \text{for every subset sum } s \text{ of } y_0, \dots, y_{n-1}. \quad (39.1)$$

There are only finitely many such sums. For each, continuity implies $s + x_k y_{n-1} \rightarrow s$, so a common choice exists. At the same time, ordered-ring positivity gives

$$y_n > 5y_{n-1} > 0. \quad (39.2)$$

For $r \in 2^{\mathbb{N}}$, define

$$\sigma_N(r) = \sum_{i < N} r(i) y_i.$$

Each σ_N is continuous, since it depends on finitely many bits. For $N \geq 1$ the distance between its next two partial sums is at most 2^{-N} , uniformly in r . Thus σ_N converges uniformly to a continuous function $r \mapsto z_r \in M$. Completeness is used at precisely this point.

If r, s differ only at finitely many positions, continuous addition implies the following *algebraic* identity in A :

$$z_s - z_r = \sum_i (s(i) - r(i))y_i. \quad (39.3)$$

For justification, add the finite removed terms to the appropriate sides of the finite partial-sum identity, then pass to the limit using addition only. Rearranging the resulting equality is an algebraic operation in A . No topology on A and no continuity of subtraction are being assumed.

Let $\bar{r}$ denote coordinatewise bit complementation and set

$$\Delta_r = z_r - z_{\bar{r}}$$

as notation for an element of the algebraic ring. Equation (39.3) yields

$$\Delta_s - \Delta_r = 2 \sum_i (s(i) - r(i))y_i \quad (39.4)$$

whenever r, s differ finitely.

39.3 Bounded complementary differences have probability zero [06]

Give $2^{\mathbb{N}}$ fair independent product measure μ . Put $S_j = \sum_{i < j} y_i$. External finite induction, using (39.2), gives

$$4S_j < y_j \quad (j \in \mathbb{N}). \quad (39.5)$$

Indeed it holds for $j = 0$, and $4S_{j+1} = 4S_j + 4y_j < 5y_j < y_{j+1}$.

For fixed n consider

$$E_n = \{r : |\Delta_r| \leq y_n\}.$$

This is a Borel set. To see this without treating Δ as a map into a topological ring, write its definition as the conjunction

$$z_r \leq z_{\bar{r}} + y_n, \quad z_{\bar{r}} \leq z_r + y_n.$$

All terms are continuous M -valued maps and the order is Borel.

Freeze all coordinates except $n+1, \dots, n+k$. If two assignments r, s on these coordinates differ last at $j > n$, then (39.4) and (39.5) imply

$$|\Delta_s - \Delta_r| \geq 2(y_j - S_j) > 2y_n.$$

For the last inequality, $S_j \geq y_n$ and $y_j > 4S_j$. Two values lying in the ordered interval $[-y_n, y_n]$ cannot be this far apart. Thus at most one of the 2^k assignments in each fiber lies in E_n . Integrating over the frozen coordinates gives

$$\mu(E_n) \leq 2^{-k}$$

for every k , and hence $\mu(E_n) = 0$. Therefore

$$C = \bigcap_{n \in \mathbb{N}} \{r : |\Delta_r| > y_n\} \quad (39.6)$$

is Borel and has measure one. In particular, complementary subset sums are unequal almost surely. No injectivity of $r \mapsto z_r$ is needed.

39.4 The sign is a tail event [06]

If r, s differ only at positions at most j , write $b = \Delta_s - \Delta_r$. Equations (39.4) and (39.5) give

$$|b| \leq 2S_{j+1} < 3y_j.$$

For an arbitrary n , take $m = \max(n, j) + 1$. Then

$$y_n + |b| < 4y_{m-1} < y_m.$$

If $r \in C$, it follows that $|\Delta_s| \geq |\Delta_r| - |b| > y_n$ and that Δ_s has the same sign as Δ_r . Since n was arbitrary, C and the sign on C are invariant under every finite change of coordinates.

Consequently the Borel event

$$B = \{r \in C : z_{\bar{r}} < z_r\}$$

belongs to the tail sigma algebra. Such an event has probability zero or one. One quick proof is that it is independent of the first N coordinates for every N ; a monotone-class argument makes it independent of the full sigma algebra, and in particular of itself. Thus $\mu(B) = \mu(B)^2$.

But $r \mapsto \bar{r}$ preserves measure and C , and reverses the sign of Δ_r . Since C has measure one, it follows that $\mu(B) = 1/2$, a contradiction. This proves Theorem 39.1. $\square$

Remark 39.3 (Exactly what the proof uses [06]). The proof uses the ordered-ring laws, discreteness, separability, complete metrizable, continuity, and ordinary external finite induction. It does not use induction *inside* M , standard division in M , internal exponentiation, saturation, or an integer-part theorem. It concludes that zero is isolated. It does not by itself imply that every standard numeral is isolated, since predecessor has not been assumed continuous.

40 Surreal and omnific realizations

40.1 The real model as an omnific subgroup

The most direct connection to the repository is the map

$$J_{\mathbb{R}} : G_{\mathbb{R}} \longrightarrow \mathbf{No}, \quad J_{\mathbb{R}}(r, n) = r\omega + n. \quad (40.1)$$

Here multiplication and addition on the right are surreal field operations. Conway normal forms compare the coefficient of the dominant monomial ω before the constant coefficient. Thus $J_{\mathbb{R}}$ is an ordered additive isomorphism onto $\mathbb{R}\omega + \mathbb{Z}$.

An omnific integer has a Conway normal form with no negative exponents and with integer constant coefficient. This description places $\mathbb{R}\omega + \mathbb{Z}$ inside $\mathbf{Oz}$. The standard normal-form facts used here and below are recalled in [23]; a general treatment of initial ordered subgroups appears in [5].

Transport the topology of $M_{\mathbb{R}}$ through $J_{\mathbb{R}}$. The nonnegative part of $\mathbb{R}\omega + \mathbb{Z}$ becomes an uncountable Polish additive model with continuous addition. This is a topology on a *set-sized substructure*, described by its coefficients. It is not the intrinsic surreal order topology.

For example, in this transported topology,

$$\frac{\omega}{j} \longrightarrow 0,$$

although ω/j exceeds every real number for every positive standard j . A coefficient limit is not a claim about order convergence of surreal numbers.

40.2 Normal forms for the entire ordinal family

Theorem 40.1 (Explicit omnific embedding; extends [theorem 9.1](#)). *Fix a countably infinite ordinal α . Regard each ordinal $\beta + 1$ as a surreal number and put*

$$\varepsilon_\beta = \frac{1}{\beta + 1} \quad (\beta < \alpha).$$

Then

$$J_\alpha(a, n) = \sum_{\beta < \alpha} a_\beta \omega^{\varepsilon_\beta} + n \quad (40.2)$$

is a well-defined ordered additive embedding of G_α into $\mathbf{Oz}$, sending $(0, 1)$ to 1. Its nonnegative image, with transported coefficient topology, has all the properties of [Theorem 35.1](#) and [Theorem 36.1](#).

Proof. The positive surreal exponents ε_β are strictly decreasing: ordinal inclusion agrees with surreal order, and reciprocation reverses strict order on positive elements of a field. Every subset of $\{\varepsilon_\beta : \beta < \alpha\}$ has a largest element, corresponding to its least index. Thus every support in (40.2) is a set-sized reverse well order, as required for a Conway normal form. The sum denotes this formal normal form, not a limit in an unspecified topology.

Uniqueness of coefficients proves injectivity. Addition of normal forms with a common set of monomials is coefficientwise, proving additivity. For unequal vectors, the least differing β supplies the largest monomial in their difference, so determines its sign. If all rational coordinates agree, the integer constant does. This proves order preservation and reflection.

All displayed exponents are positive, and the constant coefficient is an integer. Hence every value is omnific. The remaining assertions follow by transporting the topology and structure along this isomorphism onto its image. $\square$

[write] For $\alpha = \omega$, J_ω is Part I's map ι_B of [theorem 9.1](#), and $J_{\mathbb{R}}$ is $\iota_{\mathbb{R}}$. Source 06's realizations ([section 40.4](#)) use the exponents $1/(i + 1)$ for $\alpha = \omega$ and $\omega^{-\beta}$ for larger α ; both choices are strictly decreasing sequences of positive surreal exponents. Whether these images are initial in the sense of `isg:cf:main:presburger` is not checked here; the groups are split, so that theorem applies to them abstractly ([section 1.4](#)).

Remark 40.2 (Ordinals, reciprocals, and the omega-map). In (40.2), $1/(\beta + 1)$ is a reciprocal in the surreal field, not ordinal division. Likewise ω^x denotes Conway's monomial map. It should not be silently replaced by a different surreal exponential convention. For $\alpha = \omega$ the exponents are simply $1, 1/2, 1/3, \dots$

Corollary 40.3 (Omnific realization of the continuum family). *For $B_s = \mathbb{Q} + \mathbb{Q}s$ as in [Theorem 35.5](#) and any countably infinite ordinal α , the map*

$$J_{s,\alpha}(b, a, n) = b\omega^2 + \sum_{\beta < \alpha} a_\beta \omega^{1/(\beta+1)} + n$$

is an ordered additive embedding of $B_s \xrightarrow{\times} \mathbb{Q}^\alpha \xrightarrow{\times} \mathbb{Z}$ into $\mathbf{Oz}$. Consequently the continuum family, including the models in [Corollary 36.2](#), also has explicit omnific realizations with its topology transported from the coefficient space.

Proof. All tail exponents are positive and at most 1, so the extra exponent 2 lies strictly above them. The same normal-form argument applies, with the real coefficient $b \in B_s$ determining the leading component and the integer n the constant term. $\square$

40.3 Set size and formal foundations

For a fixed α , the domain $\mathbb{Q}^\alpha \times \mathbb{Z}$ is a set, and the exponents in (40.2) form a set. In the sign-sequence presentation of surreal numbers, the normal-form operation assigns a unique surreal set-code to each input. Replacement therefore makes the whole image a set. Its birthdays have an ordinal supremum; after increasing that bound, the entire image lies in a single set-sized initial segment of the surreal class. The same argument applies to the real model and to each fixed parameter s in the continuum family.

Accordingly, ordinary ZFC suffices to quantify over the model, all its subsets, its topology, and all its finite Cartesian powers. No “set of all surreals” is introduced. The proper class **No** is used as a definable ambient class of set-codes, not as an element of itself or as a domain of unrestricted set comprehension.

The images are additive ordered subgroups. We do not claim that the particular normal-form images are initial under surreal simplicity, closed under surreal multiplication, or closed under arbitrary surreal-analytic operations. Already $(\omega)^2$ is outside the rank-two group $\mathbb{R}\omega + \mathbb{Z}$. Proposition 37.5 also forbids manufacturing a different compatible semiring product internally in these examples.

40.4 Surreal realizations and a formalization route [06]

40.4.1 The same algebra at explicit surreal scales [06]

The connection with the repository’s surreal work is visible directly in Conway normal forms. Using the usual normal-form arithmetic, the map

$$(r, n) \mapsto r\omega + n \quad (40.3)$$

identifies $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ with an ordered additive subgroup of the omnific integers. The rational-sequence model has the realization

$$((q_i)_{i \in \mathbb{N}}, n) \mapsto n + \sum_{i \in \mathbb{N}} q_i \omega^{1/(i+1)}. \quad (40.4)$$

The exponents are strictly decreasing positive real numbers; every subset of this sequence is reverse well ordered. The expression is therefore a valid set-supported normal form. Its leading coefficient determines the sign, and addition is coefficientwise, exactly as in our lexicographic group. For arbitrary countable α , one can similarly use the strictly decreasing positive surreal exponents $\omega^{-\beta}$, $\beta < \alpha$.

Finally the substitution $t \mapsto \omega$ identifies $\mathbb{Z} + t\mathbb{R}[t]$ with the ring

$$\left\{ n + \sum_{i=1}^d a_i \omega^i : n \in \mathbb{Z}, a_i \in \mathbb{R}, d < \omega \right\},$$

which is explicitly studied in the repository [36]. These identifications use standard surreal normal-form facts, also documented in the cited repository reports [35, 36]; none is needed for the proofs of our Polish models.

Two distinctions matter. First, the subgroup in (40.4) is being used additively; no closure under surreal multiplication is asserted. Second, all Polish topologies here must be *transported* through the displayed identifications. They are not claimed to be inherited from a topology on the proper class **No**. Our constructions are set-sized throughout and need no class of all subsets, global enumeration, or paradox-prone quantification over all surreal numbers.

40.4.2 What the existing formal development certifies [06]

The repository's Presburger development supplies a promising implementation starting point. In the inspected snapshot, its `Syntax.lean` interprets formulas under valuations `List Int`. The central results in `Decision.lean`, including `holds_iff_quantifierEliminate`, therefore concern standard integer semantics. They do not already assert that an arbitrary $\mathbb{Z}$ -group satisfies every eliminated formula.

A faithful formalization can separate four layers.

1. **Algebraic semantics.** Define a discretely ordered abelian group with its least positive element and the standard division/remainder schema. Interpret affine terms and formulas in an arbitrary such group. Generalize the Cooper residue and interval lemmas so they no longer mention `Int` in their semantic valuation type.
2. **Concrete models.** Build the ordered groups $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$, $\mathbb{Q}^{\mathbb{N}} \overrightarrow{\times} \mathbb{Z}$, and the additive group of $\mathbb{Z} + t\mathbb{R}[t]$. Prove the explicit division identity and transfer the generic semantics to their nonnegative cones.
3. **Topology.** Supply the product or disjoint-sum topologies, prove the specified operations continuous, and certify Polishness by the explicit homeomorphisms or complete metrics. The polynomial model requires a topology on the positive cone, not an incompatible claim about a topological signed ring.
4. **Boundary results.** Formalize the cofinal-scale obstruction and the single failed induction instance. The isolated-zero theorem can be a separate later target, adding descriptive-set-theoretic and probability dependencies.

An adjacent interface requires care. The inspected file `PAHF/PASyntax.lean` contains a shallow `PA.Model` interface with induction for every Lean predicate $\alpha \rightarrow \text{Prop}$. Such full external induction excludes the nonstandard objects of interest. The appropriate route uses raw `PreModel` operations, syntactic formulas and satisfaction, and first-order induction instances [37]. For the polynomial construction, one must also keep the additive language distinct from the language with multiplication.

These are proposed formalization steps, not claims that a formalization was run in this project. The deliverable is a self-contained mathematical article; its full proofs make the remaining kernel-checking task precise and reviewable.

41 Formalization scope and a concrete development plan

41.1 What the existing repository actually proves

The repository's executable development has a specific semantic domain. In the inspected source,

```
Formula.holds : Formula -> List Int -> Prop
```

interprets quantifiers over Lean's ordinary integer type. In particular, the existential case is an existential over `Int`. The theorem `holds_iff_quantifierEliminate` proves the computed elimination correct for these integer valuations [32].

This is relevant infrastructure, but does not formally establish that an arbitrary ordered group with a divisible quotient is a Presburger model. It also does not verify a topology

on $\mathbb{R} \times \mathbb{Z}$ or $\mathbb{Q}^\alpha$. The source was inspected during this investigation; no Lean or Rocq build was run. The new theorems in this report are ordinary mathematical proofs, not claims of a successful proof-assistant execution.

[write] The claims about the formal project hold at the repository’s current state: `Syntax.lean` defines `holds : Formula -> List Int -> Prop` with an existential over `Int` (lines 192–197), and `holds_iff_quantifierEliminate` is in `Decision.lean` (line 18); the project’s Lean files are unchanged since both pins. [Section 13.5](#) names the declarations that Part I’s porting requirement concerns. **Nothing in this part is formalized.**

41.2 A small first formalization target

The best first target is Theorem [32.1](#), with the algebra and topology proved in separate modules whose assumptions meet in one final statement. A suitable sequence is:

1. Define the coordinate type $\mathbb{R} \times \mathbb{Z}$, its lexicographic order, and the nonnegative subtype $M_{\mathbb{R}}$. Prove the least-positive property and the explicit division identity ([32.4](#)).
2. Prove, or port, the semantic theorem that a $\mathbb{Z}$ -group satisfies the complete Presburger theory. Generalize formula interpretation from `List Int` to valuations in a structure. Avoid assuming the new model satisfies the theory as an unproved axiom.
3. Put the product topology on the coordinates, independently of the lexicographic order instance. Prove closure under addition, restricted continuity, and Polishness using ([32.2](#)) or the explicit metric ([32.3](#)).
4. Combine the semantic and topological theorems. State cardinality through an explicit injection from $\mathbb{R}_{\geq 0}$, so uncountability is not hidden in a typeclass declaration.

The central implementation risk is accidentally selecting the intrinsic order topology when defining a lexicographically ordered type. That topology is discrete. The product topology used here must be installed explicitly and its compatibility claims stated only for the operations that have actually been proved continuous.

41.3 Extending the verified boundary

Once the real-coordinate construction is complete, the $\alpha = \omega$ Baire model is the next manageable target. Its strict coefficient cones are open, avoiding the transfinite Borel decomposition. A coordinate enumeration, cylinder basis, and the noncompact-cylinder lemma can be implemented without any proper-class formalization.

The general cone theorem requires more library infrastructure: Baire category for closed subgroups, Borel sets’ Baire property, countable-base arguments, and transfinite recursion on closed subgroups. The oscillation theorem needs ordinal arithmetic and a general relative-boundary recursion. Those are substantive formalization tasks, not consequences of running the existing integer decision procedure.

The surreal embedding can be isolated still further. One may first work with a set-sized Hahn additive group supported on the fixed exponent set, prove its coordinate order and addition facts, and then use a separately verified normal-form map into a universe-bounded surreal type. This avoids making the core topological result depend on a full surreal field development.

Layer	Mathematical obligation	Present status
Ordinary integer formulas	Executable Cooper elimination and semantic correctness	Existing repository sources inspected
Generic additive models	$\mathbb{Z}$ -group characterization and full theory	Classical theorem used explicitly
Real Polish model	Cone, metric, continuity, uncountability	Proved in this report
Baire ordinal family	G_δ cone, cylinder argument, nonisomorphism	Proved in this report
Oscillation ranks	Relative derivatives and ordinal calculation	Proved in this report
Omnific embedding	Normal-form existence and coefficient transport	Classical normal forms plus proof here
New Lean/Rocq certificates	Kernel-checked versions of the report's new claims	Not supplied

42 Where Part III meets Parts I and II [merge]

Source 06's ring $A = \mathbb{Z} + t\mathbb{R}[t]$, with t positive infinite, becomes under $t \mapsto X$ a subring of 03's finite principal-part ring $\mathcal{A}_{\text{fin}} = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{>0}} \mathbb{R}X^q$ of [Part II](#): the elements with integer exponents. Both rings are ordered by the sign of the leading coefficient, so $P = A_{\geq 0} = A \cap \mathcal{A}_{\text{fin}, \geq 0}$. Write τ_P for source 06's topology ([38.1](#)) on P .

Proposition 42.1 (The stratum of degree one, and two topologies on P [merge]). *The following hold.*

(i) *The set of elements of degree at most one,*

$$P_{\leq 1} = \{x \in P : \deg x \leq 1\} = \mathbb{N} \cup \bigcup_{m \in \mathbb{Z}} P_{1,m},$$

is clopen in (P, τ_P) . The map $m + at \mapsto (a, m)$ is an isomorphism of ordered additive monoids and a homeomorphism from $P_{\leq 1}$ onto source 07's $M_{\mathbb{R}}^\#$ ([proposition 37.4](#)), and $t \mapsto X$ maps $P_{\leq 1}$ homeomorphically onto the stratum $(C_{\mathbb{R}}, \tau_{\text{FS}})$ of [proposition 26.1](#).

(ii) *On P , the topology τ_P is strictly coarser than the topology induced by 03's τ_{FS} .*

(iii) *Addition and multiplication are continuous on (P, τ_P) ([theorem 38.1](#)), but addition is not continuous on P with the topology induced by τ_{FS} .*

Proof. (i) $P_{\leq 1}$ is a union of components of ([38.1](#)). Each $P_{1,m} = \{m + at : a > 0\}$ carries the topology of $(0, \infty)$ in the coefficient a , and the points of $\mathbb{N}$ are isolated. These are exactly the components of $M_{\mathbb{R}}^\#$, namely the rays $(0, \infty) \times \{m\}$ and the isolated standard numerals, and of $C_{\mathbb{R}}$, namely $U_{\{1\},m}$ and $U_{\emptyset,n}$, $n \geq 0$. The algebra is the lexicographic algebra of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ in all three cases.

(ii) A component $U_{S,n}$ of τ_{FS} that lies in P has $S \subseteq \{1, \dots, d\}$ with $\max S = d$, so it lies in $P_{d,n}$, and its inclusion there, which inserts zero coefficients, is continuous. Hence every τ_P -open set meets each such component in an open set, and τ_P is coarser. It is strictly coarser: $t^2 + t/j \rightarrow t^2$ in τ_P , inside the component $P_{2,0}$, while $t^2 + t/j$ and t^2 lie in the different components $U_{\{1,2\},0}$ and $U_{\{2\},0}$ of τ_{FS} .

(iii) Put $x_j = t^2 + (1 + 1/j)t$ and $y = t^2 - t$; both lie in P , and $x_j \rightarrow x = t^2 + t$ for τ_{FS} . Then $x_j + y = 2t^2 + t/j$ lies in $U_{\{1,2\},0}$ while $x + y = 2t^2$ lies in $U_{\{2\},0}$, so $x_j + y$ does not converge to $x + y$. The pair (x, y) lies on the cancellation locus ([24.5](#)) of [theorem 24.3](#). $\square$

Glazer’s Corollary 1 inside one ring. Section 26 placed the real model as a stratum of 03’s cone $\mathcal{A}_{\text{fin}, \geq 0}$, which satisfies $Q + \text{IOpen}$ with Borel multiplication and on which no Borel-compatible Polish topology makes addition continuous (theorem 25.2). The integer-exponent subcone P sits between the two: it carries a Polish topology with both operations continuous, and Corollary 1 does not apply to it because P fails open induction at the single formula $x^2 < t$ (theorem 38.2). In $\mathcal{A}_{\text{fin}}$ the same formula is satisfied by $x = \frac{1}{2}X^{1/2}$, which is not a standard integer. Source 06’s isolated-zero theorem (theorem 39.1) explains why τ_P isolates 0, whereas the additive models of Part I are perfect.

Cone topologies versus group topologies. The same ring is source 08’s $A_* = \mathbb{Z} + X\mathbb{R}[X]$ (example 54.1) and, under $t \mapsto \omega$, the ring $\mathbb{Z} + \omega\mathbb{R}[\omega]$ of corollary 55.1 and `odg:frac:prop:` intersection of the omnific Diophantine report. By source 08’s theorem 44.1, no Hausdorff locally compact second-countable additive-group topology on A has a Borel order, so τ_P , a locally compact Polish topology on the cone, is induced by no such group topology. Source 06 already notes that the signed analogue of τ_P is not a ring topology. This is the situation of source 08’s question 57.2 and source 09’s question 73.11, which ask when a cone topology extends to the group of differences; the Hilbert’s-tenth programme’s triage of the arrival (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/recent_incoming_substrate_triage_20261003.md) makes the same caveat.

The same failed instance elsewhere. The formula $x^2 < X$ defining the standard cut in $\mathbb{Z} + X\mathbb{Q}[X]$ is used by batch 86’s manuscript 10, written into the report Algebra/Surreal1Numbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization.[write] [Batch-86 reciprocal notes, 3 October 2026.] There it is source 03, Sections 19–22. It is motivated by the same paper of Glazer but is not topological, and its obstruction precedes any topology: the additive group of the difference ring of a nonstandard model of $\text{I}\Delta_0$ has no nonzero homomorphism to $\mathbb{Z}$ (`isg:sc:thm:nochar`), so no such model has a unit-preserving additive map into **Oz** (`isg:sc:cor:omnific`), and by that report’s Presburger criterion the ordered group completion of its additive monoid is not initially realizable (`isg:sc:cor:fragment`). It answers neither of Glazer’s questions. For **Oz** itself the collection records existential failures of induction (`odg:thm:induction`, `odg:rem:ringinduction`); as source 06 says, the failure for P does not transfer to **Oz**.

43 Further research questions and topics

The following program separates an explicitly published question from questions suggested by the constructions. Unless stated otherwise, these are proposed directions for investigation; we do not assert that every formulation is historically new or unresolved in all existing literature.

[06] Source 06 lists twelve numbered projects rather than question environments. Each one that coincides with a question of source 07 is printed after it, with source 06’s number:

43.1 Published reverse-mathematical question

Question 43.1 (Glazer’s Question 1). Can the stronger topological Tennenbaum theorem in [1] be proved over ATR_0 ?

This report does not answer that question. The affirmative Presburger construction shows that a proposed route from a weaker metatheory to a negative Presburger conclusion cannot yield the latter conclusion as stated. It does not determine the metatheoretic strength of the theorem about arithmetic with multiplication.

[write] Source 06's project with the same subject:

6. **Exact strength of the isolated-zero principle.** Formalize the proof of Theorem 39.1 over weak subsystems of second-order arithmetic using codes for Polish spaces and continuous functions. Locate the strength of analytic/coanalytic separation and of the coded measure argument. This is related to, but does not resolve, Glazer's published Question 1 asking whether his full Theorem 2 is provable in ATR_0 [1].

43.2 Classification and topology

Question 43.2 (Which spaces and which additive models?). Characterize the uncountable Polish spaces that support continuous Presburger addition, and characterize which continuum-sized Presburger models admit such a presentation.

The real-coordinate model, Baire-space family, and isolated-standard variant give several positive topological types. Compact spaces are excluded. A useful first refinement is to classify locally compact presentations in terms of their components and the position of the standard numerals. A separate algebraic problem asks how much of the existence theory is captured by split $\mathbb{Z}$ -groups.

[write] Source 06's project with the same subject:

4. **Which Polish spaces support additive arithmetic?** Classify the possible underlying spaces. We have a locally compact perfect example and a nowhere locally compact example. Compact Hausdorff spaces are excluded by [2]. A useful next step is to analyze connected spaces and spaces with prescribed sets of isolated points, while specifying whether a group-completion topology is also required.

Question 43.3 (Non-split integer extensions). Which ordered extensions $0 \rightarrow \mathbb{Z} \rightarrow G \rightarrow D \rightarrow 0$ with divisible D , but without a group-theoretic splitting, have positive cones with Polish topologies making addition continuous?

The present construction starts with a splitting. General $\mathbb{Z}$ -groups need not possess it. The residue and extension invariants appearing in the repository's omnific research provide a natural algebraic language for formulating obstructions, but do not automatically decide the topological question.

[write] Source 06's project with the same subject:

2. **Which additive groups admit Polish presentations?** Find necessary and sufficient conditions for a continuum-sized $\mathbb{Z}$ -group to have a nonnegative cone with a Polish topology and continuous addition. Our split products are a substantial class, but a general $\mathbb{Z}$ -group need not split. In particular, investigate how a nonsplit extension of a divisible ordered group by $\mathbb{Z}$ affects the problem.

Question 43.4 (Connected examples). Does a connected uncountable Polish space support a continuous Presburger presentation?

The examples here have clopen residue classes and hence are disconnected. In an arbitrary presentation, residue classes are Borel by Proposition 34.3; that alone does not make them clopen. The gap between these two regularities is a concrete place to investigate.

43.3 Definability and descriptive invariants

Question 43.5 (Ranks for a fixed model). For one fixed continuous presentation, which boundary oscillation ranks occur among parameter-definable sets? How do they depend on arity and on the chosen parameter?

Theorem 36.1 varies the model and uses a parameter with positive zeroth coordinate. A full parameter spectrum requires analyzing cuts based at vectors with a transfinite first nonzero coordinate, where the boundary of the model itself affects the derivative. It should also distinguish the rank of a relation from the rank of a unary section.

Question 43.6 (Uniform definability bounds). Must every Polish Presburger presentation with continuous addition have all definable relations at some uniformly bounded Borel level? Is the Δ_2^0 bound forced, or special to the ambient-group construction?

Proposition 34.3 proves Borelness in complete generality. Theorem 34.1 proves the stronger level-two conclusion under an additional structural hypothesis. Closing that gap would turn the examples into a general regularity theorem.

[write] For the cones of locally compact signed groups with Borel order, source 08's theorem 52.1 gives the uniform Δ_2^0 bound; for an arbitrary Polish presentation the question stays open.

Question 43.7 (Interpreting the cone closure rank). Which structural features of an ordered Polish group are detected by $\rho(A, <)$? How does the rank behave under closed convex subgroups, quotients, products, and order-preserving topological embeddings?

Proposition 33.4 realizes every countable rank on real coordinate groups. A useful next goal is an intrinsic characterization of the first few ranks and precise inequalities for those operations. The rank records the interaction of order with topology; it need not agree with the algebraic Archimedean rank.

43.4 Arithmetic expansions and formal verification

Question 43.8 (Additional continuous functions). Which natural expansions of Presburger arithmetic still admit uncountable Polish presentations with all named functions continuous?

Fixed-divisor quotient and remainder are positive examples. Total truncated subtraction, minimum, and maximum are impossible individually. Predecessor depends on the presentation. Expansions by selected unary functions, or restricted coding functions, offer an intermediate program before multiplication is considered.

[write] Source 06's project with the same subject:

9. **Effective comparison and definable functions.** For the explicit computable Baire model, calibrate the effective complexity of its order, equality, minimum, and truncated predecessor. The continuous addition algorithm does not decide an infinite stream's equality to zero. A useful target is an exact classification of the discontinuity or computational strength required by each definable operation.

Question 43.9 (Removing the cofinal-multiple obstruction). Construct and analyze continuous Polish Presburger models in which no nonstandard element has cofinal standard multiples. Which of their additive structures admit an ordered-ring expansion at all?

The explicit family in this report cannot answer the latter question positively. Divisible coefficient groups with no greatest Archimedean class are natural candidates for a next stage. Any resulting multiplication must then be tested against the appropriate arithmetic axioms and Glazer’s regularity obstruction.

[write] Source 06’s project with the same subject:

3. **Presburger reducts of genuine PA models.** Can the additive reduct of a nonstandard model of PA have an uncountable Polish continuous-addition presentation if its multiplication is allowed to be non-Borel? Glazer’s published obstruction rules out a Borel multiplication under stronger arithmetic hypotheses. Our lexicographic examples do not answer this question, because their additive structure already excludes a semiring multiplication.

Question 43.10 (Weak foundations and certified models). What is the weakest useful coded metatheory for the real and Baire constructions? Can the generic Presburger semantic theorem and the Polish topology be joined in a short audited Lean development?

The explicit coordinate formulas suggest a substantially lighter proof than the general transfinite cone theorem. A careful analysis must still specify codes for Borel sets, complete metrics, countable bases, and satisfaction of first-order formulas. Section 41 gives an implementation order with a concrete first target.

[write] Source 06’s project with the same subject:

1. **Formal generic Presburger semantics.** Complete the four-layer route in Section 40.4, beginning with generic $\mathbb{Z}$ -group soundness for the repository’s Cooper eliminator. The first deliverable should be a theorem about all models of the division/remainder axioms, followed by the real cone as its first nonstandard instance. This would convert the main answer into a directly auditable formal result.

Question 43.11 (Simplicity and coefficient topologies). Which initially realized omnific additive groups admit coefficient topologies of this kind, and which initiality operations preserve Polishness and continuity of addition?

Theorem 40.1 supplies an explicit ordered embedding without asserting initiality of the displayed image. Combining a proved initial realization theorem with a transported Polish topology is straightforward as an abstract existence argument; preserving a natural coefficient description during that transport is a sharper and more informative goal.

[write] Source 06’s project with the same subject:

12. **Surreal topology without class-level assumptions.** Study the transported topologies on set-sized omnific subgroups and subrings such as (40.3) and (40.4). Ask which algebraic embeddings are continuous for these explicit topologies. Keep the support set, coefficient topology, and ambient set theory specified; the proper class of all surreal numbers is not itself an ordinary Polish space.

43.5 Further projects of source 06 [06]

The remaining projects of source 06 have no counterpart among source 07’s questions.

5. **How many isolated points are necessary for semirings?** Theorem 39.1 forces zero to be isolated, and the polynomial example isolates every standard numeral. Determine whether a Polish topological PA^- model can have zero as its only isolated point, or classify which patterns of isolated numerals are possible. Requiring continuous predecessor is an additional hypothesis and should be treated separately.
7. **Completeness and separability thresholds.** Find sharp replacements for Polishness in the isolated-zero theorem. Completeness supplies limits of all subset sums, and separability supplies descriptive regularity of the order. One can test these roles separately using incomplete metrizable models and complete nonseparable presentations.
8. **Classification beyond cardinality.** Theorem 35.8 gives the exact number of isomorphism types but no full classification. Determine the descriptive complexity of isomorphism for suitably coded Polish Presburger presentations. Convex quotients and residue extensions provide natural invariants. Compare with the countable-model Scott analysis in [25], without assuming that countable and Polish presentations have identical classification behavior.
10. **Coefficient fields and other support families.** Replace $\mathbb{R}$ in the polynomial construction by an ordered Polish topological field whose positive cone is Polish. The same degree-stratum proof applies whenever coefficient operations are continuous. More ambitiously, determine which infinite support families allow a Polish positive-cone topology with both operations continuous, and identify where leading-term cancellation or completeness obstructs such a topology.
11. **A bridge from additive induction to open induction.** The formula $x^2 < t$ isolates one exact failure in the polynomial model. Seek a useful classification of additional multiplication principles that remain compatible with a Polish continuous semiring presentation. The class must stop before the hypotheses of Glazer’s additive obstruction, but the interval between additive induction and IOpen contains many distinct theories.

[merge] Relation to Part I’s questions. Source 07’s question on spaces and models and its connected-carrier question, with source 06’s project 4, are [question 14.3](#) extended to all carriers; source 07’s question on weak foundations and source 06’s project 1 continue [questions 14.1](#) and [14.2](#); the cofinal-multiple question and project 3 continue [question 14.4](#). Source 06’s [theorem 38.1](#) answers the first sentence of source 05’s part of that question affirmatively: its cone is an explicit uncountable Polish continuous-addition Presburger model whose additive structure carries a unital semiring multiplication, continuous for the same topology (the note after [question 14.4](#) records this). It fails open induction ([theorem 38.2](#)), so the clause of that question about $Q + \text{IOpen}$, and source 04’s growth-order question, stay open. Project 6 and source 07’s comment on Glazer’s Question 1 bear on the speculation discussed in [section 1.1](#).

Part IV

Local compactness forces arithmetic rigidity

[write] This part is source 08 of batch 86, *Local Compactness Forces Arithmetic Rigidity: Borel ordered groups, Presburger models, and multiplication without regularity assumptions* (dated October 3, 2026; “prepared with ChatGPT”), pinned to repository commit 883e0b3b2. All unmarked text is source 08’s; its labels carry the prefix `pma:lc:.` Source 08 cites two earlier

manuscripts as “user-library” files: [40] is source 05, the base of [Part I](#), and [41] is source 03, [Part II](#). Its finite checks are shipped as `code/08-local-compactness-verify.py` with the record `data/08-local-compactness-verification_results.json` ([section K](#)). Its central theorem is complementary to source 09’s [theorem 62.1](#) in [Part V](#): both make an uncountable discretely ordered ring impossible under a topology, 08 assuming local compactness and no regularity of multiplication, 09 assuming Borel scalar maps and no local compactness. Neither result bears directly on Glazer’s Question 2, which concerns models without multiplication ([section 1.1](#)).

44 The intersection and the contribution

The point of contact is the topology of arithmetic presentations. In *A Topological Tennenbaum Theorem*, Elliot Glazer rules out continuous addition with Borel multiplication for uncountable Polish models of $\mathbb{Q} + \text{IOpen}$; his Question 2 asks for an uncountable Polish Presburger model with continuous addition [[1](#), Corollary 1 and Question 2]. ProveIt’s relevant interfaces are its Cooper-elimination development and its ordered surreal and omnific constructions [[38](#), [39](#)]. These suggest separating three issues: addition, the regularity of order, and the possibility of installing multiplication.

The earlier companion manuscript *Polish Presburger Arithmetic inside the Omnific Integers* already constructs the model

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (44.1)$$

with continuous addition, and gives example-specific multiplication obstructions [[40](#)]. Repeating that construction would not be an advance. The question pursued here is structural:

What does local compactness force on every Borel ordered additive-group presentation, and can multiplication evade the obstruction by being completely nonmeasurable?

Our answer is negative for discretely ordered rings. The obstruction occurs before induction, definability of multiplication, or a choice of particular normal forms enters the argument.

[write] [Equation \(44.1\)](#) is Part I’s real model ([theorem 3.1](#)); source 07’s [theorem 32.1](#) and source 06’s [theorem 32.2](#) prove it again.

44.1 The central theorem

Theorem 44.1 (Arithmetic rigidity). *Let A be a nonzero commutative unital ring with a total order such that addition preserves order, products of positive elements are positive, and 1 is the least positive element. Suppose that A has a Hausdorff locally compact second-countable topology making $(A, +, -, 0)$ a topological group, and that $A_{>0}$ is Borel. Then the topology is discrete and A is countable. No regularity assumption is imposed on the multiplication map.*

The proof is given in [section 48](#). Its two essential parts are an algebraic growth argument and a classical topological structure argument. The former produces more cosets than the latter permits. In particular, it does not attempt to pass a discontinuous multiplication through a limit.

44.2 Other results proved here

The structural conclusions have independent content even when no ring multiplication exists. For signed Presburger models, we obtain an exact classification of the *pointed topological additive groups*, the local-compactness threshold for their positive cones, and a uniform Δ_2^0 bound on all first-order definable relations. We do not claim that every order itself is the standard product lexicographic order.

There is also a second, distinct obstruction. On these same additive groups, an arbitrary integral-domain multiplication, not required to respect the specified order, has only countably many scalars whose multiplication maps are Borel. For the additive type $\mathbb{R}^d \times \mathbb{Z}$, these scalars are exactly the ordinary integers. This theorem is not vacuous: an explicit algebraic transport construction supplies such ring structures.

44.3 What is classical, and what remains uncertified

Measurable orders on locally compact abelian groups were studied by Hewitt and Koshi and by Asmar and Montgomery-Smith [26, 22]. The compact-group obstruction, finite-dimensional separation arguments, and the structure theorem for locally compact abelian groups are classical inputs, not discoveries claimed here. We supply the needed proofs except for the principal structure theorem itself. Presburger quantifier elimination is likewise classical [28]; section 49 records the version needed for nonstandard groups.

The contribution proposed for independent review is the amplification and arithmetic-rigidity package, together with its classification, scalar, and omnific consequences. A bounded literature search did not establish priority for these formulations. In particular, lack of a matching search result is not evidence of a first proof. The existing companion constructions are cited as prior work, not relabelled as new.

45 Conventions, topology, and foundations

All groups in this paper are abelian and written additively. An ordered group has a translation-invariant *total* order. The associated strict positive cone is $P = G_{>0}$, and $G_{\geq 0} = P \cup \{0\}$. For a topological group, P is Borel if and only if the binary order relation is Borel: one direction takes a section at zero, and the other uses the continuous difference map $(x, y) \mapsto y - x$.

The phrase *discretely ordered* refers to the *algebraic order*: there is a least positive element, denoted 1. It does not say that the presentation topology is discrete. In a ring, $\underline{n} = n1$ denotes the ordinary integer numeral. Throughout, n, m, N used as moduli, finite indices, or bounds are external ordinary integers, not arbitrary elements of a nonstandard model. We write $|x| = \max(x, -x)$.

A Hausdorff locally compact second-countable topological group is Polish. Here, a Polish space is a separable completely metrizable space. The notation Δ_2^0 means both F_σ and G_δ in the specified topology (usually Polish); Δ_1^0 means clopen. No order topology is silently substituted for the given topology.

Definition 45.1 (Order unit). If V is an ordered additive subgroup, an *order unit* in V is an element $u > 0$ in V such that for every $v \in V$ there exists an ordinary integer $N \geq 1$ with $|v| \leq Nu$.

This is an additive notion. It neither says that u is a ring unit nor that V is closed under multiplication. The distinction is indispensable in the main proof.

All theorems are statements about sets in ZFC. When surreals are mentioned, we work with explicitly set-sized subgroups or subrings of **No** or **Oz**. We never equip the proper class of all surreals with a Polish topology, form a set of all its subsets, or apply Haar measure to a proper class. The algebraic splitting arguments use the ordinary ZFC fact that divisible abelian groups are injective. No assertion about the minimum choice or reverse-mathematical strength of the complete proof is made.

46 An algebraic amplification theorem

The key obstruction can be proved without topology. The strongest useful hypothesis is weaker than divisibility.

Lemma 46.1 (Finite elements are ordinary integers). *In a discretely ordered ring A , every element of $[0, \underline{n}]$, for an ordinary $n \in \mathbb{N}$, is one of $\underline{0}, \dots, \underline{n}$. Consequently, for an additive subgroup $V \subseteq A$ the following are equivalent:*

1. $V \cap \mathbb{Z}1 = \{0\}$;
2. for every $v \in V \setminus \{0\}$ and every ordinary $n \geq 0$, $|v| > \underline{n}$.

Every divisible additive subgroup of A satisfies these conditions.

Proof. The interval assertion follows by external induction on n . If $0 < x \leq \underline{n+1}$, then $x \geq 1$ and $0 \leq x-1 \leq \underline{n}$; the induction hypothesis applies. This proves the equivalence because $|v| \in V$.

For the last assertion, take $v > 0$ in a divisible subgroup and an ordinary $n \geq 0$. Write $v = (n+1)w$ with $w \in V$. Ordered-group torsion-freeness and positivity imply $w > 0$, hence $w \geq 1$. Therefore $v \geq \underline{n+1} > \underline{n}$. Apply the same argument to $-v$ for negative elements. $\square$

Theorem 46.2 (Coset and polynomial amplification). *Let A be a discretely ordered ring, and let $V \subseteq A$ be a nonzero additive subgroup with $V \cap \mathbb{Z}1 = \{0\}$. Suppose $u \in V$ is an order unit. Then:*

1. $uV \cap V = \{0\}$, and the additive map

$$V \longrightarrow A/V, \quad v \longmapsto uv + V \tag{46.1}$$

is injective. In particular, $|A/V| \geq |V|$.

2. The finite-sum map

$$\Psi : \bigoplus_{j \geq 0} V \longrightarrow A, \quad (v_j)_{j \geq 0} \longmapsto \sum_{j \geq 0} u^j v_j \tag{46.2}$$

is an injective additive homomorphism. If the source is ordered by the sign of its highest-index nonzero coefficient, then Ψ is an order embedding.

3. The induced additive map $\bigoplus_{j \geq 1} V \rightarrow A/V$ is injective.

Proof. Suppose $uv \in V$ for $v \neq 0$. Since $u > 0$, the order-unit condition provides an ordinary N with

$$u|v| = |uv| \leq Nu.$$

Cancellation of the positive factor u gives $|v| \leq \underline{N}$, contrary to [lemma 46.1](#). Thus $uv \in V$ implies $v = 0$, which proves the first assertion and injectivity of (46.1).

For the second assertion, take a nonzero finite tuple and let k be its largest nonzero index. If $k = 0$, the result is immediate. For each $j < k$ choose an ordinary $N_j \geq 1$ with $|v_j| \leq N_j u$. Put $N = \sum_{j < k} N_j$. By [lemma 46.1](#), $u > 1$ and $|v_k| > \underline{N}$. The lower-degree terms satisfy

$$\left| \sum_{j < k} u^j v_j \right| \leq \sum_{j < k} u^j |v_j| \leq \sum_{j < k} N_j u^{j+1} \leq N u^k < u^k |v_k|. \quad (46.3)$$

Therefore the sum in (46.2) is nonzero and has the sign of v_k . Additivity is distributivity. Applying the sign statement to the difference of two tuples proves the claimed order embedding.

Finally, if a finite sum $\sum_{j \geq 1} u^j v_j$ belongs to V , subtract that element as the zeroth coefficient. Injectivity of Ψ makes all coefficients zero, proving the third assertion. $\square$

Remark 46.3 (What the polynomial language does not mean). The domain of Ψ is an ordered *additive* direct sum. In general V is not a coefficient ring, and the theorem does not define a ring $V[t]$ or an evaluation homomorphism from such a ring. The quotient A/V is also only an additive quotient: V need not be an ideal or convex.

Corollary 46.4 (Divisible-subgroup obstruction). *If a discretely ordered ring contains an uncountable divisible additive subgroup V with an order unit, then A/V is uncountable. More precisely, A/V contains an additive copy of a countable direct sum of copies of V .*

The strength of this result is its insensitivity to the regularity of multiplication. The map (46.1) may be arbitrarily nonmeasurable; its existence as an injection is already enough for a contradiction.

47 Borel orders on locally compact additive groups

This section assembles the classical topological ingredients in the form needed later. The measurable-order literature [26, 22] contains broader structure results. Borelness is useful here because it survives restriction to every closed subgroup.

47.1 A compact subgroup cannot carry the restricted order

Lemma 47.1 (Compact-order obstruction). *A second-countable topological abelian group with a Borel invariant total order has no nontrivial compact subgroup.*

Proof. Let K be a compact subgroup. Ordered groups are torsion-free, so a nontrivial K cannot be finite. Let μ be normalized Haar measure on an infinite K , and set $P_K = K \cap G_{>0}$. Singletons have measure zero: otherwise arbitrarily many disjoint translates of one singleton would have total measure greater than one. Since inversion preserves Haar measure, the partition

$$K = P_K \dot{\cup} (-P_K) \dot{\cup} \{0\}$$

gives $p := \mu(P_K) = 1/2$.

For every $t \in K$, the two upper rays P_K and $P_K + t$ are nested. Their measures are equal by translation invariance. Thus

$$\mu(P_K \cap (P_K + t)) = p \quad (t \in K).$$

On the other hand, the relevant indicator on $K \times K$ is Borel; second countability permits the usual product-measure Fubini theorem. Consequently

$$\begin{aligned} \int_K \mu(P_K \cap (P_K + t)) d\mu(t) &= \int_K \int_K 1_{P_K}(x) 1_{P_K}(x - t) d\mu(x) d\mu(t) \\ &= \int_K 1_{P_K}(x) p d\mu(x) = p^2. \end{aligned}$$

Hence $p = p^2$, contradicting $p = 1/2$. $\square$

47.2 The connected component is an open real vector group

Theorem 47.2 (Topological additive decomposition). *Let G be a Hausdorff locally compact second-countable abelian group with a Borel invariant total order. There are a finite $d \geq 0$ and a countable discrete torsion-free abelian group D such that*

$$G \cong_{\text{top.grp}} \mathbb{R}^d \times D. \quad (47.1)$$

The identity component $C = G^\circ$ is open and is the $\mathbb{R}^d$ factor.

Proof. The principal structure theorem for locally compact abelian groups supplies an open subgroup $H \cong \mathbb{R}^d \times K$, where K is compact; see [27, p. 4] and the classical references recorded there. The image of K is a compact subgroup of G , so lemma 47.1 forces $K = 0$. Thus $H \cong \mathbb{R}^d$ is open and connected. It is also closed, as every open subgroup is closed. Therefore $H = G^\circ = C$.

As an abstract abelian group, C is divisible and hence injective. There is a homomorphism $r : G \rightarrow C$ extending the identity of C . The restriction of r to the open neighborhood C of zero is continuous, so r is continuous at zero and hence everywhere. Let $D = \ker r$. Each coset of C meets D in exactly one point; in particular D is discrete. The maps $(c, e) \mapsto c + e$ and $g \mapsto (r(g), g - r(g))$ are continuous inverses. Second countability makes the family of disjoint open cosets countable. Finally D is torsion-free because it is a subgroup of the ordered group G . $\square$

For completeness, the injectivity fact just used can be obtained by the usual maximal-extension argument. Given a homomorphism from a subgroup B of an abelian group into a divisible group, adjoin one new element x . If a least positive n has $nx \in B$, choose an n th root of its image; otherwise assign any image to x . Each choice extends the homomorphism to $B + \mathbb{Z}x$. Zorn's lemma then gives an extension to the whole group. Its use here is purely algebraic, with continuity recovered from the openness of C .

47.3 A Borel order on the real component is a linear flag

Lemma 47.3 (Finite-dimensional Borel-order lemma). *Every Borel invariant total order on $(\mathbb{R}^d, +)$ is, after an invertible real-linear change of coordinates, the standard lexicographic order. For $d > 0$ it has a positive order unit.*

Proof. Proceed by induction on d . The assertion is trivial for $d = 0$. Let P be the strict positive cone when $d > 0$. Ordered divisibility implies that P is stable under positive rational scaling and rational convex combinations. The sets $P, -P, \{0\}$ cover $\mathbb{R}^d$, so the Baire category theorem implies that P is not meager. Since P is Borel, it has the Baire property and is comeager in some nonempty open set U .

If $z \in U + U$, then $U \cap (z - U)$ is nonempty and open. Both P and $z - P$ are comeager in this intersection, so it contains $x \in P$ with $z - x \in P$. Hence $z \in P + P \subseteq P$. This proves that P has nonempty interior.

Let $F = \overline{P}$. Rational convexity and passage to limits show that F is a closed convex real cone. It is proper: the nonempty open set $-\text{int}(P)$ is disjoint from P , and therefore from its closure. Finite-dimensional separation yields a nonzero real-linear functional ℓ with $\ell(F) \subseteq [0, \infty)$. Totality then gives

$$\{x : \ell(x) > 0\} \subseteq P \subseteq \{x : \ell(x) \geq 0\}. \quad (47.2)$$

Indeed, an element with positive ℓ cannot be zero or negative. The restriction of the order to $\ker \ell$ is again Borel. Apply the induction hypothesis there, and extend its coordinate functionals to $\mathbb{R}^d$, with ℓ as first coordinate. Equation (47.2) and the induction hypothesis give the required lexicographic rule.

In those coordinates, $u = (1, 0, \dots, 0)$ is an order unit: for any $v = (v_1, \dots, v_d)$ choose an ordinary integer $N > |v_1|$. Then $-Nu < v < Nu$. $\square$

Remark 47.4. We did not assume that the given order respects multiplication of vectors by arbitrary positive real numbers. That compatibility is a conclusion of Borelness and finite dimensionality. Applying an ordered-vector-space classification without first establishing this point would leave a gap.

48 Proof and scope of arithmetic rigidity

Proof of theorem 44.1. Apply theorem 47.2 to the additive group of A . Its identity component C is an open real vector group $\mathbb{R}^d$, and A/C is countable. If $d > 0$, then C is uncountable and divisible. By lemma 47.3, the restricted order has an order unit $u \in C$. The amplification theorem therefore gives an injection

$$C \hookrightarrow A/C, \quad c \mapsto uc + C.$$

This contradicts countability of A/C . Hence $d = 0$. The zero subgroup is then open, so the topology is discrete. A second-countable discrete space is countable. $\square$

[write] Source 09 reaches the same conclusion, discreteness and countability, without local compactness but with Borel scalar maps and a cone with the Baire property (theorem 62.1); the two hypotheses are incomparable. Source 06's theorem 39.1 is a third result of the kind, on the nonnegative cone: if addition and multiplication are both continuous for a Polish topology on the cone, then 0 is isolated.

Corollary 48.1 (No compatible ordered multiplication). *Let G be an uncountable discretely ordered abelian group with a Borel order and a locally compact Polish group topology. There is no commutative unital ring multiplication on its underlying set, with unit the named least positive element, which respects the specified order. This remains true when multiplication is allowed to be non-Borel.*

Corollary 48.2 (Positive-cone semirings). *Let G satisfy the hypotheses of corollary 48.1, and let $M = G_{\geq 0}$. There is no commutative unital semiring multiplication on M with the given addition and unit, such that the product of two nonzero elements is nonzero.*

Proof. The cancellative additive monoid M has Grothendieck group G . A commutative semiring multiplication extends by distributivity to this group: $(a-b)(c-d) = ac+bd-ad-bc$. The assumed absence of zero divisors on positive elements makes products of positive elements positive in G . It would therefore yield the ring forbidden by [corollary 48.1](#). $\square$

The group-topology hypothesis is important. A topology on a cancellative positive monoid need not automatically give its Grothendieck group the same desirable separation and local-compactness properties. Thus our result is not a theorem about every possible Polish presentation of a positive arithmetic model. Similarly, the theorem does not replace the general Polish-space result of [\[1\]](#); it is a different refinement with a stronger topological assumption and much weaker multiplicative assumptions.

The proof uses no standard-residue axiom, no square roots, no induction scheme in the ring, and no compatibility between multiplication and limits. The only induction in [lemma 46.1](#) is ordinary external induction on a finite integer. These distinctions make the theorem suitable for applications to weak arithmetic and integer parts.

49 Signed Presburger arithmetic and a uniform elimination lemma

Definition 49.1 ($\mathbb{Z}$ -group). A $\mathbb{Z}$ -group is an ordered abelian group with least positive element 1 such that for each ordinary $m \geq 1$ and each $x \in G$ there are unique $q \in G$ and $r \in \{0, \dots, m-1\}$ satisfying

$$x = mq + r1. \quad (49.1)$$

Equivalently, $\mathbb{Z}1$ is a convex subgroup and $G/\mathbb{Z}1$ is divisible. For the implication from quotient divisibility, write $x = mq + k1$, perform ordinary Euclidean division on k , and absorb its quotient into q . Uniqueness follows because $|r-s| < m$ cannot be a nonzero multiple of m in a discretely ordered group. The converse follows immediately from [\(49.1\)](#). Convexity of $\mathbb{Z}1$ follows from the finite-interval argument of [lemma 46.1](#), which only needs the group order.

We use the language

$$\mathcal{L}_{\text{Pres}} = \{+, -, 0, 1, <, \equiv_m : m = 2, 3, \dots\}, \quad x \equiv_m y \iff x - y \in mG.$$

The equivalence between $\mathbb{Z}$ -groups and models of signed Presburger arithmetic, and quantifier elimination in this language, are standard [\[28\]](#). We give the finite-witness argument to identify precisely what transfers from ordinary integers to nonstandard groups.

Lemma 49.2 (Coset-invariant finite witnesses). *Fix an ordinary $m \geq 1$. Let $P(y)$ be a predicate invariant under adding every element of mG . Let a_i, b_j be finite lists of elements of a $\mathbb{Z}$ -group G . Then*

$$\exists y \left(\bigwedge_i a_i < y \wedge \bigwedge_j y < b_j \wedge P(y) \right)$$

is decided by a finite test as follows. If lower bounds occur, let $a = \max_i a_i$ and test $y = a + s1$ for $1 \leq s \leq m$. If only upper bounds occur, let $b = \min_j b_j$ and test $y = b - s1$ for $1 \leq s \leq m$. If no bounds occur, test $y = r1$ for $0 \leq r < m$.

Proof. Only completeness of the candidate list needs proof. Suppose first that $y > a$ is a witness. Euclidean division of $y - a$ gives $y - a = mq + r1$, $0 \leq r < m$. If $r > 0$, then $q \geq 0$;

otherwise $q \leq -1$ would imply $y - a \leq -1$. Set $s = r$ in this case. If $r = 0$, positivity implies $q \geq 1$, and we set $s = m$. In either case

$$a < a + s1 \leq y, \quad y - (a + s1) \in mG.$$

Thus $P(a + s1)$ holds and every upper bound remains satisfied. With only upper bounds, apply the reflected argument to $b - y > 0$. With no bounds, use the remainder of y modulo mG . The reverse implications are immediate because every accepted candidate is itself a witness. $\square$

Theorem 49.3 (Classical Presburger elimination, in the needed form). *The axioms of $\mathbb{Z}$ -groups admit quantifier elimination in $\mathcal{L}_{\text{Pres}}$ and form the complete theory of $(\mathbb{Z}, +, -, 0, 1, <, \equiv_m)_{m \geq 2}$. Their nonnegative cones are models of the full first-order theory $\text{Th}(\mathbb{N}; 0, 1, +, <)$, including all its induction instances.*

Proof. After disjunctive normal form, it suffices to eliminate one existential variable from a conjunction. Negated inequalities and equalities become finite combinations of strict inequalities using discreteness. Negated congruences become finite disjunctions of the other ordinary residue classes. Atoms not involving the eliminated variable are carried along.

Normalize the nonzero integer coefficients of the eliminated variable x by taking an ordinary common multiple $L > 0$ of their absolute values and replacing Lx by y , with the additional condition $y \in LG$. For example, with $a \neq 0$ and $k = L/|a|$, an inequality $ax < t$ becomes $\text{sgn}(a)y < kt$, and

$$ax + t \in nG \iff \text{sgn}(a)y + kt \in nkG.$$

The equivalence uses torsion-freeness: $kz \in nkG$ if and only if $z \in nG$. Equalities with coefficient 1 or -1 permit substitution. What remains is a finite collection of upper and lower bounds on y and a Boolean combination $P(y)$ of congruences. This predicate is invariant under mG , where m is a common multiple of the finitely many moduli, including L .

Apply [lemma 49.2](#). Choosing a maximum or minimum among a finite list can itself be expressed by a finite disjunction of comparisons. Thus the output is quantifier-free. Repeating eliminates all quantifiers. A quantifier-free sentence concerns only ordinary integer numerals, their order, and their standard residues, so it has the same truth value in every $\mathbb{Z}$ -group as in $\mathbb{Z}$. This proves completeness.

Finally, the nonnegative cone is definable, and any formula in its language translates to a group formula by restricting its quantifiers to $x \geq 0$. The cone therefore satisfies exactly the first-order sentences true in $\mathbb{N}$. In particular the translated induction instances hold; they are consequences, not additional assumptions of the construction. $\square$

[write] [Lemma 49.2](#) is Part I's coset-invariant finite search, [lemma 4.7](#) (source 05) and source 04's [theorem 4.8](#), and [theorem 49.3](#) is the classical [theorem 4.9](#); source 08 proves them again in its own notation.

Remark 49.4 (The transfer condition is coset invariance). For nonstandard G , invariance under the single ordinary shift $m1$ does not by itself imply invariance under mG . The latter is what the proof needs and what congruence formulas supply. This distinction already appears in the earlier companion manuscript [40]. ProveIt's integer-specific periodicity lemmas must not be generalized merely by replacing the carrier $\mathbb{Z}$ with an arbitrary ordered group.

50 An exact pointed topological-group classification

Theorem 50.1 (Locally compact signed Presburger groups). *Let $(G, +, <, 0, 1)$ be a $\mathbb{Z}$ -group equipped with a locally compact Polish group topology and a Borel order. There exist a finite $d \geq 0$ and a countable $\mathbb{Z}$ -group $(E, +, <_E, 0, 1_E)$ such that*

$$(G, +, 1) \cong_{\text{top.grp},1} (\mathbb{R}^d \times E, +, (0, 1_E)), \quad (50.1)$$

where E is discrete. On the identity component the given order is real-linearly conjugate to the lexicographic order.

Conversely, every such pair (d, E) admits a Borel $\mathbb{Z}$ -group order on the displayed pointed topological group. The theorem classifies the pointed topological additive groups, not all possible full orders on them.

Proof. By [theorem 47.2](#), write $G = C \oplus D$ topologically, with $C \cong \mathbb{R}^d$ and D countable discrete. A divisible subgroup cannot contain a nonzero ordinary integer by [lemma 46.1](#). Hence $C \cap \mathbb{Z}1 = 0$.

Write the named element as $1 = (v, e)$ in this splitting. The element $e \in D$ has infinite order: otherwise a nonzero ordinary multiple of 1 would belong to C . Define a homomorphism on $\mathbb{Z}e$ by $ne \mapsto nv$ and extend it to $f : D \rightarrow C$ using divisibility of C . Since D is discrete, f is continuous. The shear

$$(c, x) \mapsto (c - f(x), x)$$

is a topological group automorphism taking 1 to $(0, e)$. Transport the order along it and let $E = \{0\} \times D$ with the restricted order and named unit $1_E = e$.

The element 1_E is least positive in E , because it is least positive in the whole group. If $(0, x) = m(c, y) + r(0, e)$ is standard division in G , then $mc = 0$ forces $c = 0$. Thus the quotient lies in E , proving its standard-residue property. The component order is handled by [lemma 47.3](#).

Conversely, put the usual lexicographic order on $\mathbb{R}^d$, and compare pairs first in this vector coordinate and then in E . The element $(0, 1_E)$ is least positive; vector divisibility together with division in E gives [\(49.1\)](#). The order is Borel because the vector comparison is a finite union of coordinate conditions and E is countable discrete. This constructs every asserted topological type. $\square$

Remark 50.2 (Why the order caveat matters). The component G° need not be convex in the full ordered group. For example, take the ordered additive group with coordinates $(r, q, s, n) \in \mathbb{R} \times \mathbb{Q} \times \mathbb{R} \times \mathbb{Z}$, compared in that order, and give $\mathbb{R}^2 \times (\mathbb{Q} \times \mathbb{Z})$ the product topology with the second factor discrete. This is a $\mathbb{Z}$ -group: the first three coordinates are divisible and n supplies the remainder. Its connected component has $q = n = 0$, but a positive element in the q coordinate lies between positive elements at the two real scales. Thus countable discrete scales can be interleaved with real scales.

There is a precise invariant distinguishing this example from a block lexicographic product with all real coordinates first. Define

$$B_C = \{g \in G : |g| < c \text{ for every } c \in C_{>0}\}.$$

In a block product $\mathbb{R}^d \times E$ with the vector coordinate compared first, $B_C = \{0\} \times E$, so $G = C + B_C$. In the displayed interleaved example, comparison with arbitrarily small positive vectors in each real coordinate shows that B_C consists exactly of the integer coordinate. Hence $C + B_C$ misses every element with $q \neq 0$. Both C and B_C are preserved by an ordered

topological-group isomorphism, so the example has no such block-order presentation. This does not affect the pointed additive-group decomposition proved above.

51 Positive cones: exact regularity and the dimension threshold

A useful feature of finite-dimensional lexicographic order is that even its arbitrary cuts have low descriptive complexity.

Lemma 51.1 (Cuts in finite lexicographic powers). *Every upper set in $\mathbb{R}^d$ with the lexicographic order is Δ_2^0 in the usual Euclidean topology. No definability hypothesis on the upper set is needed.*

Proof. Induct on d . For $d = 0$ there are only the empty and full sets. For $d > 0$, let U be a nonempty proper upper set. Consider the first coordinates of its elements. They form a nonempty upper subset of $\mathbb{R}$ with a finite lower endpoint a . The endpoint is finite because an excluded first coordinate gives a lower bound and an included one gives an upper bound for the infimum. If all first coordinates occurred, the upper-set condition would make U the whole space, contrary to assumption. For first coordinate $> a$ every point belongs to U , and for first coordinate $< a$ none does. Thus

$$U = \{x : x_1 > a\} \cup (\{a\} \times U'), \quad (51.1)$$

where U' is an upper set in lexicographic $\mathbb{R}^{d-1}$, possibly empty or full. By induction U' is both F_σ and G_δ . Equation (51.1) expresses U as an F_σ set. The corresponding formula for its complement does the same, so U is also G_δ . Empty and full upper sets are immediate. $\square$

Theorem 51.2 (Cone regularity and local compactness). *Let G satisfy [theorem 47.2](#), and write $G^\circ \cong \mathbb{R}^d$. Then:*

1. $G_{\geq 0}$ and the order relation are Δ_2^0 ;
2. $M = G_{\geq 0}$ with its subspace topology is Polish, and addition $M \times M \rightarrow M$ is continuous;
3. M is locally compact if and only if $d \leq 1$.

Proof. The cosets of $C = G^\circ$ are a countable family of clopen subsets of G . Translate each to C and use the linear coordinates of [lemma 47.3](#). In each coset the intersection with M is an upper set for the induced lexicographic order. By [lemma 51.1](#), each such intersection and its relative complement are F_σ in the coset, and hence in G . Taking countably many unions shows that both M and $G \setminus M$ are F_σ . The order relation follows by the continuous difference map. This proves the first assertion.

A G_δ subspace of a Polish space is Polish [8]. The addition on M is the restriction of the continuous group operation, with its image in the subspace M , giving the second assertion.

If $d = 0$, M is discrete. If $d = 1$, each nonempty coset slice is a real line or an open or closed half-line, by the proof of [lemma 51.1](#). Each is locally compact. As the slices are open in M , their union is locally compact.

Suppose now $d \geq 2$. The slice at the identity component is the nonnegative lexicographic cone in $\mathbb{R}^d$. If M had a compact neighborhood K of zero, then for some $\epsilon > 0$ its intersection with C would contain all points of this cone in the Euclidean ball of radius ϵ . In particular, for sufficiently large n ,

$$x_n = (\epsilon/n, -\epsilon/2, 0, \dots, 0) \in K.$$

But in G this sequence converges to

$$x = (0, -\epsilon/2, 0, \dots, 0) < 0.$$

The inclusion $M \hookrightarrow G$ is continuous, so K is compact and therefore closed in the Hausdorff space G . This would put x in $K \subset M$, a contradiction. $\square$

[write] Part (1) is the locally compact case of source 07's [theorem 33.1](#), proved here through [lemma 51.1](#); parts (2) and (3) are new. [Proposition 51.4](#) below is an instance of [theorem 7.4](#) and [proposition 37.1](#).

Corollary 51.3 (The locally compact positive Presburger case). *In the class of positive cones of locally compact Borel signed Presburger-group presentations, an uncountable cone is locally compact exactly when the connected component of its signed group is one-dimensional. Every cone in this class is Polish and has continuous addition.*

This is a classification within the stated presentation class. It does not assume, or prove, that every topological Presburger monoid arises from a locally compact topological group in this manner.

51.1 The order bound is genuinely rank two

Proposition 51.4 (Order is neither open nor closed). *For an uncountable model in [theorem 50.1](#), the strict order on $M = G_{\geq 0}$ is Δ_2^0 but neither open nor closed in M^2 . The same is true of the weak order.*

Proof. Here $d > 0$. Choose $c_n > 0$ in C with $c_n \rightarrow 0$, for example the points $(1/n, 0, \dots, 0)$ in the flag coordinates. The component is divisible, so $c_n > 1$ for every n by [lemma 46.1](#). The pairs $(0, c_n)$ belong to the strict order and converge to $(0, 0)$, which does not; the relation is not closed. The point $(0, 1)$ belongs to the strict order, whereas $(c_n, 1)$ does not and converges to it; the relation is not open. Its complement after exchanging coordinates is the weak order, giving both failures there as well. The upper Δ_2^0 bound follows from [theorem 51.2](#) and restriction to M^2 . $\square$

There is no contradiction between continuous addition and this exact rank-two behavior. Arithmetic comparison is a Borel relation in a topology that is intentionally different from the discrete order topology.

52 Definable relations and continuous standard division

Theorem 52.1 (Uniform definable regularity). *In every model of [theorem 50.1](#), all Presburger-definable subsets of G^k , with parameters and arbitrary finite k , are Δ_2^0 . The same holds for definable subsets of the positive cone M^k in its relative topology. For each ordinary $m \geq 1$, the quotient and remainder maps in [\(49.1\)](#) are continuous, both on G and on M .*

Proof. Use the pointed splitting $G \cong \mathbb{R}^d \times E$. Since the real factor is divisible,

$$mG = \mathbb{R}^d \times mE.$$

This is clopen, because E is discrete. Every congruence condition is therefore the inverse image of a clopen set under a continuous affine map. Every comparison is Δ_2^0 by [theorem 51.2](#).

Equality is closed and hence Δ_2^0 in a metrizable space. Affine integer-linear terms are continuous. Quantifier elimination in [theorem 49.3](#) now expresses each definable set as a finite Boolean combination of Δ_2^0 sets, which is again Δ_2^0 .

For positive-cone formulas, restrict quantifiers to $x \geq 0$, apply group quantifier elimination, and then restrict the resulting set to M^k . This preserves the stated relative Borel class.

The remainder map r_m has the finitely many clopen fibers $r1 + mG$, $0 \leq r < m$, so it is continuous into the discrete finite set of ordinary residues. Multiplication by the ordinary integer m is a homeomorphism from G onto mG : in the product splitting it is $(v, e) \mapsto (mv, me)$, whose inverse is continuous on its image. Consequently

$$q_m(x) = (m \cdot)^{-1}(x - r_m(x)1) \quad (52.1)$$

is continuous. If $x \geq 0$, the quotient $q_m(x)$ is nonnegative; otherwise $q_m(x) \leq -1$ would imply $x \leq -m + (m - 1) = -1$. Restricting these maps to M proves the final claim. $\square$

[write] This is the statement of [theorem 7.1](#) and [theorem 34.1\(iii\)](#) for a class that also admits the interleaved, non-block orders of the remark after [theorem 50.1](#). It answers source 07's question on uniform definability bounds ([section 43](#)) within that class.

52.1 A model in every finite real dimension

Take $E = \mathbb{Z}$ and order $\mathbb{R}^d \times \mathbb{Z}$ lexicographically, with the integer coordinate last. Write

$$G_d = \mathbb{R}^d \overrightarrow{\times} \mathbb{Z}, \quad M_d = (G_d)_{\geq 0}. \quad (52.2)$$

For $x = (v, n)$,

$$r_m(v, n) = n \bmod m, \quad q_m(v, n) = (v/m, \lfloor n/m \rfloor). \quad (52.3)$$

Thus standard division is explicit and continuous. Every M_d is a Polish model of full Presburger arithmetic with continuous addition. For $d = 1$, (52.2) is exactly (44.1); for $d \geq 2$ the cone is not locally compact although the signed group is.

The case $d = 1$ therefore provides the same affirmative construction for the question cited in [section 44](#), already given in [40]. The advance here is the classification and dimensional boundary, not a second priority claim for that example.

[write] M_1 is Part I's $M_{\mathbb{R}}$; for $d \geq 2$ the models M_d are new to the collection.

52.2 Why ordinary scalar operations are not ring multiplication

All maps $x \mapsto nx$ for ordinary $n \in \mathbb{Z}$ are continuous group homomorphisms. This remains compatible with the complete absence of an order-compatible ring multiplication in uncountable models. Presburger arithmetic only provides these externally indexed scalar maps, not a binary multiplication of arbitrary elements.

Likewise, low Borel complexity of every definable relation does not make all definable functions continuous. For instance, truncated subtraction on the positive cone can fail to be continuous. A claim of continuity requires an argument such as (52.1), not merely quantifier elimination.

53 Even unordered multiplication has few Borel scalars

The main theorem forbids multiplication preserving the order, without any measurability assumption. We now remove order compatibility instead and ask how much measurable multiplication can survive. In this section the specified Borel $\mathbb{Z}$ -group order and the multiplication may be unrelated; their named additive unit 1 is the same.

53.1 Automatic continuity and component rigidity

Lemma 53.1 (Classical automatic continuity). *A Borel homomorphism between Polish groups is continuous.*

Proof. We give the abelian case needed here. Let $h : G \rightarrow H$ be Borel and additive. For a neighborhood W of zero in H , choose an open neighborhood V with $V - V \subseteq W$. Separability gives countably many translates of V covering H . Their Borel preimages cover the Baire space G , so one preimage B is nonmeager.

A nonmeager set with the Baire property has $B - B$ as a neighborhood of zero. Indeed, it is comeager in some nonempty open set U ; for small t , the set $U \cap (t + U)$ is nonempty open, and the two translated comeager sets meet, putting t in $B - B$. It follows that $h^{-1}(W)$ contains a neighborhood of zero, because $h(B - B) \subseteq V - V$. Thus h is continuous at zero, and hence everywhere. This is the usual Baire-category automatic-continuity argument; see also [8] for the Polish-group setting. $\square$

[write] [Lemma 53.1](#) is the first half of source 09's [lemma 61.2](#).

Lemma 53.2 (Injective maps cannot move a component into itself from outside). *Let $G \cong \mathbb{R}^d \times E$ with E discrete, and let $C = G^\circ$. Every continuous injective additive endomorphism $h : G \rightarrow G$ satisfies*

$$h(C) = C, \quad h^{-1}(C) = C. \quad (53.1)$$

Proof. Continuity and connectedness give $h(C) \subseteq C$. The restriction to C is a continuous additive map between finite-dimensional real vector groups, hence real-linear: first use rational scalars, then continuity and rational approximation for real scalars. Injectivity makes this linear endomorphism bijective. If $h(g) \in C$, choose $c \in C$ with $h(c) = h(g)$. Injectivity gives $g = c$, proving the preimage statement. $\square$

53.2 The Borel-scalar subring

Definition 53.3. For a ring multiplication on a topological additive group, define

$$\mathcal{B}(A) = \{a \in A : L_a : A \rightarrow A, x \mapsto ax, \text{ is Borel}\}.$$

We call this the *Borel-scalar subring* once its subring property has been proved.

Theorem 53.4 (Countably many Borel scalar multipliers). *Let $(G, +, <, 0, 1)$ be a locally compact Borel Polish $\mathbb{Z}$ -group, and let A be any commutative unital integral-domain multiplication on its underlying additive group, with this same 1. Multiplication need not respect the order. Then $\mathcal{B}(A)$ is a countable closed discrete subring, and the component quotient map is injective on it. In particular:*

1. no nonzero element of $C = G^\circ$ has a Borel scalar map;

2. if the pointed topological group is $\mathbb{R}^d \times \mathbb{Z}$, then $\mathcal{B}(A) = \mathbb{Z}1$;
3. if G is uncountable, multiplication cannot be separately Borel in every scalar, and therefore cannot be jointly Borel.

Proof. We have $C \cap \mathbb{Z}1 = 0$ and $1 \notin C$ by the additive discreteness and divisibility argument used in [theorem 50.1](#). If $a \in C \setminus \{0\}$ and L_a were Borel, then it would be continuous by [lemma 53.1](#). It is injective because A is a domain. But $L_a(1) = a \in C$, contradicting [lemma 53.2](#). This proves $\mathcal{B}(A) \cap C = \{0\}$.

The maps L_0, L_1 are continuous. If $a, b \in \mathcal{B}(A)$, then $L_{a-b} = L_a - L_b$ is Borel, and $L_{ab} = L_a \circ L_b$ is Borel. Thus $\mathcal{B}(A)$ is a subring, and it contains $\mathbb{Z}1$. If two of its elements have the same coset modulo C , their difference belongs to $\mathcal{B}(A) \cap C$ and is zero. Its projection into the countable discrete quotient G/C is therefore injective.

Each open component contains at most one element of $\mathcal{B}(A)$. Within that component a singleton is closed, and its complement is open. Taking the union of these complements over all components proves that $\mathcal{B}(A)$ is closed in G ; the same component neighborhoods make it discrete. It is countable because the quotient is countable.

If the pointed quotient is $\mathbb{Z}$ generated by the image of 1, every $a \in \mathcal{B}(A)$ has the same image as $n1$ for some ordinary n . Injectivity of the projection gives $a = n1$. Finally, an uncountable G has $d > 0$ and nonzero elements in C , whose scalar maps are not Borel. Sections of a jointly Borel multiplication would all be Borel, so joint Borelness is impossible as well. $\square$

Remark 53.5 (Two independent obstructions). [Theorem 44.1](#) uses order-compatible multiplication, but no regularity of its scalar maps. [Theorem 53.4](#) does not require order compatibility, but bounds which scalar maps can be Borel. Neither statement is obtained merely by rephrasing the other.

54 Examples isolating the hypotheses

54.1 The countable conclusion is optimal

Every countable discretely ordered ring has a discrete Polish additive group topology with Borel order and continuous multiplication. Ordinary $\mathbb{Z}$ is the smallest example, but there are non-Archimedean ones, such as $\mathbb{Z}[X]$ ordered by the sign of the highest nonzero coefficient. Thus the conclusion is discreteness and countability, not that the ordered ring must be isomorphic to $\mathbb{Z}$.

54.2 Second countability cannot simply be deleted

Example 54.1 (An uncountable locally compact ordered ring presentation). Let

$$A_* = \mathbb{Z} + X\mathbb{R}[X] \tag{54.1}$$

with the leading-coefficient order, so X is positive and greater than all ordinary integers. This is a discretely ordered domain. Give its additive group the product decomposition

$$A_* = (\mathbb{R}X) \oplus D_*, \quad D_* := \mathbb{Z} \oplus X^2\mathbb{R}[X], \tag{54.2}$$

where $\mathbb{R}X$ has its usual real-line topology and D_* is discrete. Then the additive group is Hausdorff and locally compact, the order is Borel, and the ring is uncountable. The topology is not second-countable or separable.

Proof. The leading-coefficient rule defines an ordered domain, and every positive nonconstant polynomial exceeds all ordinary integers. Positive constants are positive integers, so 1 is least positive. The product in (54.2) is locally compact. Its discrete factor has cardinality $\mathfrak{c}$, making the family of open components uncountable.

For Borelness, a component with nonzero degree-at-least-two part has a constant sign, independent of the X coefficient and integer constant. The union of all such positive components is clopen. On components with no degree-at-least-two part, positivity is the ordinary condition $r > 0$, with the boundary $r = 0$ governed by the integer constant. These conditions and their complements are countable unions of closed sets. Thus the positive cone is in fact Δ_2^0 in this metrizable, nonseparable topology. $\square$

Here $V = \mathbb{R}X$ has the order unit X , and

$$|A_*/V| = |V| = \mathfrak{c}.$$

So the cardinal lower bound of [theorem 46.2](#) is attained. Multiplication by X takes the connected line $\mathbb{R}X$ injectively into the discrete $X^2\mathbb{R}$ coordinate. It is not continuous. It is not even Borel: the inverse images of open sets obtained from arbitrary subsets of that discrete coordinate give arbitrary subsets of the real line, including non-Borel ones. No topological-ring claim is made for this example.

54.3 Borelness of the order is also essential

Example 54.2 (A locally compact Polish additive presentation with wild order). Fix $d \geq 1$. As rational vector spaces, both $\mathbb{R}^d$ and $X\mathbb{R}[X]$ have dimension $\mathfrak{c}$. Choose a $\mathbb{Q}$ -linear isomorphism $\theta : \mathbb{R}^d \rightarrow X\mathbb{R}[X]$, and define

$$\Theta : \mathbb{R}^d \times \mathbb{Z} \longrightarrow A_*, \quad \Theta(v, n) = n + \theta(v).$$

Transport the ring structure and the leading-coefficient order of A_* along Θ , while retaining the usual product topology on $\mathbb{R}^d \times \mathbb{Z}$. This gives an uncountable discretely ordered domain with a locally compact Polish additive-group topology. Its order is necessarily non-Borel by [theorem 44.1](#).

The dimension assertion uses ordinary ZFC basis theory: an infinite dimensional vector space over a countable field has cardinality equal to its basis cardinality, and a countable direct sum of copies of $\mathbb{R}$ still has rational dimension $\mathfrak{c}$. This construction is not effective and is not proposed as a usable notation system.

Independently give the same carrier the *different*, standard Borel lexicographic $\mathbb{Z}$ -group order. The transported multiplication is not compatible with that order, but [theorem 53.4](#) applies to it and gives the exact identity

$$\mathcal{B}(A) = \mathbb{Z}1.$$

Thus the Borel-scalar theorem has concrete algebraic realizations rather than merely a conditional conclusion about nonexistent ring structures.

54.4 Other boundaries

Discreteness of the arithmetic order is indispensable: the ordinary real field is uncountable, locally compact, and has continuous operations and Borel order, but has no least positive

element. Totality also matters: our compact-order and finite-dimensional arguments do not apply to a partial order without additional hypotheses.

We do not present a separable non-locally-compact counterexample with all the other hypotheses of [theorem 44.1](#). The general Polish-group problem is left as a research question, rather than being called a settled sharpness result.

55 Consequences for surreal and omnific presentations

55.1 A set-sized no-go theorem for actual omnific rings

ProveIt's surreal-number guide describes its construction of the ordered surreal field, its omnific integer ring, the least-positive-element property, and the constant-term map [\[39\]](#). Only elementary finite normal forms and the discrete order are needed below; none of the repository's stronger unrefereed research claims is an input.

Corollary 55.1 (Omnific additive topologies). *Let $A \subseteq \mathbf{Oz}$ be an uncountable set-sized unital subring, with its inherited ordered-ring structure. There is no Hausdorff locally compact second-countable additive-group topology on A for which the inherited order is Borel. This holds without requiring multiplication to have any topological or measurable regularity.*

Proof. Since $1 \in A$ and 1 is least positive in $\mathbf{Oz}$, it is least positive in A . Apply [theorem 44.1](#). $\square$

[merge] Source 06 equips the nonnegative cone of the same ring $\mathbb{Z} + t\mathbb{R}[t]$ with a locally compact Polish topology in which addition and multiplication are continuous ([theorem 38.1](#)). There is no conflict with [corollary 55.1](#): that topology lives on the cone and is induced by no locally compact second-countable group topology on the ring ([section 42](#)).

This is not merely the observation that a Polish space has cardinality at most continuum. The concrete ring

$$\mathbb{Z} + \omega\mathbb{R}[\omega] = \left\{ n + \sum_{j=1}^k r_j \omega^j : n \in \mathbb{Z}, r_j \in \mathbb{R}, k < \omega \right\} \quad (55.1)$$

has cardinality exactly $\mathfrak{c}$, so its cardinality alone permits a Polish coding. Finite surreal normal forms show that it is an ordered copy of A_* in [\(54.1\)](#). The theorem obstructs the combination of Borel order, locally compact additive-group topology, and the actual ordered multiplication even at this set-sized level.

55.2 Finite additive fragments have excellent topology but no ring expansion

Corollary 55.2 (Finite surreal scales). *For every ordinary $d \geq 1$, let*

$$H_d = \{n + r_1\omega + \cdots + r_d\omega^d : n \in \mathbb{Z}, r_j \in \mathbb{R}\} \subseteq \mathbf{Oz}. \quad (55.2)$$

Its inherited ordered additive group, with coefficient topology $\mathbb{R}^d \times \mathbb{Z}$, is a locally compact Borel $\mathbb{Z}$ -group. Its nonnegative cone is Polish with continuous addition and continuous standard quotient and remainder. That cone is locally compact exactly for $d = 1$. All its Presburger-definable relations are Δ_2^0 .

No alternative commutative unital ring multiplication on H_d can respect this addition, order, and named unit. In particular, failure of closure under the actual surreal multiplication cannot be repaired by inventing a different order-compatible multiplication on the same carrier.

Proof. In descending coefficient order $(r_d, \dots, r_1, n)$, addition and order identify H_d with G_d of (52.2). Standard division is (52.3). Apply theorems 51.2 and 52.1 and corollary 48.1. Alternatively, the last assertion follows already from algebraic amplification with $V = \sum_{j=1}^d \mathbb{R}\omega^j$ and order unit $u = \omega^d$: $H_d/V \cong \mathbb{Z}$ is countable but V is not. $\square$

55.3 Implications for exact symbolic systems

These results distinguish three legitimate design choices. One may retain finite surreal additive scales and continuous addition, accepting that there is no compatible full multiplication. One may enlarge to a genuine set-sized ordered ring, abandoning local compactness or separability of an additive-group topology. Or one may transport the algebra to a convenient Polish additive space and accept a non-Borel inherited order. The examples in section 54 make the last two alternatives mathematically explicit.

The earlier support-barrier manuscript concerns a different obstruction: which exact Hahn or omnific normal-form families admit Borel coefficient observations [41]. Our obstruction requires neither unrestricted supports nor an analysis of support validity. It already appears for the finite polynomial ring (55.1).

56 A formalization plan with explicit trust boundaries

The proof should be formalized in layers. Its algebraic core is much smaller than its locally compact group input, so formalization need not begin with Haar measure or the full theory of real Lie groups.

Layer A: finite-order arithmetic. Formalize that $[0, n1]$ contains exactly the ordinary numerals in a discretely ordered group. Deduce that a subgroup disjoint from $\mathbb{Z}1$ has no nonzero element bounded by a standard integer, and that a divisible subgroup has this property. The induction index is a natural number in the metatheory of the formal development.

Layer B: amplification. For a subgroup V , an element $u \in V$, and the explicit order-unit predicate, prove $uv \in V \Rightarrow v = 0$. Package (46.1) as an injective additive map into the quotient additive group. Then formalize (46.3) for finitely supported coefficient families and derive the ordered direct-sum embedding. No topological assumption, infinite series, or proper class occurs in this layer.

Layer C: the topological interface. State a reusable interface consisting of an open connected subgroup C , a topological additive-group isomorphism $C \cong \mathbb{R}^d$, and a countable quotient. Prove arithmetic rigidity from this interface and the Borel finite-dimensional flag lemma. Separately discharge the interface using the compact-order obstruction and the principal LCA structure theorem. A conditional theorem with this interface is useful, but must not be advertised as an unconditional formalization of the whole locally compact result.

Layer D: Presburger transfer. Prove lemma 49.2 over an abstract $\mathbb{Z}$ -group with a predicate invariant on cosets of mG . Prove the group theory of standard division before connecting to syntax. ProveIt already supplies syntax, coefficient normalization, and an executable

integer decision procedure, according to its inspected guide [38]; the semantic transfer must be proved rather than inferred from that guide.

Layer E: topology of presentations. Formalize lexicographic cuts by finite dimension, prove their F_σ and G_δ descriptions, and derive the cone dimension threshold and continuity of the standard quotient and remainder maps. The scalar theorem then reuses an automatic-continuity theorem and finite-dimensional linear algebra.

No Lean files are included as if they had been checked. The companion Python script validates finite exact-arithmetic examples, not proofs in Lean or in any other kernel. In particular it does not test Haar measure, nonmeager sets, arbitrary Borel orders, cardinal injections, or the principal LCA structure theorem.

57 Further research questions and concrete next targets

Question 57.1 (Remove local compactness). Can an uncountable discretely ordered ring have a Polish additive-group topology with Borel order when multiplication has no regularity at all?

The proof here uses local compactness to obtain a finite-dimensional open component with an order unit. A non-locally-compact Polish group can have an infinite-dimensional component or be totally disconnected without being discrete. An extension would need a substitute for that finite-dimensional argument, perhaps an uncountable subgroup to which [theorem 46.2](#) applies and a suitably small quotient. Neither a proof nor a counterexample is claimed here.

[write] Source 09's [theorem 62.1](#) answers this question negatively when every scalar map $x \mapsto ax$ is Borel and the positive cone has the Baire property; without any regularity of multiplication it stays open.

Question 57.2 (Topological group completion of positive models). Which locally compact Polish cancellative Presburger monoids admit a Hausdorff locally compact Polish topology on their Grothendieck groups that induces the original topology on the positive cone?

A positive answer under usable hypotheses would convert [corollary 51.3](#) into a broader classification of positive arithmetic presentations. The relevant issue is topological embedding of the monoid into its group completion, not the elementary algebraic existence of that completion. Conditions on translations, quotient maps, and neighborhoods of zero are natural starting points.

[write] Source 09 asks the same question as [question 73.11](#). Source 06's cone of $\mathbb{Z} + t\mathbb{R}[t]$ is a data point ([section 42](#)): a locally compact Polish cone with continuous addition, whose group of differences has no locally compact second-countable group topology with Borel order, by [theorem 44.1](#).

Question 57.3 (Classify the orders, not only the pointed groups). For a fixed countable $\mathbb{Z}$ -group E , classify Borel $\mathbb{Z}$ -group orders on $\mathbb{R}^d \times E$ up to continuous additive order isomorphism fixing 1.

The real component carries a finite flag, but discrete coordinates can occur between real scales, as the example following [theorem 50.1](#) shows. The cut description in [lemma 51.1](#) and the continuous separating homomorphisms of the classical measurable-order literature

suggest finite flag data together with countable extension data. Determining which data are complete invariants is a concrete next classification problem.

Question 57.4 (Weaken Borel order carefully). Which hereditary measurability assumptions on an invariant order suffice for arithmetic rigidity? In particular, what replaces Borelness if restrictions to every compact subgroup and every closed real subspace are required to be Haar measurable or to have the Baire property?

An ambient Haar-measurable set can have uncontrolled behavior on a null closed subgroup. The present proof intentionally uses Borelness to avoid that problem. A weakening should state restriction hypotheses explicitly rather than treating ambient measurability as automatically hereditary.

Question 57.5 (Cardinal and topological size spectra). For nonseparable locally compact additive presentations of discretely ordered rings, what cardinalities of the component quotient can occur relative to the identity component and the weight of the topology?

Example 54.1 attains $|A/C| = |C| = \mathfrak{c}$ with a real component. The direct-sum strengthening in theorem 46.2 records more than one injection. It may impose additional ordered-group or module constraints on possible quotients even when it yields no stronger bare infinite-cardinal inequality.

Question 57.6 (Realize Borel-scalar subrings). For fixed $\mathbb{R}^d \times E$, which countable subrings can arise as $\mathcal{B}(A)$ for integral-domain multiplications on the additive group?

The projection of $\mathcal{B}(A)$ into E is injective, and it contains the ordinary integers. When $E = \mathbb{Z}$ the answer is exactly $\mathbb{Z}$. For larger E , continuous scalar maps act real-linearly on the component and induce additive maps on E ; compatibility of these two actions and their extension data may give a useful algebraic realization criterion.

Question 57.7 (Effective and reverse-mathematical strength). What is the exact effective content of extracting a finite real flag from a code for a Borel order, and of proving the resulting rigidity statement over weak base theories?

The current ZFC proof uses category, separation, Haar measure, and an LCA structure theorem. It is not an analysis of their strength in second-order arithmetic. Glazer's separate Question 1 about the strength of his theorem [1] remains outside the results proved here; an argument for the locally compact fragment should not be substituted for an answer to that question.

[write] Section 1.1 discusses Glazer's Question 1 and the speculation that follows his Question 2.

Question 57.8 (Certified amplification and notation design). Can the algebraic amplification theorem be kernel-checked first, and then used as an automatic obstruction to proposed finite-scale omnific or non-Archimedean ring interfaces?

A small certificate could consist of a divisible or integer-disjoint additive subgroup, an order unit, and an upper bound on the quotient's size. This would separate a universally reusable algebraic theorem from application-specific normal-form calculations. The finite-scale examples in corollary 55.2 are a natural first target, followed by the Borel-scalar theorem and the full topological interface.

58 Conclusions of Part IV

Locally compact arithmetic separates cleanly into an additive possibility and a multiplicative obstruction. Uncountable Polish Presburger models with continuous addition are available, and their locally compact signed group presentations have a precise finite-dimensional structure. Their order relations have exact rank-two regularity, and their standard quotient and remainder operations remain continuous.

But a discretely ordered ring cannot use that topology in the uncountable second-countable case. The obstruction does not ask multiplication to be Borel: it uses multiplication only to create an impossible injection from a real component into a countable component quotient. When order compatibility is dropped instead, only countably many scalar maps can remain Borel. This gives a concrete structural connection between Glazer’s arithmetic-presentation questions and ProveIt’s finite surreal, omnific, and Presburger developments, while keeping the unresolved Polish-group and formal-verification questions explicit.

Part V

Polish topologies at the Hahn–Puisseux–Levi-Civita boundary

[write] This part is source 09 of batch 86, *Polishability at the Hahn–Puisseux–Levi-Civita Boundary: Automatic order bounds, local Polish hulls, and countable omnific integer parts* (dated October 3, 2026; “prepared with AI assistance”), pinned to repository commit 4af956fe0. All unmarked text is source 09’s; its labels carry the prefix `pma:ser:.` It continues [Part II](#): source 03 asked when Hahn-series codes form a *standard Borel* space with observable coefficients, and source 09 asks when the additive group of a series field can be made *Polish* with measurable arithmetic. Its “preceding support manuscript” [44] is source 03 (Part II), and its Presburger companion [43] is source 05 (Part I). Its finite checks are shipped as `code/09-hahn-polishability-verify.py` with the record `data/09-hahn-polishability-verification.json`.

Convention [write]. In this part t is a positive infinitesimal, exponents of t that are negative give infinite elements, and the surreal realization is $t^q \mapsto \omega^{-q}$ ([equation \(71.1\)](#)). This is the reverse of [Part III](#), where t is positive infinite, and agrees with source 03’s increasing- t convention; 03’s growth variable is $X = t^{-1}$. Source 09’s macros for its fields are renamed only in the source: the Puisseux field $\mathcal{P}_k$ is printed exactly as before and is not 03’s cone $\mathcal{A}_{\text{fin}, \geq 0}$ ([section 1.3](#)).

Reading routes. Sections [61–63](#) contain the main obstruction and classification. Sections [64–67](#) explain the Puisseux/Levi-Civita contrast. Sections [68–71](#) connect the analysis to real closed fields, arithmetic, and ProveIt. The final sections separate proved conclusions from further research.

59 The research intersection and the questions answered

Elliot Glazer’s *A Topological Tennenbaum Theorem* [1] studies Borel arithmetic on uncountable Polish spaces. Its Theorem 2 excludes continuity of either arithmetic operation for Borel

models of $Q + I\Delta_0 + \text{Exp}$, and its Corollary 1 excludes continuous addition with Borel multiplication for uncountable Polish models of $Q + \text{IOpen}$. This provides a specific, documented intersection with ProveIt's work on ordered algebra, exact Hahn normal forms, omnific integers, and formal arithmetic. No inference about Glazer's present employment or endorsement is needed.

The ProveIt source collection distinguishes formalized constructions from unrefereed research reports. Its surreal documentation describes the normal-form realization of the surreal field and the integer-constant, nonnegative-growth normal forms of the omnific integers [42]. We use those as motivation and as a possible implementation interface, not as substitutes for proofs of the topological assertions below.

Two earlier manuscripts prepared for this research program already address Polish Presburger positive cones and descriptive support barriers [43, 44]. In particular, the preceding support manuscript discusses the coanalytic obstruction for unrestricted rational Hahn supports. Repeating that obstruction would not answer the next question: *even when exact series codes form a standard Borel space, can their additive group be made Polish without losing measurable arithmetic?*

The answer is sensitive to which series are admitted. Fix a countable ordered field k , and use t as a positive infinitesimal. Three natural fields are

$$\mathcal{P}_k = \bigcup_{n \geq 1} k((t^{1/n!})), \quad (59.1)$$

$$\mathcal{L}_k = \left\{ \sum_{q \in \mathbb{Q}} a_q t^q : \text{supp}(a) \cap (-\infty, B] \text{ is finite for every } B \in \mathbb{Q} \right\}, \quad (59.2)$$

$$\mathcal{H}_k = k((t^{\mathbb{Q}})). \quad (59.3)$$

They are, respectively, the Puiseux field, the left-finite Levi-Civita field, and the full rational Hahn field. Their inclusions are strict. The middle field, not the largest or the descriptively simplest one, admits the desired Polish arithmetic.

59.1 The central conclusions

The term *admissible* is made precise in Definition 60.1. It requires a Polish additive group, a Baire-measurable positive cone, and Borel multiplication by each fixed scalar. It does not initially require jointly continuous multiplication.

Main results. For a nonzero ordered abelian group Γ , the full Hahn field $k((t^\Gamma))$ has an admissible Polish topology if and only if k is countable and $\Gamma \cong \mathbb{Z}$ as an ordered group. For countable ordered k , $\mathcal{P}_k$ has no admissible Polish topology, while $\mathcal{L}_k$ has exactly one: its complete valuation topology. Every discretely ordered ring with an admissible Polish additive group is countable. Every integer part of an admissible Polish ordered field is countable and closed discrete.

There is also a sharp local converse. Every countable family of rational Hahn series lies in a complete separable subfield, and these Polish subfields have no maximal member. Exactly $2^{\aleph_0}$ of them are needed to cover the full Hahn field (Theorem 69.6). Over $\mathbb{R}_{\text{alg}}$ they can be chosen real closed and truncation-closed, with canonical countable omnific integer parts. Thus the Levi-Civita field is not the largest possible positive construction: the global

obstruction concerns simultaneous freedom of prefixes, not any one prescribed countable support pattern.

The main advance over a support-code obstruction is quantification over *all* admissible Polish topologies on the abstract field. The Hahn classification is not defeated by abandoning coefficientwise coding. The Puiseux obstruction is stronger still in a different direction: the code domain is already F_σ , but no compatible Polish group topology can support the required arithmetic.

For $k = \mathbb{R}_{\text{alg}}$, the real algebraic numbers, both $\mathcal{P}_k$ and $\mathcal{L}_k$ are real closed and share the same explicitly described countable integer part. Its nonnegative cone satisfies open induction. The integer-part algebra therefore does not distinguish these two sharply different topological situations.

59.2 What is and is not claimed as new

The article gives complete arguments for its central classifications and regularity conclusions. The category tools are classical. The construction and completeness of Levi-Civita fields belong to established non-Archimedean analysis; see, for example, Shamseddine–Berz [31]. Generalized series and Newton–Puiseux theory are classical, with a useful account in Poonen [30]. The coanalytic support obstruction and the negative-infinitesimal floor correction are already present in the preceding project materials.

The contribution developed here is their combination into a presentation-invariant classification, the Puiseux versus Levi-Civita Polishability separation, the local Polish-hull and exact field-cover theorems, and the resulting integer-part and floor-regularity statements. These are proposed research contributions relative to the inspected materials. They are not accompanied by a certified global priority claim. We answer the precisely formulated questions in this article; we do not claim to have newly resolved either of Glazer’s printed open questions.

60 Conventions: four different notions of presentation

All fields, groups, supports, and topological spaces in the main arguments are sets in ZFC. Every ordered field has its specified field-compatible linear order. A *discretely ordered ring* is a commutative ordered unital ring whose least positive element is 1. An *integer part* of an ordered field F is a discretely ordered unital subring $I \subseteq F$ such that each $x \in F$ has an $a \in I$ satisfying $a \leq x < a + 1$.

A Polish space is separable and completely metrizable. A set has the *Baire property* when it differs from an open set by a meager set. Every Borel set has this property. A Polish additive group has jointly continuous addition and continuous negation.

Definition 60.1 (Admissible Polish arithmetic). A topology τ on an ordered field or discretely ordered ring F is *admissible* when:

- (i) $(F, +, \tau)$ is a Polish topological group;
- (ii) $F_{>0}$ has the Baire property in τ ;
- (iii) for each $a \in F$, the map $L_a : F \rightarrow F$, $L_a(x) = ax$, is Borel.

An admissible Polish presentation through another carrier means a transported topology satisfying these conditions. It is therefore equivalent to an admissible topology on the

original algebraic carrier.

The following distinctions prevent several false conclusions. A standard Borel presentation supplies measurable codes and operations, but need not supply continuous addition. A *Polishable Borel group* is one whose given Borel structure comes from a Polish group topology. An admissible ordered-field topology may initially be allowed to change the Borel structure; our absolute nonexistence results exclude that possibility too. Finally, a Polish topology on a *positive cone* with continuous addition is only a topological monoid. It is not a Polish topology on the group of differences.

In particular, the ring theorem below cannot be applied by silently adjoining negative elements to a Polish model of Presburger arithmetic. Extending the algebra is easy; extending the topology with all the required properties is a separate problem.

For a Hahn series $f = \sum a_\gamma t^\gamma$, the support is well ordered in Γ , and

$$v(f) = \min \text{supp}(f) \quad (f \neq 0), \quad v(0) = +\infty.$$

The order is determined by the coefficient at the least support exponent. Positive exponents represent small quantities. In particular, if $\alpha < \beta$ and $c \in k_{>0}$, then $ct^\alpha > t^\beta$. We distinguish this formal order from the ordinary absolute value of a coefficient when k happens to be a subfield of $\mathbb{R}$.

61 Category forces local order bounds

The mechanism is a short application of classical category theory. We include the needed proofs to make the hypothesis boundary visible. Pettis's theorem, automatic continuity, and the open mapping theorem are classical; Chen [29] reviews their relationship and develops localic counterparts.

Lemma 61.1 (Pettis, additive form). *Let G be a Polish group and let $A \subseteq G$ have the Baire property and be nonmeager. Then $A - A$ contains a neighborhood of 0.*

Proof. There is a nonempty open set U on which A is comeager. Choose a nonempty open V and a neighborhood W of 0 so small that $V \subseteq U$ and $V - g \subseteq U$ for every $g \in W$. For each such g , both $A \cap V$ and $(g + A) \cap V$ are comeager in V . Their intersection is nonempty because a nonempty open subspace of a Polish space is Baire. An element x of the intersection satisfies $x \in A$ and $x - g \in A$, so $g \in A - A$. $\square$

Lemma 61.2 (Automatic continuity and open mapping). *A Borel homomorphism between Polish groups is continuous. A continuous surjective homomorphism between Polish groups is open.*

Proof. For the first assertion, let $\phi : G \rightarrow H$ be Borel and let U be a neighborhood of 0 in H . Choose an open neighborhood V with $V - V \subseteq U$. Separability gives a countable cover of H by translates of V . Their inverse images are Borel and cover G , so one inverse image A is nonmeager. Then $A - A \subseteq \phi^{-1}(U)$, and Lemma 61.1 gives continuity at 0.

For the second assertion, let U be a neighborhood of 0 in G , and choose an open neighborhood V with $V - V \subseteq U$. Countably many translates of V cover G , hence their images cover H . The set $\phi(V)$ is analytic, and therefore has the Baire property. At least one of its translates is nonmeager. Pettis gives a neighborhood of 0 inside $\phi(V) - \phi(V) \subseteq \phi(U)$. $\square$

The descriptive-set-theoretic facts about analytic sets used here are standard [8]. In particular, continuous images of Polish spaces are analytic, analytic sets have the Baire property, and an analytic set with analytic complement is Borel.

Lemma 61.3 (A Baire order is locally bounded). *Let G be a nontrivial linearly ordered abelian group with a Polish group topology. If $P = \{x : x > 0\}$ has the Baire property, some neighborhood of 0 is contained in a bounded order interval $[-a, a]$, with $a > 0$.*

Proof. For a discrete group, take the neighborhood $\{0\}$ and any $a > 0$. Suppose the topology is not discrete. Then $\{0\}$ is nowhere dense. Inversion is a homeomorphism and takes P to $-P$. If P were meager, then $G = P \cup (-P) \cup \{0\}$ would be meager in itself, contrary to the Baire category theorem. Thus P is nonmeager.

Choose a nonempty open set U on which P is comeager, and choose $p \in P \cap U$. Let V be a symmetric open neighborhood of 0 with $W = p + V \subseteq U$. The reflection $x \mapsto 2p - x$ maps W onto itself. Consequently, both $P \cap W$ and $(2p - P) \cap W$ are comeager in W . The set

$$A = W \cap P \cap (2p - P)$$

has the Baire property and is nonmeager. Every $x \in A$ satisfies $0 < x < 2p$. Therefore $A - A \subseteq (-2p, 2p)$. Pettis gives a neighborhood of 0 inside this interval. Take $a = 2p$. $\square$

Theorem 61.4 (Automatic order-topology refinement). *Every admissible Polish topology on an ordered field refines its order topology. More generally, the same conclusion holds for a Polish additive group topology with Baire positive cone and continuous fixed scalar maps.*

Proof. In the admissible case every L_a is continuous by Lemma 61.2. Choose a neighborhood U of 0 with $U \subseteq [-a, a]$ as in Lemma 61.3. For any $\varepsilon > 0$, put

$$c = \frac{a+1}{\varepsilon} > 0.$$

Then $L_c^{-1}(U)$ is a neighborhood of 0, and its elements satisfy

$$|x| \leq \frac{a}{c} = \varepsilon \frac{a}{a+1} < \varepsilon.$$

Thus every interval $(-\varepsilon, \varepsilon)$ contains a neighborhood of 0. Translating this statement at every point of any order interval proves that the interval is open in the given topology. $\square$

Corollary 61.5 (Uniqueness when the order topology is Polish). *If the order topology of an ordered field is a Polish group topology, then it is the only possible admissible Polish topology on that field.*

Proof. The identity from an admissible topology to the order topology is a continuous bijective group homomorphism between Polish groups. It is open by Lemma 61.2. $\square$

Corollary 61.6 (Order-free hypothesis for real closed fields). *Let F be real closed. If its additive group is Polish and its multiplication is jointly Borel, then its unique field order is Borel. Consequently, Theorem 61.4 applies without separately assuming measurable order.*

Proof. The positive cone is the image of $F \setminus \{0\}$ under the Borel map $x \mapsto x^2$, so is analytic. Its complement is the union of $\{0\}$ with the image of $F \setminus \{0\}$ under $x \mapsto -x^2$, hence is analytic too. The analytic/coanalytic criterion gives Borelness. The scalar maps are sections of multiplication and therefore Borel. $\square$

Remark 61.7. The last corollary uses *joint* Borel multiplication to obtain a Borel square map. Merely postulating Borelness of each fixed scalar map does not by itself justify substituting the variable into both multiplication arguments.

62 Discretely ordered rings and integer parts

Theorem 62.1 (Automatic discreteness). *An admissible Polish topology on a discretely ordered ring is discrete. In particular, the ring is countable.*

Proof. By Lemma 61.3, choose a neighborhood U of 0 contained in $[-a, a]$, where $a > 0$. Multiplication by $b = a + 1$ is a continuous additive homomorphism. Thus $W = L_b^{-1}(U)$ is a neighborhood of 0. If $x \neq 0$, discrete order gives $|x| \geq 1$, whence $|bx| \geq b > a$. Such an x cannot belong to W . Therefore $W = \{0\}$. A discrete separable space is countable. $\square$

[write] Theorem 62.1 and source 08's theorem 44.1 have the same conclusion under incomparable hypotheses: 09 needs Borel scalar maps and a cone with the Baire property but no local compactness, 08 needs local compactness but no regularity of multiplication. Together they answer source 08's question 57.1 under the Borel-scalar hypothesis only. The non-example below is the group of Part I's real model.

Corollary 62.2 (A non-Polishable Borel group criterion). *Let R be an uncountable standard Borel discretely ordered ring whose order, addition, negation, and multiplication are Borel. Its additive Borel group is not Polishable.*

Proof. A compatible Polish group topology would preserve the stipulated Borel order and scalar maps and would contradict Theorem 62.1. $\square$

This result needs no induction scheme. That does not make it a theorem with weaker hypotheses than Glazer's arithmetic result: here a Polish *group of differences* is assumed from the outset. Glazer's theorem concerns the arithmetic domain itself. These are different boundaries.

Theorem 62.3 (All integer parts are countable). *Let F have an admissible Polish topology. Every discretely ordered unital subring $I \subseteq F$ is countable and closed discrete. In particular this holds for every integer part, without any measurability assumption on I .*

Proof. By Theorem 61.4, the interval $(-1, 1)$ is open in F . Its intersection with I is $\{0\}$, so every translate of a sufficiently small order interval meets I in at most one point. For example, two distinct points of I cannot lie in an interval of radius $1/3$, because their nonzero difference has absolute value at least 1.

Each point of I therefore has an open neighborhood containing no other point of I . A countable base supplies distinct base elements for distinct points, proving countability. To see closedness, suppose $x \in \bar{I}$. The interval $(x - 1/3, x + 1/3)$ contains at most one point a of I , and must contain one by closure. If $x \neq a$, a smaller neighborhood of x avoiding a contradicts $x \in \bar{I}$. Hence $x = a \in I$. $\square$

Example 62.4 (An important non-example). The lexicographically ordered group $\mathbb{R} \times \mathbb{Z}$, with its ordinary product group topology, has a Borel discrete order: $(0, 1)$ is its least positive element. It is uncountable and Polish. This does not contradict Theorem 62.1, since no ordered-ring structure with the required scalar maps has been supplied. The scalar $a + 1$ in the proof is a genuine ring-theoretic resource.

63 A complete classification for full Hahn fields

The existence and algebraic operations of full Hahn fields are classical [30]. We use the entire well-ordered-support field, not a smaller convergence-restricted subfield.

Lemma 63.1 (A bounded increasing sequence in a noncyclic group). *Let $\Gamma \neq 0$ be an ordered abelian group. Either Γ is order-isomorphic to $\mathbb{Z}$, or there are*

$$0 < \gamma_0 < \gamma_1 < \cdots < \delta$$

with $\delta \in \Gamma_{>0}$.

Proof. Suppose first that Γ has no least positive element. Choose $0 < \varepsilon_0 < \delta$ and then recursively $0 < \varepsilon_{n+1} < \varepsilon_n$. Put $\gamma_n = \delta - \varepsilon_n$.

Suppose instead that ε is the least positive element. If every positive element is bounded by a positive integer multiple of ε , then for every $x > 0$ choose the least n with $x \leq n\varepsilon$. The difference $x - (n-1)\varepsilon$ is positive and at most ε , hence equals ε . Thus $x = n\varepsilon$ and $\Gamma = \mathbb{Z}\varepsilon$. Otherwise some δ exceeds every $n\varepsilon$, and $\gamma_n = (n+1)\varepsilon$ has the required properties. $\square$

Lemma 63.2 (Disjoint intervals in a Hahn field). *Let $H = k((t^\Gamma))$ with $\Gamma \neq 0$. If k is uncountable, its order topology has an uncountable family of pairwise disjoint nonempty open intervals. If Γ is not order-isomorphic to $\mathbb{Z}$, it has such a family of cardinality $2^{\aleph_0}$, even with centers having only 0–1 coefficients.*

Proof. For uncountable k , fix $\delta > 0$. The intervals

$$(c - t^\delta, c + t^\delta), \quad c \in k,$$

are disjoint: every positive nonzero constant dominates $2t^\delta$.

For noncyclic Γ , use Lemma 63.1. For $b \in 2^{\mathbb{N}}$ define

$$f_b = \sum_{n \geq 0} b(n)t^{\gamma_n}.$$

This is a legal Hahn series: each support is a subset of the increasing sequence (γ_n) and hence is well ordered. If $b \neq c$, let m be the first index where they differ. Then $v(f_b - f_c) = \gamma_m < \delta$, so $|f_b - f_c| > 2t^\delta$. The intervals $(f_b - t^\delta, f_b + t^\delta)$ are pairwise disjoint. $\square$

Theorem 63.3 (Full Hahn classification). *For an ordered field k and nonzero ordered abelian group Γ ,*

$$k((t^\Gamma)) \text{ admits an admissible Polish topology} \iff |k| = \aleph_0 \text{ and } \Gamma \cong \mathbb{Z} \text{ as ordered groups.}$$

In the positive case the topology is unique and is the Laurent valuation topology; all field operations, including inversion away from 0, are continuous.

Proof. An admissible topology refines the order topology by Theorem 61.4. A second-countable space has only countably many pairwise disjoint nonempty open sets, since one can choose a distinct basic open set inside each. Lemma 63.2 therefore forces k countable and $\Gamma \cong \mathbb{Z}$.

Conversely identify Γ with $\mathbb{Z}$ and use the metric

$$d(f, g) = \min\{1, 2^{-v(f-g)}\}, \quad 2^{-\infty} = 0.$$

A Cauchy sequence eventually stabilizes all coefficients of exponent at most N , for every integer N . Stabilization for $N = 0$ also gives a common finite negative principal part for its tail. The stabilized coefficients therefore define a Laurent series, and the sequence converges to it. Finite Laurent polynomials form a countable dense set because k is countable.

The valuation identities $v(fg) = v(f) + v(g)$ and $v(f + g) \geq \min\{v(f), v(g)\}$ give continuity of addition and multiplication. For $f \neq 0$, sufficiently small changes do not change $v(f)$, and

$$v(f^{-1} - g^{-1}) = v(f - g) - v(f) - v(g)$$

gives continuity of inversion. The order and valuation topologies agree: a valuation ball sufficiently far above the leading exponent of a positive ε lies inside $(-\varepsilon, \varepsilon)$, while a sufficiently small monomial interval lies inside any prescribed valuation ball. Thus the order is open and the topology is admissible. Uniqueness follows from Corollary 61.5. $\square$

[write] Source 03's [theorem 20.2](#) is the Borel counterpart: $\mathbb{R}((t^\Gamma))$ has a coefficient-observable standard Borel parametrization if and only if Γ is scattered. The two results are consistent: for $\Gamma = \mathbb{Z}^d$, $d \geq 2$, the canonical support domain is Borel (03's [theorem 19.1](#)) but no admissible Polish topology exists (the corollary below). For real coefficients, [lemma 63.2](#) excludes an admissible Polish topology on every field containing $\mathbb{R}$ and a nonzero infinitesimal, among them 03's real Puiseux field $\mathcal{P}$, which nevertheless has Borel codes ([section 23](#)).

Corollary 63.4 (The obstruction is not a coding artifact). *For any countable ordered field k , the full field $k((t^\mathbb{Q}))$ has no admissible Polish topology, regardless of how its elements are encoded. For finite lexicographic powers $\Gamma = \mathbb{Z}^d$, $d \geq 2$, the same conclusion holds even when the canonical support domain is Borel.*

Proof. Both ordered groups are noncyclic. In the second assertion the relevance of Borel support coding is only comparative; the topological conclusion uses Theorem 63.3 alone. $\square$

The excluded case $\Gamma = 0$ is deliberate. It reduces to the coefficient field itself; for example $\mathbb{R}$ with its usual topology is admissible. The nontrivial infinitesimal scale is exactly what separates distinct constants in Lemma 63.2.

For countable k and countable noncyclic Γ , the order density and cellularity of the full Hahn field are both exactly $2^{\aleph_0}$: the lemma gives the lower bound, and $|k^\Gamma| = 2^{\aleph_0}$ gives the upper bound.

64 The Puiseux obstruction: a Borel field that is not Polishable

A second category obstruction does not use an order at all.

Theorem 64.1 (Countable proper-subfield cover). *Let F be an uncountable field whose additive group has a Polish topology. Assume every fixed scalar map is Borel. Then F cannot be the union of countably many Borel proper subfields.*

Proof. Suppose $F = \bigcup_n K_n$, with each K_n a Borel proper subfield. By the Baire category theorem some K_n is nonmeager. As it is an additive subgroup, Pettis shows that $K_n = K_n - K_n$ is open. Choose $a \in F \setminus K_n$. The maps L_a and $L_{a^{-1}}$ are Borel homomorphisms, hence continuous; thus L_a is a homeomorphism and aK_n is open. But

$$K_n \cap aK_n = \{0\}.$$

Indeed, a nonzero $x = ay$ with $x, y \in K_n$ would imply $a = x/y \in K_n$. It follows that $\{0\}$ is open. The Polish group would be discrete and countable, a contradiction. $\square$

The Borel hypothesis on the covering subfields matters. A category argument cannot simply be applied to arbitrary nonmeasurable subfields. In the Puiseux application the subfields will actually be closed under *every* admissible topology, so no measurability assumption is being hidden.

Let k now be a countable ordered field, and set

$$K_n = k((t^{1/n!})), \quad \mathcal{P}_k = \bigcup_{n \geq 1} K_n.$$

The factorial convention makes the chain increasing. Each K_n is a proper subfield of $\mathcal{P}_k$, because rational monomials of denominators not dividing $n!$ occur at later stages.

Lemma 64.2 (Closed stages). *Each K_n is closed in the order topology of $\mathcal{P}_k$, and hence closed under every admissible Polish topology on $\mathcal{P}_k$.*

Proof. If $f \notin K_n$, some nonzero coefficient f_q has $q \notin (1/n!)\mathbb{Z}$. Every g agreeing with f at all exponents at most q still lies outside K_n . The set

$$\{g : v(g - f) > q\}$$

is an order-open neighborhood of f . To check order-openness, a monomial interval of exponent greater than q about any of its points lies inside it. Thus the complement of K_n is open. Theorem 61.4 transfers this closedness to any admissible topology. $\square$

Theorem 64.3 (Absolute Puiseux non-Polishability). *The Puiseux field $\mathcal{P}_k$ over a countable ordered field has no admissible Polish topology. In its canonical standard Borel presentation, even without mentioning the order, no compatible Polish additive-group topology can make all fixed scalar maps Borel.*

Proof. The field is uncountable, since $k((t))$ already contains all binary power series. For an arbitrary admissible topology, Lemma 64.2 makes the proper subfields K_n closed, and Theorem 64.1 applies. For the canonical Borel structure, K_n is defined by the vanishing of all coefficients outside $(1/n!)\mathbb{Z}$, hence is Borel. The same theorem applies directly, without an order hypothesis. $\square$

The ordinary valuation topology on $\mathcal{P}_k$ is separable and metrizable, but is not complete. More strongly, it is meager in itself: each closed stage K_n has empty interior. Given $f \in K_n$ and a valuation ball about f , add a monomial t^q of sufficiently large rational exponent with $q \notin (1/n!)\mathbb{Z}$. The resulting point is in the ball but outside K_n . Thus merely exhibiting an incomplete metric is not the reason for Theorem 64.3; the theorem rules out every admissible replacement.

65 The Levi-Civita completion: a positive and unique presentation

The classical Levi-Civita field is often defined with real coefficients. Here the coefficient field is deliberately *countable*. This changes separability decisively. Our proofs work for every countable ordered k ; the algebraic and completeness arguments also work without an order on k .

Definition 65.1 (Left-finite support). A set $S \subseteq \mathbb{Q}$ is *left-finite* if $S \cap (-\infty, B]$ is finite for every rational B . The field candidate $\mathcal{L}_k$ consists of all functions $f : \mathbb{Q} \rightarrow k$ with left-finite support, written $\sum f_q t^q$.

A nonempty left-finite set has a least element. Indeed, if $b \in S$, the finite nonempty set $S \cap (-\infty, b]$ has a least element, which is least in S . A left-finite support is either finite or an increasing sequence tending to $+\infty$. In particular, it is a legal Hahn support.

Proposition 65.2 (Field operations). *The usual coefficientwise addition and convolution multiplication turn $\mathcal{L}_k$ into a subfield of $k((t^{\mathbb{Q}}))$.*

Proof. A finite union of left-finite supports is left-finite, so addition is defined and preserves the support condition. Let S, T be nonempty left-finite sets with minima s_0, t_0 . If $s + t \leq B$, then $s \leq B - t_0$ and $t \leq B - s_0$. There are only finitely many such pairs. Thus convolution coefficients are finite sums, and the product has left-finite support. The coefficient at the least exponent of a nonzero product is the product of the two leading coefficients, so the product is nonzero.

Write a nonzero series in the form

$$f = at^q(1 + h), \quad a \in k^\times, \quad q \in \mathbb{Q}, \quad h = 0 \text{ or } v(h) = \delta > 0.$$

For $h \neq 0$, define

$$g = \sum_{j \geq 0} (-h)^j.$$

At exponents at most B , only indices j with $j\delta \leq B$ can contribute, apart from the constant term $j = 0$. There are finitely many such indices, and each corresponding power has finite support below B . Therefore g is left-finite and all coefficient manipulations are finite below any prescribed cutoff. The geometric identity gives $(1 + h)g = 1$. Hence $f^{-1} = a^{-1}t^{-q}g \in \mathcal{L}_k$. The case $h = 0$ is immediate. $\square$

Theorem 65.3 (A complete separable field topology). *For countable ordered k , the metric*

$$d(f, g) = \min\{1, 2^{-v(f-g)}\}, \quad 2^{-\infty} = 0, \quad (65.1)$$

makes $\mathcal{L}_k$ an uncountable Polish topological field. Its topology equals the order topology and is the unique admissible Polish topology on $\mathcal{L}_k$.

Proof. The valuation inequality gives the ultrametric inequality for d . Let (f_j) be Cauchy. For each integer $N \geq 0$ there is an index j_N such that all f_j , $j \geq j_N$, have exactly the same coefficients at every exponent at most N . Choose the j_N increasing. These stabilized finite truncations are compatible, and their union defines a coefficient function f . For every bound B choose an integer $N \geq \max\{0, B\}$. The support of f below B is contained in the finite stabilized support below N . Thus $f \in \mathcal{L}_k$. Stabilization also shows $f_j \rightarrow f$.

Finite rational-exponent polynomials over k form a countable set. Truncating any left-finite series above successively larger integers proves density. The binary series $\sum_{n \geq 0} b(n)t^n$ show uncountability.

Addition, multiplication, and inversion are continuous by the valuation identities used in Theorem 63.3. For completeness, continuity of multiplication at (f, g) follows from

$$(f + h)(g + u) - fg = fu + gh + hu$$

and the lower bound on the valuation of a sum. If $v(h) > v(f)$ and $f \neq 0$, then $v(f+h) = v(f)$, and the inverse-difference identity proves continuity of inversion there.

For $\varepsilon > 0$, every series of valuation strictly greater than $v(\varepsilon)$ has absolute ordered value less than ε . Conversely, an interval of radius t^{B+1} lies inside the ball $\{x : v(x) > B\}$. Translation gives equality of the two topologies. The positive cone is open, so the topology is admissible. Uniqueness follows from Corollary 61.5. $\square$

Proposition 65.4 (Completion and strict inclusions). *The valued-field completion of $\mathcal{P}_k$, and also of the rational-function field $k(t^{\mathbb{Q}})$, is $\mathcal{L}_k$. Moreover,*

$$k(t^{\mathbb{Q}}) \subseteq \mathcal{P}_k \subsetneq \mathcal{L}_k \subsetneq \mathcal{H}_k.$$

Proof. Each Puiseux support lies on a discrete rational lattice and is bounded below, so is left-finite. Both $\mathcal{P}_k$ and $k(t^{\mathbb{Q}})$ contain every finite rational-exponent polynomial. These polynomials are dense in the complete field $\mathcal{L}_k$, proving the completion assertion.

For strictness of the first displayed strict inclusion, use

$$x = \sum_{n \geq 2} t^{n+1/n}. \quad (65.2)$$

Its exponents are increasing and unbounded, so $x \in \mathcal{L}_k$. Their reduced denominators are exactly n , since $\gcd(n^2 + 1, n) = 1$. They cannot all divide one fixed factorial, so $x \notin \mathcal{P}_k$. For the other strict inclusion use

$$y = \sum_{n \geq 2} t^{1-1/n}. \quad (65.3)$$

The support is increasing of order type ω , hence well ordered, but is infinite below 1. Thus $y \in \mathcal{H}_k \setminus \mathcal{L}_k$. $\square$

The partial sums of (65.2) are valuation-Cauchy. The partial sums of (65.3) are not: their successive differences have valuations bounded above by 1. A Hahn-summable family need not be a convergent sequence of partial sums in the valuation topology. This is the precise support distinction behind the completion boundary.

Proposition 65.5 (The topology is not locally compact). *No point of $\mathcal{L}_k$ has a compact neighborhood.*

Proof. Every neighborhood of 0 contains a valuation ball $V_B = \{f : v(f) > B\}$. Fix $q > B$ and infinitely many distinct coefficients $c_j \in k$. Then $c_j t^q \in V_B$, and the mutual distances are the same positive number $\min\{1, 2^{-q}\}$. Thus no set containing V_B is totally bounded, and it cannot be compact. Translation deals with other points. $\square$

Remark 65.6 (Why real coefficients do not work). The usual real-coefficient Levi-Civita field is complete in its valuation topology but is not separable in that topology. More strongly, it has no admissible Polish topology at all: the constant-indexed disjoint intervals from Lemma 63.2 lie inside this smaller field as well. The positive construction is therefore not a claim that the conventional real-coefficient field is Polish.

66 A prefix criterion for separability

The Hahn and Levi-Civita results can be organized without referring to a particular coding of supports. Let F be any subfield of $k((t^{\mathbb{Q}}))$ that contains k and all rational monomials. For $B \in \mathbb{Q}$, let

$$\pi_B(f) = (f_q)_{q \leq B}, \quad V_B = \{f \in F : v(f) > B\}.$$

The prefix $\pi_B(f)$ may be infinite. It need not itself represent an element of F .

Theorem 66.1 (Countable prefixes characterize order separability). *The following conditions are equivalent:*

- (i) *the valuation topology of F is separable;*
- (ii) *the order topology of F is second countable;*
- (iii) *for every integer $N \geq 0$, the set $\pi_N(F) = \{\pi_N(f) : f \in F\}$ is countable.*

Every admissible Polish topology on F forces these conditions.

Proof. The order and valuation topologies agree by the monomial comparison in Theorem 65.3. The prefix classes at cutoff N are exactly the open cosets of V_N . A dense countable set meets every such coset, proving (i) $\Rightarrow$ (iii).

Conversely, choose one representative of each prefix class for every integer $N \geq 0$. The union D of these countably many countable sets is dense: every valuation neighborhood contains a coset at an integer cutoff and hence one of the chosen representatives. The cosets themselves give a countable base. This proves (iii) $\Rightarrow$ (i),(ii), and (ii) $\Rightarrow$ (i) is immediate.

Finally, in an admissible topology each V_N is open by Theorem 61.4. Its pairwise disjoint cosets are therefore countable in number. This gives (iii) even when the candidate Polish topology is strictly finer than the valuation topology. $\square$

For $\mathcal{L}_k$, each prefix is a finite rational-exponent polynomial, so there are only countably many. The same holds for $\mathcal{P}_k$. For $\mathcal{H}_k$, a binary family supported below a fixed bound gives continuum many different prefixes. Thus the prefix criterion captures the Hahn obstruction but cannot by itself capture the Puiseux obstruction: separability and complete metrizability remain distinct questions.

A useful sufficient condition follows. If F has countably many prefixes at each cutoff and is complete for the valuation metric, then its order topology is Polish and admissible; it is then the unique admissible topology. Whether all admissible intermediate-field presentations must arise this way is left as a sharply stated further question in Section 73.

67 Exact descriptive complexity of the three code domains

In this section k is a countable ordered field with the discrete topology, and all domains are regarded as subsets of the Polish product $k^{\mathbb{Q}}$. The phrase “complete” refers to continuous reductions, with zero-dimensional Polish source spaces. This is a complexity statement about coefficient codes; it is not a statement that their subspace topology is the desired group topology.

Theorem 67.1 (Puiseux codes). *The coefficient-code domain $\mathcal{P}_k \subseteq k^{\mathbb{Q}}$ is Σ_2^0 -complete.*

Proof. For positive integers n and nonnegative integers M , the set of all coefficient functions supported on

$$(1/n!)\mathbb{Z} \cap [-M, \infty)$$

is closed in $k^{\mathbb{Q}}$. Every such support defines a Puiseux series, and every Puiseux series occurs for some n, M . Hence $\mathcal{P}_k$ is F_σ .

Let $\text{FIN} \subseteq 2^{\mathbb{N}}$ be the set of binary sequences with finite support, a standard Σ_2^0 -complete set. Define a continuous map by placing the bit $b(m)$ at exponent $1 - 1/(m + 2)$ and placing zero at every other exponent. Finite support gives a polynomial. Infinite support gives infinitely many nonzero exponents in the bounded interval $[1/2, 1)$, which cannot lie in one discrete rational lattice. The map reduces FIN to $\mathcal{P}_k$. $\square$

Theorem 67.2 (Levi-Civita codes). *The coefficient-code domain $\mathcal{L}_k \subseteq k^{\mathbb{Q}}$ is Π_3^0 -complete.*

Proof. At a fixed integer cutoff $N \geq 0$, having finite support below N is F_σ : take the union over finite $S \subseteq \mathbb{Q} \cap (-\infty, N]$ of the closed condition that all coefficients below N outside S vanish. Intersect over N to obtain a Π_3^0 definition.

The set

$$\text{FIN}^{\mathbb{N}} = \{b \in 2^{\mathbb{N} \times \mathbb{N}} : \text{each row } b(n, \cdot) \text{ has finite support}\}$$

is Π_3^0 -complete. Place the coefficient $b(n, m)$ at

$$q(n, m) = n + 1 - \frac{1}{m + 2}, \quad (67.1)$$

and zero elsewhere. These rational exponents are pairwise distinct. If one row is infinite, its support is infinite below $n + 1$, so the resulting code is not left-finite. If every row is finite, only finitely many rows can contribute below any prescribed bound, and each contributes finitely many terms. The map is continuous and gives the required reduction. $\square$

[write] These are the Borel levels of source 03's [theorem 19.1](#) ($\text{WO}(\mathbb{Z})$ is Σ_2^0 -complete, $\text{WO}(\mathbb{Z}_{\text{lex}}^d)$ is Π_3^0 -complete for $d \geq 2$), attained here by different sets.

For completeness, the usual hardness inputs can be obtained directly. Given an F_σ set $A = \bigcup_n C_n$ in Baire space, take the C_n closed and increasing. At stage s , let $m_s(x)$ be the least $n \leq s$ whose closed-set tree is compatible with the length- s prefix of x , or $s + 1$ if there is none. The values m_s are nondecreasing. Output a 1 exactly when the value increases. This is a continuous map into $2^{\mathbb{N}}$; it has finitely many 1s exactly when $x \in A$. Coordinatewise application to a countable intersection of F_σ sets proves the completeness of $\text{FIN}^{\mathbb{N}}$.

Theorem 67.3 (Full rational Hahn codes; classical support obstruction). *The coefficient-code domain $\mathcal{H}_k \subseteq k^{\mathbb{Q}}$ is Π_1^1 -complete. Consequently, it has no standard Borel parametrization with Borel coefficient observations and exactly one code for each series.*

Proof. A support fails to be well ordered exactly when there is an infinite strictly decreasing sequence of rational exponents with nonzero coefficients. Existence of that sequence is an analytic condition. Thus $\mathcal{H}_k$ is coanalytic.

For hardness use the standard Π_1^1 -complete set WF of well-founded trees on $\mathbb{N}$ [8]. Give $\mathbb{N}^{<\mathbb{N}}$ the Kleene–Brouwer order: proper extensions precede their prefixes, and incompatible sequences are compared at their first differing coordinate. This is a countable linear order, so it embeds into $\mathbb{Q} \cap (0, 1)$. Fix such an embedding e . For a tree T , put coefficient 1 at $e(s)$ exactly when $s \in T$, and put zero elsewhere. The map is continuous. An infinite branch gives a descending chain in the Kleene–Brouwer order. Conversely, well-founded induction on a

well-founded tree shows that the order below each node is a countable ordered sum of the child orders, followed by the node itself; it is therefore a well-order. Hence $T \in WF$ exactly when the resulting support is well ordered.

If a standard Borel space parametrized all Hahn series uniquely and every coefficient were Borel, its coefficient map into $k^{\mathbb{Q}}$ would be a Borel injection with range exactly $\mathcal{H}_k$. The Lusin–Souslin theorem would make that range Borel, contradicting completeness. $\square$

[write] [Theorem 67.3](#) is a second proof of the non-scattered half of source 03’s [theorems 18.4](#) and [20.2](#) (via [lemma 18.2](#)) and of the collection’s lightface results `onot:neg:lem:kbtree` and `onot:neg:thm:pionesone`; source 09 credits “the preceding project report” and claims no priority for it.

The last theorem is included to complete the comparison, not to reclaim the project’s earlier coanalytic support result [\[44\]](#). It is also logically weaker than the absolute topological obstruction for a different reason: it requires coefficient-observable codes, whereas [Theorem 63.3](#) does not.

Proposition 67.4 (The Polish Levi-Civita topology preserves the Borel codes). *The valuation topology on $\mathcal{L}_k$ has the same Borel sets as the subspace Borel structure inherited from $k^{\mathbb{Q}}$.*

Proof. Each coefficient map is continuous for the valuation topology: fixing the entire prefix up to its exponent fixes that coefficient. Thus coefficient Borel sets are valuation-Borel. In the other direction, take the countable base consisting of balls centered at finite rational-exponent polynomials with nonnegative integer cutoffs. A basic ball is defined by countably many coefficient equalities. It is therefore Borel in the coefficient structure. Every valuation-open set is a countable union of these basic balls, proving the converse inclusion. $\square$

Both $\mathcal{P}_k$ and $\mathcal{L}_k$ carry Borel field operations in their coefficient structures. For $\mathcal{L}_k$ this follows from [Proposition 67.4](#) and continuous valuation arithmetic. On $\mathcal{P}_k$ the same Borel equality holds for its countable valuation base; its field operations are valuation-continuous. This does not imply that $\mathcal{P}_k$ is completely metrizable as a group.

Field	Exact coefficient-code complexity	Admissible Polish arithmetic
$\mathcal{P}_k$	Σ_2^0 -complete	None; proper-subfield cover obstruction
$\mathcal{L}_k$	Π_3^0 -complete	Exists and is unique; valuation completion
$\mathcal{H}_k$	Π_1^1 -complete	None; uncountably many separated prefixes

Table 2: Countable ordered coefficients and rational exponents throughout. A low Borel rank does not predict Polishability.

68 Real closedness and completion without a defect argument

To connect these fields to integer-part arithmetic, we specialize to a real closed coefficient field. A useful proof avoids importing a full theory of henselian defect. The sole substantial algebraic input is the classical Newton–Puiseux theorem in characteristic zero: for an algebraically closed field K of characteristic zero, $\bigcup_n K((t^{1/n!}))$ is algebraically closed. Poonen gives this as [Corollary 7 \[30\]](#).

Lemma 68.1 (Completion preserves algebraic closedness). *The metric completion of a non-Archimedean valued algebraically closed field is algebraically closed.*

Proof. Write $|\cdot|$ for the multiplicative real-valued absolute value; it is not the truncated metric. Let K be the field and $\widehat{K}$ its completion. Take a monic polynomial

$$P(X) = X^d + \sum_{i < d} a_i X^i \in \widehat{K}[X], \quad d \geq 1.$$

Choose monic polynomials $P_n \in K[X]$ of degree d whose coefficients converge to those of P , and positive real numbers $\varepsilon_n \rightarrow 0$ such that every coefficient of $P_{n+1} - P_n$ has absolute value at most ε_n . There is a common real bound $M \geq 1$ for 1 and for the absolute values of all coefficients of all P_n . Every root z of any P_n satisfies $|z| \leq M$: otherwise the monic term would strictly dominate every lower-degree term.

Choose a root z_0 of P_0 . Having chosen z_n , factor $P_{n+1}(X) = \prod_{j=1}^d (X - w_j)$ over K . Since $P_n(z_n) = 0$,

$$\prod_{j=1}^d |z_n - w_j| = |P_{n+1}(z_n)| \leq \varepsilon_n M^{d-1}.$$

Some root w_j therefore satisfies $|z_n - w_j| \leq (\varepsilon_n M^{d-1})^{1/d}$. Choose it as z_{n+1} . Consecutive distances tend to zero, and the ultrametric inequality then makes (z_n) Cauchy: the distance between any two late terms is at most the largest intervening consecutive distance. Let $z \in \widehat{K}$ be its limit. Continuity of polynomial evaluation and coefficient convergence give $P(z) = 0$. Thus every nonconstant polynomial has a root, which is equivalent to algebraic closedness. $\square$

Theorem 68.2 (Real closed Puiseux and Levi-Civita fields). *If k is a real closed ordered field, then $\mathcal{P}_k$ and $\mathcal{L}_k$ are real closed. In particular, this applies to the countable field $k = \mathbb{R}_{\text{alg}}$.*

Proof. Let $K = k(i)$, which is algebraically closed because k is real closed. Splitting coefficients into their real and imaginary parts gives

$$\mathcal{P}_K = \mathcal{P}_k(i), \quad \mathcal{L}_K = \mathcal{L}_k(i).$$

For the first identity, finitely many denominator bounds can be replaced by one common bound; for the second, a finite union of left-finite supports is left-finite. By Newton–Puiseux, $\mathcal{P}_K$ is algebraically closed. By Proposition 65.4 and Lemma 68.1, so is $\mathcal{L}_K$. The fields $\mathcal{P}_k$ and $\mathcal{L}_k$ are ordered and adjoining i makes them algebraically closed; the standard characterization of real closed fields therefore proves the assertion. $\square$

Corollary 68.3 (A ring-language separation). *For $k = \mathbb{R}_{\text{alg}}$, the field $\mathcal{P}_k$ has no Polish additive-group topology with jointly Borel multiplication, even without a separately specified measurable order. The field $\mathcal{L}_k$ does have such a topology, and it is unique.*

Proof. Real closedness makes the order Borel by Corollary 61.6. Apply Theorems 64.3 and 65.3. $\square$

Real closedness here is a classical algebraic property proved using a classical input; it is not the proposed novelty. Its role is to remove the order-measurability hypothesis from the final field-language contrast.

69 Local Polish hulls, unbounded support ranks, and an exact cover number

The failure of Polishability for the full Hahn field is global, not an obstruction to accommodating any prescribed countable collection of its elements. This section proves that much stronger local positive statement. In particular, the Levi-Civita field is a minimal completion, not a maximal Polish subfield of the Hahn field.

Lemma 69.1 (Metric completeness of the rational Hahn field). *For every field k , $k((t^{\mathbb{Q}}))$ is complete for the metric $d(f, g) = \min\{1, 2^{-v(f-g)}\}$.*

Proof. For a Cauchy sequence, every prefix up to an integer $N \geq 0$ eventually stabilizes, just as in Theorem 65.3. This time a stabilized prefix need not have finite support, but it has well-ordered support because it is a prefix of a Hahn series. The compatible stabilized prefixes define a coefficient function f . Its support is well ordered: given any nonempty subset S of that support, choose $s \in S$ and an integer $N \geq \max\{0, s\}$. The nonempty set $S \cap (-\infty, N]$ has a least element since it lies in the stabilized well-ordered prefix. That element is also least in S , as all remaining points exceed N . Thus f is a Hahn series, and stabilization shows that it is the limit of the original sequence. $\square$

Theorem 69.2 (Countable-parameter Polish hulls). *Let k be countable and ordered, $H = k((t^{\mathbb{Q}}))$, and $A \subseteq H$ countable. Then*

$$F_A = \overline{k(t^{\mathbb{Q}})(A)}^v \subseteq H \quad (69.1)$$

is a complete separable ordered subfield containing $\mathcal{L}_k \cup A$. Its valuation topology is its unique admissible Polish topology. It is the smallest valuation-closed subfield of H containing $k(t^{\mathbb{Q}}) \cup A$. Every valuation-complete separable intermediate field $\mathcal{L}_k \subseteq F \subseteq H$ arises in this way.

Proof. The algebraic field $k(t^{\mathbb{Q}})(A)$ is countable. Its closure is separable and is complete by Lemma 69.1. Closure preserves addition and multiplication by continuity, and preserves inversion at every nonzero limit by the inverse-difference identity. Thus F_A is a subfield. The closure of $k(t^{\mathbb{Q}})$ in H is exactly $\mathcal{L}_k$: it is dense in $\mathcal{L}_k$, and $\mathcal{L}_k$ is complete and hence closed in the metric space H . Therefore $\mathcal{L}_k \subseteq F_A$. The valuation and order topologies agree, and Corollary 61.5 gives uniqueness. Minimality follows directly from taking closure. Finally, for a complete separable intermediate field F , choose a countable valuation-dense subset $A \subseteq F$. Completeness makes F closed in H , and (69.1) then gives $F_A = F$. $\square$

Taking $A = \{y\}$ for the bounded infinite support in (65.3) immediately gives

$$\mathcal{L}_k \subsetneq F_{\{y\}} \subsetneq H.$$

The second strict inclusion follows from Theorem 63.3. Thus bounded infinite supports do not individually prevent Polish field arithmetic. What is impossible is allowing all independently varying bounded prefixes at once.

Theorem 69.3 (A support bound for every separable family). *Let $X \subseteq H = k((t^{\mathbb{Q}}))$ be valuation-separable, and let $D \subseteq X$ be countable and dense. Then*

$$\text{ot}(\text{supp } f) \leq \beta_D := \sup_{d \in D} \text{ot}(\text{supp } d) < \omega_1 \quad (f \in X). \quad (69.2)$$

Here the empty support has order type 0. In particular, every admissibly Polish intermediate field has a uniform countable support-order bound. There is no uniform countable bound valid for all Polish hulls F_A .

Proof. For $f \in X$ and every integer $N \geq 0$, density supplies a $d_N \in D$ with $v(f - d_N) > N$. Their prefixes through N agree. Hence

$$\text{ot}(\text{supp } f \cap (-\infty, N]) \leq \text{ot}(\text{supp } d_N) \leq \beta_D.$$

These prefixes are increasing initial segments whose union is the support of f , so their order types have supremum $\text{ot}(\text{supp } f)$. This proves (69.2). The supremum defining β_D is countable because D and all rational supports are countable.

For an admissible topology, a countable dense set remains valuation-dense by Theorem 61.4, so the preceding argument applies. For the last assertion, let α be any countable ordinal. Embed its linear order into $\mathbb{Q} \cap (0, 1)$ and let h_α have coefficient 1 on that support and 0 elsewhere. It belongs to the Polish hull $F_{\{h_\alpha\}}$ and has support order type α . $\square$

[write] Source 03's theorem 21.2 proves a uniform countable support bound for every Borel coefficient-observable family by analytic boundedness; theorem 69.3 is the elementary counterpart for separable families, as source 09 says below.

This bound uses separability and prefix stabilization, not analytic boundedness or a Borel-domain hypothesis. It is a specialized elementary counterpart of the support bounds in the preceding Borel-support manuscript. It also makes the failure of maximality explicit.

Corollary 69.4 (No maximal admissibly Polish intermediate field). *Every admissibly Polish field F with $k(t^{\mathbb{Q}}) \subseteq F \subseteq H$ is properly contained in another such field. The larger field can have an element whose support order type exceeds all support order types in F .*

Proof. Choose a countable dense set D in the admissible topology of F and let β_D be as in Theorem 69.3. Choose a Hahn series h with support order type $\beta_D + 1$. The hull $F_{D \cup \{h\}}$ contains the valuation closure of D , and hence contains F . It also contains $h \notin F$. Theorem 69.2 proves that it is admissibly Polish. $\square$

69.1 Real-closed, truncation-closed hulls

With $k = \mathbb{R}_{\text{alg}}$, a hull can be required to preserve both real closedness and all rational truncations. Full Hahn real closedness follows from the classical algebraic-closedness theorem for Hahn fields with algebraically closed coefficients and divisible exponent group [30, Corollary 4], applied after adjoining i .

Theorem 69.5 (A strengthened hull with a countable truncation core). *Let $H = \mathbb{R}_{\text{alg}}((t^{\mathbb{Q}}))$ and let $A \subseteq H$ be countable. There exist a countable real closed truncation-closed subfield $B \subseteq H$ and a field $C = \overline{B}^v$ such that*

$$k(t^{\mathbb{Q}}) \cup A \subseteq B \subseteq C,$$

C is real closed and admissibly Polish, and every rational prefix of every $f \in C$ belongs to B :

$$T_{\leq q}(C) \subseteq B \quad (q \in \mathbb{Q}). \quad (69.3)$$

In particular, C is truncation-closed and contains $\mathcal{L}_k \cup A$.

Proof. Start with the countable field $B_0 = k(t^{\mathbb{Q}})(A)$. Given B_n , adjoin $T_{\leq q}b$ for all $b \in B_n$ and all $q \in \mathbb{Q}$, then take its real closure inside the real closed field H to obtain B_{n+1} . Each step is countable: there are countably many truncations and countably many elements algebraic over a countable field. The union $B = \bigcup_n B_n$ is a countable field, is truncation-closed, and is real closed. For the last assertion, any finite list of polynomial coefficients and any positive

element occur at one stage, and the requisite odd-degree roots and positive square roots occur at the next stage.

The field $C = \overline{B}^v$ is complete and separable, hence admissibly Polish by the same argument as Theorem 69.2. For $f \in C$ and $q \in \mathbb{Q}$, choose $b \in B$ with $v(f - b) > q$. Then $T_{\leq q} f = T_{\leq q} b \in B$, proving (69.3). To prove real closedness, regard $B(i)$ inside $H(i)$. The coefficientwise valuation satisfies

$$v(a + ib) = \min\{v(a), v(b)\} \quad (a, b \in H).$$

Consequently its completion is $C(i)$. Since B is real closed, $B(i)$ is algebraically closed; Lemma 68.1 makes $C(i)$ algebraically closed. As C is ordered, it is real closed. $\square$

69.2 An exact global invariant

Let $\text{pcov}_k(H)$ be the least cardinality of a family of admissibly Polish intermediate subfields $k(t^{\mathbb{Q}}) \subseteq F \subseteq H$ whose union is H . The topology on a covering field need not initially be the inherited valuation topology.

Theorem 69.6 (The Polish field-cover number). *For every countable ordered coefficient field k ,*

$$\text{pcov}_k(k((t^{\mathbb{Q}}))) = 2^{\aleph_0}.$$

For $k = \mathbb{R}_{\text{alg}}$, the same minimum holds even if all covering fields are required to be real closed, truncation-closed, and Polish in their valuation topology.

Proof. For the upper bound, $|H| = 2^{\aleph_0}$, and the fields $F_{\{h\}}$ from Theorem 69.2, one for each $h \in H$, cover H . For real algebraic coefficients use the stronger hulls from Theorem 69.5 instead.

For the lower bound, use the continuum-sized binary family

$$h_b = \sum_{n \geq 0} b(n)t^{1-1/(n+2)}, \quad b \in 2^{\mathbb{N}}.$$

Distinct centers differ in absolute ordered value by more than $2t$. In any one admissibly Polish intermediate field, the intervals of radius t about the centers it contains are nonempty, open, and pairwise disjoint. It therefore contains only countably many members of this binary family. A union of $\kappa < 2^{\aleph_0}$ such fields contains at most $\kappa \cdot \aleph_0 < 2^{\aleph_0}$ members of the family, and cannot cover H . This cardinal calculation is in ZFC and uses no continuum hypothesis. $\square$

The local and global statements now fit together exactly. Every countable collection of Hahn data is contained in one well-behaved Polish field; no such field is maximal; nevertheless continuum many are necessary to cover all Hahn data. These conclusions are not consequences of the mere cardinality $|H| = 2^{\aleph_0}$: the lower bound uses the fixed-scale separation forced by arithmetic and order.

70 A countable omnific integer part and its exact floor regularity

Throughout this section $k = \mathbb{R}_{\text{alg}} \subseteq \mathbb{R}$. Define

$$\mathcal{I}_k = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{<0}} kt^q. \quad (70.1)$$

Thus an element has finitely many negative-exponent terms and an ordinary integer constant term. The direct sum is literal finite support; replacing it by arbitrary Hahn support would change the ring.

Theorem 70.1 (A common integer part). *The ring $\mathcal{I}_k$ is countable and discretely ordered. It is an integer part of both $\mathcal{P}_k$ and $\mathcal{L}_k$. If*

$$f = p + c + u,$$

where p is the negative principal part, $c = f_0 \in k$, and u has only positive exponents, then

$$\lfloor f \rfloor_{\mathcal{I}_k} = p + \lfloor c \rfloor - \begin{cases} 1, & c \in \mathbb{Z} \text{ and } u < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (70.2)$$

Proof. Finite rational supports with coefficients in a countable field form a countable set. Addition and multiplication preserve finite negative support and integer constant terms, so $\mathcal{I}_k$ is a countable unital subring. If a nonzero element has a negative leading exponent, its absolute ordered value exceeds every positive ordinary integer. Otherwise it is a nonzero ordinary integer. Hence 1 is its least positive element.

For $f \in \mathcal{L}_k$, left-finiteness makes p finite, and every nonzero u is smaller in absolute ordered value than every positive constant of k . If $c \notin \mathbb{Z}$, both $c - \lfloor c \rfloor$ and $\lfloor c \rfloor + 1 - c$ are positive constants; the infinitesimal u cannot change either inequality. If $c \in \mathbb{Z}$ and $u \geq 0$, the floor is $p + c$. If $c \in \mathbb{Z}$ and $u < 0$, the floor is $p + c - 1$. In every case the stated element a satisfies $a \leq f < a + 1$. Uniqueness follows because distinct elements of $\mathcal{I}_k$ differ by at least 1 in absolute ordered value. Since $\mathcal{P}_k \subseteq \mathcal{L}_k$ and contains $\mathcal{I}_k$, the same result holds in $\mathcal{P}_k$. $\square$

[write] With $t^q \mapsto \omega^{-q}$, formula (70.2) is the collection's floor rule `odg : thm : floor`, proved in Lean for every surreal, and source 03's lemma 23.1. For $k = \mathbb{R}_{\text{alg}}$, the ring $\mathcal{I}_k$ is source 03's finite principal-part ring $\mathcal{A}_{\text{fin}}$ with coefficients restricted to $\mathbb{R}_{\text{alg}}$ (with $X = t^{-1}$), a countable subring of it; it is also the Shepherdson ring used by batch 86's manuscript 10 in Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization.

Formula (70.2), particularly its correction for a negative infinitesimal at an integer constant, is a specialization of the omnific normal-form floor rule already documented in ProveIt [42]. It is not claimed as a new floor formula. The following topological consequences are the point of its use here.

Theorem 70.2 (Exact discontinuity set and Baire class). *Give $\mathcal{L}_k$ its unique admissible Polish topology and $\mathcal{I}_k$ the discrete topology. The floor map $\lfloor \cdot \rfloor_{\mathcal{I}_k} : \mathcal{L}_k \rightarrow \mathcal{I}_k$ is Baire class one. Its set of discontinuities is exactly $\mathcal{I}_k$, a countable closed nowhere-dense subset of $\mathcal{L}_k$.*

Proof. Its fibers are the half-open intervals $[a, a + 1)$ for $a \in \mathcal{I}_k$. If $f \notin \mathcal{I}_k$ and $a = \lfloor f \rfloor_{\mathcal{I}_k}$, then $a < f < a + 1$, so a sufficiently small order neighborhood of f lies in the same fiber. The floor is continuous there. At $a \in \mathcal{I}_k$, the sequence $a - t^n$ converges to a , while

$$\lfloor a - t^n \rfloor_{\mathcal{I}_k} = a - 1 \neq a = \lfloor a \rfloor_{\mathcal{I}_k}.$$

Thus the discontinuity set is exactly $\mathcal{I}_k$.

For $n \geq 0$ let $T_n f = \sum_{q \leq n} f_q t^q$, a finite polynomial. The map T_n into its discrete, countable range is locally constant: $v(f - g) > n$ implies $T_n f = T_n g$. Consequently

$$F_n(f) = \lfloor T_n f \rfloor_{\mathcal{I}_k}$$

is a continuous map $\mathcal{L}_k \rightarrow \mathcal{I}_k$. These maps converge pointwise to the floor, and indeed are eventually equal to it at each input. The negative principal part and constant are already fixed for all $n \geq 0$. When the constant is an integer and the positive tail is nonzero, the sign needed for the correction is determined by the first positive exponent, which is included once n is large enough. In all other cases no such correction can change. This proves the Baire class one assertion.

Closed discreteness follows from Theorem 62.3. Its interior is empty because every neighborhood of $a \in \mathcal{I}_k$ contains $a + t^n \notin \mathcal{I}_k$ for large n . Hence it is nowhere dense. $\square$

Example 70.3 (Boundary tests). The formulas include

$$\lfloor -t \rfloor_{\mathcal{I}_k} = -1, \quad \lfloor 7 - t^{1/3} \rfloor_{\mathcal{I}_k} = 6, \quad \lfloor 7 + t^{1/3} \rfloor_{\mathcal{I}_k} = 7,$$

and

$$\lfloor 3t^{-2} - 5t^{-1/2} + \frac{3}{2} - t^{10} \rfloor_{\mathcal{I}_k} = 3t^{-2} - 5t^{-1/2} + 1.$$

A finite-precision procedure that ignores a small negative tail at an integer constant is wrong exactly at the kind of boundary detected by the theorem. The continuous approximants above permit eventual correction; they do not provide an input-uniform stage at which correction is finished.

70.1 Floors in the larger Polish hulls

The strengthened hulls of Theorem 69.5 support the same regularity without restricting their negative principal parts to finite support.

Theorem 70.4 (Canonical integer parts in real closed Polish hulls). *Let $B \subseteq C \subseteq \mathbb{R}_{\text{alg}}((t^{\mathbb{Q}}))$ be as in Theorem 69.5. Then*

$$J_C = \{f \in C : \text{supp } f \subseteq \mathbb{Q}_{\leq 0}, f_0 \in \mathbb{Z}\}$$

is a countable closed discrete integer part of C , and $J_C \subseteq B$. Its floor is Baire class one as a map into discrete J_C , with discontinuity set exactly J_C . Under the surreal normal-form embedding it is a subring of the omnific integers.

Proof. For $f \in C$, the principal part $p = T_{\leq 0}f - f_0$ belongs to B by (69.3). Thus the formula

$$\lfloor f \rfloor_{J_C} = p + \lfloor f_0 \rfloor - \mathbf{1}_{\{f_0 \in \mathbb{Z}, f - T_{\leq 0}f < 0\}}$$

defines an element of C with no positive exponents and integer constant. The same leading-term comparisons as in Theorem 70.1 prove the integer-part inequalities and discrete order. Ring closure follows from convolution: negative exponents cannot contribute to the constant term. Every $j \in J_C$ equals $T_{\leq 0}j$, so lies in the countable set B . Closed discreteness also follows from Theorem 62.3.

For each nonnegative integer n , the map $T_{\leq n} : C \rightarrow B$, with B given the discrete topology, is locally constant. The maps $f \mapsto \lfloor T_{\leq n}f \rfloor_{J_C}$ are therefore continuous and are eventually equal to the floor at each input. Only the first nonzero positive term is needed to decide the possible correction. The interval fibers and the sequence $a - t^n \rightarrow a$ give the exact discontinuity set as before. Finally, replacing t^q by ω^{-q} turns the nonpositive support and integer constant conditions into the omnific normal-form conditions. $\square$

Remark 70.5 (The target topology matters). The Baire class one assertion is about a *discrete* integer-part target, not just its inclusion in the field. It is stronger than merely producing continuous real-valued approximants to the ordinary floor on $\mathbb{R}$. Indeed, the ordinary floor $\mathbb{R} \rightarrow \mathbb{Z}_{\text{discrete}}$ is not Baire class one: every continuous map from connected $\mathbb{R}$ to discrete $\mathbb{Z}$ is constant, and a pointwise convergent sequence of constant maps has a constant limit. The locally constant prefix maps are a genuine non-Archimedean resource.

70.2 The common integer part satisfies open induction

The classical relationship between integer parts of real closed fields and open induction is often called the integer-part characterization of open induction. For our specific ring the needed implication has a short proof.

Proposition 70.6 (Open induction). *The nonnegative cone $\mathcal{I}_{k,\geq 0}$ satisfies $Q + \text{IOpen}$ in the language of ordered arithmetic, with parameters allowed in the induction scheme.*

Proof. The elementary axioms of discretely ordered semiring arithmetic hold in the non-negative cone of any discretely ordered unital ring. Every positive $a \in \mathcal{I}_k$ has predecessor $a - 1 \geq 0$.

A quantifier-free formula in one variable with parameters from $\mathcal{I}_k$ is a Boolean combination of polynomial equations and inequalities. In the real closed field $\mathcal{L}_k$, the real roots of its finitely many nonzero polynomials partition the line into finitely many intervals and points on which the formula has constant truth value. Every nonempty trace of one of these pieces on $\mathcal{I}_{k,\geq 0}$ has a least element: the lower endpoint, when included and already in $\mathcal{I}_k$, or the first integer-part element above that endpoint, with 0 serving as a lower bound when needed. The floor property supplies this first element. Therefore every nonempty quantifier-free definable subset of $\mathcal{I}_{k,\geq 0}$ has a least element.

If a quantifier-free formula held at 0 and were preserved by successor but failed somewhere, its counterexamples would have a least element a . Then $a > 0$ and the formula holds at $a - 1$, contradicting preservation by successor. This is open induction. $\square$

[write] This is the route of `dsn:cor:openinduction` and of source 03's [theorem 23.2](#), which proves open induction for $\mathcal{A}_{\text{fin}}$; the Shepherdson model is classical [6].

Corollary 70.7 (Arbitrary countable support complexity in countable arithmetic). *For every countable ordinal α , there is a real closed Polish truncation-closed field $C \subseteq \mathbb{R}_{\text{alg}}((t^{\mathbb{Q}}))$ whose countable canonical integer part J_C contains an element with support order type α . The nonnegative cone of J_C satisfies $Q + \text{IOpen}$.*

Proof. Embed α into $\mathbb{Q} \cap (-2, -1)$ and take the Hahn series with coefficient 1 on that support. Put it in a strengthened hull using Theorem 69.5. Its support is negative and its constant is 0, so it belongs to J_C . The proof of Proposition 70.6 applies verbatim with this integer part and its real closed ambient field. $\square$

There is no conflict with Glazer's result: the arithmetic model $\mathcal{I}_{k,\geq 0}$ is countable. Its uncountable real closed ambient field $\mathcal{L}_k$ is not a discretely ordered arithmetic model. Moreover, $\mathcal{P}_k$ and $\mathcal{L}_k$ share this same integer part even though only the latter supports admissible Polish field arithmetic.

70.3 Why the finite integer part does not extend to the full Hahn field

The element

$$z = t^{-2}y = \sum_{n \geq 2} t^{-1-1/n} \in \mathcal{H}_k,$$

where y is from (65.3), has infinitely many negative terms. For every $a \in \mathcal{I}_k$, at least one negative coefficient of $z - a$ remains nonzero; its leading exponent is negative, so $|z - a| > 1$. Hence $\mathcal{I}_k$ is not an integer part of $\mathcal{H}_k$.

Replacing finite negative support in (70.1) by arbitrary well-ordered negative support gives an integer part of $\mathcal{H}_k$, with the same coefficientwise floor rule. This larger ring is uncountable: binary choices along the support of z already produce continuum many elements. Theorem 62.3 supplies another obstruction to admissible Polish arithmetic on this full Hahn field. Here that argument agrees with, but is less general than, Theorem 63.3.

71 Surreal realization and a formalization interface

71.1 A set-sized embedding, not a topology on a proper class

With $k = \mathbb{R}_{\text{alg}} \subseteq \mathbb{R}$, the normal-form assignment

$$\iota \left(\sum_{q \in \mathbb{Q}} a_q t^q \right) = \sum_{q \in \mathbb{Q}} a_q \omega^{-q} \quad (71.1)$$

realizes the three fields as set-sized subfields of the surreal numbers. Well-ordered support in the q -variable becomes reverse-well-ordered support in the Conway growth exponents $-q$. The normal-form rules preserve addition, multiplication, and order; t corresponds to ω^{-1} . The image of $\mathcal{I}_k$ has only positive growth exponents and an integer constant term, so lies in $\mathbf{Oz}$.

This is a formal normal-form embedding. It does not assert that an arbitrary infinite series can be evaluated at an infinite real or surreal argument by ordinary analytic convergence. In particular, the direction of the exponent change in (71.1) is essential. The full proper class **No** is never asserted to be a Polish space. The series carriers have cardinality continuum and can be constructed as subsets of the set $k^{\mathbb{Q}}$. Only afterwards are they identified with surreal normal forms.

The project documentation is an interface to these structures, not a proof that the new topological statements have already been formalized. The reported kernel checks in Provel concern the repository's declared constructions and theorems. No new kernel run was performed for this article.

71.2 A modular Lean development plan

A formalization need not begin with surreal arithmetic. Most of the proof is independent of it. A suitable division into modules is as follows.

Category and ordered groups. Formalize the Baire-order local-bound lemma over a linearly ordered abelian group, separately from the field specialization. Its interface should return an order bound $a > 0$ and a neighborhood inclusion, not an implicit assertion that the ambient

topology is the order topology. Import a proved Pettis lemma and Borel-homomorphism automatic continuity at clearly stated instances.

Ordered rings and fields. Prove automatic discreteness using one scalar $a + 1$, and prove the field refinement theorem using $(a + 1)/\varepsilon$. These are different lemmas: quotient scalars are not available in a general ordered ring. Then prove closed discreteness and countability of any integer-part subring without assuming its carrier is measurable.

Series and topology. Construct left-finite coefficient functions as a subtype of $\mathbb{Q} \rightarrow k$. Give finite-below-cutoff witnesses explicitly. Prove convolution finiteness, the inverse construction, valuation identities, and metric completeness. The countable dense subspace consists of finitely supported functions. A separate theorem identifies the metric and order topologies.

Obstruction certificates. Formalize the binary family from Lemma 63.2; the only infinite support needed for the negative result has order type at most ω . For Puiseux, package the general countable proper-subfield-cover theorem, then prove each lattice-support stage is closed. This separation prevents a proof of incompleteness of one metric from being mistaken for a proof of absolute non-Polishability.

Integer parts and measurable floors. First prove (70.2) in the left-finite series subtype. Then identify this ring with the corresponding set-sized subring of the repository's omnific integers. The exact discontinuity theorem and the pointwise eventually constant sequence (F_n) can be formalized independently of the full surreal field.

Descriptive complexity. The explicit maps in (67.1) and the FIN reduction are the finite combinatorial portion. The well-founded-tree reduction and Lusin–Souslin step belong in a separate descriptive-set-theory layer. They are not required for the core Hahn classification or the positive Levi-Civita construction.

This architecture leaves no proper-class quantification in the main proof. Universe parameters are needed only when identifying the set-sized series carrier with a universe-relative surreal implementation. No additional large cardinal assumption is used.

72 Consequences for exact symbolic arithmetic

An exact coefficient stream and a valuation-Cauchy name provide different information. A coefficient stream reports one coefficient at a time. A valuation-Cauchy name eventually fixes *all* coefficients below a specified cutoff. In a left-finite representation, certifying a completed prefix therefore includes a support bound, not just a list of currently observed nonzero terms.

Theorem 65.3 supports continuous arithmetic in the second representation. Proposition 67.4 says that the measurable information is the same, but it does not turn these representations into the same effective oracle. No computability claim follows merely from Borel equivalence.

For an exact rational-coefficient prototype one can represent finite truncations by dictionaries $q \mapsto a_q$ with $q, a_q \in \mathbb{Q}$. When multiplying to cutoff B , lower bounds on the two supports determine which input prefixes suffice: terms of the first factor above $B - v(g)$, and terms of the second above $B - v(f)$, cannot affect the result. For inversion of $at^q(1 + h)$, a positive

lower bound $\delta \leq v(h)$ bounds the number of geometric powers needed below any fixed cutoff. These are mathematically certified finite reductions, not numerical approximations to real coefficients.

Floor is different. Its Baire class one approximants may change once the first nonzero infinitesimal term is revealed at an integer boundary. No continuous exact floor algorithm on the whole field can exist, because the mathematical function is discontinuous there. The theorem pinpoints the necessary limitation instead of attributing it to a poor choice of code.

The accompanying Python program uses rational exponents and coefficients to check finite algebraic identities, truncated geometric inverses, the floor inequalities, and finite instances of the support witnesses. The deterministic run passed 20,116 finite assertions; its output records the counts by test family. These tests catch implementation mistakes in the supplied examples; they cannot establish Baire category, quantification over all Polish topologies, infinite support completeness, or real closedness. The conventional proofs are the basis of those assertions.

73 Further research questions and concrete targets

The following questions are proposed next steps. They are not asserted to be previously published open problems, and no novelty claim is inferred from failure to locate an answer in the inspected sources.

Question 73.1 (Intermediate-field rigidity). Let k be countable and ordered, and let $\mathcal{P}_k \subseteq F \subseteq k((t^{\mathbb{Q}}))$. If F has an admissible Polish topology, must its valuation topology be complete, and must the admissible topology equal the valuation topology?

The prefix criterion gives countably many prefixes, but it does not prove completeness of a coarser topology. A proof must close precisely that gap; one cannot appeal to the open mapping theorem until both topologies are known to be Polish.

Question 73.2 (Classification of Polish hulls). Within a fixed standard Borel family of coefficient-observable parameters, what is the descriptive complexity of equality and inclusion between the countable-parameter hulls F_A ? Can one construct useful canonical dense cores or invariants for their ordered-field isomorphism types?

The existence of proper Polish extensions of $\mathcal{L}_k$ and the absence of maximal ones are settled by Theorem 69.2 and Corollary 69.4. The remaining issue is classification, not existence. The ambient parameter space must be specified: the full Hahn coefficient domain is not standard Borel.

Question 73.3 (Minimal scalar hypotheses). How few Borel scalar maps suffice to force the order topology in the rational-exponent setting? Can a specified countable set of monomial multiplications replace all scalar maps under additional algebraic or truncation-closure assumptions?

The current proof uses a scalar depending on the local order bound and the target interval. An answer should identify which part of that freedom is essential, rather than dropping scalar regularity altogether.

Question 73.4 (Order-free non-real-closed fields). For countable ordered k that is not real closed, can the Hahn and Puiseux nonexistence statements be derived from a Polish additive group and jointly Borel multiplication alone, without requiring the displayed order to have the Baire property?

Corollary 61.6 settles the real closed case. In general a field may have several orders, and the image-of-squares argument does not identify one positive cone.

Question 73.5 (Other support completions). Extend the left-finite construction and its classification to countable ordered exponent groups of higher rank. Which support restrictions give separable complete field topologies, and which admit a countable cofinal valuation scale but still fail by a proper-subfield cover?

The inverse proof in Proposition 65.2 explicitly uses $n\delta \rightarrow +\infty$ in an Archimedean exponent group. It cannot be copied unchanged into a lexicographic higher-rank group.

Question 73.6 (Descriptive ranks beyond the trichotomy). Which Borel ranks can occur for coefficient-code domains of admissibly Polish subfields of $k((t^{\mathbb{Q}}))$? Is there a useful algebraic characterization of when a Borel support family yields a Polishable additive group?

Table 2 shows that a smaller rank is not sufficient. A productive refinement should combine support closure, completeness of prefixes, and the proper-subfield obstruction.

Question 73.7 (The least support bound of a one-parameter hull). For $y = \sum_{n \geq 2} t^{1-1/n}$, determine the least countable ordinal bounding all support order types in $F_{\{y\}}$. How does that least bound change when real closure and closure under all rational truncations are imposed before metric completion?

Theorem 69.3 proves existence of a bound but does not compute it. A solution would require quantitative control of support order types under convolution, inversion, algebraic extension, and stabilized-prefix limits. It would turn the qualitative local-hull theorem into an ordinal complexity calculation.

Question 73.8 (Other integer parts and floor discontinuities). Classify integer parts of $\mathcal{L}_k$ and the larger real closed Polish hulls up to field automorphisms. Which floors admit canonical or effective locally constant approximants for presentations not built from truncations?

Countability and closed discreteness hold for every integer part by Theorem 62.3. The discontinuity set of its floor is also its integer part in the order topology. The strengthened hulls retain explicit truncation approximants. The remaining constructive issue is canonical or effective approximation for other presentations, together with a classification of the integer parts.

Question 73.9 (A uniform effective presentation). For computably presented k , construct a certified Type-2 representation of $\mathcal{L}_k$ with computable arithmetic, explicit precision transfer for products and inverses, and a finite-mind-change floor procedure for $k = \mathbb{R}_{\text{alg}}$.

The representation must include effective support-completion information. Bare enumerations of nonzero coefficients do not suffice to know that a prefix is complete. Any claimed effective embedding into Baire space should state which oracle representation is being used.

Question 73.10 (Formal proof milestones). Can the local order-bound theorem, automatic discreteness, and full Hahn classification be formalized as a small reusable library independent of the full surreal development? What is the minimal imported descriptive-set-theory interface for the exact code-complexity theorems?

These are concrete implementation tasks. The first milestone is the category-to-order bridge; the coanalytic lower bound should not block the rest of the project.

Question 73.11 (Positive cones versus groups of differences). Characterize Borel discretely ordered semirings whose positive cones admit Polish continuous addition but whose groups of differences are not Polishable with measurable multiplication. Which hypotheses force a topology to extend from the cone to its group completion?

This returns directly to the boundary emphasized by Glazer’s arithmetic questions and the earlier Presburger constructions. Theorem 62.1 provides an obstruction on the group side, not a classification of monoid presentations.

[write] This is source 08’s [question 57.2](#); source 06’s cone of $\mathbb{Z} + t\mathbb{R}[t]$ is a data point ([section 42](#)).

Question 73.12 (Weak choice and category). Which portions of the classification survive in weaker choice settings? Separate the dependent choices used to build the bounded increasing exponent sequence, the Baire theorem, and the selections of representatives of countably many prefix classes.

The present paper works in ZFC and makes no choice-free assertion. A foundational analysis would be a distinct project rather than an automatic consequence of the set-sized constructions.

74 Conclusions of Part V

The main dividing line is not whether exact infinite series can be assigned Borel codes. It is whether their arithmetic, order, and completeness fit within a separable Polish group. A Baire-measurable order and Borel scalar arithmetic force local order bounds and hence the order topology. Full Hahn support then admits too many mutually separated prefixes, except in the countable-coefficient Laurent case. Puiseux support avoids that obstruction but fails a different category test because it is a countable union of proper measurable subfields. Left-finite completion avoids both obstructions and supplies a unique Polish topological field. Countable-parameter hulls go further: arbitrary countable Hahn data fit into a Polish subfield, even with real closedness and truncation closure over real algebraic coefficients. There is no maximal Polish intermediate field, while the exact cover number of the full field is the continuum.

The integer-part consequences connect this topological classification to arithmetic and surreal normal forms, including arbitrary prescribed countable support types inside suitable countable integer parts. A Polish ordered field cannot hide an uncountable discrete integer part, even a nonmeasurable one. The real algebraic-coefficient Levi-Civita field nevertheless supports a countable omnific integer part with open induction, and its floor has exactly controlled discontinuities. These results supply a precise next layer beyond Borel support analysis while leaving topology rigidity on intermediate fields and the broader positive-cone classification open.

A Worked examples and an audit of the hypotheses (source 05)

A.1 A negative constant term is compatible with positivity

In $M_{\mathbb{R}}$, take

$$x = (\tfrac{3}{2}, -17), \quad y = (\tfrac{1}{4}, 6).$$

Both are positive, and

$$x + y = (\tfrac{7}{4}, -11).$$

Division of x by 5 gives

$$q_5(x) = (\frac{3}{10}, -4), \quad r_5(x) = 3,$$

because $-17 = 5(-4) + 3$. The quotient is positive because its real coordinate is positive. A definition that incorrectly demanded a nonnegative integer coordinate for every element would break the predecessor and division axioms.

A.2 A nonstandard finite-candidate calculation

Let $L = (\frac{1}{2}, -5)$, $U = (\frac{2}{3}, 0)$, and require

$$y \equiv (\frac{7}{13}, 2) \pmod{3G}.$$

The real coefficient of a congruence parameter is irrelevant because $\mathbb{R}$ is divisible. Among $L, L+1, L+2$, the candidate $L+1$ has integer coordinate $-4 \equiv 2 \pmod{3}$, so it is a witness. Every candidate lies below U , since its leading coefficient is $1/2 < 2/3$. No iteration a nonstandard number of times is used: the finite search is justified by full invariance on $3G$ -cosets.

A.3 A Baire convergence that order cannot detect continuously

Let e_j be the rational unit vector at coordinate j . Then

$$(e_j, 0) \rightarrow (0, 0), \quad (e_j, 0) > n\mathbf{1} \quad \text{for every fixed standard } n.$$

The initial coefficients eventually agree with zero to every prescribed finite length. In omnific notation this sequence is

$$\omega, \quad \omega^{1/2}, \quad \omega^{1/3}, \quad \dots,$$

which converges to zero in the *specified coefficient topology*, not in the arithmetic order topology. This is a concrete illustration, not a claim about an unspecified topology on surreal numbers.

A.4 Hypothesis checklist

Potential pitfall	Resolution in the proofs
Only the division axioms are checked	Theorem 4.9 , corollary 4.11 , and proposition 4.12 establish the complete theory and every additive induction instance.
An open ray is given an incomplete metric	Section 5 provides a compatible logarithmic metric on each open ray and a complete metric on the whole sum.
The intended topology is silently replaced by the order topology	Both topologies are explicitly distinguished; the arithmetic order topology is discrete.
Standard-shift invariance is used with a nonstandard quotient	Example 4.4 exhibits the failure; lemma 4.7 uses invariance on mG -cosets instead.
The all-zero rational branch is lost	The nonnegative-cone tree in (6.2) explicitly includes it.

Potential pitfall	Resolution in the proofs
Computing an output tag secretly requires deciding equality of streams	For a negative output integer coordinate, a negative input supplies a finite bound on the first positive coefficient.
A type-theoretic “computable model” is confused with a countable one	The presentation is Type-2 on all Baire sequences; not every element is computable, and order has only a limit decision procedure.
The omnific image is assumed closed under multiplication	It is only an additive fragment. Theorem 10.4 rules out even an alternative semiring multiplication on the same carrier.
A proper class is treated as a Polish space	Only explicit set-sized images and a sufficiently large set-sized birthday cutoff are used.
Existing repository proofs are credited with new theorems	The formalization roadmap distinguishes the inspected integer theorems from the additional nonstandard and topological obligations.
Numerical testing is mistaken for a proof	The exact test scope is documented in section 13 ; no formal verification claim is made for this article.
An affirmative proof is mistaken for certified historical priority	The source question and version are identified; independent priority and refereeing remain separate from the mathematical proof.

B Reproduction and the scope of the executable checks (source 04)

The package requires a standard LaTeX installation for the article and Python 3.10 or later for the companion code. The Python program uses only the standard library and exact `Fraction` arithmetic. From the package directory, run

```
python3 verification.py --output verification_results.json
pdflatex -interaction=nonstopmode -halt-on-error polish_presburger.tex
pdflatex -interaction=nonstopmode -halt-on-error polish_presburger.tex
```

A Makefile supplies equivalent targets. The delivered run used seed 20261003 and passed 75,079 assertions. Its breakdown is:

Check category	Assertions
Rational, integer, and pairing codes	6,468
Finite group and order identities	9,955
Exact quotient and remainder	2,151
Boundary witness transfer	21,037
Finite candidate comparison	1,200
External periodicity counterexample	2,321
Positive-pivot matrix tests	18,000
Monus integer-coordinate obstruction	400
Baire-code round trips	97
Baire quotient and remainder prefixes	9,700
Baire addition prefixes	3,750
Total	75,079

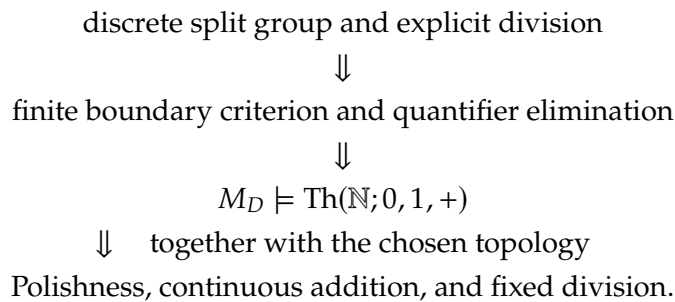
The finite group tests use rational coefficient vectors and signed integer coordinates. The boundary tests examine witnesses from a specified finite pool and compare them with the exact candidate set. They do not exhaust an infinite group. The matrix tests use finite initial pieces of matrices satisfying the positive-pivot conditions. The stream tests check finite prefixes of deterministic infinite oracles, including long zero runs. The monus tests verify exact coordinate obstructions along finite portions of the sequences used in the proof, not topological convergence by numerical approximation.

The model-theoretic, infinite-matrix, and topological statements rest on the proofs in the manuscript. Tests are useful for detecting sign errors, incorrect negative quotients, wrong coordinate offsets, and mismatched coding branches, but a successful run is not a substitute for those proofs.

[write] The commands above use source 04's delivery names. In this report its program is `code/04-polish-presburger-verification.py`, its recorded run data `data/04-polish-presburger-verification_results.json`, its Makefile `code/04-polish-presburger-Makefile` (which still names `polish_presburger.tex`, `verification.py` and `verification_results.json`), and its manuscript is merged into `this_article.tex`. The program's default output is `verification_results.json` in the working directory, so it is rerun with an explicit output path outside the report (README). The intake reran it on a copy with Python 3.14.4: 75,079 assertions passed, and the record was identical up to line endings.

C A compact proof-dependency map (source 04)

The answer can be read independently of the optional extensions:



For the real example, [section 5.4](#) completes the argument at that point. For literal Baire space, add only the two chart codings in [section 6.3](#). The computation of the monus locus,

the embedding classification, the growth-order obstruction, and the surreal realization are subsequent results; none is a hidden prerequisite for the affirmative construction.

D Source audit, contribution ledger, and reproducibility (source 03)

D.1 Repository material inspected

The repository branch was resolved to the commit printed in Section 16. The following guides were inspected in selected ranges at that revision:

```
Algebra/SurrealNumbers/README.md
Algebra/SurrealNumbers/docs/README.md
Algebra/SurrealNumbers/docs/
foundations-and-computation/omnific-notations/README.md
```

The first guide supplies the context of the omnific ring, its least positive element, and the floor boundary correction. The notations guide reports the existing Kleene–Brouwer support-validity classification and carefully states the limited scope of its finite checks. The present article does not certify all statements in either guide.

The external mathematical inputs are Newton–Puiseux, the classical Hahn-field construction, Hausdorff’s characterization of countable scattered orders, standard descriptive-set-theoretic facts, and Glazer’s Corollary 1. Analytic boundedness and the needed Kleene–Brouwer reduction are proved in the text. Shepherdson’s theorem is contextualized and its needed direction is also proved directly.

D.2 Contribution ledger

Statement	Status in this report
General coanalytic support hardness	Prior classical mechanism; already recorded for omnific notations in ProveIt. Reproved to make the reductions explicit.
Scattered/support dichotomy	Proved from Hausdorff theory and the fixed tree reduction; global priority not asserted.
Observable full-Hahn criterion	Proved, including Borel field operations on the scattered side and exclusion of even repeated observable names on the nonscattered side.
Exact $\mathbb{Z}^d$ Borel levels	Proved with explicit continuous reductions and a separate sharp support-order bound.
Uniform observable support bound	Classical analytic boundedness specialized to coefficient families; a proof is supplied.
Fixed-interval binary localization	Explicit application of the support reduction, not advertised as a new foundational lower-bound method.
Principal-part integer part and floor	Shepherdson-type classical model; negative-infinitesimal correction also already in ProveIt.

Statement	Status in this report
Exact cancellation loci	Proved for the stated Polish finite-normal-form topology, including closed-piece continuity and zero-factor cases.
No compatible continuous addition	Direct application of Glazer after all model and Borel hypotheses have been verified.
Formal verification	No Lean proof delivered. Python checks only finite exact-rational identities, order tests, floor cases, and finite embedding data.

D.3 Finite verification artifact

The archive contains `verify.py`, a Python 3.10+ standard-library-only program, and the actual result `verification.json`. The recorded run used Python 3.13.5, deterministic seed 2601003, and passed **84,388 assertions**. These include 6,000 ring-law checks, 2,000 closure checks, 2,000 positive-cone checks, 999 discreteness checks, 7,500 floor-interval checks, 63 negative-infinitesimal boundary checks, 25 cancellation checks, 65,792 rational Kleene–Brouwer embedding checks on 256 nodes, and 9 coherence and branch-approximation checks.

The finite embedding is an initial segment of one fixed midpoint construction: nodes are enumerated by the weight $\sum_i (s_i + 1)$, and each new node is inserted between its previously assigned neighbors. This checks the direction of the order reversal and the rational band. It does not test the well-foundedness of arbitrary infinite trees.

To reproduce:

```
python verify.py --output verification.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is internal, so no BibTeX run or network access is required. The finite checks are useful for catching sign, support, and boundary mistakes; they neither prove descriptive completeness nor replace the mathematical proofs or an independent review.

[write] In this report source 03’s program is `code/03-borel-presentations-verify.py`, its recorded run data `data/03-borel-presentations-verification.json`, and its provenance record `data/03-borel-presentations-provenance.json`; the record still describes 03’s own 23-page PDF, which is not shipped. Its manuscript is merged into `article.tex`. The intake reran the program on a copy with Python 3.14.4: 84,388 assertions passed, and the record differed from the delivered one only in the interpreter version and line endings.

E Provenance of this report

Sources of Parts I and II. Three manuscripts of batch 86 of the repository’s intake, all dated October 3, 2026, all AI-assisted (03: “prepared with ChatGPT”), arrived together in commit 31fdc6571 and were placed in commit 3d2177df4. Each inspected the repository at its own pin; the pins are 28 to 35 commits before the placement.

Source	Archive	Pin	Contributes
05	polish-presburger-glazer	a11efab09	base of Part I: sections 2 to 13 , its questions, sections A and 15
04	Polish_Presburger_Glazer_ProveIt	3ad5f878c	every passage marked [04] in Part I ; sections 2.3, 4.3, 4.5, 5.2, 5.4, 6.3, 7.3, 8.1, 10.1, 11.2, 11.3 and 13.4 , its questions, sections B and C
03	glazer-proveit-research	c5b4881fa	Part II except the notes and section 26 ; section D

Written for this report: [section 1](#) (except the quoted status statements), [section 13.5](#), the merged questions of [section 14.1](#), [section 26](#), the notes marked [write] or [merge], and these appendices.

Where the merge of Parts I and II chose.

- *Base.* Source 05, for the weaker hypotheses of its Polish criterion and Δ_2^0 theorem ([section 1.2](#)).
- *Printed once.* The main theorem (05's [theorem 3.1](#), with 04's statement as [remark 3.2](#)); the division algorithm ([proposition 4.2](#) = 04's Lemma 2.2); full Presburger arithmetic of the cone ([corollary 4.11](#) = 04's Corollary 3.5); the predecessor and truncated-subtraction loci ([proposition 8.2](#) and [theorem 8.3](#) = 04's Proposition 7.4 and Theorem 7.3); the omnific realization ([theorem 9.1](#) = 04's Proposition 10.1, same proof); the no-semiring theorem and its corollary ([theorem 10.4](#) and [corollary 10.5](#) = 04's Theorem 9.3 and Corollary 9.4, whose model-specific proof coincides with 05's); the real endomorphisms ([theorem 11.1](#) = 04's Theorem 8.2).
- *Second proofs from 04.* [Proposition 4.2](#) (uniqueness by coordinates), [lemma 4.7](#) (as [theorem 4.8](#)), [theorem 4.9](#) (as [theorem 4.10](#)), [theorem 6.1](#) (as [theorems 6.3 and 6.4](#)), [theorem 7.1](#) (as [theorem 7.2](#)), [proposition 8.2](#), [theorem 8.3](#) (with the nowhere-density clause), [theorem 10.4](#), [theorem 11.1](#).
- *Both printed.* The order obstructions [theorem 7.4](#) (05) and [theorem 7.5](#) (04), whose hypotheses are incomparable ([remark 7.7](#)); the counterexamples [example 4.4](#) (05) and [proposition 4.6](#) (04); the Polish criterion [theorem 5.2](#) (05) and the transfer theorem [theorem 5.4](#) (04), which alone gives continuous division for general D .
- *Questions.* Four questions asked by both 05 and 04 are merged ([section 14.1](#)); 05's question on canonical surreal cutoffs and topologies and 04's on coefficient topologies inside surreal arithmetic overlap only partly and are printed separately. 03's eight questions are printed as delivered. Part I has seventeen questions, Part II eight.
- *Not printed separately.* 04's title page, abstract (its non-claims are in [section 1.5](#)), reading routes, Definition 2.1 (05's setup), the opening paragraph of its Section 9 (Glazer's Corollary 1, in [section 1.1](#)), and the statements and proofs listed under "printed once"; 05's and 03's title pages and abstracts (their status statements are in [section 1.5](#)). The merged abstract replaces them.
- *Renamed symbols.* Listed in [section 1.3](#); no normalization was changed.
- *Added.* [Proposition 26.1](#) and the discussion in [section 26](#); the remark [remark 7.7](#); Glazer's speculation ([section 1.1](#)), which no source mentions; the notes citing the collection's

labels in [Part II](#).

Intake checks for Parts I and II. The intake ran the three delivered suites on copies (Python 3.14.4, standard library only): 03 passed 84,388 assertions in 11 groups, 04 passed 75,079 assertions in 13 categories, 05 passed 17,930 checks in 18 families. Their records agree with the shipped ones up to interpreter strings and line endings. The three delivered checksum manifests verified and were not shipped. It also checked by hand that $M_{\mathbb{R}}$ is a countable clopen sum of rays and a model of $\text{Th}(\mathbb{N}; 0, 1, +)$, the Polish clause of [theorem 11.2](#), [theorem 7.5](#), and the worked examples of [section A](#); no false statement was found. For this report the following references were checked against their publishers' records on October 3, 2026: Glazer's arXiv record (one version, no journal reference); Tserunyan's notes (the file at the cited address is dated November 26, 2025); Paran and Vo (*Israel J. Math.* 273 (2026), no. 2, 979–1000, online December 11, 2025); Enayat, Hamkins and Wcisło (*Fund. Math.* 256 (2022), 171–193); Jeřábek (*Math. Logic Quart.* 65 (2019), no. 1, 108–115); Haase (*ACM SIGLOG News* 5(3) (2018), 67–82); Alvir and Rossegger (published in *J. Symb. Log.* 85 (2020), 1079–1101, added here). The other classical references were not re-checked.

Correction. [Section 26](#) corrects the placement record's description of how 03's topology restricts to the real model.

Sources of Parts III–V. Four further manuscripts of batch 86 arrived together in commit fb8414869 and were placed in commit 0bd0e5527, with manuscript 10 of the same arrival, which went to Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization. They are numbered by archive time. Their pins are 33 to 43 commits before the placement; the repository files they inspected are unchanged between the pins and the placement. Sources 08 and 09 are AI-assisted by their own statement ("prepared with ChatGPT", "prepared with AI assistance"); 06 and 07 describe themselves as research reports prepared for Vladimir Reshetnikov.

Source	Archive	Pin	Contributes
07	polish_presburger_research	0f95145cb	base of Part III : every unmarked passage, its questions, sections G and H
06	polish_presburger_article	fa2f3e419	every passage marked [06] in Part III , among them sections 38 and 39 ; its twelve projects; sections I and J
08	glazer_proveit_local_compactness	883e0b3b2	Part IV except the notes; section K ; staged code, data, sources and proof audit (prefix 08-local-compactness-)
09	Polish_Hahn_Puisseux_Levi_Civita_Research	4af956fe0	Part V except the notes; sections L and M ; staged code, data and sources (prefix 09-hahn-polishability-)

Written for this report in Parts III–V: the introductions of the three parts, [section 42](#) with [proposition 42.1](#), the notes marked [write] or [merge], the additions to [section 1](#) (the paragraph on Parts III–V and Glazer's speculation, the second block of the notation table, the status boxes), the note after [question 14.4](#), and the additions to these appendices.

Where the merge of [Part III](#) chose.

- *Base.* Source 07, for the weaker hypotheses of its realization theorem ([Part III](#)).
- *Printed again with a flag.* Results of Part I proved again by 06 or 07 are printed with their proofs, and each title names the statement of Part I it repeats (listed in the introduction of [Part III](#)). Source 06's [theorem 35.9](#) repeats 07's [theorems 35.1](#) and [35.4](#) with a different homeomorphism argument.
- *Both printed.* The two continuum theorems ([theorems 35.5](#) and [35.8](#)), whose constructions and refinements differ.
- *Placement.* Source 06's Sections 1 to 6 and 9 are printed as subsections of source 07's Sections 1, 2, 3, 6, 6, 8 and 9 respectively, its own subsections one level lower; its Sections 7 and 8 are sections of their own. Its twelve projects follow source 07's question with the same subject, the other five in [section 43.5](#).
- *Renamed.* Source 06's M_s, G_s, H_s ($s \in 2^{\mathbb{N}}$) are M'_s, G'_s, H'_s ; macros of all four sources are mapped to this report's ([section 1.3](#)); no normalization changed.
- *Labels added* to unlabelled statements so that notes can cite them: `pma:lc:q:remove1c`, `pma:lc:q:completion`, `pma:ser:q:cones`.
- *Not printed separately.* The four title pages and abstracts (their status statements are in [section 1.5](#); the merged abstract replaces them), source 07's table-of-contents material, and the section headings of 06 that were replaced by subsection headings. Source 07's figure, source 06's map of results and source 09's reading routes are printed.

Intake checks for Parts III–V. The suites of 08 and 09 were rerun on copies (Python 3.14.4, standard library only, about one second each): 08 passed 24,470 cases in four families, 09 passed 20,116 assertions in 29 families. Their records are JSON-equal to the shipped ones and differ only in line endings. Source 08's checksum manifest verified at placement (8 of 8) and was not shipped; 06, 07 and 09 ship none, and 06 and 07 ship no code. For this report the proofs of source 06's [theorem 39.1](#) and [theorems 38.1](#) and [38.2](#) and the oscillation table after [theorem 36.1](#) were checked by hand, with no error found, and the following records were checked on October 3, 2026: Block's arXiv record (v3 of June 16, 2026, the version cited by 06); Chen's arXiv record (one version, 2021, no journal reference). The arXiv record of Ehrlich and Kaplan lists no journal reference; the journal reference that source 07 gives was not re-checked. The other references of 06–09 were not re-checked.

Reviews elsewhere in the repository. The Hilbert's-tenth programme triaged the arrival fb8414869 for its own search in Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/recent_incoming_substrate_triage_20261003.md (commit 39e03adc5), after a README-only note on manuscripts 03–05 (review_recent_incoming_glazer_scope.md in the same directory, commit 0f3b760ae). Both are routing decisions, not proof audits; the later one notes, as [section 42](#) does, that source 06's cone topology is not automatically a locally compact signed-group topology.

Stale statements and addresses. Source 07's statement that the repository's text files never mention Glazer or Tennenbaum was true at its pin and is corrected after it ([section H](#)). Source 09 cites two repository files by unversioned addresses; the note in [section M](#) gives the pinned ones. Sources 08 and 09 cite sources 05 and 03 as “user-library” PDFs; in this report they are Parts I and II.

F Limitations and non-claims of the sources

Every limitation stated by a source is kept. They are grouped by source; the items written for the merge come last.

Source 05.

- 05.1 The construction answers Glazer’s Question 2 *as written*; independent bibliographic priority has not been established. The literature search verified the question in the source PDF and checked the arXiv record; it did not establish the absence of an earlier solution.
- 05.2 Classical Presburger elimination, the $\mathbb{Z}$ -group facts and elementary topological-group facts are not claimed as new; the lexicographic algebra is classical.
- 05.3 The manuscript is AI-assisted, not independently refereed and not Lean-verified; no new Lean or Rocq formalization is included; the proposed Lean modules are a plan, not existing files.
- 05.4 The executable tests are finite instances and stream prefixes; they do not quantify over arbitrary elements of $\mathbb{Q}^{\mathbb{N}}$, and finite prefixes do not decide equality of infinite streams.
- 05.5 No answer to Glazer’s Question 1 is claimed; no uncountable Polish model of Peano arithmetic; no continuous arithmetic order on these nondiscrete spaces; no terminating decision of equality of arbitrary infinite sequences (comparison is approximated with at most one change).
- 05.6 The omnific maps are additive and order-preserving, not ring embeddings; the topologies are specified coefficient topologies, not a canonical topology on $\mathbf{Oz}$; no topology on a proper class is used; all constructions are set-sized in ZFC.
- 05.7 Neither model admits any unital semiring multiplication with the given addition and unit; nothing contradicts Glazer’s obstruction for stronger arithmetic, and the no-semiring argument does not replace it.
- 05.8 The repository’s integer Cooper lemmas are correct on $\mathbf{Int}$; the existence of the integer eliminator does not certify the theorem for a new carrier; the original integer theorem need not be weakened or retracted.
- 05.9 The questions are proposed directions, not certified as open in the literature; the user’s repository was not modified.

Source 04.

- 04.1 The construction answers Question 2 as printed; historical priority has not been established; the literature check did not establish that the construction has never appeared elsewhere.
- 04.2 The manuscript is unrefereed; no new theorem is represented as machine-checked; Lean/Rocq certification is not supplied.
- 04.3 The classical $\mathbb{Z}$ -group characterization and quantifier elimination are not claimed as new.

- 04.4 The difference between one-step periodicity and residue invariance is a porting requirement, not a reported bug in the integer proof.
- 04.5 Baire coordinates encode the arithmetic structure and are not base digits; the stream API cannot decide totality of an arbitrary oracle, and partial, negative or noninteger entries are not supported.
- 04.6 The source audit was limited to the named files at its pin; no repository files were modified and no repository-wide build was performed; the formalization stages are proposals, not assertions about existing declarations.
- 04.7 The tests are finite, exact and partial; the model-theoretic, infinite-matrix and topological statements rest on the proofs.
- 04.8 No third-party font files or paper copies are included.

Source 03.

- 03.1 A research draft, not refereed and not a Lean proof; global priority and independent correctness review are not established.
- 03.2 The repository's coanalytic support obstruction and the negative-infinitesimal floor correction are credited, not claimed as new; the integer-part model is of Shepherdson type, not new; analytic boundedness and the Hausdorff characterization are classical.
- 03.3 The full-Hahn nonpresentation results need observable coefficients and coverage of the whole field; they do not exclude unobservable Borel presentations or Borel real closed subfields.
- 03.4 Only addition is shown impossible to make continuous; Glazer's multiplication obstruction (his Theorem 2) is not transferred, since only open induction is proved for $\mathcal{A}_{\text{fin}, \geq 0}$; continuous multiplication under another compatible topology is open.
- 03.5 The zero loci are not claimed minimal in cardinality, dimension or descriptive rank; countable closed-piece continuity is not identified with the Baire-one class.
- 03.6 The dichotomy is per fixed L , not about the class of scattered orders, nor uniform in a code of L ; the Hausdorff decomposition used need not be computable; the uniform support bound need not be computable from a code.
- 03.7 No general computability over arbitrary real coefficients is claimed.
- 03.8 All results are set-sized in ZFC; no class-sized Polish topology, global choice or large cardinal is used.
- 03.9 The Python checks are finite (they do not test well-foundedness of infinite trees, completeness, or the external theorems); the formalization plan is proposed only; no repository files were changed and the whole repository was not audited.

Source 07.

- 07.1 The construction answers Question 2 as printed; the claim that it is the first published resolution is not made. The literature search was targeted, not exhaustive; the precise place in the literature of the cone theorem, the Baire family and the rank computations is left to specialist review.

- 07.2 The lexicographic construction of a $\mathbb{Z}$ -group is classical; classical model theory and standard descriptive set theory are inputs; no historical priority for the formulation of [theorem 33.1](#) is needed or claimed, measurable orders on locally compact groups having been studied by Asmar and Montgomery-Smith.
- 07.3 No new result has been checked by Lean or Rocq; no Lean or Rocq build was run; the repository was used read-only; no new formal theorem is claimed; no publication priority or external peer review is claimed.
- 07.4 Only first-order induction is claimed, not second-order induction over external subsets.
- 07.5 No computable homeomorphism is claimed for an arbitrary presentation of a countable ordinal; no classification of the pure underlying orders is claimed; the oscillation-rank convention is fixed and not identified with every convention for difference-hierarchy ranks.
- 07.6 The omnific images are not claimed initial, closed under surreal multiplication or under surreal-analytic operations; the transported topology is not the surreal order topology; every model is set-sized and the class **No** is used only as a definable ambient class.
- 07.7 Glazer's Question 1 is not answered, and the metatheoretic strength of his theorem with multiplication is not determined.
- 07.8 The LinkedIn page supplied with the brief could not be read; the repository search covered `.tex`, `.md` and `.lean` files at the pin, not every branch or binary artifact.
- 07.9 The questions are proposed directions, not asserted to be historically new or unresolved in all of the literature.
- 07.10 No finite numerical experiment is offered as evidence; compilation and visual inspection verify the document, not the mathematics; the proof checks made during preparation are not external refereeing or proof-assistant certification.

Source 06.

- 06.1 No machine-checked certification and no unrestricted publication-priority claim is made; the literature search is a bounded bibliographic finding, not a proof of universal priority.
- 06.2 The lexicographic $\mathbb{Z}$ -group construction and its model theory are classical (Jeřábek); rational polynomial Presburger examples appear in lecture notes (Cook–Pitassi); the polynomial ring has classical precedents and occurs in the repository.
- 06.3 The computability claims are Type-2 claims about streams: no computable enumeration of the uncountable model and no decidability of equality of streams are asserted, so there is no conflict with Tennenbaum's theorem; the continuum family is not claimed computable.
- 06.4 The ordinal scales are classes of magnitude, not ω_1 -sequences in one order; there is no conflict with the obstruction of Enayat, Hamkins and Wcisło.
- 06.5 The failure of open induction concerns $\mathbb{Z} + t\mathbb{R}[t]$ only and does not transfer to **Oz**; the analogous topology on the signed ring is not a ring topology, and partial subtraction on the cone is not continuous.

- 06.6 The isolated-zero theorem does not imply that every standard numeral is isolated; it uses no induction inside the model, no standard division, exponentiation, saturation or integer-part theorem.
- 06.7 The surreal embeddings are additive; no closure under surreal multiplication is asserted; the topologies are transported, not inherited from a topology on the class **No**.
- 06.8 The formalization steps are proposed, not run; the repository reports are points of contact, not premises; no full-repository audit or Lean rebuild was made.
- 06.9 The projects are not certified as unsolved in the literature.
- 06.10 The proof reviews made during preparation are mathematical checks, not formal certification or peer review.

Source 08.

- 08.1 Bibliographic priority has not been established; no independently certified breakthrough is claimed; the manuscript is AI-assisted, unrefereed and not Lean-checked; a missing search result is not evidence of a first proof.
- 08.2 The compact-group obstruction, finite-dimensional separation and the locally compact abelian structure theorem are classical; the principal structure theorem is imported, not reproved; Presburger elimination is classical; Hewitt and Koshi's paper was not read in full.
- 08.3 A Hausdorff locally compact second-countable topology on the signed additive group is assumed; the result is not a theorem about every Polish presentation of a positive arithmetic model and does not replace Glazer's general theorem.
- 08.4 The real model is credited to the companion manuscript (source 05); the classification is of pointed topological additive groups, not of all orders or of all topological monoids.
- 08.5 No separable non-locally-compact counterexample is presented; the general Polish-group case and Glazer's Question 1 are not settled.
- 08.6 The transported examples are not effective; no topological-ring claim is made for the nonseparable example.
- 08.7 The finite tests cannot establish statements about all real points, arbitrary Borel sets, local compactness, Haar measure or infinite cardinalities; no Lean file is included as if checked, and a conditional formalization with an interface must not be advertised as unconditional.
- 08.8 The companion manuscripts were not independently audited; the repository guides identify interfaces only; no repository file was modified or built.
- 08.9 The proofs are in ZFC; no claim is made about the choice or reverse-mathematical strength needed; no topology, Haar measure or power set is applied to a proper class.

Source 09.

- 09.1 The proofs are conventional and unrefereed; nothing is Lean- or Rocq-verified; historical novelty is not established; the literature review was targeted and no absence-of-result argument is used as evidence.

- 09.2 Category theorems, Hahn and Puiseux constructions, Newton–Puiseux, real-closedness inputs, the Levi-Civita construction, the coanalytic support obstruction and the omnific floor rule are credited, not claimed as new.
- 09.3 No new answer to either of Glazer’s printed questions is claimed, and Glazer is not a collaborator or endorser; the LinkedIn page was not used.
- 09.4 The conventional real-coefficient Levi-Civita field is not claimed to be Polish.
- 09.5 The 20,116 finite checks are not proofs of the category, completeness, descriptive or real-closedness statements.
- 09.6 The repository’s formalization ledger could not be fetched; no claim-by-claim audit, local checkout or build was made; the pin is a locator.
- 09.7 The proofs are in ZFC with no choice-free assertion; no computability claim follows from Borel equivalence; no Polish topology on a proper class is used.
- 09.8 The questions are proposed next steps, not previously published open problems.

The merge.

- M.1 The affirmative answer concerns Question 2 as printed. Glazer speculated the opposite; this report draws no conclusion about his Question 1 or about any variant of Question 2 with further hypotheses.
- M.2 [Proposition 26.1](#) and the observations of [section 26](#) are elementary consequences of the sources’ theorems; they are not presented as new results about Glazer’s problem.
- M.3 Citing a collection label does not certify that label: of those cited, only `odg:thm:floor` is formalized.
- M.4 Nothing in this report is formalized. The pointer from the formal project `Logic/PresburgerArithmetic` and the placement in the Surreal collection confer no formal status.
- M.5 The reading of Glazer’s speculation for Parts III–V in [section 1.1](#) is editorial; no source of Parts III–V draws it, and nothing in them answers Question 1.
- M.6 [Proposition 42.1](#) and the observations of [section 42](#) are elementary consequences of the sources’ theorems, not new results on Glazer’s problem.
- M.7 The note after [question 14.4](#) re-scopes one sentence of that question; the rest of it, and every other question of the report except where a note says otherwise, stays open.

G An explicit induction check after quantifier elimination (source 07)

For readers concerned that discrete order and division might conceal a missing induction argument, the standard quantifier-elimination input also yields a direct least-element proof.

Let G be any $\mathbb{Z}$ -group with positive cone M . After quantifier elimination, a unary definable set with parameters is a finite Boolean combination of linear inequalities and congruences. Division by fixed standard coefficients changes each inequality into an upper or lower bound

on the variable. Taking a common standard modulus and partitioning at finitely many endpoints writes the set as a finite union of cells

$$I \cap \{x \in M : x \equiv r \pmod{d}\},$$

where I is an arithmetic interval, possibly with an infinite end. Empty cells may be discarded.

Every nonempty cell has a least element. Intersect with M and replace a strict finite lower bound by its successor; if there is no effective lower bound, use zero. This gives a closed lower endpoint $a \in M$. Choose the unique standard $t \in \{0, \dots, d-1\}$ such that $a + t \equiv r \pmod{d}$. Then $a + t$ is the first element of that residue class at or above a . If the cell is nonempty, $a + t$ also satisfies its upper bound, since it precedes any other candidate in the cell. Thus it is the cell's minimum. Taking the least of finitely many minima shows that every nonempty unary definable subset of M has a minimum.

Now let a definable $D \subseteq M$ contain zero and be closed under successor. If its complement is nonempty, choose its least element b . Then $b > 0$ and its predecessor $b - 1$ lies in D . Successor closure puts b in D , a contradiction. This proves every first-order induction instance with parameters. The only imported step is classical quantifier elimination, explicitly identified in Section 31; no topological assumption enters.

[write] This is a second route to the induction clause of [corollary 4.11](#); source 06's [section I](#) and source 08's [theorem 49.3](#) prove the elimination itself again.

H Source audit, proof dependencies, and scope (source 07)

H.1 Primary source checks

The supplied LinkedIn page could not be read directly during this investigation. The selection of topic rests on Glazer's named mathematical paper, whose full text was inspected, rather than on inferred biographical details. The exact wording and number of its Presburger question were checked on its final page.

The literature check covered the current arXiv record, the earlier Enayat–Hamkins–Wcisło paper and author page, and targeted searches combining Presburger arithmetic, Polish spaces, Glazer, and continuous addition. No later resolution of the cited question was located. This is a report of a limited search, not a certification that the problem remained open until the date of this manuscript.

General Baire presentations with low-complexity definable relations are already part of the background discussed in [2]. The additional content here is continuous addition together with explicit coordinate models, their ordinal and topological invariants, and their omnific realizations.

H.2 Repository inspection and duplication boundary

At the pinned snapshot, a search of the checked-out `.tex`, `.md`, and `.lean` files found no occurrence of “Glazer” or “Tennenbaum”. The initial-subgroup report does cite the earlier topological-arithmetic paper, but the inspected material does not give the Polish Presburger constructions used here. This statement concerns the searched text formats and snapshot; it is not a search of every historical branch or every binary artifact.

[write] *Correction of a stale statement.* The search result above was true at source 07's pin 0f95145cb. It is no longer true. The word “Glazer” first occurs, in archive names, in the

Hilbert’s-tenth programme’s scope note `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_recent_incoming_glazer_scope.md` (commit `0f3b760ae`), and Glazer’s paper, whose title names Tennenbaum, is discussed in this report since its placement commit `3d2177df4`. Part I there gives the real and Baire models of sections 32 and 35 independently of source 07.

The exact formalization boundary was checked in:

- `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Syntax.lean`: the interpretation `Formula.holds` uses ordinary integer valuations.
- `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Decision.lean`: the elimination correctness theorem quantifies over `List Int`.
- `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Audit.lean`: the file requests dependency reports for the existing integer theorems.

The repository was used read-only. No claim is made that this report has been merged into it or that any new formal theorem has been accepted by its kernel.

H.3 Dependency structure of the new arguments

Conclusion	Required ingredients
Affirmative real model	Lexicographic coordinate algebra; classical $\mathbb{Z}$ -group theorem; an explicit Polish topology
General cone regularity	Baire category; regularity of Borel sets; countable bases; closure and transfinite recursion
General Polish realization	Cone regularity; group division; classical Presburger quantifier elimination
Baire-space models	Countable products; direct G_δ cone computation; noncompact cylinders; Baire-space characterization
Ordinal nonisomorphism	Leading-coordinate comparison; group completion; Archimedean classes
Oscillation calculation	Finite-coordinate neighborhoods; relative closures; countable ordinal arithmetic
No ordered-semiring expansion	Distributivity; positivity; a cofinal family of standard multiples
Omnific realization	Set-sized Conway normal forms; coefficientwise addition and leading-term order

The real example does not depend on the general cone theorem, the transfinite Baire family, or surreal normal forms. The direct proof of the Baire family’s sign-cone complexity likewise makes it independent of the general cone theorem. These independent routes reduce the chance that a single auxiliary claim could undermine the positive answer.

H.4 Review status

The coordinate model, the Borel-cone recursion, the transfinite Baire family, and the boundary oscillation calculation were subjected to separate mathematical proof checks during preparation. Those checks specifically examined induction, completeness versus the inherited metric, transfinite leading coordinates, limit stages, and the distinction between partial and

truncated subtraction. They are independent reasoning checks, not external refereeing or proof-assistant certification.

No finite numerical experiment is offered as evidence for an infinite model-theoretic theorem. The deliverable consists of the proofs, identified classical inputs, exact source references, and a reproducible LaTeX source. Compilation and visual inspection verify the document, not the mathematics.

I A self-contained Presburger elimination argument (source 06)

This appendix proves the algebraic characterization used in the main constructions. It is included to make the induction claim fully checkable without relying only on a reference to a characterization theorem.

I.1 Division in a $\mathbb{Z}$ -group

Let G be a discretely ordered abelian group with least positive element 1 and divisible quotient $G/\mathbb{Z}$. Fix a positive standard integer n . For each $g \in G$, divisibility of the quotient gives $h \in G$ with $g - nh = k \cdot 1$ for some standard integer k . Write $k = nq + r$ in ordinary $\mathbb{Z}$, where $0 \leq r < n$. Then

$$g = n(h + q \cdot 1) + r \cdot 1.$$

This representation is unique: if two remainders r, r' in the specified finite interval differ by n times a group element, discreteness forces that element to be zero. Explicitly, a positive difference would be at least n , and a negative one at most $-n$, whereas $|r - r'| < n$.

Conversely, this division property makes $G/\mathbb{Z}$ divisible. Thus we can work with the ordered-group axioms, discreteness, and this schema of standard division algorithms.

I.2 Eliminating one quantified variable

Expand the language by predicates $D_n(u)$, for standard $n \geq 1$, with the interpretation $u \in nG$. Every group term is integer linear. To eliminate an existential quantifier, first put its quantifier-free matrix in disjunctive normal form and treat each conjunction separately. Split disequalities into the two strict inequalities and use discreteness to replace weak inequalities by strict inequalities with a shift of 1. Atoms not involving the quantified variable are retained as parameter conditions.

Suppose the variable is x . Let L be the least common multiple of the nonzero absolute coefficients of x in the conjunction, or 1 if there are none. Replace Lx by y and add $D_L(y)$. For an inequality or equality involving $ax + u$, multiply by the positive integer $L/|a|$; its coefficient of y becomes 1 or -1 . For a divisibility literal, use

$$D_n(ax + u) \iff D_{nm}(\text{sign}(a)y + mu), \quad m = L/|a|. \quad (\text{I.1})$$

This is valid because, in a torsion-free group, $v \in nG$ if and only if $mv \in nmG$. The same equivalence handles negated divisibility. Thus every occurrence of y has coefficient 1 or -1 .

If an equality specifies y , substitute its value. Otherwise the conjunction consists of finitely many lower bounds $a_i < y$, upper bounds $y < b_j$, and positive or negative congruence conditions, together with parameter-only literals.

Let N be a common multiple of all congruence moduli, including L , and split into the finitely many possibilities $y \equiv r \pmod{N}$ for $0 \leq r < N$. In each branch, a condition

$D_n(\pm y + u)$ is equivalent to the parameter condition $D_n(\pm r + u)$. Only the one residue condition modulo N and the bounds now involve y .

If lower bounds are present, divide into finitely many cases choosing their greatest value a ; this is expressed by parameter comparisons. Among

$$a + 1, a + 2, \dots, a + N$$

there is exactly one member of the chosen residue class. It is the least member of that class above a . To justify minimality, if $y > a$ has the same residue and lies below that candidate, their positive difference is a multiple of N smaller than N , contradicting discreteness. Thus, when upper bounds are also present, existence is equivalent to the finite disjunction

$$\bigvee_{k=1}^N \left(D_N(a + k - r) \wedge \bigwedge_j (a + k < b_j) \right).$$

If there are no upper bounds, the candidate always works. If there are upper bounds but no lower bounds, a residue-class element exists among the N positions just below the least upper bound. If neither type of bound is present, the standard representative r itself works.

This eliminates one existential quantifier. Universal quantifiers are handled by negation, and induction on formulas gives quantifier elimination in the expanded language. The output consists of Boolean combinations of integer-linear comparisons and standard congruences.

I.3 Completeness and induction in the nonnegative cone

A closed quantifier-free formula in this language concerns only standard numerals and their congruences and comparisons. Its truth is the same in every $\mathbb{Z}$ -group and in ordinary $\mathbb{Z}$. Consequently every $\mathbb{Z}$ -group is elementarily equivalent to $\mathbb{Z}$.

The structure $G_{\geq 0}$ is interpreted in G by restricting quantifiers to $x \geq 0$. Its addition, constants, and order are the restrictions of those in G . Every sentence true in $(\mathbb{N}, 0, 1, +, <)$ therefore holds in the cone.

In particular, for every additive formula $\varphi(x, \bar{a})$ the sentence

$$\forall \bar{a} \left[(\varphi(0, \bar{a}) \wedge \forall x (\varphi(x, \bar{a}) \rightarrow \varphi(x + 1, \bar{a}))) \rightarrow \forall x \varphi(x, \bar{a}) \right]$$

holds. Parameters are covered because their universal closure is part of the sentence. Multiplication is absent from this language; the argument therefore makes no assertion about the polynomial model's formula $x^2 < t$.

[write] This is a second route to [theorem 4.9](#) and the induction clause of [corollary 4.11](#), by the finite search of [lemma 4.7](#).

J Provenance and verification record (source 06)

Mathematical dependencies. The main constructions use ordered-group algebra, elementary Presburger elimination, standard facts about Polish spaces, and explicit coordinate arguments. The isolated-zero theorem additionally uses analytic/coanalytic separation and fair product measure. Its tail zero-one step is proved in the text. The surreal realizations are explanatory embeddings and are not premises of any of the main topological theorems.

Independent scrutiny. The real and Baire models, the polynomial topology, the distinction between additive and open induction, and the isolated-zero argument were subjected to separate proof reviews during preparation. Those reviews specifically checked complete metrizability rather than mere separability, negative integer slices, the parameter version of induction, joint continuity, cancellation, and the tail-event invariance. The final proof omits an unnecessary Cantor-injectivity claim and never assumes continuous subtraction. These reviews are mathematical checks, not formal proof-assistant certification or external peer review.

Repository audit. The repository comparison used its path inventory and selected relevant files at the pinned commit. No full-repository theorem audit or Lean rebuild was performed. Existing exploratory reports are credited for their stated algebraic content, with the main results here proved independently.

Reproducibility. This source is self-contained and uses standard $\text{\LaTeX}$ packages. It can be compiled with `latexmk -pdf polish_presburger.tex`. No external image, private data, bibliography processor, or downloaded repository is needed to reproduce the PDF. The accompanying ZIP contains this source, its compiled PDF, and a short build-and-status readme.

[write] Source 06 shipped only its $\text{\LaTeX}$ source, its PDF and a short plain-text readme. None of them is shipped here (the source is merged into this report; the archive survives in commit fb8414869); there is no program to rerun.

K Contribution ledger, source audit, and reproducibility (source 08)

K.1 Contribution ledger

Statement or ingredient	Status in this manuscript
Compact-group order obstruction; finite-dimensional separation; LCA principal structure theorem	Classical material, credited in section 47 . The first two are proved in the needed form. The LCA principal structure theorem is an explicitly imported theorem.
Presburger quantifier elimination	Classical. A complete finite-witness argument is supplied to control the nonstandard-group transfer.
The real positive-cone example $M_{\mathbb{R}}$	Previously constructed in the companion manuscript [40] ; not claimed as new.
Integer-disjoint order-unit amplification and ordered direct sums	Derived and proved here, theorem 46.2 ; global priority unverified.
Arithmetic rigidity without multiplication regularity	Derived and proved here, theorem 44.1 ; global priority unverified. No induction assumption.

Statement or ingredient	Status in this manuscript
Pointed group classification; cone dimension threshold	Derived here from the classical structure inputs. Not a classification of all full orders or all topological monoids.
Uniform Δ_2^0 definability and continuous standard division	Derived for the classified presentations. Standard elimination and topological facts are not claimed as new.
Countable Borel-scalar subring; exact answer for $E = \mathbb{Z}$	Derived and proved here, theorem 53.4 ; does not assume order-compatible multiplication.
General Polish-group version; full order classification; Lean proof	Not established here. Listed as further research.

K.2 What was actually inspected

The repository snapshot used for the interfaces is

883e0b3b22263bd9d92fdbbf10376ca4b652b862.

The inspected guides were the root `README.md`, `Logic/PresburgerArithmetic/README.md`, and `Algebra/SurrealNumbers/README.md`. The first arithmetic guide identifies the syntax, normalization, Cooper, elimination, and decision layers and their stated audit scope. The surreal guide distinguishes constructed objects, mapped formal results, and unrefereed research reports. Those guides were used to identify interfaces, not as evidence that the new theorems in this manuscript have already been formalized. No repository files were modified, and no full repository build was run.

The Glazer source was the publicly available 2023 arXiv version of [1]; the supplied LinkedIn profile was not needed as mathematical evidence. Classical cross-checks used the primary papers or author notes listed below, including the measurable-order discussion in Section 3 of [22] and the LCA structural statement on page 4 of [27]. The bibliographic entry for [26] is recorded through its explicit discussion and citation in [22]; no claim of an independent full reading of that 1983 paper is made. The prior companion manuscripts were retrieved from the user's library.

K.3 Finite verification

The included `verify.py` uses Python's standard-library `fractions.Fraction`. Its output on the supplied version is:

Test family	Cases passed
Standard division in rational samples of $\mathbb{R}^d \times \mathbb{Z}$	7,595
Coset-invariant finite-witness tests	9,000
Polynomial amplification tuples	1,875
Lexicographic order triples	6,000
Total	24,470

The finite-witness tests compare the candidate construction with an independent oracle on rational lexicographic vector coordinates and integer intervals. The polynomial tests compute in $\mathbb{Q}[X]$, with $V = \mathbb{Q}X$ and u a positive rational multiple of X ; they check finite injectivity, the highest-coefficient sign rule, and lower-term domination. All comparisons are exact, with fixed random seeds for the sampled tests.

These checks can catch sign, indexing, and residue mistakes. They cannot establish any statement involving all real points, arbitrary Borel sets, local compactness, Haar measure, or infinite cardinalities. The mathematical proofs, not the tests, carry those conclusions.

K.4 Build instructions

The bundle contains the source and compiled PDF, this verification script, its JSON output, a source-provenance file, and a short build script. With a standard TeX Live installation including the New PX packages:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python3 verify.py --output verification_results.json
```

The PDF should then be rendered and visually checked. No external font files, third-party source PDFs, or unexecuted Lean proof claims are included.

[write] In this report the program is `code/08-local-compactness-verify.py`, its recorded output `data/08-local-compactness-verification_results.json`, the build script `code/08-local-compactness-build.sh`, and the source-provenance file and proof audit `08-local-compactness-sources.md` and `08-local-compactness-proof_audit.md`. They are byte-identical to the delivery and still use the delivery names `verify.py`, `verification_results.json` and `article.tex`; the build script builds a file `article.tex` in its own directory and must not be run in the report directory. The intake reran the program on a copy with Python 3.14.4: the four families and the total of 24,470 cases agree with the shipped record, which differs from the new output only in line endings.

L Contribution ledger and proof dependencies (source 09)

Statement or ingredient	Status and dependency boundary
Pettis, automatic continuity, open mapping	Classical tools; short proofs are included in Section 61. Their use is not claimed as new.
Local order bounds and forced order topology	Proved here from the category tools, with Baire order and scalar hypotheses explicit. Priority of this formulation has not been certified.
Full Hahn classification	Central presentation-invariant deduction, Theorem 63.3; depends only on the order bridge and the explicit separated binary family, plus classical existence of Hahn fields.
Proper-subfield cover and Puiseux obstruction	Proved here, Theorems 64.1 and 64.3; the measurability of covering subfields is essential and is verified in the application.

Statement or ingredient	Status and dependency boundary
Levi-Civita field and completeness	Classical construction; explicit proofs are supplied. The uniqueness and comparison with non-Polishable Puiseux arithmetic use the category bridge.
Exact code domains	Explicit FIN and row-finiteness reductions are given. The full Hahn coanalytic obstruction is credited to classical descriptive methods and the preceding project report, not reclaimed as new.
Real closedness	Classical Newton–Puisseux input plus a supplied proof that completion preserves algebraic closedness. Not a new algebraic theorem.
Polish hulls and exact cover number	Proved here using countable dense field cores, prefix stabilization, and the fixed-scale binary obstruction. The hull construction uses a standard completion mechanism; the local/global classification is explicitly developed in Section 69.
Countable integer parts	A consequence of the forced order topology, with no measurability assumption on the integer part.
Floor formula	Existing omnific normal-form principle, proved here in the restricted field for clarity.
Floor regularity	Explicit continuous approximants and exact discontinuity sets for Levi-Civita and strengthened Polish hulls, Theorems 70.2 and 70.4; not a kernel check.
Open induction	Classical integer-part implication, proved directly for the constructed ring; no uncountable arithmetic model is asserted.
Executable verification	Finite rational arithmetic regression tests only. They do not verify the set-theoretic or topological quantifiers.

The logical core can be read as

$$\begin{aligned}
 &\text{Pettis} \implies \text{local order bounds} \\
 &\implies \text{order-topology refinement} \\
 &\implies \left\{ \begin{array}{l} \text{full Hahn classification,} \\ \text{countability of integer parts,} \\ \text{uniqueness of complete order presentations.} \end{array} \right.
 \end{aligned}$$

The proper-subfield-cover theorem is a separate branch from Pettis and automatic continuity. Descriptive completeness and real closedness are independent supplementary branches; neither is needed for the main Hahn classification.

M Source audit and reproducibility (source 09)

Research scope. The research-interest overlap was based on Glazer’s mathematical paper [1], not on a speculative interpretation of a social-media profile. The article does not imply

that he reviewed, collaborated on, or endorsed this manuscript. The public LinkedIn page supplied in the request was not needed to substantiate the mathematical connection.

Repository scope. The root and surreal READMEs were inspected, together with repository metadata and the directory structure of the surreal implementation. The repository head recorded during the audit was

4af956fe015e043c5e163ee5b4d2cea23e6157ea

This pin is a reproducible repository locator, not a claim to have rebuilt the project at that commit. The main proofs here are self-contained and do not depend on assuming a repository theorem beyond the explicitly credited normal-form background. The large formalization ledger could not be fetched through the available connector; no claim-by-claim audit of that ledger is represented as completed.

Earlier reports. The Library copies of the previous Polish Presburger and Borel-support manuscripts were checked at the abstract/contribution level to avoid presenting the same results as new. The present article does not use their proposed theorems as unproved premises. References [43, 44] are unpublished project manuscripts, not independently validated literature.

Literature scope. The primary mathematical sources below were accessed during preparation. The review was targeted, not exhaustive. The sources establish the classical inputs and the research connection; the priority of the combined classification and its refinements remains open to specialist review. In particular, no absence-of-search-result argument is used as evidence of novelty.

Build and checks. The archive contains this self-contained `article.tex`, the compiled PDF, `verify.py`, its machine-readable output `verification.json`, a source record, and a build README. Compile with a standard TeX Live installation using

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
python3 verify.py --output verification.json
```

The Python program uses only the standard library. It records deterministic finite test counts and fails on a violated identity. No numerical tolerance is used: all tested coefficients and exponents are exact rational numbers. The PDF was also rendered for layout inspection. Font packages are referenced by the TeX source; no font files are distributed in the archive.

[write] In this report the program is `code/09-hahn-polishability-verify.py`, its recorded output `data/09-hahn-polishability-verification.json`, the build script `code/09-hahn-polishability-build.sh`, and the source record `09-hahn-polishability-SOURCES.md`; they are byte-identical to the delivery and use the delivery names. The intake reran the program on a copy with Python 3.14.4: 20,116 assertions in 29 families with seed 20261003, equal to the shipped record up to line endings. Source 09's bibliography and source record cite the surreal README and root README by unversioned blob/main addresses, beside the pin above; the pinned versions are <https://github.com/VladimirReshetnikov/ProveIt/blob/4af956fe015e043c5e163ee5b4d2cea23e6157ea/Algebra/SurrealNumbers/README.md>

and the root `README.md` at the same commit. Both files are unchanged between the pin and the placement commit `0bd0e5527`.

References

- [1] Elliot Glazer, *A Topological Tennenbaum Theorem*, arXiv:2311.13699v1, submitted November 22, 2023 (dated November 20, 2023 on its first page); one version, no journal reference as of October 3, 2026. Theorem 1 is on page 1, Theorem 2 and Corollary 1 on page 3; Questions 1 and 2 and the speculation after Question 2 are on page 8. The remark that Theorem 1 and Corollary 1 can be proven in ATR_0 is on page 7. <https://arxiv.org/abs/2311.13699>.
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